CITY OF CADILLAC

ADDITIONAL LOCAL LIMITS REPORT

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PREPARED FOR:

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1.0 BACKGROUND & PURPOSE

The City of Cadillac completed a Maximum Allowable Headworks Loading (MAHL) and Local Limits study from 2023-2025 for several pollutants of concern: arsenic, cadmium, total chromium, copper, available cyanide, lead, mercury, nickel, selenium, silver, zinc, BOD, TSS, ammonia, and phosphorus. The final MAHL report was submitted to EGLE in May 2025. EGLE approved the proposed local limits for those parameters in June 2025.

Numerical benzidine and beryllium effluent limits were added to the City's most recent NPDES permit; however, these parameters were not included in the May 2025 MAHL submittals, as part of the MAHL study samples were collected prior to issuance of the NPDES permit.

This Local Limits study was completed for the newly added pollutants: benzidine and beryllium. In addition, EGLE requested that the local limit for phenol also be updated as part of the Additional Local Limits study.

This report discusses the approaches used, data, and calculations used for this Additional Local Limits Study and proposed limits for the City of Cadillac.

The study was conducted in accordance with EGLE and US EPA guidelines. Detailed descriptions and rationale are provided where procedures or methods varied from those standard procedures outlined in the guidance.

2.0 DATA COLLECTION

Historic WWTP and industrial user (IU) data was used in conjunction with data collected specifically for this MAHL evaluation in accordance with the EGLE-approved sampling plan. WWTP influent, primary effluent, final effluent and representative domestic/background wastewater samples were collected in November and December 2025 to provide a basis for the MAHL evaluation. Samples were also collected at each of the permitted IUs to determine their concentrations of the four additional pollutants.

The flow weighted average for the background concentrations use respective average concentrations and domestic flowrates from two data sets. The average domestic sample concentration from the first data set is multiplied by its corresponding domestic flow rate, and the same calculation is performed for the second data set. The resulting products are then added together. This sum is divided by the combined domestic flow rates from both data sets. See the following equation:

X1 = Average domestic sample concentration #1 (refer to Appendix A)

X2 = Average domestic sample concentration #2 (refer to Appendix A)

Q1 = Domestic sample flowrate #1, 49.5 gpm

Q2 = Domestic sample flowrate #2, 500 gpm

$$Flow\ Weighted\ Average\ = \frac{(X1\ *\ Q1) + (X2\ *\ Q2)}{(\mathrm{Q1}\ + \mathrm{Q2})}$$

*Table 1 – Non-Compatibles - Data Collection Average Results**

PARAMETER	FLOW WEIGHTED BACKGROUND AVERAGE (mg/L)	WWTP INFLUENT (mg/L)	PRIMARY EFFLUENT (mg/L)	FINAL EFFLUENT (mg/L)
Phenol	0.0084	0.0051	0.0042	0.00233
Beryllium	0.0020	0.002	0.0020	0.002
Benzidine	0.00113	0.00113	0.0014	0.00027

*Note that analytical results below the method detection limit were taken as half of the detection limit when calculating the average concentration for these parameters.

The full analytical results are presented in Appendix A for reference.

3.0 MAHL METHODS AND CALCULATIONS

The proposed local limits for the four additional parameters were determined using the EGLE-developed spreadsheet model, customized by Fleis & VandenBrink Engineering, Inc. (F&V) for use at the City of Cadillac.

The following parameters are key user inputs in the non-compatibles MAHL spreadsheet:

- Influent flows, including: domestic/background, Significant Industrial Users and other non-domestic flows
- Domestic/background, industrial, and WWTP influent wastewater characteristics
- Biosolids flows and percent solids
- Treatment plant % removals

Table 2 summarizes the key flow and miscellaneous variables used in the Cadillac MAHL. In order to be consistent with May 2025 MAHL and Local Limits Study, the average total influent flow, total industrial flow, sludge flow, and sludge percent solids from the May 2025 MAHL study were also utilized for this study.

Table 2 – Variables and Values Used for Non-Compatible Pollutant MAHL

VARIABLE	VALUE	UNITS
Q_{dom}, Domestic/Background Flow (est.)	0.863	MGD
Q_{ind}, Total Industrial (SIU/CIU) Flow	0.857	MGD
Q_{POTW}, Total Avg WWTP Influent Flow	1.72	MGD
Q_{Max}, Maximum Design WWTP Influent Flow	4.5	MGD
Q_{dig}, Sludge Flow to Digester	0.07	MGD
Q_{Sludge}, Sludge Flow to Disposal	0.0025	MGD
Sludge (to disposal) % Solids	3.0	%

Table 3 summarizes the percent removals used and the source reference for the value selected to be used in the calculations.

Table 3 – Treatment Plant % Removals

PARAMETER	PRIMARY % REMOVAL	OVERALL % REMOVAL	SOURCE (Primary, Secondary)
Phenol	8%	90%	EPA LLDG App. R EPA LLDG App. R;
Beryllium	0%	61%	Site Specific, EPA Dev. For Iron & Steel (Table 11-3)
Benzidine	35%	87.5%	Site Specific, Site Specific

*Note that Site Specific % removals were calculated using a surrogate value for influent and effluent for non-detectable values (1/2 the detection limit), if necessary, as described in Table 1. If the majority of the site-specific influent and effluent values were non-detectable and a literature % removal was available, the literature value was used in accordance with EPA LLDG Section 5.1.3.

The allowable headworks loading for each parameter was calculated for the following criteria:

- NPDES Effluent Limits (beryllium, benzidine)
- Secondary treatment inhibition
- Nitrification treatment inhibition
- Anaerobic Digester Inhibition
- Water quality based on chronic toxicity pass-through
- Water quality based on acute toxicity pass-through

Note that there are no criteria for these parameters for land application of biosolids.

3.1 Additional Pollutants – Loading Calculations

The formula and a table summarizing the Allowable Headworks Loading (AHL) results for each criterion is described below.

3.2 Allowable Loading Based on NPDES Permit Limits

$$L_{NPDES} = \frac{Q_{POTW} * 8.34 * C_{NPDES}}{1 - R_{Avg}}$$

Where:

L_{NPDES} = Loading based on NPDES permit limits (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

C_{NPDES} = NPDES permit limit (mg/L)

R_{Avg} = Removal % for the WWTP (expressed as a decimal)

Table 4 – AHLs based on NPDES Permit Limits

PARAMETER	C_{NPDES} (mg/L)	AHL (lb/day)
Beryllium	0.0041	0.1508
Benzidine	0.00012	0.0138

3.3 ALLOWABLE LOADING BASED ON SECONDARY BIOLOGICAL TREATMENT INHIBITION

$$L_{INHIB,Sec} = \frac{Q_{POTW} * 8.34 * C_{INHIB,Sec}}{1 - R_{PRIM}}$$

Where:

$L_{INHIB,\ Sec}$ = Loading based on biological secondary inhibition (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

$C_{INHIB,\ Sec}$ = Concentration that inhibits biological treatment of the secondary process (mg/L), literature values from *EPA Local Limits Development Guidance, Appendix G* and *EPA Guidance Manual for Preventing Interference at POTW (Table 2-1)*

R_{PRIM} = Removal % across the primary process (expressed as a decimal)

Table 5 – AHLs based on Secondary Treatment Inhibition

PARAMETER	C_{INHIB} (mg/L)	AHL (lb/day)
Phenol	50.0	779.6

3.4 ALLOWABLE LOADING BASED ON NITRIFICATION INHIBITION

$$L_{INHIB,Nit} = \frac{Q_{POTW} * 8.34 * C_{INHIB,Nit}}{1 - R_{Pr\,im}}$$

Where:

$L_{INHIB,\ Nit}$ = Loading based on nitrification inhibition (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

$C_{INHIB,\ Nit}$ = Concentration that inhibits nitrification (mg/L), using the lowest literature value from *EPA Local Limits Development Guidance, Appendix G* and *EPA Guidance Manual for Preventing Interference at POTW (Table 2-1)*

R_{PRIM} = Removal % across the primary process (expressed as a decimal)

Table 6 – AHLs based on Nitrification Treatment Inhibition

PARAMETER	C_{INHIB} (mg/L)	AHL (lb/day)
Phenol	1.0	15.592

3.5 ALLOWABLE LOADING BASED ON ANAEROBIC DIGESTER INHIBITION

$$L_{INHIB,Dig} = \frac{Q_{Dig} * 8.34 * C_{INHIB,Dig}}{R_{Avg}}$$

Where:

$L_{INHIB,\ Dig}$ = Loading based on digester inhibition (lbs/day)

Q_{Dig} = Total sludge flow to the digester (MGD)

$C_{INHIB,\ Dig}$ = Concentration that inhibits anaerobic digestion (mg/L), literature values from *EPA Local Limits Development Guidance, Appendix G* and *EPA Guidance Manual for Preventing Interference at POTW (Table 2-1)*

R_{Avg} = Removal % across the WWTP (expressed as a decimal)

Table 7 – AHLs based on Anaerobic Digester Inhibition

PARAMETER	C_{INHIB} (mg/L)	AHL (lb/day)
Phenol	100	64.9

3.6 Allowable Loading Based on Water Quality (Chronic toxicity pass-through)

Chronic Loading Formula:

$$L_{Chronic} = Q_{POTW} * 8.34 * \frac{WQBEL_C/1000}{(1-R_{Avg})}$$

Where:

$L_{Chronic}$ = Loading based on chronic toxicity pass through (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

R_{avg} = average overall removal fraction for the treatment plant (fraction)

$WQBEL_C$ = Water quality based effluent limit for chronic toxicity protection (μg/L), provided by EGLE for Cadillac, refer to Interoffice Memo dated 9/30/2025, included in Appendix B for reference.

Table 8 – AHLs based on Chronic Toxicity Pass-through

PARAMETER	$WQBEL_C$ (mg/L)	AHL (lb/day)
Phenol	0.51	73.16
Beryllium	0.0041	0.151
Benzidine	0.00012	0.0137

3.7 Allowable Loading Based on Water Quality (Acute toxicity pass-through)

Acute Toxicity Loading Formula:

$$L_{Acute} = Q_{POTW} * 8.34 * \frac{WQBEL_A/1000}{(1-R_{Avg})}$$

Where:

L_{Acute} = Loading based on acute toxicity pass through (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

R_{avg} = average overall removal fraction for the treatment plant (expressed as a fraction)

$WQBEL_A$ = Water quality-based effluent limit for acute toxicity protection (μg/L), provided by EGLE for Cadillac, refer to Interoffice Memo dated 9/30/2025, included in Appendix B for reference.

Table 9 – AHLs based on Acute Toxicity Pass-through

PARAMETER	$WQBEL_A$ (mg/L)	AHL (lb/day)
Phenol	6.8	975
Beryllium	0.066	2.43
Benzidine	0.049	5.62

The most restrictive loading was chosen as the controlling MAHL and safety factors were then applied to determine the MAIL as shown in the following table.

Table 10 – Controlling MAHL, Criteria, MAIL, and Safety Factor

PARAMETER	Controlling Criteria	MAHL (lb/day)	Domestic Loading (lb/day)	Safety Factor (S.F.)	MAIL after SF (lb/day)
Phenol	Nitrification Inhibition	15.59	0.060	20%	**12.41**
Beryllium	Chronic Toxicity / NPDES Monthly Limit	0.1508	0.014	10%	**0.121**
Benzidine	Chronic Toxicity / NPDES Monthly Limit	0.0138	0.008	10%	**0.004**

*Safety Factor as defined in Section 6.2.3 of US EPA's *Local Limits Guidance*

A safety factor of 20% was used for phenol because all influent and final effluent values were non-detectable, making it difficult to ascertain the exact loading to the plant and its treatment performance.

4.0 ALLOCATION & LOCAL LIMIT DEVELOPMENT

After the MAHL was determined for each of the non-compatible parameters, the Maximum Allowable Industrial Loading (MAIL) was determined by subtracting out the domestic background portion of the total loading.

The City has chosen to utilize a uniform allocation of the available MAIL. The available MAIL was converted into a calculated local limit by dividing by the current total industrial flow of 0.857 MGD and the conversion factor 8.34. The calculated Local Limits for the non-compatible pollutants are summarized and compared with the City's current local limit in Table 11 below.

Table 11 – Additional Local Limits

POLLUTANT	Current Local Limit (Daily Max) (µg/L)	Calculated Local Limit (Daily Max) (µg/L)	Proposed Local Limit (Daily Max) (µg/L)
Phenol	1,236	1,737	1,737
Beryllium	NA	17	17
Benzidine	NA	0.6	0.6

APPENDIX A

City of Cadillac
Maximum Allowable Headworks Loading Evaluation
Table A1 - Flows and Data Summary

Q_{POTW}	1.72	MGD
Q_{Ind}	0.857	MGD
$Q_{AAR\ Facility\ +\ Rinse\ W}$	0.298	MGD
$Q_{AAR\ 6th\ St}$	0.006	MGD
Q_{ARVCO}	0.005	MGD
Q_{AVON}	0.004	MGD
$Q_{Akwell\ 5th\ St}$	0.066	MGD
$Q_{Akwell\ 7th\ St}$	0.178	MGD
$Q_{Borg\ Warner}$	0.017	MGD
$Q_{Cadillac\ Castings}$	0.097	MGD, est.
$Q_{Cadillac\ Renewable\ En}$	0.065	MGD
Q_{Fiamm}	0.002	MGD
$Q_{Hutchinson}$	0.008	MGD
$Q_{Michigan\ Rubber}$	0.005	MGD
Q_{Munson}	0.027	MGD
$Q_{Piranha}$	0.077	MGD
$Q_{Rec\ Boat}$	0.0003	MGD
$Q_{Spencer\ Plastics}$	0.0006	MGD
$Q_{domestic/commercial}$	0.863	MGD
$Q_{max,\ design}$	4.50	MGD
$Q_{max,\ observed\ 2021-20}$	2.85	MGD
Q_{dig}	0.070	MGD
Q_{sludge}	0.0025	MGD
Sludge % TS	3%	
95% exceedence	2.6	cfs
Clam River	2	MGD

Sampling Results (non-detect taken as 1/2 the detection limit)

Parameter	Background #1 (mg/L)	Background #2 (mg/L)	Flow-Weighted Avg Bkg (mg/L)	Influent (mg/L)	Primary Effluent (mg/L)	Final Effluent (mg/L)
Phenol	0.0084	0.0084	0.0084	0.0051	0.0042	0.00233
Beryllium	0.0020	0.0020	0.0020	0.002	0.0020	0.002
Benzidine	0.0015	0.0011	0.00113	0.0022	0.0014	0.00027

Domestic #1 flow	49.5	gpm
Domestic #2 flow	500	gpm

TABLE A2

Local Limits Determination Based on NPDES Monthly Effluent Limits

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE								MAXIMUM LOADING					INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	POTW Flow (MGD) (Qpotw)	Average Removal Efficiency (%) (Ravg)	Nonvolatile Removal Fraction (%)	NPDES Monthly Limit (mg/l) (Ccrit)		Domestic and Conc. (mg/l) (Cdom)	Commercial Flow (MGD) (Qdom)		Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Phenol	0.8568	1.720		100	-		0.0084	0.8632		-	0.0603			-	-	20
Beryllium	0.8568	1.720	61.0	100	0.0041		0.0020	0.8632		0.1508	0.0144			0.1213	0.0170	10
Benzidine	0.8568	1.720	87.5	100	0.00012		0.00113	0.8632		0.0138	0.00815			0.0042	0.0006	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.
(Q_{potw}) POTW's average influent flow in MGD.
(R_{avg}) Average removal efficiency across POTW as percent.
(C_{crit}) NPDES monthly maximum permit limit for a particular pollutant in mg/l.
(Q_{dom}) Domestic/commercial background flow in MGD.
(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.
(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).
(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).
(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.
(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.
$F_{(nonvol)}$ Fraction of overall POTW removal not due to volatilization as a percent
(SF) Safety factor as a percent.
8.34 Unit conversion factor

$$L_{hw} = \frac{8.34 * C_{crit} * Q_{potw}}{(1 - Ravg) * F_{(nonvol)}}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.34}$$

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TABLE A3

Local Limits Determination Based on Secondary Treatment Inhibition Level

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE						MAXIMUM LOADING				INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	POTW Flow (MGD) (Qpotw)	Removal Efficiency (%) (Rprim)	Sec. Treatment Inhibition Level (mg/l) (Ccrit)	Domestic and Commercial Conc. (mg/l) (Cdom)	Domestic and Commercial Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Phenol	0.8568	1.720	8	50	0.0084	0.8632	779.609	0.060			584.6	81.8	25
Beryllium	0.8568	1.720	0		0.0020	0.8632	-	0.014			-	-	10
Benzidine	0.8568	1.720			0.0011	0.8632	-	0.008			-	-	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.

(Q_{potw}) POTW's average influent flow in MGD.

(R_{prim}) Removal efficiency across across primary treatment as percent.

(C_{crit}) Activated sludge threshold inhibition level, mg/l. Source: US EPA Local Limits Development Guidance, Appendix G.

(Q_{dom}) Domestic/commercial background flow in MGD.

(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.

(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).

(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).

(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.

(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.

(SF) Safety factor as a percent.

8.34 Unit conversion factor

$$L_{hw} = \frac{8.34 * C_{crit} * Q_{potw}}{1 - R_{prim}}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.34}$$

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TABLE A4

Local Limits Determination Based on Nitrification Inhibition Level

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE						MAXIMUM LOADING				INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	POTW Flow (MGD) (Qpotw)	Removal Efficiency (%) (Rprim)	Nitrification Inhibition Level (mg/l) (Ccrit)	Domestic and Commercial Conc. (mg/l) (Cdom)	Domestic and Commercial Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Phenol	0.8568	1.720	8.0	1.0	0.0084	0.8632	15.592	0.060			12.41	1.74	20
Beryllium	0.8568	1.720	0		0.002	0.8632	-	0.014			-	-	10
Benzidine	0.8568	1.720	35		0.00113	0.8632	-	0.008			-	-	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.
(Q_{potw}) POTW's average influent flow in MGD.
(R_{prim}) Removal efficiency across primary treatment as percent.
(C_{crit}) Nitrification threshold inhibition level, mg/l.
(Q_{dom}) Domestic/commercial background flow in MGD.
(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.
(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).
(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).
(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.
(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.
(SF) Safety factor as a percent.
8.34 Unit conversion factor

$$L_{hw} = \frac{8.34 * C_{crit} * Q_{potw}}{1 - R_{sec}}$$

$$L_{ind} = L_{hw} * (1 - SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.34}$$

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TABLE A5

Local Limits Determination Based on Anaerobic Digester Inhibition Level

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE								MAXIMUM LOADING						
	IU Pollut. Flow (MGD) (Qind)	POTW Flow (MGD) (Qpotw)	Sludge Flow to Digester (MGD) (Qdig)	Sorption Removal Fraction (%)	Removal Efficiency (%) (Rpotw)	Anaerobic Digester Inhibition Level (mg/l) (Ccrit)	Domestic and Commercial Conc. (mg/l) (Cdom)	Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Phenol	0.8568	1.720	0.07	100	90.0	100	0.0084	0.8632	64.9	0.0603			51.83305	7.25	20
Beryllium	0.8568	1.720	0.07	100	61.0		0.002	0.8632	-	0.0144			-	-	10
Benzidine	0.8568	1.720	0.07	100	87.5		0.00113	0.8632	-	0.0081			-	-	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.
(Q_{potw}) POTW's average influent flow in MGD.
(Q_{dig}) Sludge flow to digester in MGD.
(R_{potw}) Removal efficiency across POTW as percent.
(C_{crit}) Anaerobic digester threshold inhibition level in mg/l.
(Q_{dom}) Domestic/commercial background flow in MGD.
(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.
(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).
(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).
(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.
(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.
$F_{(sorp)}$ Fraction of overall removal due to sorption as a percent
(SF) Safety factor as a percent.
8.34 Unit conversion factor

$$L_{hw} = \frac{8.34 * C_{crit} * Q_{dig}}{R_{potw} * F_{sorp}}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.34}$$

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TABLE A6

Local Limits Determination Based on Chronic Water Quality Standards

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE										MAXIMUM LOADING				INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	Avg POTW Flow (MGD) (Qpotw)		River 95% Exceedence Flow (MGD)	Nonvolatile Removal Fraction (%)	Average Removal Efficiency (%) (Ravg)	Chronic WQBEL (mg/l) (Ccrit)	Domestic and Commercial Conc. (mg/l) (Cdom)	Domestic and Commercial Flow (MGD) (Qdom)		Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Available Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Phenol	0.8568	1.720		1.7	100	90.0	0.51	0.00837	0.8632		73.16	0.060			58.467	8.2	20
Beryllium	0.8568	1.720		1.7	100	61.0	0.0041	0.002	0.8632		0.151	0.014			0.121	0.017	10
Benzidine	0.8568	1.720		1.7	100	87.5	0.00012	0.00113	0.8632		0.01377	0.008			0.004	0.0006	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.
(Q_{potw}) POTW's average influent flow in MGD.
(Q_{95ex}) EGLE Designated 95% exceedence flow for receiving stream (MGD)
(Ravg) Average removal efficiency across POTW as percent.
(C_{crit}) State chronic water quality standard for a particular pollutant in mg/l. Reference: EGLE Interoffice Memo 9/30/2025
(Q_{dom}) Domestic/commercial background flow in MGD.
(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.
(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).
(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).
(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.
(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.
$F_{(nonvol)}$ Fraction of overall POTW removal not due to volatilization as a percent
(SF) Safety factor as a percent.
(Q_{max}) Maximim permitted POTW flow
8.337 Unit conversion factor

$$C_{crit} = \frac{C_{std} * (Q_{max} + 25\% * Q_{95ex})}{Q_{max}}$$

$$L_{hw} = \frac{8.34 * Q_{potw} * C_{crit}}{(1 - R_{avg} * F_{nonvol})}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.34}$$

TABLE A7

Local Limits Determination Based on Acute Water Quality Standards

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE									MAXIMUM LOADING				INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	Avg POTW Flow (MGD) (Qpotw)		River 95% Exceedence Flow (MGD)	Nonvolatile Removal Fraction (%)	Average Removal Efficiency (%) (Ravg)	Acute WQBEL (mg/l) (Ccrit)	Domestic & Commercial Conc. (mg/l) (Cdom)	Domestic & Commercial Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Phenol	0.8568	1.720		1.7	100	90.0	6.800	0.0084	0.8632	975	0.0603			780.30	109.2	20
Beryllium	0.8568	1.720		1.7	100	61.0	0.066	0.002	0.8632	2.43	0.0144			2.17	0.304	10
Benzidine	0.8568	1.720		1.7	100	87.5	0.049	0.00113	0.8632	5.62	0.0081			5.05	0.707	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.
(Q_{potw}) POTW's average influent flow in MGD.
(Q_{95ex}) EGLE Designated 95% exceedence flow for receiving stream (MGD)
(Ravg) Average removal efficiency across POTW as percent.
(C_{crit}) State acute water quality standard for a particular pollutant in mg/l. Reference: EGLE Interoffice Memo 9/30/2025
(Q_{dom}) Domestic/commercial background flow in MGD.
(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.
(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).
(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).
(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.
(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.
$F_{(nonvol)}$ Fraction of overall POTW removal not due to volatilization as a percent
(SF) Safety factor as a percent.
(Q_{max}) Maximim permitted POTW flow
8.34 Unit conversion factor

$$C_{crit} = C_{std}$$

$$L_{hw} = \frac{8.34 * Q_{potw} * C_{crit}}{(1 - R_{avg} * F_{nonvol})}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.34}$$

TABLE A8

Summary of Determined Limits

				Phenol	Beryllium	Benzidine
NPDES Monthly Limits	Allowable Headworks Load	(lbs/day)	(Lhw)	-	0.151	0.014
	Domestic/Commercial	(lbs/day)	(Ldom)	0.060	0.014	0.008
	Allowable Industrial Load	(lbs/day)	(Lind)	-	0.121	0.004
	Local Limit (Cind)	(mg/l)	(Cind)	-	0.017	0.0006
Secondary Treatment Inhibition	Allowable Headworks Load	(lbs/day)	(Lhw)	779.6	-	-
	Domestic/Commercial	(lbs/day)	(Ldom)	0.060	0.014	0.008
	Allowable Industrial Load	(lbs/day)	(Lind)	584.6	-	-
	Local Limit (Cind)	(mg/l)	(Cind)	81.8	-	-
Nitrification Inhibition	Allowable Headworks Load	(lbs/day)	(Lhw)	15.6	-	-
	Domestic/Commercial	(lbs/day)	(Ldom)	0.060	0.014	0.008
	Allowable Industrial Load	(lbs/day)	(Lind)	12.4	-	-
	Local Limit (Cind)	(mg/l)	(Cind)	1.7	-	-
Digester Inhibition	Allowable Headworks Load	(lbs/day)	(Lhw)	64.867	-	-
	Domestic/Commercial	(lbs/day)	(Ldom)	0.060	0.014	0.008
	Allowable Industrial Load	(lbs/day)	(Lind)	51.833	-	-
	Local Limit (Cind)	(mg/l)	(Cind)	7.254	-	-
Chronic Water Quality Standards	Allowable Headworks Load	(lbs/day)	(Lhw)	73.16	0.151	0.014
	Domestic/Commercial	(lbs/day)	(Ldom)	0.060	0.014	0.008
	Allowable Industrial Load	(lbs/day)	(Lind)	58.47	0.12	0.004
	Local Limit (Cind)	(mg/l)	(Cind)	8.18	0.017	0.0006
Acute Water Quality Standards	Allowable Headworks Load	(lbs/day)	(Lhw)	975.4	2.43	5.62
	Domestic/Commercial	(lbs/day)	(Ldom)	0.060	0.014	0.008
	Allowable Industrial Load	(lbs/day)	(Lind)	780.3	2.17	5.05
	Local Limit (Cind)	(mg/l)	(Cind)	109.2	0.304	0.71

TABLE A9
Allowable Headworks Load

			Phenol	Beryllium	Benzidine
NPDES Monthly Limit ML	(lbs/day)	(Lhw)	-	**0.1508**	**0.0138**
Secondary Treatment Inhibition SI	(lbs/day)	(Lhw)	780	-	-
Nitrification Inhibition NI	(lbs/day)	(Lhw)	**15.59**	-	-
Anaerobic Digester Inhibition DI	(lbs/day)	(Lhw)	64.87	-	-
Chronic Water Quality Standards C	(lbs/day)	(Lhw)	73.16	**0.1508**	**0.0138**
Acute Water Quality Standards A	(lbs/day)	(Lhw)	975	2.428	5.623
MAHL	(lbs/day)		15.59	0.151	0.014
Basis			NI	ML/C	ML/C

TABLE A10
Allowable Industrial Load

			Phenol	Beryllium	Benzidine
NPDES Monthly Limit ML	(lbs/day)	(Lhw)	-	**0.121**	**0.004**
Secondary Treatment Inhibition SI	(lbs/day)	(Lind)	584.6	-	-
Nitrification Inhibition NI	(lbs/day)	(Lind)	**12.41**	-	-
Anaerobic Digester Inhibition DI	(lbs/day)	(Lind)	51.833	-	-
Chronic Water Quality Standards C	(lbs/day)	(Lind)	58.47	**0.121**	**0.004**
Acute Water Quality Standards A	(lbs/day)	(Lind)	780.3	2.17	5.05
MAIL	(lbs/day)		12.41	0.12	0.004
Basis			NI	ML/C	ML/C

TABLE A11
Calculated Local Limits - Uniform Allocation

		Phenol	Beryllium	Benzidine
Daily Maximum	(mg/L)	1.737	0.017	0.0006
	(ug/L)	1737	17	0.6
Basis		NI	ML/C	ML/C

TABLE A12 C_{crit} VALUES

Highlighted Values Were Used In This Local Limits Determination

The following tables are taken from EPA's Local Limits Development Guidance Appendixes, and Guidance Manual for Preventing Interference at POTW

	Appendix R tables				Appendix V
	Removal Efficiencies				Domestic
	Primary	Activated sludge	Trickling Fltr	Tertiary	Level
Phenol	8%	90%			
Beryllium*		61%			
Benzidine					

Appendix G and Table 2-1 tables			
Inhibition Levels			
Activated sludge	Trickling Fltr	Nitrification	Digestion
50		1	100

Site Specific Data from Cadillac WWTP Sampling		
WWTF % Removals		
	Primary	Total Plant
Phenol	18%	54.4%
Beryllium	0%	0%
Benzidine	35%	87.5%

City of Cadillac WWTP - Additional Local Limits

Table A13 -WWTP Influent

Date	Phenol	Beryllium	Benzidine
	µg/L	µg/L	µg/L
3/10/2025	<2.5		<2.7
11/19/2025	<20	<4.0	9.3
11/20/2025	<19	<4.0	0.84
11/25/2025	<5.0	<4.0	1.7
12/5/2025	<10	<4.0	2.0
12/13/2025	<10	<4.0	<0.23
12/14/2025	<5.0	<4.0	<0.22
Average	**5.11**	**2.0**	**2.202**

Table A14 - WWTP Final Effluent

Date	Phenol	Beryllium	Benzidine
	µg/L	µg/L	µg/L
3/10/2025	<2.6		<2.9
11/19/2025	<5.0	<4.0	<0.10
11/20/2025	<5.0	<4.0	0.11
11/25/2025	<5.0	<4.0	0.16
12/5/2025	<5.0	<4.0	<0.10
12/13/2025	<5.0	<4.0	<0.10
12/14/2025	<5.0	<4.0	<0.10
Average	**2.33**	**2.0**	**0.274**
Total % removals	54.4%	**0.0%**	**87.5%**

Date	Table A15 - WWTP Primary Effluent			
	Flow	Phenol	Beryllium (Be)	Benzidine
	MGD	µg/L	µg/L	µg/L
11/19/2025		<10	<4.0	3.7
11/20/2025		<5.0	<4.0	1.1
11/25/2025		<5.0	<4.0	2.4
12/5/2025		<10	<4.0	1.2
12/13/2025		<10	<4.0	<0.23
12/14/2025		<10	<4.0	<0.26
Average (ug/L)		**4.2**	**2.0**	**1.4**

Primary % removal		**18.4%**	**0.0%**	**34.6%**

Permitted Industrial User Sampling Results

Date	Cadillac Akwel C.B			
	Phenol	Beryllium (Be)	Benzidine	Bromomethane
	µg/L	µg/L	µg/L	µg/L
11/17/2025	<5.0	<4.0	<0.097	<5.0
12/3/2025	<53	<4.0	<0.42	<5.0

Date	Cadillac AAR- 6th			
	Phenol	Beryllium (Be)	Benzidine	Bromomethane
	µg/L	µg/L	µg/L	µg/L
11/17/2025	<5.0	<4.0	0.6	<5.0
12/3/2025	<5.0	<4.0	<0.23	<5.0

Date	Cadillac AAR RT			
	Phenol	Beryllium (Be)	Benzidine	Bromomethane
	µg/L	µg/L	µg/L	µg/L
11/20/2025	<5.0	<4.0	<0.098	<5.0
12/5/2025	<5.0	<4.0	<0.10	<5.0

Date	Cadillac AAR FAC			
	Phenol	Beryllium (Be)	Benzidine	Bromomethane
	µg/L	µg/L	µg/L	µg/L
11/20/2025	9.5	<5.0	<0.093	<5.0
12/5/2025	<5.1	<4.0	0.6	<5.0

Date	Cadillac Arvco			
	Phenol	Beryllium (Be)	Benzidine	Bromomethane
	µg/L	µg/L	µg/L	µg/L
11/21/2025	<9.5	<4.0	0.23	<5.0
12/9/2025	<11	<4.0	<0.39	<5.0

Date	Cadillac Borg Warner			
	Phenol	Beryllium (Be)	Benzidine	Bromomethane
	µg/L	µg/L	µg/L	µg/L
11/25/2025	<5.0	<4.0	0.14	<5.0
12/8/2025	<5.0	<4.0	<0.20	<5.0

Date	Cadillac CCI			
	Phenol	Beryllium (Be)	Benzidine	Bromomethane
	µg/L	µg/L	µg/L	µg/L
12/1/2025	8.4	<4.0	<0.11	<5.0
12/10/2025	6.1	<4.0	<0.22	<5.0

Date	Cadillac CRE			
	Phenol	Beryllium (Be)	Benzidine	Bromomethane
	µg/L	µg/L	µg/L	µg/L
11/25/2025	<5.0	<4.0	<0.099	<5.0
12/8/2025	<5.0	<4.0	<0.21	<5.0

Date	FIAMM			
	Phenol	Beryllium (Be)	Benzidine	Bromomethane
	µg/L	µg/L	µg/L	µg/L
12/2/2025	<5.0	<4.0	<0.11	<5.0
12/11/2025	<5.0	<4.0	<0.24	<5.0

Date	Cadillac Hutchinson			
	Phenol	Beryllium (Be)	Benzidine	Bromomethane
	µg/L	µg/L	µg/L	µg/L
12/1/2025	<5.0	<4.0	<0.23	<5.0
12/10/2025	36	<4.0	<0.21	<5.0

Date	Cadillac Piranha			
	Phenol	Beryllium (Be)	Benzidine	Bromomethane
	µg/L	µg/L	µg/L	µg/L
11/21/2025	<5.0	<4.0	<0.098	<5.0
12/9/2025	<5.0	<4.0	<0.21	<5.0

Date	MI Rubber			
	Phenol	Beryllium (Be)	Benzidine	Bromomethane
	µg/L	µg/L	µg/L	µg/L
12/2/2025	<5.0	<4.0	1.1	<5.0
12/11/2025	<5.0	<4.0	<0.24	<5.0

APPENDIX B

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY (EGLE)

INTEROFFICE COMMUNICATION

TO: Claire Lowe, Cadillac District Office, Water Resources Division

FROM: Glen Schmitt, Permits Section, Water Resources Division

DATE: September 30, 2025

SUBJECT: Cadillac Wastewater Treatment Plant (WWTP)
Local Limits Review
Submission Number: HQF-TQCJ-GZWEB
National Pollutant Discharge Elimination System (NPDES) Permit Number MI0020257

Pursuant to a request dated September 23, 2025, we are providing theoretical water quality-based effluent limitations (WQBELs) for the Cadillac WWTP facility. The facility is currently authorized to continuously discharge treated municipal wastewater from Monitoring Point 001A through Outfall 001. Outfall 001 discharges to the Clam River at Latitude 44.2657, Longitude -85.3984. A design flow of 3.2 million gallons per day (MGD) was used in determining final effluent limitations and monitoring requirements in the current NPDES permit, but this flow is not to be considered a limitation or actual capacity.

The Clam River has a lowest 95 percent exceedance flow, harmonic mean flow, and 90dQ10 low flow of 2.6, 13, and 3.4 cubic feet per second (cfs), respectively (EGLE Low Flow Database File Number 10194 [Wexford County]). The Clam River is designated as warmwater where Outfall 001 discharges into and is not protected as a public drinking water supply source.

A total hardness value of 70 milligrams per liter (mg/l) was used for this review. This total hardness value was collected by EGLE staff upstream of Outfall 001 on August 19, 2021 at Latitude 44.263376, Longitude -85.4011 (STORET Number 830054).

The theoretical water quality-based effluent limits (WQBELs) for the parameters you requested are provided below in micrograms per liter (ug/l) and pounds per day (lbs./day).

Parameter	CAS Number	Daily Maximum WQBEL (ug/L)	Daily Maximum Load (lbs./day)	Monthly Average WQBEL (ug/l)	Monthly Average Load (lbs./day)
Beryllium	7440417	66	1.8	4.1	0.11
Benzidine #	92875	49	1.3	0.12	0.0032
Bromomethane	74839	75	2.0	4.8	0.13
Phenol	108952	6,800	180	510	14

= carcinogen.

The Preliminary Effluent Limitations (PEL) file is saved in the MiEnviro Portal under the following filename: IPP Local Limits PEL_MI0020257_9-30-2025.xls

Please contact me at (517) 290-6424 if you have any questions regarding the recommendations above.

PERMIT NO. MI0020257

STATE OF MICHIGAN
DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

AUTHORIZATION TO DISCHARGE UNDER THE NATIONAL POLLUTANT DISCHARGE ELIMINATION SYSTEM

In compliance with the provisions of the federal Clean Water Act (federal Water Pollution Control Act, 33 U.S.C., Section 1251 *et seq*., as amended); Part 31, Water Resources Protection, of the Natural Resources and Environmental Protection Act, 1994 PA 451, as amended (NREPA); Part 41, Sewerage Systems, of the NREPA; and Michigan Executive Order 2019-06,

City of Cadillac
200 North Lake Street
Cadillac, MI 49601

is authorized to discharge from the **Cadillac Wastewater Treatment Plant** located at

1121 Plett Road
Cadillac, MI 49601

designated as **Cadillac WWTP**

to the receiving water named the Clam River in accordance with effluent limitations, monitoring requirements, and other conditions set forth in this permit.

This permit is based on a complete application submitted on September 21, 2022.

This permit takes effect on January 1, 2024. The provisions of this permit are severable. After notice and opportunity for a hearing, this permit may be modified, suspended, or revoked in whole or in part during its term in accordance with applicable laws and rules. On its effective date, this permit shall supersede National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257 (expiring October 1, 2022).

This permit and the authorization to discharge shall expire at midnight on **October 1, 2027**. In order to receive authorization to discharge beyond the date of expiration, the permittee shall submit an application that contains such information, forms, and fees as are required by the Michigan Department of Environment, Great Lakes, and Energy (Department) by **April 4, 2027**.

Issued: November 29, 2023.

Original signed by Christine Alexander
Christine Alexander, Manager
Permits Section
Water Resources Division

PERMIT FEE REQUIREMENTS

In accordance with Section 324.3120 of the NREPA, the permittee shall make payment of an annual permit fee to the Department for each October 1 the permit is in effect regardless of occurrence of discharge. The permittee shall submit the fee in response to the Department's annual notice. Payment may be made electronically via the Department's MiEnviro Portal system. The MiEnviro Portal website is located at https://mienviro.michigan.gov/ncore/. Payment shall be submitted or postmarked by January 15 for notices mailed by December 1. Payment shall be submitted or postmarked no later than 45 days after receiving the notice for notices mailed after December 1.

Annual Permit Fee Classification: Municipal Major, less than 10 MGD (Individual Permit)

In accordance with Section 324.3132 of the NREPA, the permittee shall make payment of an annual biosolids land application fee to the Department if the permittee land applies biosolids. The permittee shall submit the fee in response to the Department's annual notice. Payment may be made electronically via the Department's MiEnviro Portal system. The MiEnviro Portal website is located at https://mienviro.michigan.gov/ncore/. Payment shall be submitted or postmarked no later than January 31 of each year for notices mailed by December 15. Payment shall be submitted or postmarked no later than 45 days after receiving the notice for notices mailed after December 15.

CONTACT INFORMATION

Unless specified otherwise, all contact with the Department required by this permit shall be made to the Cadillac District Office of the Water Resources Division. The Cadillac District Office is located at 120 West Chapin Street, Cadillac, MI 49601-2158, Telephone: 231-775-3960, Fax: 231-775-1511.

CONTESTED CASE INFORMATION

Any person who is aggrieved by this permit may file a sworn petition with the Michigan Administrative Hearing System within the Michigan Department of Licensing and Regulatory Affairs, c/o the Michigan Department of Environment, Great Lakes, and Energy, setting forth the conditions of the permit which are being challenged and specifying the grounds for the challenge. The Department of Licensing and Regulatory Affairs may reject any petition filed more than 60 days after issuance as being untimely.

PART I

Section A. Limitations and Monitoring Requirements

1. Final Effluent Limitations, Monitoring Point 001A

During the period beginning on the effective date of this permit and lasting until the expiration date of this permit, the permittee is authorized to discharge treated municipal wastewater from Monitoring Point 001A through Outfall 001. Outfall 001 discharges to the Clam River at Latitude 44.2657, Longitude -85.3984. Such discharge shall be limited and monitored by the permittee as specified below.

	Maximum Limits for Quantity or Loading				Maximum Limits for Quality or Concentration					
Parameter	**Monthly**	**7-Day**	**Daily**	**Units**	**Monthly**	**7-Day**	**Daily**	**Units**	**Monitoring Frequency**	**Sample Type**
Flow	(report)	---	(report)	MGD	---	---	---	---	Daily	Report Total Daily Flow
Carbonaceous Biochemical Oxygen Demand (CBOD5)										
June – September	190	270	(report)	lbs/day	7	---	10	mg/l	5x Weekly	24-Hr Composite
October – November	290	450	(report)	lbs/day	11	---	17	mg/l	5x Weekly	24-Hr Composite
December – March	610	910	(report)	lbs/day	23	---	34	mg/l	5x Weekly	24-Hr Composite
April – May	450	670	(report)	lbs/day	17	---	25	mg/l	5x Weekly	24-Hr Composite
Total Suspended Solids (TSS)										
June – September	530	800	(report)	lbs/day	20	30	(report)	mg/l	5x Weekly	24-Hr Composite
October – May	800	1,200	(report)	lbs/day	30	45	(report)	mg/l	5x Weekly	24-Hr Composite
Ammonia Nitrogen (as N)										
Through September 30, 2026										
June – September	24	53	(report)	lbs/day	0.9	---	2.0	mg/l	5x Weekly	24-Hr Composite
October – November	43	75	(report)	lbs/day	1.6	---	2.8	mg/l	5x Weekly	24-Hr Composite
December – March	99	---	(report)	lbs/day	3.7	---	(report)	mg/l	5x Weekly	24-Hr Composite
April – May	130	---	(report)	lbs/day	4.8	---	(report)	mg/l	5x Weekly	24-Hr Composite
Beginning October 1, 2026										
June – September	24	53	(report)	lbs/day	0.9	---	2.0	mg/l	5x Weekly	24-Hr Composite
October – November	43	75	(report)	lbs/day	1.6	---	2.8	mg/l	5x Weekly	24-Hr Composite
December – March	99	---	(report)	lbs/day	3.7	---	(report)	mg/l	5x Weekly	24-Hr Composite
April – May	80	---	(report)	lbs/day	3.0	---	(report)	mg/l	5x Weekly	24-Hr Composite

PART I

Section A. Limitations and Monitoring Requirements

Parameter	Maximum Limits for Quantity or Loading				Maximum Limits for Quality or Concentration				Monitoring Frequency	Sample Type
	Monthly	7-Day	Daily	Units	Monthly	7-Day	Daily	Units		
Total Phosphorus (as P)										
May – March	13	---	(report)	lbs/day	0.5	---	(report)	mg/l	5x Weekly	24-Hr Composite
April	17	---	(report)	lbs/day	0.65	---	(report)	mg/l	5x Weekly	24-Hr Composite
Total Copper	---	---	---	---	---	---	(report)	ug/l	2x Monthly	24-Hr Composite
Total Arsenic	0.43	---	(report)	lbs/day	16	---	(report)	ug/l	2x Monthly	24-Hr Composite
Total Beryllium	0.11	---	(report)	lbs/day	4.1	---	(report)	ug/l	2x Monthly	24-Hr Composite
Total Cadmium	0.11	0.32	(report)	lbs/day	4.1	---	12	ug/l	2x Monthly	24-Hr Composite
Total Silver	0.037	---	(report)	lbs/day	1.4	---	(report)	ug/l	2x Monthly	24-Hr Composite
Chloride	---	---	---	---	---	---	(report)	mg/l	Monthly	24-Hr Composite
Sulfate	---	---	---	---	---	---	(report)	mg/l	Monthly	24-Hr Composite
Acute Toxicity – Fathead Minnow	---	---	---	---	---	---	(report)	TU_A	Quarterly	24-Hr Composite
Acute Toxicity – *C. dubia*	---	---	---	---	---	---	(report)	TU_A	Quarterly	24-Hr Composite
							Individual Chronic Value			
Chronic Toxicity – Fathead Minnow	---	---	---	---	1.1	---	(report)	TU_C	Quarterly	24-Hr Composite
Chronic Toxicity – *C. dubia*	---	---	---	---	1.1	---	(report)	TU_C	Quarterly	24-Hr Composite
							Maximum Daily			
Fecal Coliform Bacteria	---	---	---	---	200	400	(report)	cts/ 100 ml	5x Weekly	Grab
Perfluorooctanesulfonic Acid (PFOS)	---	---	---	---	---	---	(report)	ng/l	3x Annual	Grab
Available Cyanide	0.16	---	(report)	lbs/day	5.9	---	(report)	ug/l	Monthly	Grab
Total Benzidine	0.0032	---	(report)	lbs/day	0.12	---	(report)	ug/l	2x Monthly	Grab
Total Bromomethane	---	---	---	---	---	---	(report)	ug/l	Quarterly	Grab

PART I

Section A. Limitations and Monitoring Requirements

Parameter	Maximum Limits for Quantity or Loading				Maximum Limits for Quality or Concentration				Monitoring Frequency	Sample Type
	Monthly	7-Day	Daily	Units	Monthly	7-Day	Daily	Units		
Total Mercury										
Corrected	(report)	---	(report)	lbs/day	(report)	---	(report)	ng/l	Quarterly	Calculation
Uncorrected	---	---	---	---	---	---	(report)	ng/l	Quarterly	Grab
Field Duplicate	---	---	---	---	---	---	(report)	ng/l	Quarterly	Grab
Field Blank	---	---	---	---	---	---	(report)	ng/l	Quarterly	Preparation
Laboratory Method Blank	---	---	---	---	---	---	(report)	ng/l	Quarterly	Preparation
	12-Month Rolling Avg				**12-Month Rolling Avg**					
Total Mercury	0.000053	---	---	lbs/day	2.0	---	---	ng/l	Quarterly	Calculation
					Minimum % Monthly		**Minimum % Daily**			
TSS Minimum % Removal										
October – May	---	---	---	---	85	---	(report)	%	Monthly	Calculation
					Minimum Daily		**Maximum Daily**			
pH	---	---	---	---	6.5	---	9.0	S.U.	5x Weekly	Grab
Dissolved Oxygen										
May – November	---	---	---	---	6.0	---	---	mg/l	5x Weekly	Grab
December – April	---	---	---	---	5.0	---	---	mg/l	5x Weekly	Grab

The following design flow was used in determining the above limitations, but is not to be considered a limitation or actual capacity: 3.2 MGD.

a. Narrative Standard
The receiving water shall contain no turbidity, color, oil films, floating solids, foams, settleable solids, or deposits as a result of this discharge in unnatural quantities which are or may become injurious to any designated use.

b. Sampling Locations
Samples for CBOD5, TSS, Ammonia Nitrogen, Total Phosphorus, Total Copper, Total Arsenic, Total Beryllium, Total Cadmium, Total Silver, Chloride, Sulfate, Acute Toxicity, and Chronic Toxicity shall be taken prior to disinfection. Samples for Fecal Coliform Bacteria, PFOS, Available Cyanide, Total Benzidine, Total Bromomethane, Total Mercury, pH, and Dissolved Oxygen shall be taken after disinfection. The Department may approve alternate sampling locations that are demonstrated by the permittee to be representative of the effluent.

c. Quarterly Monitoring and 3x Annual Monitoring
Quarterly samples shall be taken during the months of January, April, July, and October. 3x Annual samples shall be taken during the months of January, May, and September. If the facility does not discharge during these months, the permittee shall sample the next discharge occurring during the period in question. If the facility does not discharge during the period in question, a sample is not required for that period. For any month in which a sample is not taken, the permittee shall enter "*G" on the Discharge Monitoring Report (DMR). (For purposes of reporting on the Daily tab of the DMR, the permittee shall enter “*G” on the first day of the month only).

Section A. Limitations and Monitoring Requirements

d. Ultraviolet Disinfection
It is understood that ultraviolet light will be used to achieve compliance with the fecal coliform limitations. If disinfection other than ultraviolet light will be used, the permittee shall notify the Department in accordance with Part II.C.12. of this permit.

e. Percent Removal Requirements
Monthly percent removal shall be calculated based on the monthly average effluent TSS concentrations and the monthly average influent concentrations for approximately the same period. Daily percent removal shall be calculated based on the daily effluent TSS concentrations and the daily influent concentrations for the same day. Reporting of Daily percent removal is only required on days on which an influent sample is obtained. The calculation shall be made as follows for each parameter: Percent removal = (influent concentration - effluent concentration) / influent concentration x 100.

f. Monitoring Frequency Reduction for Available Cyanide, Total Arsenic, Total Benzidine, Total Beryllium, Total Bromomethane, Total Cadmium, and/or Total Silver
After the submittal of 12 months of data, the permittee may request, in writing, Department approval for a reduction in monitoring frequency for Available Cyanide, Total Arsenic, Total Benzidine, Total Beryllium, Total Bromomethane, Total Cadmium, and/or Total Silver. This request shall contain an explanation as to why the reduced monitoring is appropriate. Upon receipt of written approval and consistent with such approval, the permittee may reduce the monitoring frequency indicated in Part I.A.1. of this permit. The monitoring frequency for Available Cyanide, Total Benzidine, and Total Bromomethane shall not be reduced to less than annually. The monitoring frequency for Total Arsenic, Total Beryllium, Total Cadmium, and Total Silver shall not be reduced to less than quarterly. The Department may revoke the approval for reduced monitoring at any time upon notification to the permittee.

g. Monitoring Frequency Reduction for Perfluorooctanesulfonic Acid (PFOS)
After the submittal of at least 10 equally spaced sample results obtained over a minimum of three (3) months, the permittee may request, in writing, Department approval of a reduction in monitoring frequency for PFOS. This request shall contain an explanation as to why the reduced monitoring is appropriate. Upon receipt of written approval and consistent with such approval, the permittee may reduce the monitoring frequency indicated in Part I.A.1. of this permit. The monitoring frequency for PFOS shall not be reduced to less than annually. The Department may revoke the approval for reduced monitoring at any time upon notification to the permittee.

h. Final Effluent Limitation for Total Mercury
The final limit for total mercury is the Discharge Specific Level Currently Achievable (LCA) based on a multiple discharger variance from the WQBEL of 1.3 ng/l, pursuant to Rule 1103(9) of the Water Quality Standards. Compliance with the LCA shall be determined as a 12-month rolling average, the calculation of which may be done using blank-corrected sample results. The 12-month rolling average shall be determined by adding the present monthly average result to the preceding 11 monthly average results then dividing the sum by 12. For facilities with quarterly monitoring requirements for total mercury, quarterly monitoring shall be equivalent to three (3) months of monitoring in calculating the 12-month rolling average. Facilities that monitor more frequently than monthly for total mercury must determine the monthly average result, which is the sum of the results of all data obtained in a given month divided by the total number of samples taken, in order to calculate the 12-month rolling average. If the 12-month rolling average for any quarter is less than or equal to the LCA, the permittee will be considered to be in compliance for total mercury for that quarter, provided the permittee is also in full compliance with the Pollutant Minimization Program for Total Mercury, set forth in Part I.A.4. of this permit.

PART I

Section A. Limitations and Monitoring Requirements

i. Total Mercury Testing and Additional Reporting Requirements
The analytical protocol for total mercury shall be in accordance with EPA Method 1631, Revision E, "Mercury in Water by Oxidation, Purge and Trap, and Cold Vapor Atomic Fluorescence Spectrometry," EPA-821-R-02-019, August 2002. The quantification level for total mercury shall be 0.5 ng/l, unless a higher level is appropriate because of sample matrix interference. Justification for higher quantification levels shall be submitted to the Department within 30 days of such determination.

The use of clean technique sampling procedures is required unless the permittee can demonstrate to the Department that an alternate sampling procedure is representative of the discharge. Guidance for clean technique sampling is contained in EPA Method 1669, "Sampling Ambient Water for Trace Metals at EPA Water Quality Criteria Levels (Sampling Guidance)," EPA-821-R96-001, July 1996. Information and data documenting the permittee's sampling and analytical protocols and data acceptability shall be submitted to the Department upon request.

In order to demonstrate compliance with EPA Method 1631E and EPA Method 1669, the permittee shall report, on the daily sheet, the analytical results of all field blanks and field duplicates collected in conjunction with each sampling event, as well as laboratory method blanks when used for blank correction. The permittee shall collect at least one (1) field blank and at least one (1) field duplicate per sampling event. If more than 10 samples are collected during a sampling event, the permittee shall collect at least one (1) additional field blank AND field duplicate for every 10 samples collected. A "sampling event" shall be defined herein as all sampling for total mercury conducted on the same day, provided the same sampling team collected all samples using the same sampling methods, procedures, and equipment on that day. Only field blanks or laboratory method blanks may be used to calculate a concentration lower than the actual sample analytical results (i.e., a blank correction). Only one (1) blank (field OR laboratory method) may be used for blank correction of a given sample result, and only if the blank meets the quality control acceptance criteria. If blank correction is not performed on a given sample analytical result, the permittee shall report under "Total Mercury – Corrected" the same value reported under "Total Mercury – Uncorrected." The field duplicate is for quality control purposes only; its analytical result shall not be averaged with the sample result.

PART I

Section A. Limitations and Monitoring Requirements

j. Whole Effluent Toxicity Final Requirements

Test species shall include fathead minnow **and** *Ceriodaphnia dubia*. Testing and reporting procedures shall follow procedures contained in EPA-821-R-02-013, "Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms" (Fourth Edition). When the effluent ammonia nitrogen (as N) concentration is greater than 3 mg/l, the pH of the toxicity test shall be maintained at a pH of 8 Standard Units. The acute toxic unit (TU_A) value and chronic toxic unit (TU_C) value for **each species tested** shall be reported on the DMR. If multiple chronic toxicity tests for the same species are performed during the month, the maximum TU_A value and monthly average TU_C value for the species shall be reported. For **each species not tested**, the permittee shall enter **"*W"** on the DMR. (For purposes of reporting on the Daily tab of the DMR, the permittee shall enter "*W" on the first day of the month only). Completed toxicity test reports for each test conducted shall be retained by the permittee in accordance with the requirements of Part II.B.5. of this permit and shall be available for review by the Department upon request. After 24 months of toxicity testing and upon approval from the Department, the monitoring frequency may be reduced if the test data indicate that the toxicity requirements of R 323.1219 of the Michigan Administrative Code are consistently being met. After one (1) year of toxicity testing and upon approval from the Department, the chronic toxicity tests may be performed using the more sensitive species identified in the chronic toxicity results collected to date. If a more sensitive species cannot be identified, the chronic toxicity tests shall be performed with both species. Toxicity test data acceptability is contingent upon validation of the test method by the testing laboratory. Such validation shall be submitted to the Department upon request.

1) When monitoring shows persistent exceedance of the 1.1 TU_C limit for effluent toxicity or data indicate persistent exceedance of the acute toxicity requirements of R 323.1219, the Department will determine whether the permittee must implement the toxicity control program requirements specified in 2), below.

2) Upon written notification by the Department, the following conditions apply. Within 90 days of the notification, the permittee shall implement a Toxicity Reduction Evaluation (TRE). The objective of the TRE shall be to reduce the toxicity of the final effluent from Monitoring Point 001A to <1.1 TU_C and <1.0 TU_A. The following documents are available as guidance to reduce toxicity to acceptable levels: Phase I, EPA/600/6-91/005F (chronic), EPA/600/6-91/003 (acute); Phase II, EPA/600/R-92/080 (acute and chronic); Phase III, EPA/600/R-92/081 (acute and chronic); and Publicly Owned Treatment Works (POTWs), EPA/833B-99/002. The TRE shall include quarterly chronic toxicity tests of the discharge from Monitoring Point 001A for the duration of the TRE. The tests shall be conducted and reported as specified above. Annual reports shall be submitted to the Department within 30 days of the completion of the last test of each annual cycle.

3) This permit may be modified in accordance with applicable laws and rules to include additional whole effluent toxicity control requirements as necessary.

PART I

Section A. Limitations and Monitoring Requirements

2. Quantification Levels and Analytical Methods for Selected Parameters

Maximum acceptable quantification levels (QLs) are specified for selected parameters in the table below. These QLs apply to all monitoring conducted in compliance with this permit if and when the parameters specified herein are monitored. This includes monitoring conducted to meet the requirements of the application for permit reissuance. These QLs shall be considered the maximum acceptable unless a higher QL is appropriate because of sample matrix interference. Justification for higher QLs shall be submitted to the Department within 30 days of such determination.

Where necessary to help ensure that the QLs specified herein can be achieved, analytical methods may also be specified in the table below. The sampling procedures, preservation and handling, and analytical protocol for all monitoring conducted in compliance with this permit, including monitoring conducted to meet the requirements of the application for permit reissuance, shall be in accordance with the methods specified herein, or in accordance with Part II.B.2. of this permit if no method is specified herein, unless an alternate method is approved by the Department. The Department will consider only alternate methods that meet the requirements of Part II.B.2. and whose QLs are at least as sensitive (i.e., low) as those specified herein. **Not all QLs are expressed in the same units in the table below**. The table is continued on the following page:

Parameter	QL	Units	Analytical Method
1,2-Diphenylhydrazine (as Azobenzene)	3.0	ug/l	
2,4,6-Trichlorophenol	5.0	ug/l	
2,4-Dinitrophenol	19	ug/l	
3,3’-Dichlorobenzidine	1.5	ug/l	
4-Chloro-3-Methylphenol	7.0	ug/l	
4,4’-DDD	0.01	ug/l	
4,4’-DDE	0.01	ug/l	
4,4’-DDT	0.01	ug/l	
Acrylonitrile	1.0	ug/l	
Aldrin	0.01	ug/l	
Alpha-Endosulfan	0.01	ug/l	
Alpha-Hexachlorocyclohexane	0.01	ug/l	
Antimony, Total	1	ug/l	
Arsenic, Total	1	ug/l	
Barium, Total	5	ug/l	
Benzidine	0.1	ug/l	
Beryllium, Total	1	ug/l	
Beta-Endosulfan	0.01	ug/l	
Beta-Hexachlorocyclohexane	0.01	ug/l	
Bis (2-Chloroethyl) Ether	1.0	ug/l	
Bis (2-Ethylhexyl) Phthalate	5.0	ug/l	
Boron, Total	20	ug/l	
Bromomethane, Total	1.0	ug/l	
Cadmium, Total	0.2	ug/l	
Chlordane	0.01	ug/l	
Chloride	1.0	mg/l	
Chromium, Hexavalent	5	ug/l	
Chromium, Total	10	ug/l	
Copper, Total	1	ug/l	

PART I

Section A. Limitations and Monitoring Requirements

Parameter	QL	Units	Analytical Method
Cyanide, Available	2	ug/l	EPA Method OIA 1677
Cyanide, Total	5	ug/l	
Delta-Hexachlorocyclohexane	0.01	ug/l	
Dieldrin	0.01	ug/l	
Di-N-Butyl Phthalate	9.0	ug/l	
Endosulfan Sulfate	0.01	ug/l	
Endrin	0.01	ug/l	
Endrin Aldehyde	0.01	ug/l	
Fluoranthene	1.0	ug/l	
Heptachlor	0.01	ug/l	
Heptachlor Epoxide	0.01	ug/l	
Hexachlorobenzene	0.01	ug/l	
Hexachlorobutadiene	0.01	ug/l	
Hexachlorocyclopentadiene	0.01	ug/l	
Hexachloroethane	5.0	ug/l	
Lead, Total	1	ug/l	
Lindane	0.01	ug/l	
Lithium, Total	10	ug/l	
Mercury, Total	0.5	ng/l	EPA Method 1631E
Nickel, Total	5	ug/l	
PCB-1016	0.1	ug/l	
PCB-1221	0.1	ug/l	
PCB-1232	0.1	ug/l	
PCB-1242	0.1	ug/l	
PCB-1248	0.1	ug/l	
PCB-1254	0.1	ug/l	
PCB-1260	0.1	ug/l	
Pentachlorophenol	1.8	ug/l	
Perfluorooctanesulfonic acid (PFOS)	2.0	ng/l	While EPA Method 1633 remains draft, analyses may be performed using that method, or ASTM D7979, or an isotope dilution method (sometimes referred to as Method 537 modified). Once EPA Method 1633 is promulgated, only that method may be used.
Perfluorooctanoic acid (PFOA)			
Perfluorobutanesulfonic acid (PFBS)			
Phenanthrene	1.0	ug/l	
Selenium, Total	1.0	ug/l	
Silver, Total	0.5	ug/l	
Strontium, Total	1000	ug/l	
Sulfate	2.0	mg/l	
Sulfide, Total	20	ug/l	
Thallium, Total	1	ug/l	
Toxaphene	0.1	ug/l	
Vinyl Chloride	1.0	ug/l	
Zinc, Total	10	ug/l	

PART I

Section A. Limitations and Monitoring Requirements

3. Additional Monitoring Requirements

As a condition of this permit, the permittee shall monitor the discharge from monitoring point 001A for the constituents identified below. This monitoring is an application requirement of 40 CFR 122.21(j), effective December 2, 1999. Testing shall be conducted in August 2024, May 2025, March 2026, and October 2026. Grab samples shall be collected for total phenols and the Perfluoroalkyl and Polyfluoroalkyl Substances (PFAS) and Volatile Organic Compounds identified below. For all other parameters, 24-hour composite samples shall be collected.

The results of such additional monitoring shall be submitted with the application for reissuance (see the cover page of this permit for the application due date). The permittee shall notify the Department within 14 days of completing the monitoring for each month specified above in accordance with Part II.C.5. Additional reporting requirements are specified in Part II.C.11. If, upon review of the analysis, it is determined that additional requirements are needed to protect the receiving waters in accordance with applicable water quality standards, the permit may then be modified by the Department in accordance with applicable laws and rules.

Hardness
calcium carbonate

Perfluoroalkyl and Polyfluoroalkyl Substances

Perfluorooctanoic acid (PFOA)	Perfluorobutanesulfonic acid (PFBS)

Metals (Total Recoverable), Cyanide and Total Phenols

antimony	chromium	lead	nickel
selenium	thallium	total phenols	zinc

Volatile Organic Compounds

acrolein	acrylonitrile	benzene	bromoform
carbon tetrachloride	chlorobenzene	chlorodibromomethane	chloroethane
2-chloroethylvinyl ether	chloroform	dichlorobromomethane	1,1-dichloroethane
1,2-dichloroethane	trans-1,2-dichloroethylene	1,1-dichloroethylene	1,2-dichloropropane
1,3-dichloropropylene	ethylbenzene	methyl chloride	
methylene chloride	1,1,2,2-tetrachloroethane	tetrachloroethylene	toluene
1,1,1-trichloroethane	1,1,2-trichloroethane	trichloroethylene	vinyl chloride

Acid-Extractable Compounds

4-chloro-3-methylphenol	2-chlorophenol	2,4-dichlorophenol	2,4-dimethylphenol
4,6-dinitro-o-cresol	2,4-dinitrophenol	2-nitrophenol	4-nitrophenol
pentachlorophenol	phenol	2,4,6-trichlorophenol	

Base/Neutral Compounds

acenaphthene	acenaphthylene	anthracene	benzo(a)anthracene
benzo(a)pyrene	3,4-benzofluoranthene	benzo(ghi)perylene	benzo(k)fluoranthene
bis(2-chloroethoxy)methane	bis(2-chloroethyl)ether	bis(2-chloroisopropyl)ether	
bis(2-ethylhexyl)phthalate	4-bromophenyl phenyl ether	butylbenzyl phthalate	2-chloronaphthalene
4-chlorophenyl phenyl ether	chrysene	di-n-butyl phthalate	di-n-octyl phthalate
dibenzo(a,h)anthracene	1,2-dichlorobenzene	1,3-dichlorobenzene	1,4-dichlorobenzene
3,3'-dichlorobenzidine	diethyl phthalate	dimethyl phthalate	2,4-dinitrotoluene
2,6-dinitrotoluene	1,2-diphenylhydrazine	fluoranthene	fluorene
hexachlorobenzene	hexachlorobutadiene	hexachlorocyclopentadiene	hexachloroethane
indeno(1,2,3-cd)pyrene	isophorone	naphthalene	nitrobenzene
n-nitrosodi-n-propylamine	n-nitrosodimethylamine	n-nitrosodiphenylamine	phenanthrene
pyrene	1,2,4-trichlorobenzene		

PART I

Section A. Limitations and Monitoring Requirements

4. Pollutant Minimization Program for Total Mercury

The goal of the Pollutant Minimization Program is to maintain the effluent concentration of total mercury at or below 1.3 ng/l. The permittee shall continue to implement the Pollutant Minimization Program approved on May 17, 2005, and modifications thereto, to proceed toward the goal. The Pollutant Minimization Program includes the following:

a. an annual review and semi-annual monitoring of potential sources of mercury entering the wastewater collection system;

b. a program for quarterly monitoring of influent and periodic monitoring of sludge for mercury; and

c. implementation of reasonable cost-effective control measures when sources of mercury are discovered. Factors to be considered include significance of sources, economic considerations, and technical and treatability considerations.

On or before March 31 of each year, the permittee shall submit a status report to the Department for the previous calendar year that includes 1) the monitoring results for the previous year, 2) an updated list of potential mercury sources, and 3) a summary of all actions taken to reduce or eliminate identified sources of mercury.

Any information generated as a result of the Pollutant Minimization Program set forth in this permit may be used to support a request to modify the approved program or to demonstrate that the Pollutant Minimization Program requirement has been completed satisfactorily.

A request for modification of the approved program and supporting documentation shall be submitted in writing to the Department for review and approval. The Department may approve modifications to the approved program (approval of a program modification does not require a permit modification), including a reduction in the frequency of the requirements under items a. and b. above.

This permit may be modified in accordance with applicable laws and rules to include additional mercury conditions and/or limitations as necessary.

PART I

Section A. Limitations and Monitoring Requirements

5. Pollutant Minimization and Source Evaluation Program for Perfluorooctanesulfonic Acid (PFOS), Perfluorooctanoic Acid (PFOA), and Perfluorobutanesulfonic Acid (PFBS)

The goal of the Pollutant Minimization and Source Evaluation Program is to identify and address sources of PFOS, PFOA, and PFBS and to reduce and maintain the effluent concentrations of PFOS, PFOA, and PFBS at or below the water quality-based effluent limitations (WQBELs). The WQBELs are 12 ng/l for PFOS, 0.28 ug/l for PFOA, and 1,100 ug/l for PFBS.

a. Within 90 days of written notification from the Department or after the permittee notifies the Department that final effluent concentrations of PFOS, PFOA, and/or PFBS have exceeded the WQBELs, the permittee shall submit to the Department an approvable Pollutant Minimization and Source Evaluation Program for the PFAS analyte(s) in exceedance of the WQBEL(s) to proceed toward the goal. The Pollutant Minimization and Source Evaluation Program shall include the following:

   1) identification of and strategies to identify any potential and probable PFOS, PFOA, and PFBS sources, as applicable;

   2) monitoring plan for the permitted facility's influent and effluent, as well as effluent from suspected sources;

   3) implemented measures thus far to eliminate, reduce, and/or control sources, and an assessment of the degree of success and the strategies used to measure success; and

   4) proposed measures and implementation schedules for elimination, control, and/or reduction of the identified sources (prioritizing highest loadings and concentrations), and the strategies that will be used to measure success.

b. Two times annually, on or before February 1 and August 1 of each year following Department approval of the Pollutant Minimization and Source Evaluation Program, the permittee shall submit to the Department a status report for the prior six (6) month period. Status reports shall include:

   1) complete list of known or suspected PFOS, PFOA, and PFBS sources, as applicable;

   2) summary of influent and effluent monitoring data;

   3) summary of monitoring data from known or suspected sources;

   4) history and compliance status for sources;

   5) implemented measures to eliminate, reduce, or control sources, (prioritizing highest loadings and concentrations), and an assessment of the degree of success and the strategies used to measure success;

   6) proposed measures and schedules for elimination, control, or reduction of any newly identified PFOS, PFOA, and PFBS sources (prioritizing highest loadings and concentrations), as applicable, and the strategies that will be used to measure success;

   7) barriers to implementation and revisions to the implementation schedule; and

   8) any laboratory reports not previously supplied.

Any information generated as a result of the Pollutant Minimization and Source Evaluation Program set forth in this permit may be used to support a request to modify the Pollutant Minimization and Source Evaluation Program or to demonstrate that the requirement has been completed satisfactorily.

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A request for modification of the approved Pollutant Minimization and Source Evaluation Program shall be submitted in writing to the Department along with supporting documentation for review and approval. The Department may approve modifications to the approved Pollutant Minimization and Source Evaluation Program, including a reduction in the frequency of the influent and known or potential source monitoring requirements. Approval of a Pollutant Minimization and Source Evaluation Program modification does not require a permit modification.

This permit may be modified in accordance with applicable laws and rules to include additional PFOS, PFOA, and PFBS conditions and/or limitations as necessary.

6. Untreated or Partially Treated Sewage Discharge Reporting and Testing Requirements

In accordance with Section 324.3112a of the NREPA, if untreated or partially treated sewage is directly or indirectly discharged from a sewer system onto land or into the waters of the state, the permittee shall immediately, but not more than 24 hours after the discharge begins, notify local health departments, a daily newspaper of general circulation in the county in which the permittee is located, and a daily newspaper of general circulation in the county or counties in which the municipalities whose waters may be affected by the discharge are located, that the discharge is occurring. The permittee shall also notify the Department via its MiEnviro Portal system on the form entitled “Report of Discharge (CSO\SSO\RTB).” The MiEnviro Portal website is located at https://mienviro.michigan.gov/ncore/. At the conclusion of the discharge, the permittee shall make all such notifications specified in, and in accordance with, Section 324.3112a of the NREPA, and shall notify the Department via its MiEnviro Portal system on the form entitled “Report of Discharge (CSO\SSO\RTB).”

The permittee shall also annually contact municipalities, including the superintendent of a public drinking water supply with potentially affected intakes, whose waters may be affected by the permittee's discharge of untreated or partially treated sewage, and if those municipalities wish to be notified in the same manner as specified above, the permittee shall provide such notification.

Additionally, in accordance with Section 324.3112a of the NREPA, each time a discharge of untreated or partially treated sewage occurs, the permittee shall test the affected waters for *Escherichia coli* to assess the risk to the public health as a result of the discharge and shall provide the test results to the affected local county health departments and to the Department. The results of this testing shall be submitted to the Department via MiEnviro Portal as part of the notification specified above, or, if the results are not yet available, submitted as soon as they become available. This testing is not required if it has been waived by the local health department, or if the discharge(s) did not affect surface waters. The testing shall be done at locations specified by each affected local county health department but shall not exceed 10 tests for each separate discharge event. The affected local county health department may waive this testing requirement if it determines that such testing is not needed to assess the risk to the public health as a result of the discharge event.

Permittees accepting sanitary or municipal sewage from other sewage collection systems are encouraged to notify the owners of those systems of the above reporting and testing requirements.

PART I

Section A. Limitations and Monitoring Requirements

7. Facility Contact

The “Facility Contact” was specified in the application. The permittee may replace the facility contact at any time, and shall notify the Department in writing within 10 days after replacement (including the name, address and telephone number of the new facility contact).

a. The facility contact shall be (or a duly authorized representative of this person):
- for a corporation, a principal executive officer of at least the level of vice president; or a designated representative if the representative is responsible for the overall operation of the facility from which the discharge originates, as described in the permit application or other NPDES form,
- for a partnership, a general partner,
- for a sole proprietorship, the proprietor, or
- for a municipal, state, or other public facility, either a principal executive officer, the mayor, village president, city or village manager or other duly authorized employee.

b. A person is a duly authorized representative only if:
- the authorization is made in writing to the Department by a person described in paragraph a. of this section; and
- the authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the facility (a duly authorized representative may thus be either a named individual or any individual occupying a named position).

Nothing in this section releases the permittee from properly submitting reports and forms as required by law.

8. Monthly Operating Reports

Part 41 of Act 451 of 1994 as amended, specifically Section 324.4106 and associated R 299.2953, requires that the permittee file with the Department, on forms prescribed by the Department, operating reports showing the effectiveness of the treatment facility operation and the quantity and quality of liquid wastes discharged into waters of the state. Applicable forms and guidance are available on the Department’s web site at https://www.michigan.gov/egle/about/Organization/Water-Resources/wastewater-construction. The permittee may use alternate forms if they are consistent with the approved treatment facility monitoring program. Unless the Department provides written notification to the permittee that monthly submittal of operating reports is required, operating reports that result from implementation of the approved treatment facility monitoring program shall be maintained on site for a minimum of three (3) years and shall be made available to the Department for review upon request.

Within 30 days of the effective date of this permit, the permittee shall submit to the Department either a revised treatment facility monitoring program to address monitoring requirement changes reflected in this permit, or justification explaining why monitoring requirement changes reflected in this permit do not necessitate revisions to the treatment facility monitoring program. Upon receipt of approval from the Department and consistent with such approval, the permittee shall implement the approved revised treatment facility monitoring program.

PART I

Section A. Limitations and Monitoring Requirements

9. Asset Management

The permittee shall at all times properly operate and maintain all facilities (i.e., the sewer system and treatment works as defined in Part 41 of the NREPA), and control systems installed or used by the permittee to operate the sewer system and treatment works and achieve and maintain compliance with the conditions of this permit (also see Part II.D.3 of this permit). The requirements of an Asset Management Program function to achieve the goals of effective performance, adequate funding, and adequate operator staffing and training. Asset management is a planning process for ensuring that optimum value is gained for each asset and that financial resources are available to rehabilitate and replace those assets when necessary. Asset management is centered on a framework of five (5) core elements: the current state of the assets; the required sustainable level of service; the assets critical to sustained performance; the minimum life-cycle costs; and the best long-term funding strategy.

a. Asset Management Program Requirements

On or before May 1, 2014 (received August 12, 2019), the permittee shall submit to the Department an Asset Management Plan for review and approval. An approvable Asset Management Plan shall contain a schedule for the development and implementation of an Asset Management Program that meets the requirements outlined below in 1) – 4). A copy of any Asset Management Program requirements already completed by the permittee should be submitted as part of the Asset Management Plan. Upon approval by the Department the permittee shall implement the Asset Management Plan. (The permittee may choose to include the Operation and Maintenance Manual required under Part II.C.14. of this permit as part of their Asset Management Program).

1) *Maintenance Staff.* The permittee shall provide an adequate staff to carry out the operation, maintenance, repair, and testing functions required to ensure compliance with the terms and conditions of this permit. The level of staffing needed shall be determined by taking into account the work involved in operating the sewer system and treatment works, planning for and conducting maintenance, and complying with this permit.

2) *Collection System Map.* The permittee shall complete a map of the sewer collection system it owns and operates. The map shall be of sufficient detail and at a scale to allow easy interpretation. The collection system information shown on the map shall be based on current conditions and shall be kept up-to-date and available for review by the Department. **Note: Items below referencing combined sewer systems are not applicable to separate sewer systems.** Such map(s) shall include but not be limited to the following:

a) all sanitary sewer lines and related manholes;

b) all combined sewer lines, related manholes, catch basins and CSO regulators;

c) all known or suspected connections between the sanitary sewer or combined sewer and storm drain systems;

d) all outfalls, including the treatment plant outfall(s), combined sewer treatment facility outfalls, untreated CSOs, and any known SSOs;

e) all pump stations and force mains;

f) the wastewater treatment facility(ies), including all treatment processes;

g) all surface waters (labeled);

h) other major appurtenances such as inverted siphons and air release valves;

Section A. Limitations and Monitoring Requirements

i) a numbering system which uniquely identifies manholes, catch basins, overflow points, regulators and outfalls;

j) the scale and a north arrow;

k) the pipe diameter, date of installation, type of material, distance between manholes, and the direction of flow; and

l) the manhole interior material, rim elevation (optional), and invert elevations.

3) *Inventory and assessment of fixed assets.* The permittee shall complete an inventory and assessment of operations-related fixed assets including portions of the collection system owned and operated by the permittee. Fixed assets are assets that are normally stationary (e.g., pumps, blowers, buildings, manholes, and sewer lines). The inventory and assessment shall be based on current conditions and shall be kept up-to-date and available for review by the Department.

a) The fixed asset inventory shall include the following:

(1) a brief description of the fixed asset, its design capacity (e.g., pump: 120 gallons per minute), its level of redundancy, and its tag number if applicable;

(2) the location of the fixed asset;

(3) the year the fixed asset was installed;

(4) the present condition of the fixed asset (e.g., excellent, good, fair, poor); and

(5) the current fixed asset (replacement) cost in dollars for year specified in accordance with approved schedules;

b) The fixed asset assessment shall include a “Business Risk Evaluation” that combines the probability of failure of the fixed asset and the criticality of the fixed asset, as follows:

(1) Rate the probability of failure of the fixed asset on a scale of 1-5 (low to high) using criteria such as maintenance history, failure history, and remaining percentage of useful life (or years remaining);

(2) Rate the criticality of the fixed asset on a scale of 1-5 (low to high) based on the consequence of failure versus the desired level of service for the facility; and

(3) Compute the Business Risk Factor of the fixed asset by multiplying the failure rating from (1) by the criticality rating from (2).

4) *Operation, Maintenance & Replacement (OM&R) Budget and Rate Sufficiency for the Sewer System and Treatment Works.* The permittee shall complete an assessment of its user rates and replacement fund, including the following:

a) beginning and end dates of fiscal year;

b) name of the department, committee, board, or other organization that sets rates for the operation of the sewer system and treatment works;

PART I

Section A. Limitations and Monitoring Requirements

c) amount in the permittee's replacement fund in dollars for year specified in accordance with approved schedules;

d) replacement fund strategy of all assets with a useful life of 20 years or less;

e) expenditures for maintenance, corrective action and capital improvement taken during the fiscal year;

f) OM&R budget for the fiscal year; and

g) rate calculation demonstrating sufficient revenues to cover OM&R expenses. If the rate calculation shows there are insufficient revenues to cover OM&R expenses, the permittee shall document, within three (3) fiscal years after submittal of the Asset Management Plan, that there is at least one rate adjustment that reduces the revenue gap by at least 10 percent. The permittee may prepare and submit an alternate plan, subject to Department approval, for addressing the revenue gap. The ultimate goal of the Asset Management Program is to ensure sufficient revenues to cover OM&R expenses.

b. Annual Reporting
The permittee shall develop a written report that summarizes asset management activities completed during the previous year and planned for the upcoming year. The written report shall be submitted to the Department on or before <u>July 31 of each year</u>. The written report shall include:

1) a description of the staffing levels maintained during the year;

2) a description of inspections and maintenance activities conducted and corrective actions taken during the previous year;

3) expenditures for collection system maintenance activities, treatment works maintenance activities, corrective actions, and capital improvement during the previous year;

4) a summary of assets/areas identified for inspection/action (including capital improvement) in the upcoming year based on the five (5) core elements and the Business Risk Factors computed in accordance with condition a.3)b)(3) above;

5) a maintenance budget and capital improvement budget for the upcoming year that take into account implementation of an effective Asset Management Program that meets the five (5) core elements;

6) an updated asset inventory based on the original submission; and

7) an updated OM&R budget with an updated rate schedule that includes the amount of insufficient revenues, if any.

Section A. Limitations and Monitoring Requirements

10. Discharge Monitoring Report – Quality Assurance Study Program

The permittee shall participate in the Discharge Monitoring Report – Quality Assurance (DMR-QA) Study Program. The purpose of the DMR-QA Study Program is to annually evaluate the proficiency of all in-house and/or contract laboratory(ies) that perform, on behalf of the facility authorized to discharge under this permit, the analytical testing required under this permit. In accordance with Section 308 of the Clean Water Act (33 U.S.C. § 1318); and R 323.2138 and R 323.2154 of Part 21, Wastewater Discharge Permits, promulgated under Part 31 of the NREPA, participation in the DMR-QA Study Program is required for all major facilities, and for minor facilities selected for participation by the Department.

Annually and in accordance with DMR-QA Study Program requirements and submittal due dates, the permittee shall submit to the Michigan DMR-QA Study Program state coordinator all documentation required by the DMR-QA Study. DMR-QA Study Program participation is required only for the analytes required under this permit and only when those analytes are also identified in the DMR-QA Study.

If the permitted facility's status as a major facility should change, participation in the DMR-QA Study Program may be reevaluated. Questions concerning participation in the DMR-QA Study Program should be directed to the Michigan DMR-QA Study Program state coordinator.

All forms and instructions required for participation in the DMR-QA Study Program, including submittal due dates and state coordinator contact information, can be found at https://www.epa.gov/compliance/discharge-monitoring-report-quality-assurance-study-program.

11. Continuous Monitoring

If continuous monitoring equipment is used and becomes temporarily inoperable, the permittee shall manually obtain a minimum of three (3) equally spaced grab samples/readings within each 24-hour period for the affected parameter(s). On such days, in the comment field on the Daily tab of the DMR, the permittee shall indicate "continuous monitoring system inoperable," the date on which the system is expected to become operable again, and the number of samples/readings obtained during each 24-hour period.

12. PFAS Data Reporting Requirements

The permittee shall submit the complete laboratory analysis of each PFAS sample. This may be in addition to required DMR reporting. Submittals shall be made via the MiEnviro Portal using the PFAS POTW Effluent Monitoring Report form. Until EPA Method 1633 is promulgated, the complete laboratory analysis of each PFAS sample shall constitute the first 28 analytes identified on the PFAS POTW Effluent Monitoring Report form. Following promulgation of EPA Method 1633, the complete laboratory analysis of each PFAS sample shall constitute all 40 analytes identified on the PFAS POTW Effluent Monitoring Report form.

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Section B. Storm Water Pollution Prevention

Section B. Storm Water Pollution Prevention is not required for this permit.

Section C. Industrial Waste Pretreatment Program

1. Michigan Industrial Pretreatment Program

a. The permittee shall implement the Michigan Industrial Pretreatment Program (MIPP) approved on July 23, 1985, and any subsequent modifications approved up to the issuance of this permit.

b. The permittee shall comply with R 323.2301 through R 323.2317 of the Michigan Administrative Code (Part 23 Rules) and the approved MIPP.

c. The permittee shall have the legal authority and necessary interjurisdictional agreements that provide the basis for the implementation and enforcement of the approved MIPP throughout the service area. The legal authority and necessary interjurisdictional agreements shall include, at a minimum, the authority to carry out the activities specified in R 323.2306(a).

d. The permittee shall develop procedures which describe, in sufficient detail, program commitments which enable implementation of the approved MIPP and the Part 23 Rules in accordance with R 323.2306(c).

e. The permittee shall establish an interjurisdictional agreement (or comparable document) with all tributary governmental jurisdictions. Each interjurisdictional agreement shall contain, at a minimum, the following:

1) identification of the agency responsible for the implementation and enforcement of the approved MIPP within the tributary governmental jurisdiction's boundaries; and

2) the provision of the legal authority which provides the basis for the implementation and enforcement of the approved MIPP within the tributary governmental jurisdiction's boundaries

f. The permittee shall prohibit discharges that:

1) cause, in whole or in part, the permittee's failure to comply with any condition of this permit or the NREPA;

2) restrict, in whole or in part, the permittee's management of biosolids;

3) cause, in whole or in part, operational problems at the treatment facility or in its collection system;

4) violate any of the general or specific prohibitions identified in R 323.2303(1) and (2);

5) violate categorical standards identified in R 323.2311; and

6) violate local limits established in accordance with R 323.2303(4).

g. The permittee shall maintain a list of its nondomestic users that meet the criteria of a significant industrial user as identified in R 323.2302(cc).

h. The permittee shall develop an enforcement response plan which describes, in sufficient detail, program commitments which will enable the enforcement of the approved MIPP and the Part 23 Rules in accordance with R 323.2306(g).

i. The Department may require modifications to the approved MIPP which are necessary to ensure compliance with the Part 23 Rules in accordance with R 323.2309.

PART I

Section C. Industrial Waste Pretreatment Program

j. The permittee shall not implement changes or modifications to the approved MIPP without notification to the Department.

k. The permittee shall maintain an adequate revenue structure and staffing level for effective implementation of the approved MIPP.

l. The permittee shall develop and maintain, for a minimum of three (3) years, all records and information necessary to determine nondomestic user compliance with the Part 23 Rules and the approved MIPP. This period of retention shall be extended during the course of any unresolved enforcement action or litigation regarding a nondomestic user or when requested by the Department or the United States Environmental Protection Agency. All of the aforementioned records and information shall be made available upon request for inspection and copying by the Department and the United States Environmental Protection Agency.

m. The permittee shall evaluate the approved MIPP for compliance with the Part 23 Rules and the prohibitions set forth in item f. above. Based upon this evaluation, the permittee shall propose to the Department all necessary changes or modifications to the approved MIPP no later than the next Industrial Pretreatment Program Annual Report due date (see item p. below).

n. The permittee shall develop and enforce local limits to implement the prohibitions set forth in item f. above. Local limits shall be based upon data representative of actual conditions demonstrated in a maximum allowable headworks loading analysis. A technical evaluation of whether local limits need to be developed for the following pollutants shall be submitted to the Department by January 1, 2025: Total Arsenic, Total Beryllium, Total Cadmium, Total Silver, Available Cyanide, and Total Benzidine. The submittal shall provide a technical evaluation of the basis upon which this determination was made which includes information regarding the maximum allowable headworks loading, collection system protection criteria, and worker health and safety, based upon data collected since the last local limits review.

o. The permittee is required under this permit and R 323.2303(4) of the Michigan Administrative Code to review and update their local limits when:

1) new pollutants are introduced;

2) new pollutants that were previously unevaluated are identified;

3) new water quality or biosolids standards are established or additional information becomes available about the nature of pollutants, such as removal rates and accumulation in biosolids; or

4) substantial increases of pollutants are proposed as required in the notification of new or increased uses in accordance with the provisions of 40 CFR 122.42.

p. On or before April 1 of each year, the permittee shall submit to the Department, as required by R 323.2310(8), an Industrial Pretreatment Program Annual Report on the status of program implementation and enforcement activities. The reporting period shall begin on January 1 and end on December 31. At a minimum, the Industrial Pretreatment Program Annual Report shall include:

1) the Pretreatment Program Reports data identified in Appendix A to 40 CFR Part 127 – NPDES Electronic Reporting;

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Section C. Industrial Waste Pretreatment Program

2) a summary of changes to the approved MIPP that have not been previously reported to the Department;

3) a summary of results of all the sampling and analyses performed of the wastewater treatment plant's influent, effluent, and biosolids conducted in accordance with approved methods during the reporting period. The summary shall include the monthly average, daily maximum, quantification level, and number of samples analyzed for each pollutant. At a minimum, the results of analyses for all locally limited parameters for at least one monitoring event that tests influent, effluent and biosolids during the reporting period shall be submitted with each report, unless otherwise required by the Department. Sample collection shall be at intervals sufficient to provide pollutant removal rates, unless the pollutant is not measurable; and

4) any other relevant information requested by the Department.

q. Survey of Significant Industrial Users
On or before January 1, 2025, the permittee shall submit to the Department for review, the results of a survey which identifies all significant industrial users of the waste collection system. For significant industrial users the following information shall be included:

1) the user's name and mailing address which shall include both the local facility address and the main office address, if different;

2) the principal enterprise(s) of the user; the product(s) produced and raw material(s) processed; the facility's production rate(s); and the Standard Industrial Classification (SIC) code(s);

3) the quantity of process wastewater discharged (in GPD) and an indication of whether the discharge is continuous or batch;

4) a description of any pretreatment provided prior to discharge to the waste collection system; and

5) a description of the wastewater characteristics in terms of pollutant parameters and concentrations.

PART I

Section D. Residuals Management Program

1. Residuals Management Program for Land Application of Biosolids

The permittee is authorized to land-apply bulk biosolids or prepare bulk biosolids for land application in accordance with the permittee's approved Residuals Management Program (RMP) approved on May 6, 2002, and approved modifications thereto, and the requirements established in R 323.2401 through R 323.2418 of the Michigan Administrative Code (Part 24 Rules). The approved RMP, and any approved modifications thereto, are enforceable requirements of this permit. Incineration, landfilling and other residual disposal activities shall be conducted in accordance with Part II.D.7. of this permit. The Part 24 Rules can be obtained via the internet at https://www.michigan.gov/egle/about/organization/water-resources/biosolids/laws-and-rules.

a. Annual Report
On or before October 30 of each year, the permittee shall submit an annual report to the Department for the previous fiscal year of October 1 through September 30. The report shall be submitted electronically via the Department's MiEnviro Portal system at https://mienviro.michigan.gov/ncore/. At a minimum, the report shall contain:

1) a certification that current residuals management practices are in accordance with the approved RMP, or a proposal for modification to the approved RMP; and

2) a completed Annual Report Form for Reporting Biosolids, available at https://mienviro.michigan.gov/ncore/.

b. Modifications to the Approved RMP
Prior to implementation of modifications to the RMP, the permittee shall submit proposed modifications to the Department for approval. The approved modification shall become effective upon the date of approval. Upon written notification, the Department may impose additional requirements and/or limitations to the approved RMP as necessary to protect public health and the environment from any adverse effect of a pollutant in the biosolids.

c. Record Keeping
Records required by the Part 24 Rules shall be kept for a minimum of five (5) years. However, the records documenting cumulative loading for sites subject to cumulative pollutant loading rates shall be kept as long as the site receives biosolids.

d. Contact Information
RMP-related submittals shall be made to the Department.

PART II

Section A. Definitions

Part II may include terms and /or conditions not applicable to discharges covered under this permit.

Acute toxic unit (TUA) means 100/LC50 where the LC50 is determined from a whole effluent toxicity (WET) test which produces a result that is statistically or graphically estimated to be lethal to 50% of the test organisms.

Annual monitoring frequency refers to a calendar year beginning on January 1 and ending on December 31. When required by this permit, an analytical result, reading, value or observation shall be reported for that period if a discharge occurs during that period.

Authorized public agency means a state, local, or county agency that is designated pursuant to the provisions of Section 9110 of Part 91, Soil and Sedimentation Control, of the NREPA, to implement soil erosion and sedimentation control requirements with regard to construction activities undertaken by that agency.

Best management practices (BMPs) means structural devices or nonstructural practices that are designed to prevent pollutants from entering into storm water, to direct the flow of storm water, or to treat polluted storm water.

Bioaccumulative chemical of concern (BCC) means a chemical which, upon entering the surface waters, by itself or as its toxic transformation product, accumulates in aquatic organisms by a human health bioaccumulation factor of more than 1000 after considering metabolism and other physiochemical properties that might enhance or inhibit bioaccumulation. The human health bioaccumulation factor shall be derived according to R 323.1057(5). Chemicals with half-lives of less than 8 weeks in the water column, sediment, and biota are not BCCs. The minimum bioaccumulation concentration factor (BAF) information needed to define an organic chemical as a BCC is either a field-measured BAF or a BAF derived using the biota-sediment accumulation factor (BSAF) methodology. The minimum BAF information needed to define an inorganic chemical as a BCC, including an organometal, is either a field-measured BAF or a laboratory-measured bioconcentration factor (BCF). The BCCs to which these rules apply are identified in Table 5 of R 323.1057 of the Water Quality Standards.

Biosolids are the solid, semisolid, or liquid residues generated during the treatment of sanitary sewage or domestic sewage in a treatment works. This includes, but is not limited to, scum or solids removed in primary, secondary, or advanced wastewater treatment processes and a derivative of the removed scum or solids.

Bulk biosolids means biosolids that are not sold or given away in a bag or other container for application to a lawn or home garden.

CAFO means concentrated animal feeding operation.

Certificate of Coverage (COC) is a document, issued by the Department, which authorizes a discharge under a general permit.

Chronic toxic unit (TUC) means 100/MATC or 100/IC25, where the maximum acceptable toxicant concentration (MATC) and IC25 are expressed as a percent effluent in the test medium.

Class B biosolids refers to material that has met the Class B pathogen reduction requirements or equivalent treatment by a Process to Significantly Reduce Pathogens (PSRP) in accordance with the Part 24 Rules, Land Application of Biosolids, promulgated under Part 31 of the NREPA. Processes include aerobic digestion, composting, anaerobic digestion, lime stabilization and air drying.

Combined sewer system is a sewer system in which storm water runoff is combined with sanitary wastes.

PART II

Section A. Definitions

Composite sample is a sample collected over time, either by continuous sampling or by mixing discrete samples. A composite sample represents the average wastewater characteristics present during the compositing period. Various methods for compositing are available and are based on either time or flow-proportioning, the choice of which will depend on the permit requirements.

Continuous monitoring refers to sampling/readings that occur at regular and consistent intervals throughout a 24-hour period and at a frequency sufficient to capture data that are representative of the discharge. The maximum acceptable interval between samples/readings shall be one (1) hour.

Daily concentration
FOR PARAMETERS OTHER THAN pH, DISSOLVED OXYGEN, TEMPERATURE, AND CONDUCTIVITY – Daily concentration is the sum of the concentrations of the individual samples of a parameter taken within a calendar day divided by the number of samples taken within that calendar day. The daily concentration will be used to determine compliance with any maximum and minimum daily concentration limitations. For guidance and examples showing how to report and perform calculations using results below quantification levels, see the document entitled “Reporting Results Below Quantification,” available at https://www.michigan.gov/-/media/Project/Websites/egle/Documents/Programs/WRD/MiEnviro/results-below-quantification.pdf.

FOR pH, DISSOLVED OXYGEN, TEMPERATURE, AND CONDUCTIVITY – The daily concentration used to determine compliance with maximum daily pH, temperature, and conductivity limitations is the highest pH, temperature, and conductivity readings obtained within a calendar day. The daily concentration used to determine compliance with minimum daily pH and dissolved oxygen limitations is the lowest pH and dissolved oxygen readings obtained within a calendar day.

Daily loading is the total discharge by weight of a parameter discharged during any calendar day. This value is calculated by multiplying the daily concentration by the total daily flow and by the appropriate conversion factor. The daily loading will be used to determine compliance with any maximum daily loading limitations. When required by the permit, report the maximum calculated daily loading for the month in the “MAXIMUM” column under “QUANTITY OR LOADING” on the DMRs.

Daily monitoring frequency refers to a 24-hour day. When required by this permit, an analytical result, reading, value or observation shall be reported for that period if a discharge occurs during that period.

Department means the Michigan Department of Environment, Great Lakes, and Energy.

Detection level means the lowest concentration or amount of the target analyte that can be determined to be different from zero by a single measurement at a stated level of probability.

Discharge means the addition of any waste, waste effluent, wastewater, pollutant, or any combination thereof to any surface water of the state.

EC_{50} means a statistically or graphically estimated concentration that is expected to cause 1 or more specified effects in 50% of a group of organisms under specified conditions.

Fecal coliform bacteria monthly
FOR WWSLs THAT COLLECT AND STORE WASTEWATER AND ARE AUTHORIZED TO DISCHARGE ONLY IN THE SPRING AND/OR FALL ON AN INTERMITTENT BASIS – Fecal coliform bacteria monthly is the geometric mean of all daily concentrations determined during a discharge event. Days on which no daily concentration is determined shall not be used to determine the calculated monthly value. The calculated monthly value will be used to determine compliance with the maximum monthly fecal coliform bacteria limitations. When required by the permit, report the calculated monthly value in the “AVERAGE” column under “QUALITY OR CONCENTRATION” on the DMR. If the period in which the discharge event occurred was partially in each of two months, the calculated monthly value shall be reported on the DMR of the month in which the last day of discharge occurred.

PART II

Section A. Definitions

FOR ALL OTHER DISCHARGES – Fecal coliform bacteria monthly is the geometric mean of all daily concentrations determined during a reporting month. Days on which no daily concentration is determined shall not be used to determine the calculated monthly value. The calculated monthly value will be used to determine compliance with the maximum monthly fecal coliform bacteria limitations. When required by the permit, report the calculated monthly value in the "AVERAGE" column under "QUALITY OR CONCENTRATION" on the DMR.

Fecal coliform bacteria 7-day
FOR WWSLs THAT COLLECT AND STORE WASTEWATER AND ARE AUTHORIZED TO DISCHARGE ONLY IN THE SPRING AND/OR FALL ON AN INTERMITTENT BASIS – Fecal coliform bacteria 7-day is the geometric mean of the daily concentrations determined during any 7 consecutive days of discharge during a discharge event. If the number of daily concentrations determined during the discharge event is less than 7 days, the number of actual daily concentrations determined shall be used for the calculation. Days on which no daily concentration is determined shall not be used to determine the value. The calculated 7-day value will be used to determine compliance with the maximum 7-day fecal coliform bacteria limitations. When required by the permit, report the maximum calculated 7-day geometric mean value for the month in the "MAXIMUM" column under "QUALITY OR CONCENTRATION" on the DMRs. If the 7-day period was partially in each of two months, the value shall be reported on the DMR of the month in which the last day of discharge occurred.

FOR ALL OTHER DISCHARGES – Fecal coliform bacteria 7-day is the geometric mean of the daily concentrations determined during any 7 consecutive days in a reporting month. If the number of daily concentrations determined is less than 7, the actual number of daily concentrations determined shall be used for the calculation. Days on which no daily concentration is determined shall not be used to determine the value. The calculated 7-day value will be used to determine compliance with the maximum 7-day fecal coliform bacteria limitations. When required by the permit, report the maximum calculated 7-day geometric mean for the month in the "MAXIMUM" column under "QUALITY OR CONCENTRATION" on the DMRs. The first calculation shall be made on day 7 of the reporting month, and the last calculation shall be made on the last day of the reporting month.

Flow-proportioned composite sample is a composite sample in which either a) the volume of each portion of the composite is proportional to the effluent flow rate at the time that portion is obtained; or b) a constant sample volume is obtained at varying time intervals proportional to the effluent flow rate.

General permit means an NPDES permit authorizing a category of similar discharges.

Geometric mean is the average of the logarithmic values of a base 10 data set, converted back to a base 10 number.

Grab sample is a single sample taken at neither a set time nor flow.

IC_{25} means the toxicant concentration that would cause a 25% reduction in a nonquantal biological measurement for the test population.

Illicit connection means a physical connection to a municipal separate storm sewer system that primarily conveys non-storm water discharges other than uncontaminated groundwater into the storm sewer; or a physical connection not authorized or permitted by the local authority, where a local authority requires authorization or a permit for physical connections.

Illicit discharge means any discharge to, or seepage into, a municipal separate storm sewer system that is not composed entirely of storm water or uncontaminated groundwater. Illicit discharges include non-storm water discharges through pipes or other physical connections; dumping of motor vehicle fluids, household hazardous wastes, domestic animal wastes, or litter; collection and intentional dumping of grass clippings or leaf litter; or unauthorized discharges of sewage, industrial waste, restaurant wastes, or any other non-storm water waste directly into a separate storm sewer.

PART II

Section A. Definitions

Individual permit means a site-specific NPDES permit.

Inlet means a catch basin, roof drain, conduit, drain tile, retention pond riser pipe, sump pump, or other point where storm water or wastewater enters into a closed conveyance system prior to discharge off site or into waters of the state.

Interference is a discharge which, alone or in conjunction with a discharge or discharges from other sources, both: 1) inhibits or disrupts a POTW, its treatment processes or operations, or its sludge processes, use or disposal; and 2) therefore, is a cause of a violation of any requirement of the POTW's NPDES permit (including an increase in the magnitude or duration of a violation) or, of the prevention of sewage sludge use or disposal in compliance with the following statutory provisions and regulations or permits issued thereunder (or more stringent state or local regulations): Section 405 of the Clean Water Act, the Solid Waste Disposal Act (SWDA) (including Title II, more commonly referred to as the Resource Conservation and Recovery Act (RCRA), and including state regulations contained in any state sludge management plan prepared pursuant to Subtitle D of the SWDA), the Clean Air Act, the Toxic Substances Control Act, and the Marine Protection, Research and Sanctuaries Act. [This definition does not apply to sample matrix interference].

Land application means spraying or spreading biosolids or a biosolids derivative onto the land surface, injecting below the land surface, or incorporating into the soil so that the biosolids or biosolids derivative can either condition the soil or fertilize crops or vegetation grown in the soil.

LC_{50} means a statistically or graphically estimated concentration that is expected to be lethal to 50% of a group of organisms under specified conditions.

Maximum acceptable toxicant concentration (MATC) means the concentration obtained by calculating the geometric mean of the lower and upper chronic limits from a chronic test. A lower chronic limit is the highest tested concentration that did not cause the occurrence of a specific adverse effect. An upper chronic limit is the lowest tested concentration which did cause the occurrence of a specific adverse effect and above which all tested concentrations caused such an occurrence.

Maximum extent practicable means implementation of best management practices by a public body to comply with an approved storm water management program as required by a national permit for a municipal separate storm sewer system, in a manner that is environmentally beneficial, technically feasible, and within the public body’s legal authority.

MBTU/hr means million British Thermal Units per hour.

MGD means million gallons per day.

Monthly concentration is the sum of the daily concentrations determined during a reporting period divided by the number of daily concentrations determined. The calculated monthly concentration will be used to determine compliance with any maximum monthly concentration limitations. Days with no discharge shall not be used to determine the value. When required by the permit, report the calculated monthly concentration in the “AVERAGE” column under “QUALITY OR CONCENTRATION” on the DMR.

For minimum percent removal requirements, the monthly influent concentration and the monthly effluent concentration shall be determined. The calculated monthly percent removal, which is equal to 100 times the quantity [1 minus the quantity (monthly effluent concentration divided by the monthly influent concentration)], shall be reported in the "MINIMUM" column under "QUALITY OR CONCENTRATION" on the DMRs.

PART II

Section A. Definitions

Monthly loading is the sum of the daily loadings of a parameter divided by the number of daily loadings determined during a reporting period. The calculated monthly loading will be used to determine compliance with any maximum monthly loading limitations. Days with no discharge shall not be used to determine the value. When required by the permit, report the calculated monthly loading in the "AVERAGE" column under "QUANTITY OR LOADING" on the DMR.

Monthly monitoring frequency refers to a calendar month. When required by this permit, an analytical result, reading, value or observation shall be reported for that period if a discharge occurs during that period.

Municipal separate storm sewer means a conveyance or system of conveyances designed or used for collecting or conveying storm water which is not a combined sewer and which is not part of a POTW as defined in the Code of Federal Regulations at 40 CFR 122.2.

Municipal separate storm sewer system (MS4) means all separate storm sewers that are owned or operated by the United States, a state, city, village, township, county, district, association, or other public body created by or pursuant to state law, having jurisdiction over disposal of sewage, industrial wastes, storm water, or other wastes, including special districts under state law, such as a sewer district, flood control district, or drainage district, or similar entity, or a designated or approved management agency under Section 208 of the Clean Water Act that discharges to the waters of the state. This term includes systems similar to separate storm sewer systems in municipalities, such as systems at military bases, large hospital or prison complexes, and highways and other thoroughfares. The term does not include separate storm sewers in very discrete areas, such as individual buildings.

National Pretreatment Standards are the regulations promulgated by or to be promulgated by the Federal Environmental Protection Agency pursuant to Section 307(b) and (c) of the Clean Water Act. The standards establish nationwide limits for specific industrial categories for discharge to a POTW.

No observed adverse effect level (NOAEL) means the highest tested dose or concentration of a substance which results in no observed adverse effect in exposed test organisms where higher doses or concentrations result in an adverse effect.

Noncontact cooling water is water used for cooling which does not come into direct contact with any raw material, intermediate product, by-product, waste product or finished product.

Nondomestic user is any discharger to a POTW that discharges wastes other than or in addition to water-carried wastes from toilet, kitchen, laundry, bathing or other facilities used for household purposes.

Nonstructural controls are practices or procedures implemented by employees at a facility to manage storm water or to prevent contamination of storm water.

NPDES means National Pollutant Discharge Elimination System.

Outfall is the location at which a point source discharge first enters a surface water of the state.

Part 91 agency means an agency that is designated by a county board of commissioners pursuant to the provisions of Section 9105 of Part 91 of the NREPA; an agency that is designated by a city, village, or township in accordance with the provisions of Section 9106 of Part 91 of the NREPA; or the Department for soil erosion and sedimentation control activities under Part 615, Supervisor of Wells; Part 631, Reclamation of Mining Lands; or Part 632, Nonferrous Metallic Mineral Mining, of the NREPA, pursuant to the provisions of Section 9115 of Part 91 of the NREPA.

Part 91 permit means a soil erosion and sedimentation control permit issued by a Part 91 agency pursuant to the provisions of Part 91 of the NREPA.

PART II

Section A. Definitions

Partially treated sewage is any sewage, sewage and storm water, or sewage and wastewater, from domestic or industrial sources that is treated to a level less than that required by the permittee's NPDES permit, or that is not treated to national secondary treatment standards for wastewater, including discharges to surface waters from retention treatment facilities.

PFAS means perfluoroalkyl and polyfluoroalkyl substances.

Point of discharge is the location of a point source discharge where storm water is discharged directly into a separate storm sewer system.

Point source discharge means a discharge from any discernible, confined, discrete conveyance, including but not limited to any pipe, ditch, channel, tunnel, conduit, well, discrete fissure, container, or rolling stock. Changing the surface of land or establishing grading patterns on land will result in a point source discharge where the runoff from the site is ultimately discharged to waters of the state.

Polluting material means any material, in solid or liquid form, identified as a polluting material under the Part 5 Rules, Spillage of Oil and Polluting Materials, promulgated under Part 31 of the NREPA (R 324.2001 through R 324.2009 of the Michigan Administrative Code).

POTW is a publicly owned treatment work.

Predevelopment is the last land use prior to the planned new development or redevelopment.

Pretreatment is reducing the amount of pollutants, eliminating pollutants, or altering the nature of pollutant properties to a less harmful state prior to discharge into a public sewer. The reduction or alteration can be by physical, chemical, or biological processes, process changes, or by other means. Dilution is not considered pretreatment unless expressly authorized by an applicable National Pretreatment Standard for a particular industrial category.

Public (as used in the MS4 individual permit) means all persons who potentially could affect the authorized storm water discharges, including, but not limited to, residents, visitors to the area, public employees, businesses, industries, and construction contractors and developers.

Public body means the United States; the state of Michigan; a city, village, township, county, school district, public college or university, or single-purpose governmental agency; or any other body which is created by federal or state statute or law.

Qualified Personnel means an individual who meets qualifications acceptable to the Department and who is authorized by an Industrial Storm Water Certified Operator to collect the storm water sample.

Qualifying storm event means a storm event causing greater than 0.1 inch of rainfall and occurring at least 72 hours after the previous measurable storm event that also caused greater than 0.1 inch of rainfall. Upon request, the Department may approve an alternate definition meeting the condition of a qualifying storm event.

Quantification level means the measurement of the concentration of a contaminant obtained by using a specified laboratory procedure calculated at a specified concentration above the detection level. It is considered the lowest concentration at which a particular contaminant can be quantitatively measured using a specified laboratory procedure for monitoring of the contaminant.

Quarterly monitoring frequency refers to a three-month period, defined as January through March, April through June, July through September, and October through December (or otherwise defined in the permit). When required by this permit, an analytical result, reading, value or observation shall be reported for that period if a discharge occurs during that period.

PART II

Section A. Definitions

Regional Administrator is the Region 5 Administrator, U.S. EPA, located at R-19J, 77 W. Jackson Blvd., Chicago, Illinois 60604.

Regulated area means the permittee's urbanized area, where urbanized area is defined as a place and its adjacent densely populated territory that together have a minimum population of 50,000 people as defined by the United States Bureau of the Census and as determined by the latest available decennial census.

Secondary containment structure means a unit, other than the primary container, in which significant materials are packaged or held, which is required by state or federal law to prevent the escape of significant materials by gravity into sewers, drains, or otherwise directly or indirectly into any sewer system or to the surface waters or groundwaters of the state.

Separate storm sewer system means a system of drainage, including, but not limited to, roads, catch basins, curbs, gutters, parking lots, ditches, conduits, pumping devices, or man-made channels, which is not a combined sewer where storm water mixes with sanitary wastes, and is not part of a POTW.

Significant industrial user is a nondomestic user that: 1) is subject to Categorical Pretreatment Standards under 40 CFR 403.6 and 40 CFR Chapter I, Subchapter N; or 2) discharges an average of 25,000 gallons per day or more of process wastewater to a POTW (excluding sanitary, noncontact cooling and boiler blowdown wastewater); contributes a process waste stream which makes up five (5) percent or more of the average dry weather hydraulic or organic capacity of the POTW treatment plant; or is designated as such by the permittee as defined in 40 CFR 403.12(a) on the basis that the industrial user has a reasonable potential for adversely affecting the POTW's treatment plant operation or violating any pretreatment standard or requirement (in accordance with 40 CFR 403.8(f)(6)).

Significant materials means any material which could degrade or impair water quality, including but not limited to: raw materials; fuels; solvents, detergents, and plastic pellets; finished materials such as metallic products; hazardous substances designated under Section 101(14) of the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) (see 40 CFR 372.65); any chemical the facility is required to report pursuant to Section 313 of Emergency Planning and Community Right-to-Know Act (EPCRA); polluting materials as identified under the Part 5 Rules (R 324.2001 through R 324.2009 of the Michigan Administrative Code); Hazardous Wastes as defined in Part 111, Hazardous Waste Management, of the NREPA; fertilizers; pesticides; and waste products such as ashes, slag, and sludge that have the potential to be released with storm water discharges.

Significant spills and significant leaks means any release of a polluting material reportable under the Part 5 Rules (R 324.2001 through R 324.2009 of the Michigan Administrative Code).

Special-use area means storm water discharges for which the Department has determined that additional monitoring is needed from: secondary containment structures required by state or federal law; lands on Michigan's List of Sites of Environmental Contamination pursuant to Part 201, Environmental Remediation, of the NREPA; and/or areas with other activities that may contribute pollutants to the storm water.

Stoichiometric means the quantity of a reagent calculated to be necessary and sufficient for a given chemical reaction.

Storm water means storm water runoff, snowmelt runoff, surface runoff and drainage, and non-storm water included under the conditions of this permit.

Storm water discharge point is the location where the point source discharge of storm water is directed to surface waters of the state or to a separate storm sewer. It includes the location of all point source discharges where storm water exits the facility, including outfalls which discharge directly to surface waters of the state, and points of discharge which discharge directly into separate storm sewer systems.

PART II

Section A. Definitions

Structural controls are physical features or structures used at a facility to manage or treat storm water.

SWPPP means the Storm Water Pollution Prevention Plan prepared in accordance with this permit.

Tier I value means a value for aquatic life, human health or wildlife calculated under R 323.1057 of the Water Quality Standards using a tier I toxicity database.

Tier II value means a value for aquatic life, human health or wildlife calculated under R 323.1057 of the Water Quality Standards using a tier II toxicity database.

Total maximum daily loads (TMDLs) are required by the Clean Water Act for waterbodies that do not meet water quality standards. TMDLs represent the maximum daily load of a pollutant that a waterbody can assimilate and meet water quality standards, and an allocation of that load among point sources, nonpoint sources, and a margin of safety.

Toxicity reduction evaluation (TRE) means a site-specific study conducted in a stepwise process designed to identify the causative agents of effluent toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in effluent toxicity.

Water Quality Standards means the Part 4 Water Quality Standards promulgated pursuant to Part 31 of the NREPA, being R 323.1041 through R 323.1117 of the Michigan Administrative Code.

Weekly monitoring frequency refers to a calendar week which begins on Sunday and ends on Saturday. When required by this permit, an analytical result, reading, value, or observation shall be reported for that period if a discharge occurs during that period. If the calendar week begins in one month and ends in the following month, the analytical result, reading, value, or observation shall be reported in the month in which monitoring was conducted.

WWSL is a wastewater stabilization lagoon.

WWSL discharge event is a discrete occurrence during which effluent is discharged to the surface water up to 10 days of a consecutive 14-day period.

3-portion composite sample is a sample consisting of three equal-volume grab samples collected at equal intervals over an 8-hour period.

7-day concentration
FOR WWSLs THAT COLLECT AND STORE WASTEWATER AND ARE AUTHORIZED TO DISCHARGE ONLY IN THE SPRING AND/OR FALL ON AN INTERMITTENT BASIS – The 7-day concentration is the sum of the daily concentrations determined during any 7 consecutive days of discharge during a WWSL discharge event divided by the number of daily concentrations determined. If the number of daily concentrations determined during the WWSL discharge event is less than 7 days, the number of actual daily concentrations determined shall be used for the calculation. The calculated 7-day concentration will be used to determine compliance with any maximum 7-day concentration limitations. When required by the permit, report the maximum calculated 7-day concentration for the WWSL discharge event in the “MAXIMUM” column under “QUALITY OR CONCENTRATION” on the DMR. If the WWSL discharge event was partially in each of two months, the value shall be reported on the DMR of the month in which the last day of discharge occurred.

FOR ALL OTHER DISCHARGES – The 7-day concentration is the sum of the daily concentrations determined during any 7 consecutive days in a reporting month divided by the number of daily concentrations determined. If the number of daily concentrations determined is less than 7, the actual number of daily concentrations determined shall be used for the calculation. The calculated 7-day concentration will be used to determine compliance with any maximum 7-day concentration limitations in the reporting month. When required by the permit, report the maximum calculated 7-day concentration for the month in the “MAXIMUM” column under

Section A. Definitions

"QUALITY OR CONCENTRATION" on the DMR. The first 7-day calculation shall be made on day 7 of the reporting month, and the last calculation shall be made on the last day of the reporting month.

7-day loading
FOR WWSLs THAT COLLECT AND STORE WASTEWATER AND ARE AUTHORIZED TO DISCHARGE ONLY IN THE SPRING AND/OR FALL ON AN INTERMITTENT BASIS – The 7-day loading is the sum of the daily loadings determined during any 7 consecutive days of discharge during a WWSL discharge event divided by the number of daily loadings determined. If the number of daily loadings determined during the WWSL discharge event is less than 7 days, the number of actual daily loadings determined shall be used for the calculation. The calculated 7-day loading will be used to determine compliance with any maximum 7-day loading limitations. When required by the permit, report the maximum calculated 7-day loading for the WWSL discharge event in the "MAXIMUM" column under "QUANTITY OR LOADING" on the DMR. If the WWSL discharge event was partially in each of two months, the value shall be reported on the DMR of the month in which the last day of discharge occurred.

FOR ALL OTHER DISCHARGES – The 7-day loading is the sum of the daily loadings determined during any 7 consecutive days in a reporting month divided by the number of daily loadings determined. If the number of daily loadings determined is less than 7, the actual number of daily loadings determined shall be used for the calculation. The calculated 7-day loading will be used to determine compliance with any maximum 7-day loading limitations in the reporting month. When required by the permit, report the maximum calculated 7-day loading for the month in the "MAXIMUM" column under "QUANTITY OR LOADING" on the DMR. The first 7-day calculation shall be made on day 7 of the reporting month, and the last calculation shall be made on the last day of the reporting month.

24-hour composite sample is a flow-proportioned composite sample consisting of hourly or more frequent portions that are taken over a 24-hour period and in which the volume of each portion is proportional to the discharge flow rate at the time that portion is taken. A time-proportioned composite sample may be used upon approval from the Department if the permittee demonstrates it is representative of the discharge.

PART II

Section B. Monitoring Procedures

1. Representative Samples

Samples and measurements taken as required herein shall be representative of the volume and nature of the monitored discharge.

2. Test Procedures

Test procedures for the analysis of pollutants shall conform to regulations promulgated pursuant to Section 304(h) of the Clean Water Act (40 CFR Part 136 – Guidelines Establishing Test Procedures for the Analysis of Pollutants), unless specified otherwise in this permit. **Test procedures used shall be sufficiently sensitive to determine compliance with applicable effluent limitations**. For lists of approved test methods, go to https://www.epa.gov/cwa-methods. Requests to use test procedures not promulgated under 40 CFR Part 136 for pollutant monitoring required by this permit shall be made in accordance with the Alternate Test Procedures regulations specified in 40 CFR 136.4. These requests shall be submitted to the Manager of the Permits Section, Water Resources Division, Michigan Department of Environment, Great Lakes, and Energy, P.O. Box 30458, Lansing, Michigan, 48909-7958. The permittee may use such procedures upon approval.

The permittee shall periodically calibrate and perform maintenance procedures on all analytical instrumentation at intervals to ensure accuracy of measurements. The calibration and maintenance shall be performed as part of the permittee's laboratory Quality Assurance/Quality Control program.

3. Instrumentation

The permittee shall periodically calibrate and perform maintenance procedures on all monitoring instrumentation at intervals to ensure accuracy of measurements.

4. Recording Results

For each measurement or sample taken pursuant to the requirements of this permit, the permittee shall record the following information: 1) the exact place, date, and time of measurement or sampling; 2) the person(s) who performed the measurement or sample collection; 3) the dates the analyses were performed; 4) the person(s) who performed the analyses; 5) the analytical techniques or methods used; 6) the date of and person responsible for equipment calibration; and 7) the results of all required analyses.

5. Records Retention

All records and information resulting from the monitoring activities required by this permit, including all records of analyses performed, calibration and maintenance of instrumentation, and recordings from continuous monitoring instrumentation, shall be retained for a minimum of three (3) years, or longer if requested by the Regional Administrator or the Department.

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Section C. Reporting Requirements

1. Start-Up Notification
The permittee shall notify the Department of start-up if one of the following conditions applies and in accordance with the applicable condition:

a. Non-CAFOs

1) **If this is an individual permit** and the permittee will not discharge during the first 60 days following the effective date of this permit, the permittee shall notify the Department via MiEnviro Portal within 14 days following the effective date of this permit, and then again 60 days prior to commencement of the discharge.

2) **If this is a general permit** and the permittee will not discharge during the first 60 days following the effective date of the Certificate of Coverage (COC) issued under this general permit, the permittee shall notify the Department via MiEnviro Portal within 14 days following the effective date of the COC, and then again 60 days prior to commencement of the discharge.

b. CAFOs

1) **If this is an individual permit** and the permittee will not populate with animals during the first 60 days following the effective date of this permit, the permittee shall notify the Department via MiEnviro Portal within 14 days following the effective date of this permit, and then again 60 days prior to populating with animals.

2) **If this is a general permit** and the permittee will not populate with animals during 60 days following the effective date of the Certificate of Coverage (COC) issued under this general permit, the permittee shall notify the Department via MiEnviro Portal within 14 days following the effective date of the COC, and then again 60 days prior to populating with animals.

2. Submittal Requirements for Self-Monitoring Data
Part 31 of the NREPA (specifically Section 324.3110(7)); and R 323.2155(2) of Part 21, Wastewater Discharge Permits, promulgated under Part 31 of the NREPA, allow the Department to specify the forms to be utilized for reporting the required self-monitoring data. Unless instructed on the effluent limitations page to conduct "Retained Self-Monitoring," the permittee shall submit self-monitoring data via the Department's MiEnviro Portal system.

The permittee shall utilize the information provided on the MiEnviro Portal website, located at https://mienviro.michigan.gov/ncore/, to access and submit the electronic forms. Both monthly summary and daily data shall be submitted to the Department no later than the 20th day of the month following each month of the authorized discharge period(s). The permittee may be allowed to submit the electronic forms after this date if the Department has granted an extension to the submittal date.

3. Retained Self-Monitoring Requirements
If instructed on the effluent limits page (or otherwise authorized by the Department in accordance with the provisions of this permit) to conduct retained self-monitoring, the permittee shall maintain a year-to-date log of retained self-monitoring results and, upon request, provide such log for inspection to the staff of the Department. Retained self-monitoring results are public information and shall be promptly provided to the public upon request.

The permittee shall certify, in writing, to the Department, on or before January 10 (April 1 for animal feeding operation facilities) of each year, that: 1) all retained self-monitoring requirements have been complied with and a year-to-date log has been maintained; and 2) the application on which this permit is based still accurately describes the discharge. With this annual certification, the permittee shall submit a summary of the previous

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Section C. Reporting Requirements

year's monitoring data. The summary shall include maximum values for samples to be reported as daily maximums and/or monthly maximums and minimum values for any daily minimum samples.

Retained self-monitoring may be denied to a permittee by notification in writing from the Department. In such cases, the permittee shall submit self-monitoring data in accordance with Part II.C.2., above. Such a denial may be rescinded by the Department upon written notification to the permittee. Reissuance or modification of this permit or reissuance or modification of an individual permittee's authorization to discharge shall not affect previous approval or denial for retained self-monitoring unless the Department provides notification in writing to the permittee.

4. Additional Monitoring by Permittee

If the permittee monitors any pollutant at the location(s) designated herein more frequently than required by this permit, using approved analytical methods as specified above, the results of such monitoring shall be included in the calculation and reporting of the values required in the Discharge Monitoring Report. Such increased frequency shall also be indicated.

Monitoring required pursuant to Part 41 of the NREPA or Rule 35 of the Mobile Home Park Commission Act, 1987 PA 96, as amended, for assurance of proper facility operation, shall be submitted as required by the Department.

5. Compliance Dates Notification

Within 14 days of every compliance date specified in this permit, the permittee shall submit a written notification to the Department via MiEnviro Portal (https://mienviro.michigan.gov/ncore/) indicating whether or not the particular requirement was accomplished. If the requirement was not accomplished, the notification shall include an explanation of the failure to accomplish the requirement, actions taken or planned by the permittee to correct the situation, and an estimate of when the requirement will be accomplished. If a written report is required to be submitted by a specified date and the permittee accomplishes this, a separate written notification is not required.

6. Noncompliance Notification

Compliance with all applicable requirements set forth in the Clean Water Act, Parts 31 and 41 of the NREPA, and related regulations and rules is required. All instances of noncompliance shall be reported as follows:

a. 24-Hour Reporting
Any noncompliance which may endanger health or the environment (including maximum and/or minimum daily concentration discharge limitation exceedances) shall be reported, verbally, within 24 hours from the time the permittee becomes aware of the noncompliance by calling the Department at the number indicated on the second page of this permit (or, if this is a general permit, on the COC). A written submission shall also be provided via MiEnviro Portal (https://mienviro.michigan.gov/ncore/) within five (5) days.

b. Other Reporting
The permittee shall report, in writing via MiEnviro Portal (https://mienviro.michigan.gov/ncore/), all other instances of noncompliance not described in a. above at the time monitoring reports are submitted; or, in the case of retained self-monitoring, within five (5) days from the time the permittee becomes aware of the noncompliance.

Reporting shall include: 1) a description of the discharge and cause of noncompliance; and 2) the period of noncompliance, including exact dates and times, or, if not yet corrected, the anticipated time the noncompliance is expected to continue, and the steps taken to reduce, eliminate and prevent recurrence of the noncomplying discharge.

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Section C. Reporting Requirements

7. Spill Notification

The permittee shall immediately report any release of any polluting material which occurs to the surface waters or groundwaters of the state, unless the permittee has determined that the release is not in excess of the threshold reporting quantities specified in the Part 5 Rules (R 324.2001 through R 324.2009 of the Michigan Administrative Code), by calling the Department at the number indicated on the second page of this permit (or, if this is a general permit, on the COC); or, if the notice is provided after regular working hours, by calling the Department's 24-hour Pollution Emergency Alerting System telephone number, 1-800-292-4706.

Within 10 days of the release, the permittee shall submit to the Department via MiEnviro Portal (https://mienviro.michigan.gov/ncore/) a full written explanation as to the cause of the release, the discovery of the release, response measures (clean-up and/or recovery) taken, and preventive measures taken or a schedule for completion of measures to be taken to prevent reoccurrence of similar releases.

8. Upset Noncompliance Notification

If a process "upset" (defined as an exceptional incident in which there is unintentional and temporary noncompliance with technology-based permit effluent limitations because of factors beyond the reasonable control of the permittee) has occurred, the permittee who wishes to establish the affirmative defense of upset shall notify the Department by telephone within 24 hours of becoming aware of such conditions; and within five (5) days, provide in writing, the following information:

a. that an upset occurred and that the permittee can identify the specific cause(s) of the upset;

b. that the permitted wastewater treatment facility was, at the time, being properly operated and maintained (note that an upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation); and

c. that the permittee has specified and taken action on all responsible steps to minimize or correct any adverse impact in the environment resulting from noncompliance with this permit.

No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review.

In any enforcement proceedings, the permittee, seeking to establish the occurrence of an upset, has the burden of proof.

9. Bypass Prohibition and Notification

a. Bypass Prohibition
Bypass is prohibited, and the Department may take an enforcement action, unless:

1) bypass was unavoidable to prevent loss of life, personal injury, or severe property damage;

2) there were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate backup equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass; and

3) the permittee submitted notices as required under 9.b. or 9.c. below.

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Section C. Reporting Requirements

b. Notice of Anticipated Bypass
If the permittee knows in advance of the need for a bypass, the permittee shall submit written notification to the Department before the anticipated date of the bypass. This notification shall be submitted at least 10 days before the date of the bypass; however, the Department will accept fewer than 10 days advance notice if adequate explanation for this is provided. The notification shall provide information about the anticipated bypass as required by the Department. The Department may approve an anticipated bypass, after considering its adverse effects, if it will meet the three (3) conditions specified in a. above.

c. Notice of Unanticipated Bypass
As soon as possible but no later than 24 hours from the time the permittee becomes aware of the unanticipated bypass, the permittee shall notify the Department by calling the number indicated on the second page of this permit (or, if this is a general permit, on the COC); or, if notification is provided after regular working hours, call the Department's 24-hour Pollution Emergency Alerting System telephone number, 1-800-292-4706.

d. Written Report of Bypass
A written submission shall be provided within five (5) working days of commencing any bypass to the Department, and at additional times as directed by the Department. The written submission shall contain a description of the bypass and its cause; the period of bypass, including exact dates and times, and if the bypass has not been corrected, the anticipated time it is expected to continue; steps taken or planned to reduce, eliminate, and prevent reoccurrence of the bypass; and other information as required by the Department.

e. Bypass Not Exceeding Limitations
The permittee may allow any bypass to occur which does not cause effluent limitations to be exceeded, but only if it also is for essential maintenance to ensure efficient operation. These bypasses are not subject to the provisions of 9.a., 9.b., 9.c., and 9.d., above. This provision does not relieve the permittee of any notification responsibilities under Part II.C.11. of this permit.

f. Definitions

1) Bypass means the intentional diversion of waste streams from any portion of a treatment facility.

2) Severe property damage means substantial physical damage to property, damage to the treatment facilities which causes them to become inoperable, or substantial and permanent loss of natural resources which can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production.

10. Bioaccumulative Chemicals of Concern (BCC)
Consistent with the requirements of R 323.1098 and R 323.1215 of the Michigan Administrative Code, the permittee is prohibited from undertaking any action that would result in a lowering of water quality from an increased loading of a BCC unless an increased use request and antidegradation demonstration have been submitted and approved by the Department.

11. Notification of Changes in Discharge
The permittee shall notify the Department via MiEnviro Portal (https://mienviro.michigan.gov/ncore/), as soon as possible but within no more than 10 days of knowing, or having reason to believe, that any activity or change has occurred or will occur which would result in the discharge of: 1) detectable levels of chemicals on the current Michigan Critical Materials Register, priority pollutants or hazardous substances set forth in 40 CFR 122.21, Appendix D, or the Pollutants of Initial Focus in the Great Lakes Water Quality Initiative specified in 40 CFR 132.6, Table 6, which were not acknowledged in the application or listed in the application at less than

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Section C. Reporting Requirements

detectable levels; 2) detectable levels of any other chemical not listed in the application or listed at less than detection, for which the application specifically requested information; or 3) any chemical at levels greater than five times the average level reported in the complete application (see the first page of this permit, for the date(s) the complete application was submitted). Any other monitoring results obtained as a requirement of this permit shall be reported in accordance with the compliance schedules.

12. Changes in Facility Operations

Any anticipated action or activity, including but not limited to facility expansion, production increases, or process modification, which will result in new or increased loadings of pollutants to the receiving waters must be reported to the Department by a) submission of an increased use request (application) and all information required under R 323.1098 (Antidegradation) of the Water Quality Standards or b) by written notice if the following conditions are met: 1) the action or activity will not result in a change in the types of wastewater discharged or result in a greater quantity of wastewater than currently authorized by this permit; 2) the action or activity will not result in violations of the effluent limitations specified in this permit; 3) the action or activity is not prohibited by the requirements of Part II.C.10.; and 4) the action or activity will not require notification pursuant to Part II.C.11. Following such written notice, the permit or, if applicable, the facility's COC, may be modified according to applicable laws and rules to specify and limit any pollutant not previously limited.

13. Transfer of Ownership or Control

In the event of any change in ownership or control of facilities from which the authorized discharge emanates, the following requirements apply: Not less than 30 days prior to the actual transfer of ownership or control – for non-CAFOs, or within 30 days of the actual transfer of ownership or control – for CAFOs, the permittee shall submit to the Department via MiEnviro Portal (https://mienviro.michigan.gov/ncore/) a written agreement between the current permittee and the new permittee containing: 1) the legal name and address of the new owner; 2) a specific date for the effective transfer of permit responsibility, coverage and liability; and 3) a certification of the continuity of or any changes in operations, wastewater discharge, or wastewater treatment.

If the new permittee is proposing changes in operations, wastewater discharge, or wastewater treatment, the Department may propose modification of this permit in accordance with applicable laws and rules.

14. Operations and Maintenance Manual

For wastewater treatment facilities that serve the public (and are thus subject to Part 41 of the NREPA), Section 4104 of Part 41 and associated Rule 2957 of the Michigan Administrative Code allow the Department to require an Operations and Maintenance (O&M) Manual from the facility. An up-to-date copy of the O&M Manual shall be kept at the facility and shall be provided to the Department upon request. The Department may review the O&M Manual in whole or in part at its discretion and require modifications to it if portions are determined to be inadequate.

At a minimum, the O&M Manual shall include the following information: permit standards; descriptions and operation information for all equipment; staffing information; laboratory requirements; record keeping requirements; a maintenance plan for equipment; an emergency operating plan; safety program information; and copies of all pertinent forms, as-built plans, and manufacturer's manuals.

Certification of the existence and accuracy of the O&M Manual shall be submitted to the Department at least sixty days prior to start-up of a new wastewater treatment facility. Recertification shall be submitted sixty days prior to start-up of any substantial improvements or modifications made to an existing wastewater treatment facility.

PART II

Section C. Reporting Requirements

15. Signatory Requirements

All applications, reports, or information submitted to the Department in accordance with the conditions of this permit and that require a signature shall be signed and certified as described in the Clean Water Act and the NREPA.

The Clean Water Act provides that any person who knowingly makes any false statement, representation, or certification in any record or other document submitted or required to be maintained under this permit, including monitoring reports or reports of compliance or noncompliance, shall, upon conviction, be punished by a fine of not more than $10,000 per violation, or by imprisonment for not more than 6 months per violation, or by both.

The NREPA (Section 3115(2)) provides that a person who at the time of the violation knew or should have known that he or she discharged a substance contrary to this part, or contrary to a permit, COC, or order issued or rule promulgated under this part, or who intentionally makes a false statement, representation, or certification in an application for or form pertaining to a permit or COC or in a notice or report required by the terms and conditions of an issued permit or COC, or who intentionally renders inaccurate a monitoring device or record required to be maintained by the Department, is guilty of a felony and shall be fined not less than $2,500.00 or more than $25,000.00 for each violation. The court may impose an additional fine of not more than $25,000.00 for each day during which the unlawful discharge occurred. If the conviction is for a violation committed after a first conviction of the person under this subsection, the court shall impose a fine of not less than $25,000.00 per day and not more than $50,000.00 per day of violation. Upon conviction, in addition to a fine, the court in its discretion may sentence the defendant to imprisonment for not more than 2 years or impose probation upon a person for a violation of this part. With the exception of the issuance of criminal complaints, issuance of warrants, and the holding of an arraignment, the circuit court for the county in which the violation occurred has exclusive jurisdiction. However, the person shall not be subject to the penalties of this subsection if the discharge of the effluent is in conformance with and obedient to a rule, order, permit, or COC of the Department. In addition to a fine, the attorney general may file a civil suit in a court of competent jurisdiction to recover the full value of the injuries done to the natural resources of the state and the costs of surveillance and enforcement by the state resulting from the violation.

16. Electronic Reporting

Upon notice by the Department that electronic reporting tools are available for specific reports or notifications, the permittee shall submit electronically via MiEnviro Portal (https://mienviro.michigan.gov/ncore/) all such reports or notifications as required by this permit, on forms provided by the Department.

PART II

Section D. Management Responsibilities

1. Duty to Comply
All discharges authorized herein shall be consistent with the terms and conditions of this permit. The discharge of any pollutant identified in this permit, more frequently than, or at a level in excess of, that authorized, shall constitute a violation of the permit.

It is the duty of the permittee to comply with all the terms and conditions of this permit. Any noncompliance with the Effluent Limitations, Special Conditions, or terms of this permit constitutes a violation of the NREPA and/or the Clean Water Act and constitutes grounds for enforcement action; for permit or COC termination, revocation and reissuance, or modification; or denial of an application for permit or COC renewal.

It shall not be a defense for a permittee in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this permit.

2. Operator Certification
The permittee shall have the waste treatment facilities under direct supervision of an operator certified at the appropriate level for the facility certification by the Department, as required by Sections 3110 and 4104 of the NREPA. Permittees authorized to discharge storm water shall have the storm water treatment and/or control measures under direct supervision of a storm water operator certified by the Department, as required by Section 3110 of the NREPA.

3. Facilities Operation
The permittee shall, at all times, properly operate and maintain all treatment or control facilities or systems installed or used by the permittee to achieve compliance with the terms and conditions of this permit. Proper operation and maintenance includes adequate laboratory controls and appropriate quality assurance procedures.

4. Power Failures
In order to maintain compliance with the effluent limitations of this permit and prevent unauthorized discharges, the permittee shall either:

a. provide an alternative power source sufficient to operate facilities utilized by the permittee to maintain compliance with the effluent limitations and conditions of this permit; or

b. upon the reduction, loss, or failure of one or more of the primary sources of power to facilities utilized by the permittee to maintain compliance with the effluent limitations and conditions of this permit, the permittee shall halt, reduce or otherwise control production and/or all discharge in order to maintain compliance with the effluent limitations and conditions of this permit.

5. Adverse Impact
The permittee shall take all reasonable steps to minimize or prevent any adverse impact to the surface waters or groundwaters of the state resulting from noncompliance with any effluent limitation specified in this permit including, but not limited to, such accelerated or additional monitoring as necessary to determine the nature and impact of the discharge in noncompliance.

6. Containment Facilities
The permittee shall provide facilities for containment of any accidental losses of polluting materials in accordance with the requirements of the Part 5 Rules (R 324.2001 through R 324.2009 of the Michigan Administrative Code). For a POTW, these facilities shall be approved under Part 41 of the NREPA.

PART II

Section D. Management Responsibilities

7. Waste Treatment Residues

Residuals (i.e., solids, sludges, biosolids, filter backwash, scrubber water, ash, grit, or other pollutants or wastes) removed from or resulting from treatment or control of wastewaters, including those that are generated during treatment or left over after treatment or control has ceased, shall be disposed of in an environmentally compatible manner and according to applicable laws and rules. These laws may include, but are not limited to, the NREPA, Part 31 for protection of water resources, Part 55 for air pollution control, Part 111 for hazardous waste management, Part 115 for solid waste management, Part 121 for liquid industrial wastes, Part 301 for protection of inland lakes and streams, and Part 303 for wetlands protection. Such disposal shall not result in any unlawful pollution of the air, surface waters or groundwaters of the state.

8. Right of Entry

The permittee shall allow the Department, any agent appointed by the Department, or the Regional Administrator, upon the presentation of credentials and, for animal feeding operation facilities, following appropriate biosecurity protocols:

a. to enter upon the permittee's premises where an effluent source is located or any place in which records are required to be kept under the terms and conditions of this permit; and

b. at reasonable times to have access to and copy any records required to be kept under the terms and conditions of this permit; to inspect process facilities, treatment works, monitoring methods and equipment regulated or required under this permit; and to sample any discharge of pollutants.

9. Availability of Reports

Except for data determined to be confidential under Section 308 of the Clean Water Act and Rule 2128 (R 323.2128 of the Michigan Administrative Code), all reports prepared in accordance with the terms of this permit and required to be submitted to the Department shall be available for public inspection via MiEnviro Portal (https://mienviro.michigan.gov/ncore/). As required by the Clean Water Act, effluent data shall not be considered confidential. Knowingly making any false statement on any such report may result in the imposition of criminal penalties as provided for in Section 309 of the Clean Water Act and Sections 3112, 3115, 4106 and 4110 of the NREPA.

10. Duty to Provide Information

The permittee shall furnish to the Department via MiEnviro Portal (https://mienviro.michigan.gov/ncore/), <u>within a reasonable time</u>, any information which the Department may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this permit or the facility's COC, or to determine compliance with this permit. The permittee shall also furnish to the Department, upon request, copies of records required to be kept by this permit.

Where the permittee becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Department, it shall promptly submit such facts or information.

PART II

Section E. Activities Not Authorized by This Permit

1. Discharge to the Groundwaters
This permit does not authorize any discharge to the groundwaters. Such discharge may be authorized by a groundwater discharge permit issued pursuant to the NREPA.

2. POTW Construction
This permit does not authorize or approve the construction or modification of any physical structures or facilities at a POTW. Approval for the construction or modification of any physical structures or facilities at a POTW shall be by permit issued under Part 41 of the NREPA.

3. Civil and Criminal Liability
Except as provided in permit conditions on "Bypass" (Part II.C.9. pursuant to 40 CFR 122.41(m)), nothing in this permit shall be construed to relieve the permittee from civil or criminal penalties for noncompliance, whether or not such noncompliance is due to factors beyond the permittee's control, such as accidents, equipment breakdowns, or labor disputes.

4. Oil and Hazardous Substance Liability
Nothing in this permit shall be construed to preclude the institution of any legal action or relieve the permittee from any responsibilities, liabilities, or penalties to which the permittee may be subject under Section 311 of the Clean Water Act except as are exempted by federal regulations.

5. State Laws
Nothing in this permit shall be construed to preclude the institution of any legal action or relieve the permittee from any responsibilities, liabilities, or penalties established pursuant to any applicable state law or regulation under authority preserved by Section 510 of the Clean Water Act.

6. Property Rights
The issuance of this permit does not convey any property rights in either real or personal property, or any exclusive privileges, nor does it authorize violation of any federal, state or local laws or regulations, nor does it obviate the necessity of obtaining such permits, including any other Department of Environment, Great Lakes, and Energy permits, or approvals from other units of government as may be required by law.

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

March 25, 2025

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: IPP Monitoring
Pace Project No.: 50395860

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on March 11, 2025. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

CERTIFICATIONS

Project:	IPP Monitoring
Pace Project No.:	50395860

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268
Illinois Accreditation #: 200074
Indiana Drinking Water Laboratory #: C-49-06
Kansas/TNI Certification #: E-10177
Kentucky UST Agency Interest #: 80226
Kentucky WW Laboratory ID #: 98019
Louisiana Certification #: 04076
Michigan Drinking Water Laboratory #9050
Oklahoma Laboratory #: 9204
Texas Certification #: T104704355
Washington Dept of Ecology #: C1081
Wisconsin Laboratory #: 999788130
USDA Foreign Soil Permit #: 525-23-13-23119
USDA Compliance Agreement #: IN-SL-22-001

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

SAMPLE SUMMARY

Project: IPP Monitoring
Pace Project No.: 50395860

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50395860001	**Influent**	Water	03/10/25 07:00	03/11/25 10:15
50395860002	**Effluent**	Water	03/10/25 07:00	03/11/25 10:15
50395860003	**Influent**	Water	03/10/25 11:37	03/11/25 10:15
50395860004	**Effluent**	Water	03/10/25 11:17	03/11/25 10:15

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

SAMPLE ANALYTE COUNT

Project: IPP Monitoring
Pace Project No.: 50395860

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50395860001	**Influent**	EPA 608.3	DMP1	8	PASI-I
		EPA 608.3	ADM	2	PASI-I
50395860002	**Effluent**	EPA 608.3	DMP1	8	PASI-I
		EPA 608.3	ADM	2	PASI-I
50395860003	**Influent**	EPA 625.1	FIP	8	PASI-I
		EPA 624.1	TAY	17	PASI-I
		EPA 420.4	KCS	1	PASI-I
		SM 4500-CN-E	JTR	1	PASI-I
50395860004	**Effluent**	EPA 625.1	FIP	8	PASI-I
		EPA 624.1	TAY	17	PASI-I
		EPA 420.4	KCS	1	PASI-I
		SM 4500-CN-E	JTR	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

SUMMARY OF DETECTION

Project: IPP Monitoring
Pace Project No.: 50395860

Lab Sample ID / Method	Client Sample ID / Parameters	Result	Units	Report Limit	Analyzed	Qualifiers
50395860003	**Influent**					
EPA 624.1	Toluene	3.0J	ug/L	5.0	03/13/25 13:48	
EPA 624.1	Xylene (Total)	0.37J	ug/L	10.0	03/13/25 13:48	
EPA 420.4	Phenolics, Total Recoverable	306	ug/L	200	03/21/25 09:27	
SM 4500-CN-E	Cyanide	0.0058	mg/L	0.0050	03/20/25 12:41	

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

ANALYTICAL RESULTS

Project: IPP Monitoring
Pace Project No.: 50395860

Sample: Influent **Lab ID: 50395860001** Collected: 03/10/25 07:00 Received: 03/11/25 10:15 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
608.3 PCB	Analytical Method: EPA 608.3 Preparation Method: EPA 608.3 Pace Analytical Services - Indianapolis								
PCB-1016 (Aroclor 1016)	**<0.060**	ug/L	0.10	0.060	1	03/13/25 18:59	03/14/25 13:47	12674-11-2	
PCB-1221 (Aroclor 1221)	**<0.060**	ug/L	0.10	0.060	1	03/13/25 18:59	03/14/25 13:47	11104-28-2	
PCB-1232 (Aroclor 1232)	**<0.060**	ug/L	0.10	0.060	1	03/13/25 18:59	03/14/25 13:47	11141-16-5	
PCB-1242 (Aroclor 1242)	**<0.060**	ug/L	0.10	0.060	1	03/13/25 18:59	03/14/25 13:47	53469-21-9	
PCB-1248 (Aroclor 1248)	**<0.060**	ug/L	0.10	0.060	1	03/13/25 18:59	03/14/25 13:47	12672-29-6	
PCB-1254 (Aroclor 1254)	**<0.060**	ug/L	0.10	0.060	1	03/13/25 18:59	03/14/25 13:47	11097-69-1	
PCB-1260 (Aroclor 1260)	**<0.053**	ug/L	0.10	0.053	1	03/13/25 18:59	03/14/25 13:47	11096-82-5	
Surrogates									
Tetrachloro-m-xylene (S)	39	%.	1-112		1	03/13/25 18:59	03/14/25 13:47	877-09-8	
608.3 Pesticides	Analytical Method: EPA 608.3 Preparation Method: EPA 608.3 Pace Analytical Services - Indianapolis								
gamma-BHC (Lindane)	**<0.036**	ug/L	0.26	0.036	5	03/13/25 18:59	03/21/25 10:41	58-89-9	ED
Surrogates									
Decachlorobiphenyl (S)	80	%.	1-133		5	03/13/25 18:59	03/21/25 10:41	2051-24-3	

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

ANALYTICAL RESULTS

Project: IPP Monitoring
Pace Project No.: 50395860

Sample: Effluent **Lab ID: 50395860002** Collected: 03/10/25 07:00 Received: 03/11/25 10:15 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
608.3 PCB	Analytical Method: EPA 608.3 Preparation Method: EPA 608.3								
	Pace Analytical Services - Indianapolis								
PCB-1016 (Aroclor 1016)	**<0.059**	ug/L	0.10	0.059	1	03/13/25 18:59	03/14/25 14:48	12674-11-2	
PCB-1221 (Aroclor 1221)	**<0.059**	ug/L	0.10	0.059	1	03/13/25 18:59	03/14/25 14:48	11104-28-2	
PCB-1232 (Aroclor 1232)	**<0.059**	ug/L	0.10	0.059	1	03/13/25 18:59	03/14/25 14:48	11141-16-5	
PCB-1242 (Aroclor 1242)	**<0.059**	ug/L	0.10	0.059	1	03/13/25 18:59	03/14/25 14:48	53469-21-9	
PCB-1248 (Aroclor 1248)	**<0.059**	ug/L	0.10	0.059	1	03/13/25 18:59	03/14/25 14:48	12672-29-6	
PCB-1254 (Aroclor 1254)	**<0.059**	ug/L	0.10	0.059	1	03/13/25 18:59	03/14/25 14:48	11097-69-1	
PCB-1260 (Aroclor 1260)	**<0.052**	ug/L	0.10	0.052	1	03/13/25 18:59	03/14/25 14:48	11096-82-5	
Surrogates									
Tetrachloro-m-xylene (S)	41	%.	1-112		1	03/13/25 18:59	03/14/25 14:48	877-09-8	
608.3 Pesticides	Analytical Method: EPA 608.3 Preparation Method: EPA 608.3								
	Pace Analytical Services - Indianapolis								
gamma-BHC (Lindane)	**<0.0071**	ug/L	0.051	0.0071	1	03/13/25 18:59	03/21/25 09:59	58-89-9	
Surrogates									
Decachlorobiphenyl (S)	64	%.	1-133		1	03/13/25 18:59	03/21/25 09:59	2051-24-3	

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

ANALYTICAL RESULTS

Project: IPP Monitoring
Pace Project No.: 50395860

Sample: Influent **Lab ID: 50395860003** Collected: 03/10/25 11:37 Received: 03/11/25 10:15 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
625.1 Semivolatile Organics	Analytical Method: EPA 625.1 Preparation Method: EPA 625.1 Pace Analytical Services - Indianapolis								
Benzidine	**<2.7**	ug/L	47.6	2.7	1	03/12/25 17:31	03/13/25 14:18	92-87-5	
Phenol	**<2.5**	ug/L	9.5	2.5	1	03/12/25 17:31	03/13/25 14:18	108-95-2	
Surrogates									
2-Fluorophenol (S)	33	%.	1-102		1	03/12/25 17:31	03/13/25 14:18	367-12-4	
Phenol-d5 (S)	28	%.	8-424		1	03/12/25 17:31	03/13/25 14:18	4165-62-2	
Nitrobenzene-d5 (S)	34	%.	15-314		1	03/12/25 17:31	03/13/25 14:18	4165-60-0	
2-Fluorobiphenyl (S)	30	%.	2-103		1	03/12/25 17:31	03/13/25 14:18	321-60-8	
2,4,6-Tribromophenol (S)	83	%.	20-155		1	03/12/25 17:31	03/13/25 14:18	118-79-6	
p-Terphenyl-d14 (S)	86	%.	1-168		1	03/12/25 17:31	03/13/25 14:18	1718-51-0	
624.1 Volatile Organics	Analytical Method: EPA 624.1 Pace Analytical Services - Indianapolis								
Benzene	**<0.23**	ug/L	5.0	0.23	1		03/13/25 13:48	71-43-2	
Bromomethane	**<2.1**	ug/L	5.0	2.1	1		03/13/25 13:48	74-83-9	
Chloroform	**<0.19**	ug/L	4.8	0.19	1		03/13/25 13:48	67-66-3	
1,2-Dichlorobenzene	**<0.27**	ug/L	5.0	0.27	1		03/13/25 13:48	95-50-1	
1,1-Dichloroethane	**<0.22**	ug/L	5.0	0.22	1		03/13/25 13:48	75-34-3	
1,2-Dichloroethane	**<0.26**	ug/L	5.0	0.26	1		03/13/25 13:48	107-06-2	
1,2-Dichloroethene (Total)	**<0.31**	ug/L	10.0	0.31	1		03/13/25 13:48	540-59-0	
1,1-Dichloroethene	**<0.33**	ug/L	5.0	0.33	1		03/13/25 13:48	75-35-4	
cis-1,2-Dichloroethene	**<0.31**	ug/L	5.0	0.31	1		03/13/25 13:48	156-59-2	
trans-1,2-Dichloroethene	**<0.27**	ug/L	4.8	0.27	1		03/13/25 13:48	156-60-5	
Ethylbenzene	**<0.25**	ug/L	5.0	0.25	1		03/13/25 13:48	100-41-4	
Styrene	**<0.26**	ug/L	5.0	0.26	1		03/13/25 13:48	100-42-5	N2
Toluene	**3.0J**	ug/L	5.0	0.26	1		03/13/25 13:48	108-88-3	
Xylene (Total)	**0.37J**	ug/L	10.0	0.29	1		03/13/25 13:48	1330-20-7	
Surrogates									
Dibromofluoromethane (S)	98	%.	91-114		1		03/13/25 13:48	1868-53-7	
4-Bromofluorobenzene (S)	99	%.	85-120		1		03/13/25 13:48	460-00-4	
Toluene-d8 (S)	101	%.	85-117		1		03/13/25 13:48	2037-26-5	
420.4 Phenolics, Total	Analytical Method: EPA 420.4 Preparation Method: EPA 420.4 Pace Analytical Services - Indianapolis								
Phenolics, Total Recoverable	**306**	ug/L	200	39.0	1	03/20/25 11:23	03/21/25 09:27	64743-03-9	
4500CNE Cyanide	Analytical Method: SM 4500-CN-E Preparation Method: SM 4500-CN-C Pace Analytical Services - Indianapolis								
Cyanide	**0.0058**	mg/L	0.0050	0.0030	1	03/17/25 12:20	03/20/25 12:41	57-12-5	

REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

ANALYTICAL RESULTS

Project: IPP Monitoring
Pace Project No.: 50395860

Sample: Effluent **Lab ID: 50395860004** Collected: 03/10/25 11:17 Received: 03/11/25 10:15 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
625.1 Semivolatile Organics	Analytical Method: EPA 625.1 Preparation Method: EPA 625.1 Pace Analytical Services - Indianapolis								
Benzidine	**<2.9**	ug/L	50.0	2.9	1	03/12/25 17:31	03/13/25 12:22	92-87-5	
Phenol	**<2.6**	ug/L	10.0	2.6	1	03/12/25 17:31	03/13/25 12:22	108-95-2	
Surrogates									
2-Fluorophenol (S)	34	%.	1-102		1	03/12/25 17:31	03/13/25 12:22	367-12-4	
Phenol-d5 (S)	26	%.	8-424		1	03/12/25 17:31	03/13/25 12:22	4165-62-2	
Nitrobenzene-d5 (S)	34	%.	15-314		1	03/12/25 17:31	03/13/25 12:22	4165-60-0	
2-Fluorobiphenyl (S)	16	%.	2-103		1	03/12/25 17:31	03/13/25 12:22	321-60-8	
2,4,6-Tribromophenol (S)	73	%.	20-155		1	03/12/25 17:31	03/13/25 12:22	118-79-6	
p-Terphenyl-d14 (S)	92	%.	1-168		1	03/12/25 17:31	03/13/25 12:22	1718-51-0	
624.1 Volatile Organics	Analytical Method: EPA 624.1 Pace Analytical Services - Indianapolis								
Benzene	**<0.23**	ug/L	5.0	0.23	1		03/13/25 13:20	71-43-2	
Bromomethane	**<2.1**	ug/L	5.0	2.1	1		03/13/25 13:20	74-83-9	
Chloroform	**<0.19**	ug/L	4.8	0.19	1		03/13/25 13:20	67-66-3	
1,2-Dichlorobenzene	**<0.27**	ug/L	5.0	0.27	1		03/13/25 13:20	95-50-1	
1,1-Dichloroethane	**<0.22**	ug/L	5.0	0.22	1		03/13/25 13:20	75-34-3	
1,2-Dichloroethane	**<0.26**	ug/L	5.0	0.26	1		03/13/25 13:20	107-06-2	
1,2-Dichloroethene (Total)	**<0.31**	ug/L	10.0	0.31	1		03/13/25 13:20	540-59-0	
1,1-Dichloroethene	**<0.33**	ug/L	5.0	0.33	1		03/13/25 13:20	75-35-4	
cis-1,2-Dichloroethene	**<0.31**	ug/L	5.0	0.31	1		03/13/25 13:20	156-59-2	
trans-1,2-Dichloroethene	**<0.27**	ug/L	4.8	0.27	1		03/13/25 13:20	156-60-5	
Ethylbenzene	**<0.25**	ug/L	5.0	0.25	1		03/13/25 13:20	100-41-4	
Styrene	**<0.26**	ug/L	5.0	0.26	1		03/13/25 13:20	100-42-5	N2
Toluene	**<0.26**	ug/L	5.0	0.26	1		03/13/25 13:20	108-88-3	
Xylene (Total)	**<0.29**	ug/L	10.0	0.29	1		03/13/25 13:20	1330-20-7	
Surrogates									
Dibromofluoromethane (S)	97	%.	91-114		1		03/13/25 13:20	1868-53-7	
4-Bromofluorobenzene (S)	99	%.	85-120		1		03/13/25 13:20	460-00-4	
Toluene-d8 (S)	100	%.	85-117		1		03/13/25 13:20	2037-26-5	
420.4 Phenolics, Total	Analytical Method: EPA 420.4 Preparation Method: EPA 420.4 Pace Analytical Services - Indianapolis								
Phenolics, Total Recoverable	**<3.9**	ug/L	20.0	3.9	1	03/20/25 11:23	03/21/25 09:28	64743-03-9	
4500CNE Cyanide	Analytical Method: SM 4500-CN-E Preparation Method: SM 4500-CN-C Pace Analytical Services - Indianapolis								
Cyanide	**<0.0030**	mg/L	0.0050	0.0030	1	03/17/25 12:20	03/20/25 12:42	57-12-5	

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: IPP Monitoring
Pace Project No.: 50395860

QC Batch: 834173 | Analysis Method: EPA 624.1
QC Batch Method: EPA 624.1 | Analysis Description: 624.1 MSV
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50395860003, 50395860004

METHOD BLANK: 3816892 | Matrix: Water
Associated Lab Samples: 50395860003, 50395860004

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
1,1-Dichloroethane	ug/L	<0.22	5.0	0.22	03/13/25 12:52	
1,1-Dichloroethene	ug/L	<0.33	5.0	0.33	03/13/25 12:52	
1,2-Dichlorobenzene	ug/L	<0.27	5.0	0.27	03/13/25 12:52	
1,2-Dichloroethane	ug/L	<0.26	5.0	0.26	03/13/25 12:52	
1,2-Dichloroethene (Total)	ug/L	<0.31	10.0	0.31	03/13/25 12:52	
Benzene	ug/L	<0.23	5.0	0.23	03/13/25 12:52	
Bromomethane	ug/L	<2.1	5.0	2.1	03/13/25 12:52	
Chloroform	ug/L	<0.19	4.8	0.19	03/13/25 12:52	
cis-1,2-Dichloroethene	ug/L	<0.31	5.0	0.31	03/13/25 12:52	
Ethylbenzene	ug/L	<0.25	5.0	0.25	03/13/25 12:52	
Styrene	ug/L	<0.26	5.0	0.26	03/13/25 12:52	N2
Toluene	ug/L	<0.26	5.0	0.26	03/13/25 12:52	
trans-1,2-Dichloroethene	ug/L	<0.27	4.8	0.27	03/13/25 12:52	
Xylene (Total)	ug/L	<0.29	10.0	0.29	03/13/25 12:52	
4-Bromofluorobenzene (S)	%.	98	85-120		03/13/25 12:52	
Dibromofluoromethane (S)	%.	98	91-114		03/13/25 12:52	1d
Toluene-d8 (S)	%.	100	85-117		03/13/25 12:52	

LABORATORY CONTROL SAMPLE: 3816893

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethane	ug/L	20	20.6	103	70-130	
1,1-Dichloroethene	ug/L	20	17.1	86	50-150	
1,2-Dichlorobenzene	ug/L	20	20.1	101	65-135	
1,2-Dichloroethane	ug/L	20	20.3	102	70-130	
1,2-Dichloroethene (Total)	ug/L	40	40.6	101		
Benzene	ug/L	20	21.4	107	65-135	
Bromomethane	ug/L	20	20.9	104	15-185	
Chloroform	ug/L	20	21.2	106	70-135	
cis-1,2-Dichloroethene	ug/L	20	20.7	103	72-125	
Ethylbenzene	ug/L	20	20.0	100	60-140	
Styrene	ug/L	20	20.8	104	77-124	N2
Toluene	ug/L	20	20.9	104	70-130	
trans-1,2-Dichloroethene	ug/L	20	19.9	99	70-130	
Xylene (Total)	ug/L	60	61.5	102	76-119	
4-Bromofluorobenzene (S)	%.			100	85-120	
Dibromofluoromethane (S)	%.			97	91-114	
Toluene-d8 (S)	%.			103	85-117	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: IPP Monitoring
Pace Project No.: 50395860

QC Batch:	834268	Analysis Method:	EPA 608.3
QC Batch Method:	EPA 608.3	Analysis Description:	608.3 PCB
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50395860001, 50395860002

METHOD BLANK: 3817265 Matrix: Water

Associated Lab Samples: 50395860001, 50395860002

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
PCB-1016 (Aroclor 1016)	ug/L	<0.058	0.10	0.058	03/14/25 11:47	
PCB-1221 (Aroclor 1221)	ug/L	<0.058	0.10	0.058	03/14/25 11:47	
PCB-1232 (Aroclor 1232)	ug/L	<0.058	0.10	0.058	03/14/25 11:47	
PCB-1242 (Aroclor 1242)	ug/L	<0.058	0.10	0.058	03/14/25 11:47	
PCB-1248 (Aroclor 1248)	ug/L	<0.058	0.10	0.058	03/14/25 11:47	
PCB-1254 (Aroclor 1254)	ug/L	<0.058	0.10	0.058	03/14/25 11:47	
PCB-1260 (Aroclor 1260)	ug/L	<0.051	0.10	0.051	03/14/25 11:47	
Tetrachloro-m-xylene (S)	%.	50	1-112		03/14/25 11:47	

LABORATORY CONTROL SAMPLE: 3817266

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
PCB-1016 (Aroclor 1016)	ug/L	0.5	0.56	111	50-140	
PCB-1260 (Aroclor 1260)	ug/L	0.5	0.49	97	8-140	
Tetrachloro-m-xylene (S)	%.			54	1-112	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3817267 3817268

Parameter	Units	50396076002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
PCB-1016 (Aroclor 1016)	ug/L	ND	0.5	0.5	0.40	0.54	80	109	50-140	30	36	
PCB-1260 (Aroclor 1260)	ug/L	ND	0.5	0.5	0.12	0.15	24	31	8-140	26	38	
Tetrachloro-m-xylene (S)	%.						39	52	1-112			

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: IPP Monitoring
Pace Project No.: 50395860

QC Batch:	834269	Analysis Method:	EPA 608.3
QC Batch Method:	EPA 608.3	Analysis Description:	608.3 Pesticides
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50395860001, 50395860002

METHOD BLANK: 3817269 Matrix: Water

Associated Lab Samples: 50395860001, 50395860002

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
gamma-BHC (Lindane)	ug/L	<0.0070	0.050	0.0070	03/18/25 10:44	
Decachlorobiphenyl (S)	%.	65	1-133		03/18/25 10:44	

LABORATORY CONTROL SAMPLE: 3817270

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
gamma-BHC (Lindane)	ug/L	0.1	0.086	86	32-140	
Decachlorobiphenyl (S)	%.			63	1-133	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3817271 3817272

Parameter	Units	50395786001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
gamma-BHC (Lindane)	ug/L	ND	0.1	0.1	0.10	0.084	102	84	32-140	20	39	
Decachlorobiphenyl (S)	%.						72	63	1-133			

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QUALITY CONTROL DATA

Project: IPP Monitoring
Pace Project No.: 50395860

QC Batch: 834011
QC Batch Method: EPA 625.1
Analysis Method: EPA 625.1
Analysis Description: 625.1 Semivolatile Organics
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50395860003, 50395860004

METHOD BLANK: 3816154 Matrix: Water
Associated Lab Samples: 50395860003, 50395860004

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Benzidine	ug/L	<2.9	50.0	2.9	03/13/25 08:14	
Phenol	ug/L	<2.6	10.0	2.6	03/13/25 08:14	
2,4,6-Tribromophenol (S)	%.	52	20-155		03/13/25 08:14	
2-Fluorobiphenyl (S)	%.	40	2-103		03/13/25 08:14	
2-Fluorophenol (S)	%.	46	1-102		03/13/25 08:14	
Nitrobenzene-d5 (S)	%.	49	15-314		03/13/25 08:14	
p-Terphenyl-d14 (S)	%.	90	1-168		03/13/25 08:14	
Phenol-d5 (S)	%.	30	8-424		03/13/25 08:14	

LABORATORY CONTROL SAMPLE: 3816155

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Benzidine	ug/L	50	<2.9	1	1-44	
Phenol	ug/L	50	26.1	52	5-120	
2,4,6-Tribromophenol (S)	%.			69	20-155	
2-Fluorobiphenyl (S)	%.			48	2-103	
2-Fluorophenol (S)	%.			57	1-102	
Nitrobenzene-d5 (S)	%.			57	15-314	
p-Terphenyl-d14 (S)	%.			78	1-168	
Phenol-d5 (S)	%.			39	8-424	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3816156 3816157

Parameter	Units	50395538003 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Benzidine	ug/L	ND	45.5	50	<65.5	<72.0	0	0	1-29		40	M1
Phenol	ug/L	1.4 mg/L	45.5	50	1310	1690	-279	508	5-120	25	64	D3,M1, P1,P2
2,4,6-Tribromophenol (S)	%.						119	121	20-155			
2-Fluorobiphenyl (S)	%.						31	78	2-103			
2-Fluorophenol (S)	%.						43	85	1-102			
Nitrobenzene-d5 (S)	%.						33	81	15-314			
p-Terphenyl-d14 (S)	%.						0	143	1-168			S0
Phenol-d5 (S)	%.						0	90	8-424			S0

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: IPP Monitoring
Pace Project No.: 50395860

QC Batch: 835240
QC Batch Method: EPA 420.4
Analysis Method: EPA 420.4
Analysis Description: 420.4 Phenolics
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50395860003, 50395860004

METHOD BLANK: 3821651 Matrix: Water
Associated Lab Samples: 50395860003, 50395860004

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Phenolics, Total Recoverable	ug/L	<3.9	20.0	3.9	03/21/25 09:24	

LABORATORY CONTROL SAMPLE: 3821652

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Phenolics, Total Recoverable	ug/L	50	47.9	96	90-110	

MATRIX SPIKE SAMPLE: 3821655

Parameter	Units	50396120002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Phenolics, Total Recoverable	ug/L	<3.9	50	48.0	96	90-110	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3821656 3821657

Parameter	Units	50396119002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Phenolics, Total Recoverable	ug/L	<3.9	50	50	47.1	47.7	94	95	90-110	1	20	

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: IPP Monitoring
Pace Project No.: 50395860

QC Batch: 834585 — Analysis Method: SM 4500-CN-E
QC Batch Method: SM 4500-CN-C — Analysis Description: 4500CNE Cyanide
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50395860003, 50395860004

METHOD BLANK: 3819075 — Matrix: Water
Associated Lab Samples: 50395860003, 50395860004

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Cyanide	mg/L	<0.0030	0.0050	0.0030	03/20/25 12:34	

LABORATORY CONTROL SAMPLE: 3819076

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Cyanide	mg/L	0.1	0.092	92	90-110	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3819077 3819078

Parameter	Units	50395860004 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Cyanide	mg/L	<0.0030	0.1	0.1	0.11	0.11	106	105	90-110	1	20	

MATRIX SPIKE SAMPLE: 3819079

Parameter	Units	50396347001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Cyanide	mg/L	ND	0.2	0.20	97	90-110	

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALIFIERS

Project:	IPP Monitoring
Pace Project No.:	50395860

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

ANALYTE QUALIFIERS

1d	Neither matrix spike nor matrix precision data could be provided for this analytical batch due to insufficient sample volume.
D3	Sample was diluted due to the presence of high levels of non-target analytes or other matrix interference.
ED	Due to the extract's physical characteristics, the analysis was performed at dilution.
M1	Matrix spike recovery exceeded QC limits. Batch accepted based on laboratory control sample (LCS) recovery.
N2	The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
P1	Routine initial sample volume or weight was not used for extraction, resulting in elevated reporting limits.
P2	Re-extraction or re-analysis could not be performed due to insufficient sample amount.
S0	Surrogate recovery outside laboratory control limits.

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: IPP Monitoring
Pace Project No.: 50395860

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50395860001	**Influent**	EPA 608.3	834268	EPA 608.3	834370
50395860002	**Effluent**	EPA 608.3	834268	EPA 608.3	834370
50395860001	**Influent**	EPA 608.3	834269	EPA 608.3	834465
50395860002	**Effluent**	EPA 608.3	834269	EPA 608.3	834465
50395860003	**Influent**	EPA 625.1	834011	EPA 625.1	834092
50395860004	**Effluent**	EPA 625.1	834011	EPA 625.1	834092
50395860003	**Influent**	EPA 624.1	834173		
50395860004	**Effluent**	EPA 624.1	834173		
50395860003	**Influent**	EPA 420.4	835240	EPA 420.4	835436
50395860004	**Effluent**	EPA 420.4	835240	EPA 420.4	835436
50395860003	**Influent**	SM 4500-CN-C	834585	SM 4500-CN-E	835339
50395860004	**Effluent**	SM 4500-CN-C	834585	SM 4500-CN-E	835339

Pace® Location Requested (City/State):
Pace Analytical Grand Rapids
4171 40th Street SE, Grand Rapids, MI 49512

CHAIN-OF-CUSTODY Analytical Request Document

Chain-of-Custody is a LEGAL DOCUMENT - Complete all relevant fields

LAB USE ONLY - Affix Workorder/Login Label Here

WO# : 50395860

50395860

Company Name:	City of Cadillac	Contact/Report To:	Cindy Tomaszewski
Street Address:	1121 Plett Rd, Cadillac, MI 49601	Phone #:	231-775-2368
		E-Mail:	ctomaszewski@cadillac-mi.net
		Cc E-Mail:	
Customer Project #:			
Project Name:	IPP Monitoring	Invoice To:	Cindy Tomaszewski
		Invoice E-Mail:	ctomaszewski@cadillac-mi.net
Site Collection Info/Facility ID (as applicable):		Purchase Order # (if applicable):	
		Quote #:	
Time Zone Collected: [ ] AK [ ] PT [ ] MT [ ] CT [ ] ET		County / State origin of sample(s):	Michigan

Data Deliverables: [ ] Level II [ ] Level III [ ] Level IV [ ] EQUIS [ ] Other

Regulatory Program (DW, RCRA, etc.) as applicable: Reportable [ ] Yes [ ] No

Rush (Pre-approval required): [ ] Same Day [ ] 1 Day [ ] 2 Day [ ] 3 Day [ ] Other ________

DW PWSID # or WW Permit # as applicable:

Date Results Requested:

Field Filtered (if applicable): [ ] Yes [ ] No
Analysis:

* Matrix Codes (Insert in Matrix box below): Drinking Water (DW), Ground Water (GW), Waste Water (WW), Product (P), Soil/Solid (SS), Oil (OL), Wipe (WP), Tissue (TS), Bioassay (B), Vapor (V), Surface Water (SW), Sediment (SED), Sludge (SL), Caulk (CK), Leachate (LL), Biosolid (BS), Other (OT)

Specify Container Size **: 3, 3, 1, 1, 6, 5

Identify Container Preservative Type***: 3, 5, 1, 1, 1, 4

**Container Size: (1) 1L, (2) 500mL, (3) 250mL, (4) 125mL, (5) 100mL, (6) 40mL vial, (7) EnCore, (8) TerraCore, (9) 90mL, (10) Other

*** Preservative Types: (1) None, (2) HNO3, (3) H2SO4, (4) HCl, (5) NaOH, (6) Zn Acetate, (7) NaHSO4, (8) Sod. Thiosulfate, (9) Ascorbic Acid, (10) MeOH, (11) Other

Analysis Requested

Customer Sample ID	Matrix *	Comp / Grab	Composite Start Date	Composite Start Time	Collected or Composite End Date	Collected or Composite End Time	# Cont.	Res. Chlorine Results	Res. Chlorine Units	~~200.8 Metals, Total - Arsenic~~ CAT 3/5/25	420.4 Phenolics, Total	4500CNE Cyanide	608.3 PCB	608.3 Pesticides	624.1 Volatile Organics	625.1 SVOC - Phenol, Benzidine
Influent	WT	C	3-09-25	0700	3-10-25	0700	4			X	X	X	X	X	X	X
Effluent	WT	C	~~3-10-25~~ 3-09-25	0700	3-10-25	0700	4			X	X	X	X	X	X	X
influent	WT	G	—		3-10-25	1137	8				X	X			X	X
Effluent	WT	G	—		3-10-25	1117	8				X	X			X	X

Lab Use Only:
Proj. Mgr: **Brian Hall**
AcctNum / Client ID:
Table #:
Profile / Template: **7675**
Prelog / Bottle Ord. ID: **EZ 3234069**

Sample Comment

Preservation non-conformance identified for sample.

Additional Instructions from Pace®:

Collected By: (Printed Name) Taysha Laczko
Signature: T. Laczko

Customer Remarks / Special Conditions / Possible Hazards:

# Coolers:	Thermometer ID:	Correction Factor (°C):	Obs. Temp. (°C)	Corrected Temp. (°C)	On Ice:

Relinquished by/Company: (Signature)	Date/Time:	Received by/Company: (Signature)	Date/Time:
T. Laczko/City of Cadillac	3-10-25 / 12:30	UPS Overnight	
		[signature] / Pace	3/11/25 1015

Tracking Number:

Delivered by: [ ] In-Person [ ] Courier [ ] FedEX [ ] UPS [ ] Other

Page: 1 of 1

Submitting a sample via this chain of custody constitutes acknowledgment and acceptance of the Pace® Terms and Conditions found at https://www.pacelabs.com/resource-library/resource/pace-terms-and-conditions/

ENV-FRM-CORQ-0019_v02_110123 ©

Pace

Sample Conditions Upon Receipt Form (SCUR)

Date/Time: 3/11 1555	Evaluated By: DAD
Client: City of Cadillac	PM: BJH
Lab Notified of Rush or Short Holds: YES NO NA	

WO# : 50395860
PM: BJH Due Date: 03/25/25
CLIENT: GR-Cadillac

Project Received Via: FedEx UPS Client Pace Courier Other: ________				Comments:
Custody Seal Present and Intact:	YES	NO	NA	
Received Sample Information Form (SIF): Drinking Waters Only	YES	NO	NA	
Short Hold Present (≤ 48 Hours):	YES	NO		
Sample Received in Hold:	YES	NO		
Custody Signature Present:	YES	NO		
Collector Signature Present:	YES	NO		
Sample Collected Today and On Ice:	YES	NO	N/A	
IR Gun #: 350 351 Therm #: 282 283	Temp. should be 0°C - 6°C (Initial/Corrected)			
Ice Type: WET Bagged / WET Loose BLUE NONE	1. Cooler Temp. Upon Receipt: 1.6 / 1.3 °C			
Ice Location: TOP BOTTOM MIDDLE DISPERSED	2. Cooler Temp. Upon Receipt: 1.4 / 1.1 °C			
Temp Blank Received:	YES	NO		
Sample Label Matches COC (ID/Date/Time):	YES	NO		
Container Intact:	YES	NO		
Correct Container:	YES	NO		
Sufficient Volume:	YES	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with red dot on container lid if unacceptable. pH Strip Lot #: HC457151 *Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl*	YES	NO	N/A	
Residual Chlorine Absent: Cl₂ Strip Lot #: HC460768 *Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide*	YES	NO	N/A	
VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable.	YES	NO	N/A	
Trip Blank Received: HCl MeOH Other: ________	YES	NO	ON HOLD	
Comments:	3. Cooler Temp. Upon Receipt: ________ °C			
	4. Cooler Temp. Upon Receipt: ________ °C			
	Non-Conformance Form Required: YES NO			

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673

TRACE
ANALYTICAL LABORATORIES, INC.

231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

December 02, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25K0885
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

PJLA Testing Accreditation# 1031

TNI EL V1:2016

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673

TRACE ANALYTICAL LABORATORIES, INC.

231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

SAMPLE SUMMARY

Trace Project ID: 25K0885
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25K0885-01 Sample type: Grab	Akwel C.B	Wastewater	DG	11/17/25 07:25	11/18/25 10:44
25K0885-02 Sample type: Grab	AAR- 6th	Wastewater	DG	11/17/25 07:35	11/18/25 10:44

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ANALYTICAL RESULTS

Trace Project ID: 25K0885
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K0885-01 | Matrix: Wastewater | Date Collected: 11/17/25 07:25
Sample ID: Akwel C.B | Date Received: 11/18/25 10:44

PARAMETERS		RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL											
Analysis Method: EPA 200.8 Rev. 5.4											
Batch: T176318											
Beryllium		<0.0040	mg/L	0.0040	1	11/21/25	ejb	12/01/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 625.1											
Batch: T176295											
Phenol		<5.0	ug/L	5.0	1	11/20/25	kbc	11/21/25	jlh		
Surrogates:											
2-Fluorophenol	*	**0.8**	%	20-53	1	11/20/25	kbc	11/21/25	jlh	S12	
Phenol-d5	*	**0.7**	%	11-40	1	11/20/25	kbc	11/21/25	jlh	S12	
Nitrobenzene-d5		81	%	36-103	1	11/20/25	kbc	11/21/25	jlh		
2-Fluorobiphenyl		76	%	36-119	1	11/20/25	kbc	11/21/25	jlh	N	
2,4,6-Tribromophenol	*	**24**	%	30-105	1	11/20/25	kbc	11/21/25	jlh	S12, N	
Terphenyl-d14		88	%	37-109	1	11/20/25	kbc	11/21/25	jlh		
Analysis Method: EPA 625.1 SIM											
Batch: T176421											
Benzidine		<0.097	ug/L	0.097	1	11/24/25	av	11/24/25	jlh		
3,3'-Dichlorobenzidine		<0.97	ug/L	0.97	1	11/24/25	av	11/24/25	jlh		
Surrogates:											
Nitrobenzene-d5		71	%	38-130	1	11/24/25	av	11/24/25	jlh		
2-Fluorobiphenyl		59	%	39-118	1	11/24/25	av	11/24/25	jlh	N	
Terphenyl-d14		36	%	14-96	1	11/24/25	av	11/24/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 624.1											
Batch: T176203											
Bromomethane		<5.0	ug/L	5.0	1	11/18/25	sb	11/18/25	sb		
Surrogates:											
1,2-Dichloroethane-d4		106	%	68-133	1	11/18/25	sb	11/18/25	sb		
Toluene-d8		97	%	75-120	1	11/18/25	sb	11/18/25	sb		
4-Bromofluorobenzene		105	%	69-119	1	11/18/25	sb	11/18/25	sb		
1,2-Dichlorobenzene-d4		102	%	72-127	1	11/18/25	sb	11/18/25	sb		

ANALYTICAL RESULTS

Trace Project ID: 25K0885
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K0885-01 Matrix: Wastewater Date Collected: 11/17/25 07:25
Sample ID: Akwel C.B Date Received: 11/18/25 10:44

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25K0885
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K0885-02 Matrix: Wastewater Date Collected: 11/17/25 07:35
Sample ID: AAR- 6th Date Received: 11/18/25 10:44

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176318									
Beryllium	<0.0040 mg/L	0.0040	1	11/21/25	ejb	12/01/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176295									
Phenol	<5.0 ug/L	5.0	1	11/20/25	kbc	11/21/25	jlh		
Surrogates:									
2-Fluorophenol	38 %	20-53	1	11/20/25	kbc	11/21/25	jlh		
Phenol-d5	23 %	11-40	1	11/20/25	kbc	11/21/25	jlh		
Nitrobenzene-d5	92 %	36-103	1	11/20/25	kbc	11/21/25	jlh		
2-Fluorobiphenyl	83 %	36-119	1	11/20/25	kbc	11/21/25	jlh	N	
2,4,6-Tribromophenol	84 %	30-105	1	11/20/25	kbc	11/21/25	jlh	N	
Terphenyl-d14	96 %	37-109	1	11/20/25	kbc	11/21/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176421									
Benzidine	**0.60 ug/L**	**0.098**	**1**	**11/24/25**	**av**	**11/24/25**	**jlh**		
3,3'-Dichlorobenzidine	<0.98 ug/L	0.98	1	11/24/25	av	11/24/25	jlh		
Surrogates:									
Nitrobenzene-d5	84 %	38-130	1	11/24/25	av	11/24/25	jlh		
2-Fluorobiphenyl	61 %	39-118	1	11/24/25	av	11/24/25	jlh	N	
Terphenyl-d14	34 %	14-96	1	11/24/25	av	11/24/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176203									
Bromomethane	<5.0 ug/L	5.0	1	11/18/25	sb	11/18/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	107 %	68-133	1	11/18/25	sb	11/18/25	sb		
Toluene-d8	95 %	75-120	1	11/18/25	sb	11/18/25	sb		
4-Bromofluorobenzene	98 %	69-119	1	11/18/25	sb	11/18/25	sb		
1,2-Dichlorobenzene-d4	102 %	72-127	1	11/18/25	sb	11/18/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25K0885
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K0885-02 | Matrix: Wastewater | Date Collected: 11/17/25 07:35
Sample ID: AAR- 6th | | Date Received: 11/18/25 10:44

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

CERTIFICATE OF ANALYSIS

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QUALITY CONTROL RESULTS

Trace Project ID: 25K0885
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176318	Analysis Description: Beryllium, Total
QC Batch Method: EPA 200.2	Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T176318-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.0040	0.0040	

LABORATORY CONTROL SAMPLE: T176318-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.183	91	85-115	

Trace Project ID: 25K0885
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176295	Analysis Description: TTO Semi-Volatiles
QC Batch Method: EPA 625 Extraction for BNAs	Analysis Method: EPA 625.1

METHOD BLANK: T176295-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	%	32	20-53	
Phenol-d5 (S)	%	19	11-40	
Nitrobenzene-d5 (S)	%	67	36-103	
2-Fluorobiphenyl (S)	%	58	36-119	
2,4,6-Tribromophenol (S)	%	65	30-105	
Terphenyl-d14 (S)	%	80	37-109	

LABORATORY CONTROL SAMPLE: T176295-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	19.9	20	17-120	
2-Fluorophenol (S)	%	100	34.8	35	20-53	
Phenol-d5 (S)	%	100	20.7	21	11-40	
Nitrobenzene-d5 (S)	%	100	86.0	86	36-103	
2-Fluorobiphenyl (S)	%	100	79.1	79	36-119	
2,4,6-Tribromophenol (S)	%	100	79.7	80	30-105	
Terphenyl-d14 (S)	%	100	90.6	91	37-109	

Trace Project ID: 25K0885
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176421
QC Batch Method: GC/MS No Prep
Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T176421-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	42	38-130	
2-Fluorobiphenyl (S)	**%**	**34**	**39-118**	**S10**
Terphenyl-d14 (S)	%	21	14-96	

LABORATORY CONTROL SAMPLE: T176421-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	0.998	50	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	1.62	81	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.839	84	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.736	74	39-118	
Terphenyl-d14 (S)	%	1.00	0.456	46	14-96	

Trace Project ID: 25K0885
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176203
QC Batch Method: EPA 624 Purge and Trap
Analysis Description: TTO Volatiles
Analysis Method: EPA 624.1

METHOD BLANK: T176203-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	115	68-133	
Toluene-d8 (S)	%	92	75-120	
4-Bromofluorobenzene (S)	%	104	69-119	
1,2-Dichlorobenzene-d4 (S)	%	104	72-127	

LABORATORY CONTROL SAMPLE: T176203-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	49.0	98	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	33.0	110	68-133	

LABORATORY CONTROL SAMPLE: T176203-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Toluene-d8 (S)	%	30.0	28.0	93	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.2	97	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	30.0	100	72-127	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25K0885-01

Analysis: EPA 625.1

2,4,6-Tribromophenol	Note S12 : The acid surrogate recoveries were out of control. The results for the acid compounds must be considered estimated.
2-Fluorophenol	Note S12 : The acid surrogate recoveries were out of control. The results for the acid compounds must be considered estimated.
Phenol-d5	Note S12 : The acid surrogate recoveries were out of control. The results for the acid compounds must be considered estimated.

Trace ID: T176421-BLK1

Analysis: EPA 625.1 SIM

2-Fluorobiphenyl	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

TRACE ANALYTICAL LABORATORIES, INC.

CHAIN-OF-CUSTODY RECORD

Trace Analytical Laboratories, Inc.
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Page 1 of 1

Trace ID No. 25K0885

Report Results To:

Company Name:	City of Cadillac
Report To:	Cindy Tomaszewski
Mailing Address:	1121 Plett Rd
City, State, Zip Code:	Cadillac MI 49601
Office Phone:	231-775-2368
Cell Phone:	231-878-6230
Email Address:	lab@cadillac-mi.net; pmiller@cadillac-mi.net

Bill To:

PO #:	
Contact Name:	Cindy Tomaszewski
Billing Address (if different):	200 N Lake St
City, State, Zip Code:	Cadillac MI 49601
Phone Number:	
Billing Email Address:	lab@cadillac-mi.net

Trace Use:

Logged By:	JLB
Checked By:	KC
Sample Collection Time (Hrs):	

Requested Turnaround Times (TAT)

- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:

WW = Wastewater; AQ = Aqueous; A = Air
DW = Drinking Water; O = Oil; WI = Wipes
GW = Groundwater; S = Solid; U = Unknown
LW = Liquid Waste; SL = Sludge

Project Name: 2025 IPP/MAHL Additional

Sampled By (print): DEREK GOODRICH

Analysis Requested

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Composite	Grab	Metals Field Filterd (Y or N)	Matrix - see above	Number of Containers	Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes	Possible Health Hazards?
1	11-17	7.25	AKwel C.B		✓	N	WW	6	x		x		x		x		X	X	X	X		
2	11-17	7.31	AAR-6th		✓	N	WW	6	x		x		x		x		X	X	X	X		

Please Sign

	Released By	Received By	Date	Time		Released By	Received By	Date	Time
1)	CATomaszewski	UPS Ground	11-17-25	1330	2)	UPS	[signature]	11/18/25	1044
3)					4)				

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

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25K0885
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

	Original Observation	Corrected Temperature	IR-9 (CF: 0.0°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.5°C)	IR-14 (CF: -1.1°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Date: 11/18/25										
Time: 1016										
Initials: MCR										
Package Description: cooler										
Package Temp °C	1.0	0.5			✓					
Representative Sample Temp °C	3.8	3.3			✓					✓

Sample Receipt

Yes No
- [x] [ ] Received on ice or other coolant
- [x] [ ] Ice still present upon receipt
- [ ] [x] Custody seals present

[ ] Yes [ ] No Custody seals intact (if applicable)

[ ] Trace Courier [ ] Client Drop-off [x] UPS [ ] Fed Ex [ ] US Mail [ ] Other

Sample Condition

Yes No N/A
- [x] [ ] [ ] All sample containers arrived unbroken and labeled
- [x] [ ] [ ] Sufficient sample to run requested analyses
- [x] [ ] [ ] Correct chemical preservative added to samples
- [ ] [x] [ ] Samples preserved at Trace ______________________
- [x] [ ] [ ] Chemical preservation verified, check EMD pH test strip used (if applicable)
  - [x] pH 0-2.5 (Lot: HC461264) [ ] pH 11.0-13.0 (Lot: HC022540) [ ] Other
- [x] [ ] [ ] Air bubbles absent from VOAs (no bubble grater than 6mm) ______________________

Chain of Custody (COC)

Yes No
- [x] [ ] All bottle labels agree with COC
- [x] [ ] COC filled out properly
- [x] [ ] COC signed by client

Notes:

Form 70-A.59
Effective 8/20/25

TRACE Analytical Laboratories, Inc.

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December 08, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25K1027
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

PJLA Testing Accreditation# 1031

TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25K1027-01	Domestic #1	Wastewater	DG/CM	11/19/25	11/20/25 10:00
Sample type: Composite					
25K1027-02	Domestic #2	Wastewater	DG/CM	11/19/25	11/20/25 10:00
Sample type: Composite					
25K1027-03	WWTP Raw Influent	Wastewater	DG/CM	11/19/25	11/20/25 10:00
Sample type: Composite					
25K1027-04	WWTP Primary Effluent	Wastewater	DG/CM	11/19/25	11/20/25 10:00
Sample type: Composite					
25K1027-05	WWTP Final Effluent	Wastewater	DG/CM	11/19/25	11/20/25 10:00
Sample type: Composite					

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ANALYTICAL RESULTS

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1027-01 Matrix: Wastewater Date Collected: 11/19/25
Sample ID: Domestic #1 Date Received: 11/20/25 10:00

PARAMETERS		RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL										
Analysis Method: EPA 200.8 Rev. 5.4										
Batch: T176382										
Beryllium		<0.0040 mg/L	0.0040	1	11/24/25	ejb	12/05/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 625.1										
Batch: T176359										
Phenol		**13 ug/L**	**5.0**	**2**	**11/21/25**	**jj**	**11/22/25**	**jlh**		
Surrogates:										
2-Fluorophenol		38 %	20-53	2	11/21/25	jj	11/22/25	jlh		
Phenol-d5		26 %	11-40	2	11/21/25	jj	11/22/25	jlh		
Nitrobenzene-d5		72 %	36-103	2	11/21/25	jj	11/22/25	jlh		
2-Fluorobiphenyl		66 %	36-119	2	11/21/25	jj	11/22/25	jlh	N	
2,4,6-Tribromophenol		70 %	30-105	2	11/21/25	jj	11/22/25	jlh	N	
Terphenyl-d14		67 %	37-109	2	11/21/25	jj	11/22/25	jlh		
Analysis Method: EPA 625.1 SIM										
Batch: T176421										
Benzidine		**3.9 ug/L**	**1.0**	**10**	**11/24/25**	**av**	**11/24/25**	**jlh**		
3,3'-Dichlorobenzidine		<10 ug/L	10	10	11/24/25	av	11/24/25	jlh		
Surrogates:										
Nitrobenzene-d5	*	**206** %	38-130	10	11/24/25	av	11/24/25	jlh	S10	
2-Fluorobiphenyl		56 %	39-118	10	11/24/25	av	11/24/25	jlh	N	
Terphenyl-d14		64 %	14-96	10	11/24/25	av	11/24/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 624.1										
Batch: T176325										
Bromomethane		<5.0 ug/L	5.0	1	11/20/25	sb	11/21/25	sb		
Surrogates:										
1,2-Dichloroethane-d4		116 %	68-133	1	11/20/25	sb	11/21/25	sb		
Toluene-d8		93 %	75-120	1	11/20/25	sb	11/21/25	sb		
4-Bromofluorobenzene		102 %	69-119	1	11/20/25	sb	11/21/25	sb		
1,2-Dichlorobenzene-d4		105 %	72-127	1	11/20/25	sb	11/21/25	sb		

CERTIFICATE OF ANALYSIS

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ANALYTICAL RESULTS

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1027-01 Matrix: Wastewater Date Collected: 11/19/25
Sample ID: Domestic #1 Date Received: 11/20/25 10:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673

ANALYTICAL LABORATORIES, INC.

231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

ANALYTICAL RESULTS

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1027-02 | Matrix: Wastewater | Date Collected: 11/19/25
Sample ID: Domestic #2 | Date Received: 11/20/25 10:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176382									
Beryllium	<0.0040 mg/L	0.0040	1	11/24/25	ejb	12/05/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176359									
Phenol	**17 ug/L**	**5.1**	**2**	**11/21/25**	**jj**	**11/22/25**	**jlh**		
Surrogates:									
2-Fluorophenol	45 %	20-53	2	11/21/25	jj	11/22/25	jlh		
Phenol-d5	30 %	11-40	2	11/21/25	jj	11/22/25	jlh		
Nitrobenzene-d5	81 %	36-103	2	11/21/25	jj	11/22/25	jlh		
2-Fluorobiphenyl	70 %	36-119	2	11/21/25	jj	11/22/25	jlh	N	
2,4,6-Tribromophenol	74 %	30-105	2	11/21/25	jj	11/22/25	jlh	N	
Terphenyl-d14	72 %	37-109	2	11/21/25	jj	11/22/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176421									
Benzidine	**3.8 ug/L**	**1.0**	**10**	**11/24/25**	**av**	**11/24/25**	**jlh**		
3,3'-Dichlorobenzidine	<10 ug/L	10	10	11/24/25	av	11/24/25	jlh		
Surrogates:									
Nitrobenzene-d5	110 %	38-130	10	11/24/25	av	11/24/25	jlh		
2-Fluorobiphenyl	55 %	39-118	10	11/24/25	av	11/24/25	jlh	N	
Terphenyl-d14	60 %	14-96	10	11/24/25	av	11/24/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176325									
Bromomethane	<5.0 ug/L	5.0	1	11/20/25	sb	11/21/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	119 %	68-133	1	11/20/25	sb	11/21/25	sb		
Toluene-d8	91 %	75-120	1	11/20/25	sb	11/21/25	sb		
4-Bromofluorobenzene	103 %	69-119	1	11/20/25	sb	11/21/25	sb		
1,2-Dichlorobenzene-d4	107 %	72-127	1	11/20/25	sb	11/21/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1027-02 Matrix: Wastewater Date Collected: 11/19/25
Sample ID: Domestic #2 Date Received: 11/20/25 10:00

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1027-03 | Matrix: Wastewater | Date Collected: 11/19/25
Sample ID: WWTP Raw Influent | Date Received: 11/20/25 10:00

PARAMETERS		RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL										
Analysis Method: EPA 200.8 Rev. 5.4										
Batch: T176382										
Beryllium		<0.0040 mg/L	0.0040	1	11/24/25	ejb	12/05/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									DL03	
Analysis Method: EPA 625.1										
Batch: T176359										
Phenol		<20 ug/L	20	8	11/21/25	jj	11/25/25	jlh		
Surrogates:										
2-Fluorophenol	*	**17** %	20-53	8	11/21/25	jj	11/25/25	jlh	S09	
Phenol-d5		16 %	11-40	8	11/21/25	jj	11/25/25	jlh		
Nitrobenzene-d5		49 %	36-103	8	11/21/25	jj	11/25/25	jlh		
2-Fluorobiphenyl		43 %	36-119	8	11/21/25	jj	11/25/25	jlh	N	
2,4,6-Tribromophenol		36 %	30-105	8	11/21/25	jj	11/25/25	jlh	N	
Terphenyl-d14		45 %	37-109	8	11/21/25	jj	11/25/25	jlh		
Analysis Method: EPA 625.1 SIM										
Batch: T176421										
Benzidine		**9.3 ug/L**	**2.0**	**2**	**11/24/25**	**av**	**11/24/25**	**jlh**		
3,3'-Dichlorobenzidine		<20 ug/L	20	2	11/24/25	av	11/24/25	jlh		
Surrogates:										
Nitrobenzene-d5		81 %	38-130	2	11/24/25	av	11/24/25	jlh		
2-Fluorobiphenyl		57 %	39-118	2	11/24/25	av	11/24/25	jlh	N	
Terphenyl-d14		16 %	14-96	2	11/24/25	av	11/24/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 624.1										
Batch: T176325										
Bromomethane		<5.0 ug/L	5.0	1	11/20/25	sb	11/21/25	sb		
Surrogates:										
1,2-Dichloroethane-d4		116 %	68-133	1	11/20/25	sb	11/21/25	sb		
Toluene-d8		91 %	75-120	1	11/20/25	sb	11/21/25	sb		
4-Bromofluorobenzene		102 %	69-119	1	11/20/25	sb	11/21/25	sb		
1,2-Dichlorobenzene-d4		102 %	72-127	1	11/20/25	sb	11/21/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1027-03 Matrix: Wastewater Date Collected: 11/19/25
Sample ID: WWTP Raw Influent Date Received: 11/20/25 10:00

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1027-04 Matrix: Wastewater Date Collected: 11/19/25
Sample ID: WWTP Primary Effluent Date Received: 11/20/25 10:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176382									
Beryllium	<0.0040 mg/L	0.0040	1	11/24/25	ejb	12/05/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS								DL03	
Analysis Method: EPA 625.1									
Batch: T176359									
Phenol	<10 ug/L	10	4	11/21/25	jj	11/25/25	jlh		
Surrogates:									
2-Fluorophenol	29 %	20-53	4	11/21/25	jj	11/25/25	jlh		
Phenol-d5	21 %	11-40	4	11/21/25	jj	11/25/25	jlh		
Nitrobenzene-d5	53 %	36-103	4	11/21/25	jj	11/25/25	jlh		
2-Fluorobiphenyl	47 %	36-119	4	11/21/25	jj	11/25/25	jlh	N	
2,4,6-Tribromophenol	52 %	30-105	4	11/21/25	jj	11/25/25	jlh	N	
Terphenyl-d14	43 %	37-109	4	11/21/25	jj	11/25/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176421									
Benzidine	**3.7 ug/L**	**0.81**	**8**	**11/24/25**	**av**	**11/24/25**	**jlh**		
3,3'-Dichlorobenzidine	<8.1 ug/L	8.1	8	11/24/25	av	11/24/25	jlh		
Surrogates:									
Nitrobenzene-d5	100 %	38-130	8	11/24/25	av	11/24/25	jlh		
2-Fluorobiphenyl	48 %	39-118	8	11/24/25	av	11/24/25	jlh	N	
Terphenyl-d14	34 %	14-96	8	11/24/25	av	11/24/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176325									
Bromomethane	<5.0 ug/L	5.0	1	11/20/25	sb	11/21/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	117 %	68-133	1	11/20/25	sb	11/21/25	sb		
Toluene-d8	92 %	75-120	1	11/20/25	sb	11/21/25	sb		
4-Bromofluorobenzene	102 %	69-119	1	11/20/25	sb	11/21/25	sb		
1,2-Dichlorobenzene-d4	104 %	72-127	1	11/20/25	sb	11/21/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1027-04 Matrix: Wastewater Date Collected: 11/19/25
Sample ID: WWTP Primary Effluent Date Received: 11/20/25 10:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1027-05 | Matrix: Wastewater | Date Collected: 11/19/25
Sample ID: WWTP Final Effluent | Date Received: 11/20/25 10:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176382									
Beryllium	<0.0040 mg/L	0.0040	1	11/24/25	ejb	12/05/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176359									
Phenol	<5.0 ug/L	5.0	1	11/21/25	jj	11/25/25	jlh		
Surrogates:									
2-Fluorophenol	37 %	20-53	1	11/21/25	jj	11/25/25	jlh		
Phenol-d5	25 %	11-40	1	11/21/25	jj	11/25/25	jlh		
Nitrobenzene-d5	70 %	36-103	1	11/21/25	jj	11/25/25	jlh		
2-Fluorobiphenyl	64 %	36-119	1	11/21/25	jj	11/25/25	jlh	N	
2,4,6-Tribromophenol	69 %	30-105	1	11/21/25	jj	11/25/25	jlh	N	
Terphenyl-d14	77 %	37-109	1	11/21/25	jj	11/25/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176421									
Benzidine	<0.10 ug/L	0.10	1	11/24/25	av	11/24/25	jlh		
3,3'-Dichlorobenzidine	<1.0 ug/L	1.0	1	11/24/25	av	11/24/25	jlh		
Surrogates:									
Nitrobenzene-d5	70 %	38-130	1	11/24/25	av	11/24/25	jlh		
2-Fluorobiphenyl	59 %	39-118	1	11/24/25	av	11/24/25	jlh	N	
Terphenyl-d14	30 %	14-96	1	11/24/25	av	11/24/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176325									
Bromomethane	<5.0 ug/L	5.0	1	11/20/25	sb	11/21/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	120 %	68-133	1	11/20/25	sb	11/21/25	sb		
Toluene-d8	92 %	75-120	1	11/20/25	sb	11/21/25	sb		
4-Bromofluorobenzene	105 %	69-119	1	11/20/25	sb	11/21/25	sb		
1,2-Dichlorobenzene-d4	105 %	72-127	1	11/20/25	sb	11/21/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1027-05 Matrix: Wastewater Date Collected: 11/19/25
Sample ID: WWTP Final Effluent Date Received: 11/20/25 10:00

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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QUALITY CONTROL RESULTS

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176382	Analysis Description: Beryllium, Total
QC Batch Method: EPA 200.2	Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T176382-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.0040	0.0040	

LABORATORY CONTROL SAMPLE: T176382-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.200	100	85-115	

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176359	Analysis Description: TTO Semi-Volatiles
QC Batch Method: EPA 625 Extraction for BNAs	Analysis Method: EPA 625.1

METHOD BLANK: T176359-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	%	40	20-53	
Phenol-d5 (S)	%	26	11-40	
Nitrobenzene-d5 (S)	%	69	36-103	
2-Fluorobiphenyl (S)	%	63	36-119	
2,4,6-Tribromophenol (S)	%	62	30-105	
Terphenyl-d14 (S)	%	78	37-109	

LABORATORY CONTROL SAMPLE: T176359-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	26.7	27	17-120	
2-Fluorophenol (S)	%	100	42.2	42	20-53	
Phenol-d5 (S)	%	100	27.3	27	11-40	
Nitrobenzene-d5 (S)	%	100	78.6	79	36-103	
2-Fluorobiphenyl (S)	%	100	71.8	72	36-119	
2,4,6-Tribromophenol (S)	%	100	71.7	72	30-105	
Terphenyl-d14 (S)	%	100	87.4	87	37-109	

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MATRIX SPIKE: T176359-MS1 Original: **25K1027-01**

Parameter	Units	Original Result	Spike Conc.	MS Result	MS % Rec	% Rec Unit	Notes
Phenol	ug/L	13.0	101	32.7	20	5-120	
2-Fluorophenol (S)	%		101	32.4	32	20-53	
Phenol-d5 (S)	%		101	23.4	23	11-40	
Nitrobenzene-d5 (S)	%		101	61.5	61	36-103	
2-Fluorobiphenyl (S)	%		101	58.0	57	36-119	
2,4,6-Tribromophenol (S)	%		101	58.3	58	30-105	
Terphenyl-d14 (S)	%		101	60.0	59	37-109	

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176421
QC Batch Method: GC/MS No Prep
Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T176421-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	42	38-130	
2-Fluorobiphenyl (S)	**%**	**34**	**39-118**	**S10**
Terphenyl-d14 (S)	%	21	14-96	

LABORATORY CONTROL SAMPLE: T176421-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	0.998	50	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	1.62	81	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.839	84	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.736	74	39-118	
Terphenyl-d14 (S)	%	1.00	0.456	46	14-96	

Trace Project ID: 25K1027
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176325
QC Batch Method: EPA 624 Purge and Trap
Analysis Description: TTO Volatiles
Analysis Method: EPA 624.1

METHOD BLANK: T176325-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes

METHOD BLANK: T176325-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	118	68-133	
Toluene-d8 (S)	%	92	75-120	
4-Bromofluorobenzene (S)	%	104	69-119	
1,2-Dichlorobenzene-d4 (S)	%	104	72-127	

LABORATORY CONTROL SAMPLE: T176325-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	41.7	83	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	33.1	110	68-133	
Toluene-d8 (S)	%	30.0	27.9	93	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.0	97	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	30.3	101	72-127	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25K1027-01

Analysis: EPA 625.1 SIM

Nitrobenzene-d5	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Trace ID: 25K1027-03

Analysis: EPA 625.1

	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.
2-Fluorophenol	Note S09 : One of the acid surrogate recoveries was outside the control limit. Since the other two acid surrogates were within the control limits, no data requires qualification.

Trace ID: 25K1027-04

Analysis: EPA 625.1

	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Trace ID: T176421-BLK1

Analysis: EPA 625.1 SIM

2-Fluorobiphenyl	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

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CHAIN-OF-CUSTODY RECORD

TRACE ANALYTICAL LABORATORIES, INC.

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Page 1 of ~~1~~ 2

Trace ID No. 25K1027

Report Results To:

Company Name: City of Cadillac
Report To: Cindy Tomaszewski
Mailing Address: 1121 Plett Rd
City, State, Zip Code: Cadillac MI 49601
Office Phone: 231-775-2368 Cell Phone: 231-878-6230
Email Address: lab@cadillac-mi.net; pmiller@cadillac-mi.net

Bill To:

PO #:
Contact Name: Cindy Tomaszewski
Billing Address (if different): 200 N Lake St
City, State, Zip Code: Cadillac MI 49601
Phone Number:
Billing Email Address: lab@cadillac-mi.net

Trace Use:

Logged By: NC
Checked By: AC
Sample Collection Time (Hrs):

Requested Turnaround Times (TAT)

- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:

WW = Wastewater; AQ = Aqueous; A = Air
DW = Drinking Water; O = Oil; WI = Wipes
GW = Groundwater; S = Solid; U = Unknown
LW = Liquid Waste; SL = Sludge

Project Name: 2025 IPP/MAHL Additional

Sampled By (print): DG / CH

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Sample Type: Composite	Sample Type: Grab	Metals Field Filterd (Y or N)	Matrix - see above	Number of Containers	Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes
1	11/18/25	—	Domestic #1	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	
2	11/18/25	—	Domestic #2	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	grab VOCs to
3	11/18/25	—	WWTP Raw Influent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	be composited
4	11/18/25	—	WWTP Primary Effluent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	at the lab
5	11/18/25	—	WWTP Final Effluent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	

24 hr. Comp. Started on "Sample Collection Date"

Possible Health Hazards?

3.2

Please Sign

	Released By	Received By	Date	Time	Released By	Received By	Date	Time
	1) [signature]	UPS Overnite	11-19-25	1140	2) UPS	[signature]	11/20/25	10:00
	3)				4)			

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

CERTIFICATE OF ANALYSIS

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673

TRACE
ANALYTICAL LABORATORIES, INC.

231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

25K1027
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

	Original Observation	Corrected Temperature	IR-9 (CF: 0.0°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.5°C)	IR-14 (CF: -1.1°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Date: 11/20/25										
Time: 10:00										
Initials: MJ										
Package Description: Cooler										
Package Temp °C	0.0	0.0	✓							
Representative Sample Temp °C	3.2	3.2	✓							✓

Sample Receipt

Yes No
- [x] Yes [ ] No Received on ice or other coolant
- [x] Yes [ ] No Ice still present upon receipt
- [ ] Yes [x] No Custody seals present
- [ ] Yes [ ] No Custody seals intact (if applicable)
- [ ] Trace Courier [ ] Client Drop-off [x] UPS [ ] Fed Ex [ ] US Mail [ ] Other

Sample Condition

11/20/25 MJ

Yes	No	N/A	
~~☑~~	☑	☐	All sample containers arrived unbroken and labeled
☑	☐	☐	Sufficient sample to run requested analyses
☑	☐	☐	Correct chemical preservative added to samples
☐	☑	☐	Samples preserved at Trace
☑	☐	☐	Chemical preservation verified, check EMD pH test strip used (if applicable)
☑	☐	☐	Air bubbles absent from VOAs (no bubble grater than 6mm)

[ ] pH 0-2.5 (Lot: HC461264) [ ] pH 11.0-13.0 (Lot: HC022540) [ ] Other

Chain of Custody (COC)

Yes No
- [ ] Yes [ ] No All bottle labels agree with COC
- [ ] Yes [ ] No COC filled out properly
- [ ] Yes [ ] No COC signed by client

Notes:

One Amber arrived broken from Sample point 5.

Form 70-A.59
Effective 8/20/25

TRACE Analytical Laboratories, Inc

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TRACE ANALYTICAL LABORATORIES, INC.

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888-979-4469 Fax
www.trace-labs.com

December 08, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25K1054
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

PJLA Testing Accreditation# 1031

TNI EL V1:2016

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673

TRACE ANALYTICAL LABORATORIES, INC.

231-773-5998 Phone
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www.trace-labs.com

SAMPLE SUMMARY

Trace Project ID: 25K1054
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25K1054-01 Sample type: Grab	AAR RT	Wastewater	DG/CM	11/20/25 07:40	11/20/25 12:35
25K1054-02 Sample type: Grab	AAR FAC	Wastewater	DG/CM	11/20/25 07:55	11/20/25 12:35

Trace Analytical Laboratories, Inc.
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ANALYTICAL RESULTS

Trace Project ID: 25K1054
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1054-01 | Matrix: Wastewater | Date Collected: 11/20/25 07:40
Sample ID: AAR RT | Date Received: 11/20/25 12:35

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176382									
Beryllium	<0.0040 mg/L	0.0040	1	11/24/25	ejb	12/05/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176415									
Phenol	<5.0 ug/L	5.0	1	11/24/25	kbc	11/25/25	jlh		
Surrogates:									
2-Fluorophenol	30 %	20-53	1	11/24/25	kbc	11/25/25	jlh		
Phenol-d5	19 %	11-40	1	11/24/25	kbc	11/25/25	jlh		
Nitrobenzene-d5	85 %	36-103	1	11/24/25	kbc	11/25/25	jlh		
2-Fluorobiphenyl	79 %	36-119	1	11/24/25	kbc	11/25/25	jlh	N	
2,4,6-Tribromophenol	82 %	30-105	1	11/24/25	kbc	11/25/25	jlh	N	
Terphenyl-d14	98 %	37-109	1	11/24/25	kbc	11/25/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176421									
Benzidine	<0.098 ug/L	0.098	1	11/24/25	av	11/24/25	jlh		
3,3'-Dichlorobenzidine	<0.98 ug/L	0.98	1	11/24/25	av	11/24/25	jlh		
Surrogates:									
Nitrobenzene-d5	86 %	38-130	1	11/24/25	av	11/24/25	jlh		
2-Fluorobiphenyl	66 %	39-118	1	11/24/25	av	11/24/25	jlh	N	
Terphenyl-d14	39 %	14-96	1	11/24/25	av	11/24/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176376									
Bromomethane	<5.0 ug/L	5.0	1	11/21/25	sb	11/21/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	118 %	68-133	1	11/21/25	sb	11/21/25	sb		
Toluene-d8	93 %	75-120	1	11/21/25	sb	11/21/25	sb		
4-Bromofluorobenzene	105 %	69-119	1	11/21/25	sb	11/21/25	sb		
1,2-Dichlorobenzene-d4	104 %	72-127	1	11/21/25	sb	11/21/25	sb		

CERTIFICATE OF ANALYSIS

Trace Analytical Laboratories, Inc.
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Muskegon, MI 49444-2673

TRACE ANALYTICAL LABORATORIES, INC.

231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

ANALYTICAL RESULTS

Trace Project ID: 25K1054
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1054-01 | Matrix: Wastewater | Date Collected: 11/20/25 07:40
Sample ID: AAR RT | Date Received: 11/20/25 12:35

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25K1054
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1054-02 | Matrix: Wastewater | Date Collected: 11/20/25 07:55
Sample ID: AAR FAC | Date Received: 11/20/25 12:35

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176382									
Beryllium	<0.0050 mg/L	0.0050	5	11/24/25	ejb	12/05/25	drm	DL02	
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176415									
Phenol	**9.5 ug/L**	**5.0**	**1**	**11/24/25**	**kbc**	**11/25/25**	**jlh**		
Surrogates:									
2-Fluorophenol	34 %	20-53	1	11/24/25	kbc	11/25/25	jlh		
Phenol-d5	30 %	11-40	1	11/24/25	kbc	11/25/25	jlh		
Nitrobenzene-d5	72 %	36-103	1	11/24/25	kbc	11/25/25	jlh		
2-Fluorobiphenyl	69 %	36-119	1	11/24/25	kbc	11/25/25	jlh	N	
2,4,6-Tribromophenol	70 %	30-105	1	11/24/25	kbc	11/25/25	jlh	N	
Terphenyl-d14	66 %	37-109	1	11/24/25	kbc	11/25/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176421									
Benzidine	<0.093 ug/L	0.093	1	11/24/25	av	11/24/25	jlh		
3,3'-Dichlorobenzidine	<0.93 ug/L	0.93	1	11/24/25	av	11/24/25	jlh		
Surrogates:									
Nitrobenzene-d5	76 %	38-130	1	11/24/25	av	11/24/25	jlh		
2-Fluorobiphenyl	61 %	39-118	1	11/24/25	av	11/24/25	jlh	N	
Terphenyl-d14	33 %	14-96	1	11/24/25	av	11/24/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176376									
Bromomethane	<5.0 ug/L	5.0	1	11/21/25	sb	11/21/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	106 %	68-133	1	11/21/25	sb	11/21/25	sb		
Toluene-d8	97 %	75-120	1	11/21/25	sb	11/21/25	sb		
4-Bromofluorobenzene	105 %	69-119	1	11/21/25	sb	11/21/25	sb		
1,2-Dichlorobenzene-d4	103 %	72-127	1	11/21/25	sb	11/21/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25K1054
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1054-02 Matrix: Wastewater Date Collected: 11/20/25 07:55
Sample ID: AAR FAC Date Received: 11/20/25 12:35

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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QUALITY CONTROL RESULTS

Trace Project ID: 25K1054
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176382
QC Batch Method: EPA 200.2
Analysis Description: Beryllium, Total
Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T176382-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.0040	0.0040	

LABORATORY CONTROL SAMPLE: T176382-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.200	100	85-115	

Trace Project ID: 25K1054
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176415
QC Batch Method: EPA 625 Extraction for BNAs
Analysis Description: TTO Semi-Volatiles
Analysis Method: EPA 625.1

METHOD BLANK: T176415-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	**%**	**63**	**20-53**	**S03**
Phenol-d5 (S)	**%**	**45**	**11-40**	**S03**
Nitrobenzene-d5 (S)	%	89	36-103	
2-Fluorobiphenyl (S)	%	82	36-119	
2,4,6-Tribromophenol (S)	%	86	30-105	
Terphenyl-d14 (S)	**%**	**113**	**37-109**	**S03**

LABORATORY CONTROL SAMPLE: T176415-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	22.5	23	17-120	
2-Fluorophenol (S)	%	100	39.4	39	20-53	
Phenol-d5 (S)	%	100	25.5	25	11-40	
Nitrobenzene-d5 (S)	%	100	96.9	97	36-103	
2-Fluorobiphenyl (S)	%	100	91.9	92	36-119	
2,4,6-Tribromophenol (S)	%	100	94.8	95	30-105	
Terphenyl-d14 (S)	**%**	**100**	**115**	**115**	**37-109**	**S10**

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Trace Project ID: 25K1054
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176421
QC Batch Method: GC/MS No Prep
Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T176421-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	42	38-130	
2-Fluorobiphenyl (S)	**%**	**34**	**39-118**	**S10**
Terphenyl-d14 (S)	%	21	14-96	

LABORATORY CONTROL SAMPLE: T176421-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	0.998	50	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	1.62	81	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.839	84	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.736	74	39-118	
Terphenyl-d14 (S)	%	1.00	0.456	46	14-96	

Trace Project ID: 25K1054
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176376
QC Batch Method: EPA 624 Purge and Trap
Analysis Description: TTO Volatiles
Analysis Method: EPA 624.1

METHOD BLANK: T176376-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	119	68-133	
Toluene-d8 (S)	%	91	75-120	
4-Bromofluorobenzene (S)	%	103	69-119	
1,2-Dichlorobenzene-d4 (S)	%	105	72-127	

LABORATORY CONTROL SAMPLE: T176376-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	39.5	79	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	34.0	113	68-133	

LABORATORY CONTROL SAMPLE: T176376-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Toluene-d8 (S)	%	30.0	28.2	94	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.2	97	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	30.2	101	72-127	

AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25K1054-02

Analysis: EPA 200.8 Rev. 5.4

Beryllium	Note DL02 : The reporting limit was raised due to a dilution required because of chromatographic/matrix interference with the internal standards.

Trace ID: T176415-BLK1

Analysis: EPA 625.1

2-Fluorophenol	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.
Phenol-d5	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.
Terphenyl-d14	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.

Trace ID: T176415-BS1

Analysis: EPA 625.1

Terphenyl-d14	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Trace ID: T176421-BLK1

Analysis: EPA 625.1 SIM

2-Fluorobiphenyl	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

CHAIN-OF-CUSTODY RECORD

TRACE ANALYTICAL LABORATORIES, INC.

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673

Phone 231.773.5998
Fax 888.979.4469
www.trace-labs.com

Page 1 of 1

Trace ID No. 25K1054

Report Results To:

Company Name:	City of Cadillac
Report To:	Cindy Tomaszewski
Mailing Address:	1121 Plett Rd
City, State, Zip Code:	Cadillac MI 49601
Office Phone:	231-775-2368
Cell Phone:	231-878-6230
Email Address:	lab@cadillac-mi.net; pmiller@cadillac-mi.net

Bill To:

PO #:	
Contact Name:	Cindy Tomaszewski
Billing Address (if different):	200 N Lake St
City, State, Zip Code:	Cadillac MI 49601
Phone Number:	
Billing Email Address:	lab@cadillac-mi.net

Trace Use:

Logged By:	MS
Checked By:	MCK
Sample Collection Time (Hrs):	

Requested Turnaround Times (TAT)

- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:

WW = Wastewater, DW = Drinking Water, GW = Groundwater, LW = Liquid Waste, AQ = Aqueous, O = Oil, S = Solid, SL = Sludge, A = Air, WI = Wipes, U = Unknown

Project Name: 2025 IPP/MAHL Additional

Sampled By (print): DG/CM

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Sample Type: Composite	Sample Type: Grab	Metals Field Filterd (Y or N)	Matrix - see above →	Number of Containers	Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes	Possible Health Hazards?
1	11/20/25	0740	AAR RT		✓	N	WW	6	x		x		x		x		X	X	X	X		
2	11/20/25	0755	AAR FAC		✓	N	WW	6	x		x		x		x		X	X	X	X		

Please Sign

	Released By	Received By	Date	Time		Released By	Received By	Date	Time
1)	[signature]	[signature]	11/20/25	12:35	2)	[signature]	[signature]	11/20/25	15:28
3)					4)				

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

CERTIFICATE OF ANALYSIS

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673

TRACE
ANALYTICAL LABORATORIES, INC.

231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

25K1054
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

	Original Observation	Corrected Temperature	IR-9 (CF: 0.0°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.5°C)	IR-14 (CF: -1.1°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Date: 11/20/25										
Time: 15:28										
Initials: AL										
Package Description: Cooler										
Package Temp °C	-0.1	-0.1		/						
Representative Sample Temp °C	1.9	1.7					/		/	/

Sample Receipt

Yes	No	
☑	☐	Received on ice or other coolant
☑	☐	Ice still present upon receipt
☐	☑	Custody seals present

☐ Yes ☐ No Custody seals intact (if applicable)

☑ Trace Courier ☐ Client Drop-off ☐ UPS ☐ Fed Ex ☐ US Mail ☐ Other

Sample Condition

Yes	No	N/A	
☑	☐	☐	All sample containers arrived unbroken and labeled
☑	☐	☐	Sufficient sample to run requested analyses
☑	☐	☐	Correct chemical preservative added to samples
☐	☐	☑	Samples preserved at Trace
☐	☐	☐	Chemical preservation verified, check EMD pH test strip used (if applicable)

☑ pH 0-2.5 (Lot: HC461264) ☐ pH 11.0-13.0 (Lot: HC022540) ☐ Other

Yes	No	N/A	
~~☑~~	☐	☑	Air bubbles absent from VOAs (no bubble grater than 6mm)

11/20/25 AL

Chain of Custody (COC)

Yes	No	
☑	☐	All bottle labels agree with COC
☑	☐	COC filled out properly
☑	☐	COC signed by client

Notes:

Form 70-A.59
Effective 8/20/25

TRACE Analytical Laboratories, Inc.

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673

TRACE ANALYTICAL LABORATORIES, INC.

231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

December 08, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25K1055
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

PJLA Testing Accreditation# 1031

TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 25K1055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25K1055-01	Domestic #1	Wastewater	DG/CM	11/20/25	11/20/25 12:35
Sample type: Composite					
25K1055-02	Domestic #2	Wastewater	DG/CM	11/20/25	11/20/25 12:35
Sample type: Composite					
25K1055-03	WWTP Raw Influent	Wastewater	DG/CM	11/20/25	11/20/25 12:35
Sample type: Composite					
25K1055-04	WWTP Primary Effluent	Wastewater	DG/CM	11/20/25	11/20/25 12:35
Sample type: Composite					
25K1055-05	WWTP Final Effluent	Wastewater	DG/CM	11/20/25	11/20/25 12:35
Sample type: Composite					

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ANALYTICAL RESULTS

Trace Project ID: 25K1055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1055-01 Matrix: Wastewater Date Collected: 11/20/25
Sample ID: Domestic #1 Date Received: 11/20/25 12:35

PARAMETERS		RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL											
Analysis Method: EPA 200.8 Rev. 5.4											
Batch: T176382											
Beryllium		<0.0040	mg/L	0.0040	1	11/24/25	ejb	12/05/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS										DL03	
Analysis Method: EPA 625.1											
Batch: T176415											
Phenol		<19	ug/L	19	8	11/24/25	kbc	11/26/25	jlh		
Surrogates:											
2-Fluorophenol		26	%	20-53	8	11/24/25	kbc	11/26/25	jlh		
Phenol-d5		20	%	11-40	8	11/24/25	kbc	11/26/25	jlh		
Nitrobenzene-d5		79	%	36-103	8	11/24/25	kbc	11/26/25	jlh		
2-Fluorobiphenyl		71	%	36-119	8	11/24/25	kbc	11/26/25	jlh	N	
2,4,6-Tribromophenol		79	%	30-105	8	11/24/25	kbc	11/26/25	jlh	N	
Terphenyl-d14		64	%	37-109	8	11/24/25	kbc	11/26/25	jlh		
Analysis Method: EPA 625.1 SIM											
Batch: T176508											
Benzidine		**1.0**	**ug/L**	**0.10**	**1**	**11/26/25**	**als**	**11/26/25**	**jlh**		
3,3'-Dichlorobenzidine		<1.0	ug/L	1.0	1	11/26/25	als	11/26/25	jlh		
Surrogates:											
Nitrobenzene-d5	*	**256**	%	38-130	1	11/26/25	als	11/26/25	jlh	S01	
2-Fluorobiphenyl	*	**34**	%	39-118	1	11/26/25	als	11/26/25	jlh	S02, N	
Terphenyl-d14		18	%	14-96	1	11/26/25	als	11/26/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 624.1											
Batch: T176376											
Bromomethane		<5.0	ug/L	5.0	1	11/21/25	sb	11/21/25	sb		
Surrogates:											
1,2-Dichloroethane-d4		117	%	68-133	1	11/21/25	sb	11/21/25	sb		
Toluene-d8		93	%	75-120	1	11/21/25	sb	11/21/25	sb		
4-Bromofluorobenzene		102	%	69-119	1	11/21/25	sb	11/21/25	sb		
1,2-Dichlorobenzene-d4		104	%	72-127	1	11/21/25	sb	11/21/25	sb		

CERTIFICATE OF ANALYSIS

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ANALYTICAL RESULTS

Trace Project ID: 25K1055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1055-01 Matrix: Wastewater Date Collected: 11/20/25
Sample ID: Domestic #1 Date Received: 11/20/25 12:35

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25K1055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1055-02 | Matrix: Wastewater | Date Collected: 11/20/25
Sample ID: Domestic #2 | Date Received: 11/20/25 12:35

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176382									
Beryllium	<0.0040 mg/L	0.0040	1	11/24/25	ejb	12/05/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176415									
Phenol	**6.9 ug/L**	**5.0**	**2**	**11/24/25**	**kbc**	**11/26/25**	**jlh**		
Surrogates:									
2-Fluorophenol	21 %	20-53	2	11/24/25	kbc	11/26/25	jlh		
Phenol-d5	15 %	11-40	2	11/24/25	kbc	11/26/25	jlh		
Nitrobenzene-d5	57 %	36-103	2	11/24/25	kbc	11/26/25	jlh		
2-Fluorobiphenyl	54 %	36-119	2	11/24/25	kbc	11/26/25	jlh	N	
2,4,6-Tribromophenol	62 %	30-105	2	11/24/25	kbc	11/26/25	jlh	N	
Terphenyl-d14	49 %	37-109	2	11/24/25	kbc	11/26/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176508									
Benzidine	**0.58 ug/L**	**0.11**	**1**	**11/26/25**	**als**	**11/26/25**	**jlh**		
3,3'-Dichlorobenzidine	<1.1 ug/L	1.1	1	11/26/25	als	11/26/25	jlh		
Surrogates:									
Nitrobenzene-d5	44 %	38-130	1	11/26/25	als	11/26/25	jlh		
2-Fluorobiphenyl	* **37** %	39-118	1	11/26/25	als	11/26/25	jlh	S10, N	
Terphenyl-d14	16 %	14-96	1	11/26/25	als	11/26/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176376									
Bromomethane	<5.0 ug/L	5.0	1	11/21/25	sb	11/21/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	117 %	68-133	1	11/21/25	sb	11/21/25	sb		
Toluene-d8	93 %	75-120	1	11/21/25	sb	11/21/25	sb		
4-Bromofluorobenzene	105 %	69-119	1	11/21/25	sb	11/21/25	sb		
1,2-Dichlorobenzene-d4	106 %	72-127	1	11/21/25	sb	11/21/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25K1055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1055-02 | Matrix: Wastewater | Date Collected: 11/20/25
Sample ID: Domestic #2 | Date Received: 11/20/25 12:35

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25K1055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1055-03 Matrix: Wastewater Date Collected: 11/20/25
Sample ID: WWTP Raw Influent Date Received: 11/20/25 12:35

PARAMETERS		RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL											
Analysis Method: EPA 200.8 Rev. 5.4											
Batch: T176382											
Beryllium		<0.0040	mg/L	0.0040	1	11/24/25	ejb	12/05/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS										DL03	
Analysis Method: EPA 625.1											
Batch: T176415											
Phenol		<19	ug/L	19	8	11/24/25	kbc	11/26/25	jlh		
Surrogates:											
2-Fluorophenol		32	%	20-53	8	11/24/25	kbc	11/26/25	jlh		
Phenol-d5		24	%	11-40	8	11/24/25	kbc	11/26/25	jlh		
Nitrobenzene-d5		82	%	36-103	8	11/24/25	kbc	11/26/25	jlh		
2-Fluorobiphenyl		72	%	36-119	8	11/24/25	kbc	11/26/25	jlh	N	
2,4,6-Tribromophenol		83	%	30-105	8	11/24/25	kbc	11/26/25	jlh	N	
Terphenyl-d14		55	%	37-109	8	11/24/25	kbc	11/26/25	jlh		
Analysis Method: EPA 625.1 SIM											
Batch: T176508											
Benzidine		**0.84**	**ug/L**	**0.11**	**1**	**11/26/25**	**als**	**11/26/25**	**jlh**		
3,3'-Dichlorobenzidine		<1.1	ug/L	1.1	1	11/26/25	als	11/26/25	jlh		
Surrogates:											
Nitrobenzene-d5		41	%	38-130	1	11/26/25	als	11/26/25	jlh		
2-Fluorobiphenyl	*	**32**	%	39-118	1	11/26/25	als	11/26/25	jlh	S02, N	
Terphenyl-d14	*	**13**	%	14-96	1	11/26/25	als	11/26/25	jlh	S02	
VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 624.1											
Batch: T176376											
Bromomethane		<5.0	ug/L	5.0	1	11/21/25	sb	11/21/25	sb		
Surrogates:											
1,2-Dichloroethane-d4		115	%	68-133	1	11/21/25	sb	11/21/25	sb		
Toluene-d8		92	%	75-120	1	11/21/25	sb	11/21/25	sb		
4-Bromofluorobenzene		100	%	69-119	1	11/21/25	sb	11/21/25	sb		
1,2-Dichlorobenzene-d4		100	%	72-127	1	11/21/25	sb	11/21/25	sb		

ANALYTICAL RESULTS

Trace Project ID: 25K1055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1055-03 Matrix: Wastewater Date Collected: 11/20/25
Sample ID: WWTP Raw Influent Date Received: 11/20/25 12:35

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25K1055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1055-04 Matrix: Wastewater Date Collected: 11/20/25
Sample ID: WWTP Primary Effluent Date Received: 11/20/25 12:35

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176382									
Beryllium	<0.0040 mg/L	0.0040	1	11/24/25	ejb	12/05/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS								DL03	
Analysis Method: EPA 625.1									
Batch: T176415									
Phenol	<5.0 ug/L	5.0	2	11/24/25	kbc	11/26/25	jlh		
Surrogates:									
2-Fluorophenol	27 %	20-53	2	11/24/25	kbc	11/26/25	jlh		
Phenol-d5	18 %	11-40	2	11/24/25	kbc	11/26/25	jlh		
Nitrobenzene-d5	69 %	36-103	2	11/24/25	kbc	11/26/25	jlh		
2-Fluorobiphenyl	64 %	36-119	2	11/24/25	kbc	11/26/25	jlh	N	
2,4,6-Tribromophenol	67 %	30-105	2	11/24/25	kbc	11/26/25	jlh	N	
Terphenyl-d14	53 %	37-109	2	11/24/25	kbc	11/26/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176508									
Benzidine	**1.1 ug/L**	**0.11**	**1**	**11/26/25**	**als**	**11/26/25**	**jlh**		
3,3'-Dichlorobenzidine	<1.1 ug/L	1.1	1	11/26/25	als	11/26/25	jlh		
Surrogates:									
Nitrobenzene-d5	42 %	38-130	1	11/26/25	als	11/26/25	jlh		
2-Fluorobiphenyl	* **36** %	39-118	1	11/26/25	als	11/26/25	jlh	S10, N	
Terphenyl-d14	14 %	14-96	1	11/26/25	als	11/26/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176376									
Bromomethane	<5.0 ug/L	5.0	1	11/21/25	sb	11/21/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	118 %	68-133	1	11/21/25	sb	11/21/25	sb		
Toluene-d8	92 %	75-120	1	11/21/25	sb	11/21/25	sb		
4-Bromofluorobenzene	102 %	69-119	1	11/21/25	sb	11/21/25	sb		
1,2-Dichlorobenzene-d4	104 %	72-127	1	11/21/25	sb	11/21/25	sb		

ANALYTICAL RESULTS

Trace Project ID: 25K1055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1055-04 Matrix: Wastewater Date Collected: 11/20/25
Sample ID: WWTP Primary Effluent Date Received: 11/20/25 12:35

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25K1055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1055-05 Matrix: Wastewater Date Collected: 11/20/25
Sample ID: WWTP Final Effluent Date Received: 11/20/25 12:35

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176382									
Beryllium	<0.0040 mg/L	0.0040	1	11/24/25	ejb	12/05/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176415									
Phenol	<5.0 ug/L	5.0	1	11/24/25	kbc	11/26/25	jlh		
Surrogates:									
2-Fluorophenol	31 %	20-53	1	11/24/25	kbc	11/26/25	jlh		
Phenol-d5	21 %	11-40	1	11/24/25	kbc	11/26/25	jlh		
Nitrobenzene-d5	79 %	36-103	1	11/24/25	kbc	11/26/25	jlh		
2-Fluorobiphenyl	73 %	36-119	1	11/24/25	kbc	11/26/25	jlh	N	
2,4,6-Tribromophenol	83 %	30-105	1	11/24/25	kbc	11/26/25	jlh	N	
Terphenyl-d14	75 %	37-109	1	11/24/25	kbc	11/26/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176508									
Benzidine	**0.11 ug/L**	**0.10**	**1**	**11/26/25**	**als**	**11/26/25**	**jlh**		
3,3'-Dichlorobenzidine	<1.0 ug/L	1.0	1	11/26/25	als	11/26/25	jlh		
Surrogates:									
Nitrobenzene-d5	89 %	38-130	1	11/26/25	als	11/26/25	jlh		
2-Fluorobiphenyl	70 %	39-118	1	11/26/25	als	11/26/25	jlh	N	
Terphenyl-d14	31 %	14-96	1	11/26/25	als	11/26/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176376									
Bromomethane	<5.0 ug/L	5.0	1	11/21/25	sb	11/21/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	119 %	68-133	1	11/21/25	sb	11/21/25	sb		
Toluene-d8	90 %	75-120	1	11/21/25	sb	11/21/25	sb		
4-Bromofluorobenzene	103 %	69-119	1	11/21/25	sb	11/21/25	sb		
1,2-Dichlorobenzene-d4	104 %	72-127	1	11/21/25	sb	11/21/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25K1055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1055-05 | Matrix: Wastewater | Date Collected: 11/20/25
Sample ID: WWTP Final Effluent | Date Received: 11/20/25 12:35

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

CERTIFICATE OF ANALYSIS

QUALITY CONTROL RESULTS

Trace Project ID: 25K1055
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176382	Analysis Description: Beryllium, Total
QC Batch Method: EPA 200.2	Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T176382-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.0040	0.0040	

LABORATORY CONTROL SAMPLE: T176382-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.200	100	85-115	

MATRIX SPIKE: T176382-MS2 Original: **25K1055-04**

Parameter	Units	Original Result	Spike Conc.	MS Result	MS % Rec	% Rec Unit	Notes
Beryllium	mg/L	0	0.200	0.203	102	70-130	

Trace Project ID: 25K1055
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176415	Analysis Description: TTO Semi-Volatiles
QC Batch Method: EPA 625 Extraction for BNAs	Analysis Method: EPA 625.1

METHOD BLANK: T176415-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	**%**	**63**	**20-53**	**S03**
Phenol-d5 (S)	**%**	**45**	**11-40**	**S03**
Nitrobenzene-d5 (S)	%	89	36-103	
2-Fluorobiphenyl (S)	%	82	36-119	
2,4,6-Tribromophenol (S)	%	86	30-105	
Terphenyl-d14 (S)	**%**	**113**	**37-109**	**S03**

LABORATORY CONTROL SAMPLE: T176415-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	22.5	23	17-120	
2-Fluorophenol (S)	%	100	39.4	39	20-53	
Phenol-d5 (S)	%	100	25.5	25	11-40	

LABORATORY CONTROL SAMPLE: T176415-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Nitrobenzene-d5 (S)	%	100	96.9	97	36-103	
2-Fluorobiphenyl (S)	%	100	91.9	92	36-119	
2,4,6-Tribromophenol (S)	%	100	94.8	95	30-105	
Terphenyl-d14 (S)	**%**	**100**	**115**	**115**	**37-109**	**S10**

MATRIX SPIKE: T176415-MS1 Original: **25K1055-01**

Parameter	Units	Original Result	Spike Conc.	MS Result	MS % Rec	% Rec Unit	Notes
Phenol	ug/L	12.0	94.5	28.2	17	5-120	
2-Fluorophenol (S)	%		94.5	26.7	28	20-53	
Phenol-d5 (S)	%		94.5	19.8	21	11-40	
Nitrobenzene-d5 (S)	%		94.5	69.4	73	36-103	
2-Fluorobiphenyl (S)	%		94.5	66.7	71	36-119	
2,4,6-Tribromophenol (S)	%		94.5	71.6	76	30-105	
Terphenyl-d14 (S)	%		94.5	62.5	66	37-109	

Trace Project ID: 25K1055
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176508
QC Batch Method: GC/MS No Prep
Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T176508-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	80	38-130	
2-Fluorobiphenyl (S)	%	82	39-118	
Terphenyl-d14 (S)	%	50	14-96	

LABORATORY CONTROL SAMPLE: T176508-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	1.01	51	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	1.69	85	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.631	63	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.734	73	39-118	
Terphenyl-d14 (S)	%	1.00	0.446	45	14-96	

QC Batch: T176376	Analysis Description: TTO Volatiles
QC Batch Method: EPA 624 Purge and Trap	Analysis Method: EPA 624.1

METHOD BLANK: T176376-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	119	68-133	
Toluene-d8 (S)	%	91	75-120	
4-Bromofluorobenzene (S)	%	103	69-119	
1,2-Dichlorobenzene-d4 (S)	%	105	72-127	

LABORATORY CONTROL SAMPLE: T176376-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	39.5	79	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	34.0	113	68-133	
Toluene-d8 (S)	%	30.0	28.2	94	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.2	97	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	30.2	101	72-127	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25K1055-01

Analysis: EPA 625.1

Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Analysis: EPA 625.1 SIM

2-Fluorobiphenyl — Note S02 : The surrogate recovery was out of control low when compared to control limits. All results and reporting limits must be considered estimated.

Nitrobenzene-d5 — Note S01 : The surrogate recovery was out of control high when compared to control limits. The results must be considered estimated.

Trace ID: 25K1055-02

Analysis: EPA 625.1 SIM

2-Fluorobiphenyl — Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Trace ID: 25K1055-03

Analysis: EPA 625.1

Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Analysis: EPA 625.1 SIM

2-Fluorobiphenyl — Note S02 : The surrogate recovery was out of control low when compared to control limits. All results and reporting limits must be considered estimated.

Terphenyl-d14 — Note S02 : The surrogate recovery was out of control low when compared to control limits. All results and reporting limits must be considered estimated.

Trace ID: 25K1055-04

Analysis: EPA 625.1

	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Analysis: EPA 625.1 SIM

2-Fluorobiphenyl	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Trace ID: T176415-BLK1

Analysis: EPA 625.1

2-Fluorophenol	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.
Phenol-d5	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.
Terphenyl-d14	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.

Trace ID: T176415-BS1

Analysis: EPA 625.1

Terphenyl-d14	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

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CHAIN-OF-CUSTODY RECORD

TRACE ANALYTICAL LABORATORIES, INC.

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Page 1 of 1

Trace ID No. 25K1055

Report Results To:	Bill To:
Company Name: City of Cadillac	PO #:
Report To: Cindy Tomaszewski	Contact Name: Cindy Tomaszewski
Mailing Address: 1121 Plett Rd	Billing Address (if different): 200 N Lake St
City, State, Zip Code: Cadillac MI 49601	City, State, Zip Code: Cadillac MI 49601
Office Phone: 231-775-2368 Cell Phone: 231-878-6230	Phone Number:
Email Address: lab@cadillac-mi.net; pmiller@cadillac-mi.net	Billing Email Address: lab@cadillac-mi.net

Trace Use:
Logged By: MS
Checked By: MCR
Sample Collection Time (Hrs):

Requested Turnaround Times (TAT)
- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:
WW = Wastewater, AQ = Aqueous, A = Air
DW = Drinking Water, O = Oil, WI = Wipes
GW = Groundwater, S = Solid, U = Unknown
LW = Liquid Waste, SL = Sludge

Project Name: 2025 IPP/MAHL Additional

Sampled By (print): DG/CM

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Composite	Grab	Metals Field Filtered (Y or N)	Matrix - see above →	Number of Containers	Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes	Possible Health Hazards?
1	11/19/25	–	Domestic #1	✓	✓	N	WW	8	x		x		x		x		X	X	X	X		
2	11/19/25	–	Domestic #2	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	* grab VOCs to	
3	11/19/25	–	WWTP Raw Influent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	be composited	
4	11/19/25	–	WWTP Primary Effluent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	at the lab	
5	11/19/25	–	WWTP Final Effluent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X		

* All composites started on "Sample Collection Date"

Please Sign	Released By	Received By	Date	Time	Released By	Received By	Date	Time
	1) [signature]	[signature]	11/20/25	12:35	2) [signature]	[signature]	11/21/25	8:26
	3)				4)			

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

CERTIFICATE OF ANALYSIS

Trace Analytical Laboratories, Inc.
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www.trace-labs.com

25K1055
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

		Original Observation	Corrected Temperature	IR-9 (CF: 0.0°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.5°C)	IR-14 (CF: -1.1°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Date:	11/21/25										
Time:	8:26										
Initials:	SAB										
Package Description:	Cooler										
Package Temp °C		0.5	0.5		/						
Representative Sample Temp °C		1.7	1.4						/	/	

Sample Receipt

Yes	No	
[x]	[ ]	Received on ice or other coolant
[x]	[ ]	Ice still present upon receipt
[ ]	[x]	Custody seals present

[ ] Yes [ ] No Custody seals intact (if applicable)

[x] Trace Courier [ ] Client Drop-off [ ] UPS [ ] Fed Ex [ ] US Mail [ ] Other

Sample Condition

Yes	No	N/A	
[x]	[ ]	[ ]	All sample containers arrived unbroken and labeled
[x]	[ ]	[ ]	Sufficient sample to run requested analyses
[x]	[ ]	[ ]	Correct chemical preservative added to samples
[ ]	[x]	[ ]	Samples preserved at Trace __________
[x]	[ ]	[ ]	Chemical preservation verified, check EMD pH test strip used (if applicable)
			[x] pH 0-2.5 (Lot: HC461264) [ ] pH 11.0-13.0 (Lot: HC022540) [ ] Other
[x]	[ ]	[ ]	Air bubbles absent from VOAs (no bubble grater than 6mm) __________

Chain of Custody (COC)

Yes	No	
[x]	[ ]	All bottle labels agree with COC
[x]	[ ]	COC filled out properly
[x]	[ ]	COC signed by client

Notes:

Form 70-A.59
Effective 8/20/25

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December 08, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25K1111
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

PJLA Testing Accreditation# 1031

TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 25K1111
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25K1111-01 Sample type: Grab	Piranha	Wastewater	Client	11/21/25 07:45	11/21/25 10:10
25K1111-02 Sample type: Grab	Arvco	Wastewater	Client	11/21/25 08:10	11/21/25 10:10

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ANALYTICAL RESULTS

Trace Project ID: 25K1111
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1111-01 Matrix: Wastewater Date Collected: 11/21/25 07:45
Sample ID: Piranha Date Received: 11/21/25 10:10

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176442									
Beryllium	<0.0040 mg/L	0.0040	1	11/25/25	ejb	12/05/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176415									
Phenol	<5.0 ug/L	5.0	1	11/24/25	kbc	11/26/25	jlh		
Surrogates:									
2-Fluorophenol	34 %	20-53	1	11/24/25	kbc	11/26/25	jlh		
Phenol-d5	21 %	11-40	1	11/24/25	kbc	11/26/25	jlh		
Nitrobenzene-d5	90 %	36-103	1	11/24/25	kbc	11/26/25	jlh		
2-Fluorobiphenyl	81 %	36-119	1	11/24/25	kbc	11/26/25	jlh	N	
2,4,6-Tribromophenol	85 %	30-105	1	11/24/25	kbc	11/26/25	jlh	N	
Terphenyl-d14	88 %	37-109	1	11/24/25	kbc	11/26/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176508									
Benzidine	<0.098 ug/L	0.098	1	11/26/25	als	11/26/25	jlh		
3,3'-Dichlorobenzidine	<0.98 ug/L	0.98	1	11/26/25	als	11/26/25	jlh		
Surrogates:									
Nitrobenzene-d5	85 %	38-130	1	11/26/25	als	11/26/25	jlh		
2-Fluorobiphenyl	67 %	39-118	1	11/26/25	als	11/26/25	jlh	N	
Terphenyl-d14	33 %	14-96	1	11/26/25	als	11/26/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176388									
Bromomethane	<5.0 ug/L	5.0	1	11/21/25	ats	11/21/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	103 %	68-133	1	11/21/25	ats	11/21/25	ats		
Toluene-d8	98 %	75-120	1	11/21/25	ats	11/21/25	ats		
4-Bromofluorobenzene	101 %	69-119	1	11/21/25	ats	11/21/25	ats		
1,2-Dichlorobenzene-d4	101 %	72-127	1	11/21/25	ats	11/21/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25K1111
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1111-01 Matrix: Wastewater Date Collected: 11/21/25 07:45
Sample ID: Piranha Date Received: 11/21/25 10:10

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25K1111
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1111-02 Matrix: Wastewater Date Collected: 11/21/25 08:10
Sample ID: Arvco Date Received: 11/21/25 10:10

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176442									
Beryllium	<0.0040 mg/L	0.0040	1	11/25/25	ejb	12/05/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS								DL03	
Analysis Method: EPA 625.1									
Batch: T176415									
Phenol	<9.5 ug/L	9.5	4	11/24/25	kbc	11/26/25	jlh		
Surrogates:									
2-Fluorophenol	30 %	20-53	4	11/24/25	kbc	11/26/25	jlh		
Phenol-d5	21 %	11-40	4	11/24/25	kbc	11/26/25	jlh		
Nitrobenzene-d5	76 %	36-103	4	11/24/25	kbc	11/26/25	jlh		
2-Fluorobiphenyl	71 %	36-119	4	11/24/25	kbc	11/26/25	jlh	N	
2,4,6-Tribromophenol	78 %	30-105	4	11/24/25	kbc	11/26/25	jlh	N	
Terphenyl-d14	68 %	37-109	4	11/24/25	kbc	11/26/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176508									
Benzidine	**0.23 ug/L**	**0.11**	**1**	**11/26/25**	**als**	**11/26/25**	**jlh**		
3,3'-Dichlorobenzidine	<1.1 ug/L	1.1	1	11/26/25	als	11/26/25	jlh		
Surrogates:									
Nitrobenzene-d5	87 %	38-130	1	11/26/25	als	11/26/25	jlh		
2-Fluorobiphenyl	74 %	39-118	1	11/26/25	als	11/26/25	jlh	N	
Terphenyl-d14	30 %	14-96	1	11/26/25	als	11/26/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176388									
Bromomethane	<5.0 ug/L	5.0	1	11/21/25	ats	11/21/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	105 %	68-133	1	11/21/25	ats	11/21/25	ats		
Toluene-d8	97 %	75-120	1	11/21/25	ats	11/21/25	ats		
4-Bromofluorobenzene	100 %	69-119	1	11/21/25	ats	11/21/25	ats		
1,2-Dichlorobenzene-d4	101 %	72-127	1	11/21/25	ats	11/21/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25K1111
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1111-02 Matrix: Wastewater Date Collected: 11/21/25 08:10
Sample ID: Arvco Date Received: 11/21/25 10:10

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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QUALITY CONTROL RESULTS

Trace Project ID: 25K1111
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176442
QC Batch Method: EPA 200.2
Analysis Description: Beryllium, Total
Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T176442-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.00025	0.00025	

LABORATORY CONTROL SAMPLE: T176442-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.182	91	85-115	

Trace Project ID: 25K1111
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176415
QC Batch Method: EPA 625 Extraction for BNAs
Analysis Description: TTO Semi-Volatiles
Analysis Method: EPA 625.1

METHOD BLANK: T176415-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	**%**	**63**	**20-53**	**S03**
Phenol-d5 (S)	**%**	**45**	**11-40**	**S03**
Nitrobenzene-d5 (S)	%	89	36-103	
2-Fluorobiphenyl (S)	%	82	36-119	
2,4,6-Tribromophenol (S)	%	86	30-105	
Terphenyl-d14 (S)	**%**	**113**	**37-109**	**S03**

LABORATORY CONTROL SAMPLE: T176415-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	22.5	23	17-120	
2-Fluorophenol (S)	%	100	39.4	39	20-53	
Phenol-d5 (S)	%	100	25.5	25	11-40	
Nitrobenzene-d5 (S)	%	100	96.9	97	36-103	
2-Fluorobiphenyl (S)	%	100	91.9	92	36-119	
2,4,6-Tribromophenol (S)	%	100	94.8	95	30-105	
Terphenyl-d14 (S)	**%**	**100**	**115**	**115**	**37-109**	**S10**

Trace Project ID: 25K1111
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176508	Analysis Description: 625 Benzidines SIM
QC Batch Method: GC/MS No Prep	Analysis Method: EPA 625.1 SIM

METHOD BLANK: T176508-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	80	38-130	
2-Fluorobiphenyl (S)	%	82	39-118	
Terphenyl-d14 (S)	%	50	14-96	

LABORATORY CONTROL SAMPLE: T176508-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	1.01	51	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	1.69	85	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.631	63	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.734	73	39-118	
Terphenyl-d14 (S)	%	1.00	0.446	45	14-96	

Trace Project ID: 25K1111
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176388	Analysis Description: TTO Volatiles
QC Batch Method: EPA 5030B Purge-and-Trap for Aqueous Samples	Analysis Method: EPA 624.1

METHOD BLANK: T176388-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	106	68-133	
Toluene-d8 (S)	%	96	75-120	
4-Bromofluorobenzene (S)	%	100	69-119	
1,2-Dichlorobenzene-d4 (S)	%	101	72-127	

LABORATORY CONTROL SAMPLE: T176388-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	48.2	96	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	31.2	104	68-133	

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LABORATORY CONTROL SAMPLE: T176388-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Toluene-d8 (S)	%	30.0	29.0	96	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.5	98	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	30.3	101	72-127	

AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25K1111-02

Analysis: EPA 625.1

Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Trace ID: T176415-BLK1

Analysis: EPA 625.1

2-Fluorophenol	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.
Phenol-d5	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.
Terphenyl-d14	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.

Trace ID: T176415-BS1

Analysis: EPA 625.1

Terphenyl-d14	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

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CHAIN-OF-CUSTODY RECORD

Page 1 of 1

TRACE ANALYTICAL LABORATORIES, INC.

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Trace ID No. 25K1111

Report Results To:		Bill To:
Company Name: City of Cadillac		PO #:
Report To: Cindy Tomaszewski		Contact Name: Cindy Tomaszewski
Mailing Address: 1121 Plett Rd		Billing Address (if different): 200 N Lake St
City, State, Zip Code: Cadillac MI 49601		City, State, Zip Code: Cadillac MI 49601
Office Phone: 231-775-2368	Cell Phone: 231-878-6230	Phone Number:
Email Address: lab@cadillac-mi.net; pmiller@cadillac-mi.net		Billing Email Address: lab@cadillac-mi.net

Trace Use:
Logged By: NC
Checked By: AC
Sample Collection Time (Hrs):

Requested Turnaround Times (TAT)
- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:
WW = Wastewater, AQ = Aqueous, A = Air
DW = Drinking Water, O = Oil, WI = Wipes
GW = Groundwater, S = Solid, U = Unknown
LW = Liquid Waste, SL = Sludge

Project Name: 2025 IPP/MAHL Additional

Sampled By (print):

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Composite	Grab	Metals Field Filtered (Y or N)	Matrix - see above	Number of Containers	Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes	Possible Health Hazards?
1	11-21-25	7:45	Piranha		✓	N	WW	6	x		x		x		x		X	X	X	X		
2	" "	8:10	Aruco		✓	N	WW	6	x		x		x		x		X	X	X	X		

Please Sign

Released By	Received By	Date	Time	Released By	Received By	Date	Time
1) [signature]	[signature]	11/21/25	10:10	2) [signature]	[signature]	11/21/25	12:38
3)				4)			

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

CERTIFICATE OF ANALYSIS

Trace Analytical Laboratories, Inc.
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25K1111
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

Date: 11/21/25	Original Observation	Corrected Temperature	IR-9 (CF: 0.0°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.5°C)	IR-14 (CF: -1.1°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Time: 12:38										
Initials: AL										
Package Description: Cooler										
Package Temp °C	0.0	0.0		/						
Representative Sample Temp °C	4.3	4.3		/						/

Sample Receipt

Yes	No	
[x]	[ ]	Received on ice or other coolant
[x]	[ ]	Ice still present upon receipt
[ ]	[x]	Custody seals present

[ ] Yes [ ] No Custody seals intact (if applicable)

[x] Trace Courier [ ] Client Drop-off [ ] UPS [ ] Fed Ex [ ] US Mail [ ] Other

Sample Condition

Yes	No	N/A	
[x]	[ ]	[ ]	All sample containers arrived unbroken and labeled
[x]	[ ]	[ ]	Sufficient sample to run requested analyses
[x]	[ ]	[ ]	Correct chemical preservative added to samples
[ ]	[x]	[ ]	Samples preserved at Trace
[x]	[ ]	[ ]	Chemical preservation verified, check EMD pH test strip used (if applicable)
			[x] pH 0-2.5 (Lot: HC461264) [ ] pH 11.0-13.0 (Lot: HC022540) [ ] Other
[x]	[ ]	[ ]	Air bubbles absent from VOAs (no bubble grater than 6mm)

Chain of Custody (COC)

Yes	No	
[x]	[ ]	All bottle labels agree with COC
[x]	[ ]	COC filled out properly
[x]	[ ]	COC signed by client

Notes:

Form 70-A.59
Effective 8/20/25

TRACE Analytical Laboratories, Inc.

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December 09, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25K1338
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

TNI EL V1:2016

Trace Analytical Laboratories, Inc.
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SAMPLE SUMMARY

Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25K1338-01	Domestic #1	Wastewater	DG	11/25/25	11/26/25 10:00
Sample type: Composite					
25K1338-02	Domestic #2	Wastewater	DG	11/25/25	11/26/25 10:00
Sample type: Composite					
25K1338-03	WWTP Raw Influent	Wastewater	DG	11/25/25	11/26/25 10:00
Sample type: Composite					
25K1338-04	WWTP Primary Effluent	Wastewater	DG	11/25/25	11/26/25 10:00
Sample type: Composite					
25K1338-05	WWTP Final Effluent	Wastewater	DG	11/25/25	11/26/25 10:00
Sample type: Composite					

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ANALYTICAL RESULTS

Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1338-01 Matrix: Wastewater Date Collected: 11/25/25
Sample ID: Domestic #1 Date Received: 11/26/25 10:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176600									
Beryllium	<0.0040 mg/L	0.0040	1	12/01/25	ejb	12/08/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176622									
Phenol	**5.9 ug/L**	**5.0**	**1**	**12/01/25**	**kbc**	**12/02/25**	**jlh**		
Surrogates:									
2-Fluorophenol	31 %	20-53	1	12/01/25	kbc	12/02/25	jlh		
Phenol-d5	20 %	11-40	1	12/01/25	kbc	12/02/25	jlh		
Nitrobenzene-d5	84 %	36-103	1	12/01/25	kbc	12/02/25	jlh		
2-Fluorobiphenyl	79 %	36-119	1	12/01/25	kbc	12/02/25	jlh	N	
2,4,6-Tribromophenol	85 %	30-105	1	12/01/25	kbc	12/02/25	jlh	N	
Terphenyl-d14	81 %	37-109	1	12/01/25	kbc	12/02/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176645									
Benzidine	**2.1 ug/L**	**0.11**	**1**	**12/01/25**	**als**	**12/03/25**	**jlh**		
3,3'-Dichlorobenzidine	<1.1 ug/L	1.1	1	12/01/25	als	12/03/25	jlh		
Surrogates:									
Nitrobenzene-d5	* **328** %	38-130	1	12/01/25	als	12/03/25	jlh	S10	
2-Fluorobiphenyl	44 %	39-118	1	12/01/25	als	12/03/25	jlh	N	
Terphenyl-d14	22 %	14-96	1	12/01/25	als	12/03/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176611									
Bromomethane	<5.0 ug/L	5.0	1	11/26/25	ats	11/27/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	98 %	68-133	1	11/26/25	ats	11/27/25	ats		
Toluene-d8	102 %	75-120	1	11/26/25	ats	11/27/25	ats		
4-Bromofluorobenzene	99 %	69-119	1	11/26/25	ats	11/27/25	ats		
1,2-Dichlorobenzene-d4	99 %	72-127	1	11/26/25	ats	11/27/25	ats		

ANALYTICAL RESULTS

Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1338-01 Matrix: Wastewater Date Collected: 11/25/25
Sample ID: Domestic #1 Date Received: 11/26/25 10:00

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1338-02 Matrix: Wastewater Date Collected: 11/25/25
Sample ID: Domestic #2 Date Received: 11/26/25 10:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176600									
Beryllium	<0.0040 mg/L	0.0040	1	12/01/25	ejb	12/08/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176622									
Phenol	<5.0 ug/L	5.0	1	12/01/25	kbc	12/02/25	jlh		
Surrogates:									
2-Fluorophenol	27 %	20-53	1	12/01/25	kbc	12/02/25	jlh		
Phenol-d5	17 %	11-40	1	12/01/25	kbc	12/02/25	jlh		
Nitrobenzene-d5	80 %	36-103	1	12/01/25	kbc	12/02/25	jlh		
2-Fluorobiphenyl	70 %	36-119	1	12/01/25	kbc	12/02/25	jlh	N	
2,4,6-Tribromophenol	73 %	30-105	1	12/01/25	kbc	12/02/25	jlh	N	
Terphenyl-d14	70 %	37-109	1	12/01/25	kbc	12/02/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176645									
Benzidine	**1.4 ug/L**	**0.11**	**1**	**12/01/25**	**als**	**12/03/25**	**jlh**		
3,3'-Dichlorobenzidine	<1.1 ug/L	1.1	1	12/01/25	als	12/03/25	jlh		
Surrogates:									
Nitrobenzene-d5	74 %	38-130	1	12/01/25	als	12/03/25	jlh		
2-Fluorobiphenyl	72 %	39-118	1	12/01/25	als	12/03/25	jlh	N	
Terphenyl-d14	35 %	14-96	1	12/01/25	als	12/03/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176611									
Bromomethane	<5.0 ug/L	5.0	1	11/26/25	ats	11/27/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	99 %	68-133	1	11/26/25	ats	11/27/25	ats		
Toluene-d8	103 %	75-120	1	11/26/25	ats	11/27/25	ats		
4-Bromofluorobenzene	101 %	69-119	1	11/26/25	ats	11/27/25	ats		
1,2-Dichlorobenzene-d4	100 %	72-127	1	11/26/25	ats	11/27/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1338-02 Matrix: Wastewater Date Collected: 11/25/25
Sample ID: Domestic #2 Date Received: 11/26/25 10:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1338-03 | Matrix: Wastewater | Date Collected: 11/25/25
Sample ID: WWTP Raw Influent | Date Received: 11/26/25 10:00

PARAMETERS		RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL										
Analysis Method: EPA 200.8 Rev. 5.4										
Batch: T176600										
Beryllium		<0.0040 mg/L	0.0040	1	12/01/25	ejb	12/08/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 625.1										
Batch: T176622										
Phenol		<5.0 ug/L	5.0	1	12/01/25	kbc	12/02/25	jlh		
Surrogates:										
2-Fluorophenol		23 %	20-53	1	12/01/25	kbc	12/02/25	jlh		
Phenol-d5		15 %	11-40	1	12/01/25	kbc	12/02/25	jlh		
Nitrobenzene-d5		67 %	36-103	1	12/01/25	kbc	12/02/25	jlh		
2-Fluorobiphenyl		57 %	36-119	1	12/01/25	kbc	12/02/25	jlh	N	
2,4,6-Tribromophenol		62 %	30-105	1	12/01/25	kbc	12/02/25	jlh	N	
Terphenyl-d14		59 %	37-109	1	12/01/25	kbc	12/02/25	jlh		
Analysis Method: EPA 625.1 SIM										
Batch: T176645										
Benzidine		**1.7 ug/L**	**0.12**	**1**	**12/01/25**	**als**	**12/03/25**	**jlh**		
3,3'-Dichlorobenzidine		<1.2 ug/L	1.2	1	12/01/25	als	12/03/25	jlh		
Surrogates:										
Nitrobenzene-d5		44 %	38-130	1	12/01/25	als	12/03/25	jlh		
2-Fluorobiphenyl	*	**36** %	39-118	1	12/01/25	als	12/03/25	jlh	S10, N	
Terphenyl-d14		15 %	14-96	1	12/01/25	als	12/03/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 624.1										
Batch: T176611										
Bromomethane		<5.0 ug/L	5.0	1	11/26/25	ats	11/27/25	ats		
Surrogates:										
1,2-Dichloroethane-d4		98 %	68-133	1	11/26/25	ats	11/27/25	ats		
Toluene-d8		102 %	75-120	1	11/26/25	ats	11/27/25	ats		
4-Bromofluorobenzene		98 %	69-119	1	11/26/25	ats	11/27/25	ats		
1,2-Dichlorobenzene-d4		100 %	72-127	1	11/26/25	ats	11/27/25	ats		

CERTIFICATE OF ANALYSIS

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ANALYTICAL RESULTS

Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1338-03 Matrix: Wastewater Date Collected: 11/25/25
Sample ID: WWTP Raw Influent Date Received: 11/26/25 10:00

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1338-04 | Matrix: Wastewater | Date Collected: 11/25/25
Sample ID: WWTP Primary Effluent | Date Received: 11/26/25 10:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176600									
Beryllium	<0.0040 mg/L	0.0040	1	12/01/25	ejb	12/08/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176622									
Phenol	<5.0 ug/L	5.0	1	12/01/25	kbc	12/02/25	jlh		
Surrogates:									
2-Fluorophenol	32 %	20-53	1	12/01/25	kbc	12/02/25	jlh		
Phenol-d5	21 %	11-40	1	12/01/25	kbc	12/02/25	jlh		
Nitrobenzene-d5	85 %	36-103	1	12/01/25	kbc	12/02/25	jlh		
2-Fluorobiphenyl	76 %	36-119	1	12/01/25	kbc	12/02/25	jlh	N	
2,4,6-Tribromophenol	83 %	30-105	1	12/01/25	kbc	12/02/25	jlh	N	
Terphenyl-d14	85 %	37-109	1	12/01/25	kbc	12/02/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176645									
Benzidine	**2.4 ug/L**	**0.12**	**1**	**12/01/25**	**als**	**12/03/25**	**jlh**		
3,3'-Dichlorobenzidine	<1.2 ug/L	1.2	1	12/01/25	als	12/03/25	jlh		
Surrogates:									
Nitrobenzene-d5	49 %	38-130	1	12/01/25	als	12/03/25	jlh		
2-Fluorobiphenyl	42 %	39-118	1	12/01/25	als	12/03/25	jlh	N	
Terphenyl-d14	19 %	14-96	1	12/01/25	als	12/03/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176611									
Bromomethane	<5.0 ug/L	5.0	1	11/26/25	ats	11/27/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	98 %	68-133	1	11/26/25	ats	11/27/25	ats		
Toluene-d8	103 %	75-120	1	11/26/25	ats	11/27/25	ats		
4-Bromofluorobenzene	101 %	69-119	1	11/26/25	ats	11/27/25	ats		
1,2-Dichlorobenzene-d4	101 %	72-127	1	11/26/25	ats	11/27/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1338-04 Matrix: Wastewater Date Collected: 11/25/25
Sample ID: WWTP Primary Effluent Date Received: 11/26/25 10:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1338-05 | Matrix: Wastewater | Date Collected: 11/25/25
Sample ID: WWTP Final Effluent | Date Received: 11/26/25 10:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176600									
Beryllium	<0.0040 mg/L	0.0040	1	12/01/25	ejb	12/08/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176622									
Phenol	<5.0 ug/L	5.0	1	12/01/25	kbc	12/02/25	jlh		
Surrogates:									
2-Fluorophenol	28 %	20-53	1	12/01/25	kbc	12/02/25	jlh		
Phenol-d5	18 %	11-40	1	12/01/25	kbc	12/02/25	jlh		
Nitrobenzene-d5	79 %	36-103	1	12/01/25	kbc	12/02/25	jlh		
2-Fluorobiphenyl	71 %	36-119	1	12/01/25	kbc	12/02/25	jlh	N	
2,4,6-Tribromophenol	74 %	30-105	1	12/01/25	kbc	12/02/25	jlh	N	
Terphenyl-d14	78 %	37-109	1	12/01/25	kbc	12/02/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176645									
Benzidine	**0.16 ug/L**	**0.10**	**1**	**12/01/25**	**als**	**12/03/25**	**jlh**		
3,3'-Dichlorobenzidine	<1.0 ug/L	1.0	1	12/01/25	als	12/03/25	jlh		
Surrogates:									
Nitrobenzene-d5	90 %	38-130	1	12/01/25	als	12/03/25	jlh		
2-Fluorobiphenyl	69 %	39-118	1	12/01/25	als	12/03/25	jlh	N	
Terphenyl-d14	32 %	14-96	1	12/01/25	als	12/03/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176611									
Bromomethane	<5.0 ug/L	5.0	1	11/26/25	ats	11/27/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	98 %	68-133	1	11/26/25	ats	11/27/25	ats		
Toluene-d8	102 %	75-120	1	11/26/25	ats	11/27/25	ats		
4-Bromofluorobenzene	99 %	69-119	1	11/26/25	ats	11/27/25	ats		
1,2-Dichlorobenzene-d4	100 %	72-127	1	11/26/25	ats	11/27/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1338-05 Matrix: Wastewater Date Collected: 11/25/25
Sample ID: WWTP Final Effluent Date Received: 11/26/25 10:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

QUALITY CONTROL RESULTS

Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176600	Analysis Description: Beryllium, Total
QC Batch Method: EPA 200.2	Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T176600-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.0040	0.0040	

LABORATORY CONTROL SAMPLE: T176600-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.204	102	85-115	

Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176622	Analysis Description: TTO Semi-Volatiles
QC Batch Method: EPA 625 Extraction for BNAs	Analysis Method: EPA 625.1

METHOD BLANK: T176622-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	%	22	20-53	
Phenol-d5 (S)	%	14	11-40	
Nitrobenzene-d5 (S)	%	62	36-103	
2-Fluorobiphenyl (S)	%	56	36-119	
2,4,6-Tribromophenol (S)	%	62	30-105	
Terphenyl-d14 (S)	%	79	37-109	

LABORATORY CONTROL SAMPLE: T176622-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	18.6	19	17-120	
2-Fluorophenol (S)	%	100	31.4	31	20-53	
Phenol-d5 (S)	%	100	25.5	25	11-40	
Nitrobenzene-d5 (S)	%	100	78.5	78	36-103	
2-Fluorobiphenyl (S)	%	100	69.3	69	36-119	
2,4,6-Tribromophenol (S)	%	100	79.0	79	30-105	
Terphenyl-d14 (S)	%	100	78.6	79	37-109	

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Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176645
QC Batch Method: GC/MS No Prep
Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T176645-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	89	38-130	
2-Fluorobiphenyl (S)	%	75	39-118	
Terphenyl-d14 (S)	%	46	14-96	

LABORATORY CONTROL SAMPLE: T176645-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	1.22	61	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	2.05	102	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.985	98	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.813	81	39-118	
Terphenyl-d14 (S)	%	1.00	0.497	50	14-96	

Trace Project ID: 25K1338
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176611
QC Batch Method: EPA 624 Purge and Trap
Analysis Description: TTO Volatiles
Analysis Method: EPA 624.1

METHOD BLANK: T176611-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	98	68-133	
Toluene-d8 (S)	%	101	75-120	
4-Bromofluorobenzene (S)	%	98	69-119	
1,2-Dichlorobenzene-d4 (S)	%	103	72-127	

LABORATORY CONTROL SAMPLE: T176611-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	48.2	96	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	28.4	95	68-133	

LABORATORY CONTROL SAMPLE: T176611-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Toluene-d8 (S)	%	30.0	31.4	105	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.7	99	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	30.2	101	72-127	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25K1338-01

Analysis: EPA 625.1 SIM

Nitrobenzene-d5 — Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Trace ID: 25K1338-03

Analysis: EPA 625.1 SIM

2-Fluorobiphenyl — Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

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CHAIN-OF-CUSTODY RECORD

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Page 1 of 1

Trace ID No. 25K1338

Report Results To:

Company Name: City of Cadillac	
Report To: Cindy Tomaszewski	
Mailing Address: 1121 Plett Rd	
City, State, Zip Code: Cadillac MI 49601	
Office Phone: 231-775-2368	Cell Phone: 231-878-6230
Email Address: lab@cadillac-mi.net; pmiller@cadillac-mi.net	

Bill To:

PO #:
Contact Name: Cindy Tomaszewski
Billing Address (if different): 200 N Lake St
City, State, Zip Code: Cadillac MI 49601
Phone Number:
Billing Email Address: lab@cadillac-mi.net

Trace Use:

Logged By: NC
Checked By: AL
Sample Collection Time (Hrs):

Requested Turnaround Times (TAT)

- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:

WW = Wastewater	AQ = Aqueous	A = Air
DW = Drinking Water	O = Oil	WI = Wipes
GW = Groundwater	S = Solid	U = Unknown
LW = Liquid Waste	SL = Sludge	

Project Name: 2025 IPP/MAHL Additional

Sampled By (print): Derek Grodzicki

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Sample Type (check one): Composite	Grab	Metals Field Filtered (Y or N)	Matrix - see above →	Number of Containers	Preservation: Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes	Possible Health Hazards?
1	11/24/25	–	Domestic #1	✓	✓	N	WW	8	x		x		x		x		X	X	X	X		
2	11/24/25	–	Domestic #2	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	grab VOCs to	
3	11/24/25	–	WWTP Raw Influent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	be composited	
4	11/24/25	–	WWTP Primary Effluent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	at the lab	
5	11/24/25	–	WWTP Final Effluent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X		

* All composites started on "Sample Collection Date"

Please Sign

	Released By	Received By	Date	Time		Released By	Received By	Date	Time
1)	[signature]	UPS Ground	11-25-25	1300	2)	UPS	[signature]	11/26/25	1000
3)					4)				

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

CERTIFICATE OF ANALYSIS

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25K1338
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

Date: 11/26/25
Time: 10:00
Initials: SAB
Package Description: Cooler

	Original Observation	Corrected Temperature	IR-9 (CF: +0.5°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.9°C)	IR-14 (CF: -1.3°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Package Temp °C	0.5	1.0	/							
Representative Sample Temp °C	7.0	7.5	/							/

Sample Receipt

Yes	No	
☑	☐	Received on ice or other coolant
☑	☐	Ice still present upon receipt
☐	☑	Custody seals present

☐ Yes ☐ No Custody seals intact (if applicable)

☐ Trace Courier ☐ Client Drop-off ☑ UPS ☐ Fed Ex ☐ US Mail ☐ Other

Sample Condition

Yes	No	N/A	
☑	☐	☐	All sample containers arrived unbroken and labeled
☑	☐	☐	Sufficient sample to run requested analyses
☑	☐	☐	Correct chemical preservative added to samples
☐	☑	☐	Samples preserved at Trace ______
☑	☐	☐	Chemical preservation verified, check EMD pH test strip used (if applicable)

☑ pH 0-2.5 (Lot: HC461264) ☐ pH 11.0-13.0 (Lot: HC022540) ☐ Other

Yes	No	N/A	
☑	☐	☐	Air bubbles absent from VOAs (no bubble grater than 6mm) ______

Chain of Custody (COC)

Yes	No	
☑	☐	All bottle labels agree with COC
☑	☐	COC filled out properly
☑	☐	COC signed by client

Notes:

Form 70-A.60
Effective 11/25/25

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December 09, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25K1341
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

PJLA Testing Accreditation# 1031

TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 25K1341
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25K1341-01 Sample type: Grab	Borg Warner	Wastewater	DG	11/25/25 09:15	11/26/25 10:40
25K1341-02 Sample type: Grab	CRE	Wastewater	DG	11/25/25 09:25	11/26/25 10:40

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ANALYTICAL RESULTS

Trace Project ID: 25K1341
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1341-01 | Matrix: Wastewater | Date Collected: 11/25/25 09:15
Sample ID: Borg Warner | Date Received: 11/26/25 10:40

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176600									
Beryllium	<0.0040 mg/L	0.0040	1	12/01/25	ejb	12/08/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176622									
Phenol	<5.0 ug/L	5.0	1	12/01/25	kbc	12/02/25	jlh		
Surrogates:									
2-Fluorophenol	32 %	20-53	1	12/01/25	kbc	12/02/25	jlh		
Phenol-d5	20 %	11-40	1	12/01/25	kbc	12/02/25	jlh		
Nitrobenzene-d5	91 %	36-103	1	12/01/25	kbc	12/02/25	jlh		
2-Fluorobiphenyl	77 %	36-119	1	12/01/25	kbc	12/02/25	jlh	N	
2,4,6-Tribromophenol	80 %	30-105	1	12/01/25	kbc	12/02/25	jlh	N	
Terphenyl-d14	78 %	37-109	1	12/01/25	kbc	12/02/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176645									
Benzidine	**0.14 ug/L**	**0.098**	**1**	**12/01/25**	**als**	**12/03/25**	**jlh**		
3,3'-Dichlorobenzidine	<0.98 ug/L	0.98	1	12/01/25	als	12/03/25	jlh		
Surrogates:									
Nitrobenzene-d5	85 %	38-130	1	12/01/25	als	12/03/25	jlh		
2-Fluorobiphenyl	67 %	39-118	1	12/01/25	als	12/03/25	jlh	N	
Terphenyl-d14	31 %	14-96	1	12/01/25	als	12/03/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176611									
Bromomethane	<5.0 ug/L	5.0	1	11/26/25	ats	11/27/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	99 %	68-133	1	11/26/25	ats	11/27/25	ats		
Toluene-d8	101 %	75-120	1	11/26/25	ats	11/27/25	ats		
4-Bromofluorobenzene	98 %	69-119	1	11/26/25	ats	11/27/25	ats		
1,2-Dichlorobenzene-d4	101 %	72-127	1	11/26/25	ats	11/27/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25K1341
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1341-01 Matrix: Wastewater Date Collected: 11/25/25 09:15
Sample ID: Borg Warner Date Received: 11/26/25 10:40

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

CERTIFICATE OF ANALYSIS

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ANALYTICAL RESULTS

Trace Project ID: 25K1341
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1341-02 | Matrix: Wastewater | Date Collected: 11/25/25 09:25
Sample ID: CRE | Date Received: 11/26/25 10:40

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176600									
Beryllium	<0.0040 mg/L	0.0040	1	12/01/25	ejb	12/08/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176622									
Phenol	<5.0 ug/L	5.0	1	12/01/25	kbc	12/02/25	jlh		
Surrogates:									
2-Fluorophenol	22 %	20-53	1	12/01/25	kbc	12/02/25	jlh		
Phenol-d5	14 %	11-40	1	12/01/25	kbc	12/02/25	jlh		
Nitrobenzene-d5	64 %	36-103	1	12/01/25	kbc	12/02/25	jlh		
2-Fluorobiphenyl	55 %	36-119	1	12/01/25	kbc	12/02/25	jlh	N	
2,4,6-Tribromophenol	57 %	30-105	1	12/01/25	kbc	12/02/25	jlh	N	
Terphenyl-d14	64 %	37-109	1	12/01/25	kbc	12/02/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176645									
Benzidine	<0.099 ug/L	0.099	1	12/01/25	als	12/03/25	jlh		
3,3'-Dichlorobenzidine	<0.99 ug/L	0.99	1	12/01/25	als	12/03/25	jlh		
Surrogates:									
Nitrobenzene-d5	96 %	38-130	1	12/01/25	als	12/03/25	jlh		
2-Fluorobiphenyl	74 %	39-118	1	12/01/25	als	12/03/25	jlh	N	
Terphenyl-d14	42 %	14-96	1	12/01/25	als	12/03/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176611									
Bromomethane	<5.0 ug/L	5.0	1	11/26/25	ats	11/27/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	98 %	68-133	1	11/26/25	ats	11/27/25	ats		
Toluene-d8	101 %	75-120	1	11/26/25	ats	11/27/25	ats		
4-Bromofluorobenzene	100 %	69-119	1	11/26/25	ats	11/27/25	ats		
1,2-Dichlorobenzene-d4	100 %	72-127	1	11/26/25	ats	11/27/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25K1341
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25K1341-02 Matrix: Wastewater Date Collected: 11/25/25 09:25
Sample ID: CRE Date Received: 11/26/25 10:40

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

QUALITY CONTROL RESULTS

Trace Project ID: 25K1341
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176600	Analysis Description: Beryllium, Total
QC Batch Method: EPA 200.2	Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T176600-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.0040	0.0040	

LABORATORY CONTROL SAMPLE: T176600-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.204	102	85-115	

Trace Project ID: 25K1341
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176622	Analysis Description: TTO Semi-Volatiles
QC Batch Method: EPA 625 Extraction for BNAs	Analysis Method: EPA 625.1

METHOD BLANK: T176622-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	%	22	20-53	
Phenol-d5 (S)	%	14	11-40	
Nitrobenzene-d5 (S)	%	62	36-103	
2-Fluorobiphenyl (S)	%	56	36-119	
2,4,6-Tribromophenol (S)	%	62	30-105	
Terphenyl-d14 (S)	%	79	37-109	

LABORATORY CONTROL SAMPLE: T176622-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	18.6	19	17-120	
2-Fluorophenol (S)	%	100	31.4	31	20-53	
Phenol-d5 (S)	%	100	25.5	25	11-40	
Nitrobenzene-d5 (S)	%	100	78.5	78	36-103	
2-Fluorobiphenyl (S)	%	100	69.3	69	36-119	
2,4,6-Tribromophenol (S)	%	100	79.0	79	30-105	
Terphenyl-d14 (S)	%	100	78.6	79	37-109	

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Trace Project ID: 25K1341
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176645
QC Batch Method: GC/MS No Prep

Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T176645-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	89	38-130	
2-Fluorobiphenyl (S)	%	75	39-118	
Terphenyl-d14 (S)	%	46	14-96	

LABORATORY CONTROL SAMPLE: T176645-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	1.22	61	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	2.05	102	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.985	98	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.813	81	39-118	
Terphenyl-d14 (S)	%	1.00	0.497	50	14-96	

Trace Project ID: 25K1341
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176611
QC Batch Method: EPA 624 Purge and Trap

Analysis Description: TTO Volatiles
Analysis Method: EPA 624.1

METHOD BLANK: T176611-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	98	68-133	
Toluene-d8 (S)	%	101	75-120	
4-Bromofluorobenzene (S)	%	98	69-119	
1,2-Dichlorobenzene-d4 (S)	%	103	72-127	

LABORATORY CONTROL SAMPLE: T176611-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	48.2	96	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	28.4	95	68-133	

LABORATORY CONTROL SAMPLE: T176611-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Toluene-d8 (S)	%	30.0	31.4	105	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.7	99	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	30.2	101	72-127	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

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CHAIN-OF-CUSTODY RECORD

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Page 1 of 1

Trace ID No. 25K1341

Report Results To:

Company Name: City of Cadillac	
Report To: Cindy Tomaszewski	
Mailing Address: 1121 Plett Rd	
City, State, Zip Code: Cadillac MI 49601	
Office Phone: 231-775-2368	Cell Phone: 231-878-6230
Email Address: lab@cadillac-mi.net; pmiller@cadillac-mi.net	

Bill To:

PO #:
Contact Name: Cindy Tomaszewski
Billing Address (if different): 200 N Lake St
City, State, Zip Code: Cadillac MI 49601
Phone Number:
Billing Email Address: lab@cadillac-mi.net

Trace Use:

Logged By: NC
Checked By: AL
Sample Collection Time (Hrs):

Requested Turnaround Times (TAT)

- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:

WW = Wastewater	AQ = Aqueous	A = Air
DW = Drinking Water	O = Oil	WI = Wipes
GW = Groundwater	S = Solid	U = Unknown
LW = Liquid Waste	SL = Sludge	

Project Name: 2025 IPP/MAHL Additional

Sampled By (print): DEREK GOODRICH

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Sample Type: Composite	Sample Type: Grab	Metals Field Filterd (Y or N)	Matrix - see above →	Number of Containers	Preservation: Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes	Possible Health Hazards?
1	11/25/25	0915	BORG WARNER		✓	N	WW	6	x		x		x		x		X	X	X	X		
2	11/25/25	0925	CRE		✓	N	WW	6	x		x		x		x		X	X	X	X		

Please Sign

	Released By	Received By	Date	Time		Released By	Received By	Date	Time
1)	CATomaszewski	UPS Ground	11-25-25	1300	2)	UPS	[signature]	11/26/25	10:40
3)					4)				

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

CERTIFICATE OF ANALYSIS

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25K1341
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

Date: 11/26/25	Original Observation	Corrected Temperature	IR-9 (CF: +0.5°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.9°C)	IR-14 (CF: -1.3°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Time: 10:40										
Initials: SAB										
Package Description:										
Package Temp °C	0.5	1.0	/							
Representative Sample Temp °C	5.9	6.4	/							/

Sample Receipt

Yes No
- [x] Yes [ ] No — Received on ice or other coolant
- [x] Yes [ ] No — Ice still present upon receipt
- [ ] Yes [x] No — Custody seals present

[ ] Yes [ ] No Custody seals intact (if applicable)

[ ] Trace Courier [ ] Client Drop-off [x] UPS [ ] Fed Ex [ ] US Mail [ ] Other

Sample Condition

Yes	No	N/A	
[ ]	[x]	[ ]	All sample containers arrived unbroken and labeled
[x]	[ ]	[ ]	Sufficient sample to run requested analyses
[x]	[ ]	[ ]	Correct chemical preservative added to samples
[ ]	[x]	[ ]	Samples preserved at Trace ______
[x]	[ ]	[ ]	Chemical preservation verified, check EMD pH test strip used (if applicable)
[x]	[ ]	[ ]	Air bubbles absent from VOAs (no bubble grater than 6mm) ______

[x] pH 0-2.5 (Lot: HC461264) [ ] pH 11.0-13.0 (Lot: HC022540) [ ] Other

Chain of Custody (COC)

Yes	No	
[x]	[ ]	All bottle labels agree with COC
[x]	[ ]	COC filled out properly
[x]	[ ]	COC signed by client

Notes:

1 VOA Broken (2F)

Form 70-A.60
Effective 11/25/25

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December 11, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25L0055
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

PJLA Testing Accreditation# 1031

TNI EL V1:2016

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TRACE ANALYTICAL LABORATORIES, INC.

SAMPLE SUMMARY

Trace Project ID: 25L0055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25L0055-01	CCI	Wastewater	DG	12/01/25 07:50	12/02/25 10:15
Sample type: Composite					
25L0055-02	Hutchinson	Wastewater	DG	12/01/25 08:10	12/02/25 10:15
Sample type: Composite					

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ANALYTICAL RESULTS

Trace Project ID: 25L0055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0055-01 | Matrix: Wastewater | Date Collected: 12/01/25 07:50
Sample ID: CCI | Date Received: 12/02/25 10:15

PARAMETERS		RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL											
Analysis Method: EPA 200.8 Rev. 5.4											
Batch: T176729											
Beryllium		<0.0040	mg/L	0.0040	1	12/03/25	ejb	12/08/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 625.1											
Batch: T176800											
Phenol		**8.4**	**ug/L**	**5.0**	**1**	**12/04/25**	**kbc**	**12/05/25**	**jlh**		
Surrogates:											
2-Fluorophenol		38	%	20-53	1	12/04/25	kbc	12/05/25	jlh		
Phenol-d5		23	%	11-40	1	12/04/25	kbc	12/05/25	jlh		
Nitrobenzene-d5		98	%	36-103	1	12/04/25	kbc	12/05/25	jlh		
2-Fluorobiphenyl		81	%	36-119	1	12/04/25	kbc	12/05/25	jlh	N	
2,4,6-Tribromophenol		79	%	30-105	1	12/04/25	kbc	12/05/25	jlh	N	
Terphenyl-d14		88	%	37-109	1	12/04/25	kbc	12/05/25	jlh		
Analysis Method: EPA 625.1 SIM											
Batch: T176834											
Benzidine		<0.11	ug/L	0.11	1	12/05/25	als	12/10/25	jlh		
3,3'-Dichlorobenzidine		<1.1	ug/L	1.1	1	12/05/25	als	12/10/25	jlh		
Surrogates:											
Nitrobenzene-d5	*	**261**	%	38-130	1	12/05/25	als	12/10/25	jlh	S10	
2-Fluorobiphenyl		61	%	39-118	1	12/05/25	als	12/10/25	jlh	N	
Terphenyl-d14		36	%	14-96	1	12/05/25	als	12/10/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 624.1											
Batch: T176717											
Bromomethane		<5.0	ug/L	5.0	1	12/02/25	sb	12/02/25	sb		
Surrogates:											
1,2-Dichloroethane-d4		133	%	68-133	1	12/02/25	sb	12/02/25	sb		
Toluene-d8		86	%	75-120	1	12/02/25	sb	12/02/25	sb		
4-Bromofluorobenzene		101	%	69-119	1	12/02/25	sb	12/02/25	sb		
1,2-Dichlorobenzene-d4		117	%	72-127	1	12/02/25	sb	12/02/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25L0055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0055-01 Matrix: Wastewater Date Collected: 12/01/25 07:50
Sample ID: CCI Date Received: 12/02/25 10:15

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

CERTIFICATE OF ANALYSIS

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ANALYTICAL RESULTS

Trace Project ID: 25L0055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0055-02 | Matrix: Wastewater | Date Collected: 12/01/25 08:10
Sample ID: Hutchinson | Date Received: 12/02/25 10:15

PARAMETERS		RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL											
Analysis Method: EPA 200.8 Rev. 5.4											
Batch: T176729											
Beryllium		<0.0040	mg/L	0.0040	1	12/03/25	ejb	12/08/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 625.1											
Batch: T176800											
Phenol		<5.0	ug/L	5.0	1	12/04/25	kbc	12/05/25	jlh		
Surrogates:											
2-Fluorophenol		30	%	20-53	1	12/04/25	kbc	12/05/25	jlh		
Phenol-d5		19	%	11-40	1	12/04/25	kbc	12/05/25	jlh		
Nitrobenzene-d5		79	%	36-103	1	12/04/25	kbc	12/05/25	jlh		
2-Fluorobiphenyl		72	%	36-119	1	12/04/25	kbc	12/05/25	jlh	N	
2,4,6-Tribromophenol		72	%	30-105	1	12/04/25	kbc	12/05/25	jlh	N	
Terphenyl-d14		79	%	37-109	1	12/04/25	kbc	12/05/25	jlh		
Analysis Method: EPA 625.1 SIM											
Batch: T176834											
Benzidine		<0.23	ug/L	0.23	2	12/05/25	als	12/10/25	jlh		
3,3'-Dichlorobenzidine		<2.3	ug/L	2.3	2	12/05/25	als	12/10/25	jlh		
Surrogates:											
Nitrobenzene-d5	*	**29**	%	38-130	2	12/05/25	als	12/10/25	jlh	S02	
2-Fluorobiphenyl	*	**22**	%	39-118	2	12/05/25	als	12/10/25	jlh	S02, N	
Terphenyl-d14		14	%	14-96	2	12/05/25	als	12/10/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 624.1											
Batch: T176717											
Bromomethane		<5.0	ug/L	5.0	1	12/02/25	sb	12/02/25	sb		
Surrogates:											
1,2-Dichloroethane-d4	*	**134**	%	68-133	1	12/02/25	sb	12/02/25	sb	S03	
Toluene-d8		87	%	75-120	1	12/02/25	sb	12/02/25	sb		
4-Bromofluorobenzene		100	%	69-119	1	12/02/25	sb	12/02/25	sb		
1,2-Dichlorobenzene-d4		109	%	72-127	1	12/02/25	sb	12/02/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25L0055
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0055-02 Matrix: Wastewater Date Collected: 12/01/25 08:10
Sample ID: Hutchinson Date Received: 12/02/25 10:15

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
VOLATILE ORGANIC COMPOUNDS BY GC-MS										

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QUALITY CONTROL RESULTS

Trace Project ID: 25L0055
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176729	Analysis Description: Beryllium, Total
QC Batch Method: EPA 200.2	Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T176729-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.0010	0.0010	

LABORATORY CONTROL SAMPLE: T176729-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.190	95	85-115	

Trace Project ID: 25L0055
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176800	Analysis Description: TTO Semi-Volatiles
QC Batch Method: EPA 625 Extraction for BNAs	Analysis Method: EPA 625.1

METHOD BLANK: T176800-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<5.0	5.0	
2-Fluorophenol (S)	%	37	20-53	
Phenol-d5 (S)	%	24	11-40	
Nitrobenzene-d5 (S)	%	81	36-103	
2-Fluorobiphenyl (S)	%	73	36-119	
2,4,6-Tribromophenol (S)	%	81	30-105	
Terphenyl-d14 (S)	%	99	37-109	

LABORATORY CONTROL SAMPLE: T176800-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	21.5	22	17-120	
2-Fluorophenol (S)	%	100	38.6	39	20-53	
Phenol-d5 (S)	**%**	**100**	**3.57**	**4**	**11-40**	**S09**
Nitrobenzene-d5 (S)	%	100	102	102	36-103	
2-Fluorobiphenyl (S)	%	100	93.4	93	36-119	
2,4,6-Tribromophenol (S)	%	100	105	105	30-105	
Terphenyl-d14 (S)	**%**	**100**	**126**	**126**	**37-109**	**S10**

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Trace Project ID: 25L0055
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176834
QC Batch Method: GC/MS No Prep
Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T176834-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	70	38-130	
2-Fluorobiphenyl (S)	%	69	39-118	
Terphenyl-d14 (S)	%	41	14-96	

LABORATORY CONTROL SAMPLE: T176834-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	0.730	37	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	<2.0	70	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.666	67	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.636	64	39-118	
Terphenyl-d14 (S)	%	1.00	0.398	40	14-96	

Trace Project ID: 25L0055
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176717
QC Batch Method: EPA 624 Purge and Trap
Analysis Description: TTO Volatiles
Analysis Method: EPA 624.1

METHOD BLANK: T176717-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	125	68-133	
Toluene-d8 (S)	%	86	75-120	
4-Bromofluorobenzene (S)	%	106	69-119	
1,2-Dichlorobenzene-d4 (S)	%	106	72-127	

LABORATORY CONTROL SAMPLE: T176717-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	39.5	79	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	35.8	119	68-133	

LABORATORY CONTROL SAMPLE: T176717-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Toluene-d8 (S)	%	30.0	26.2	88	75-120	
4-Bromofluorobenzene (S)	%	30.0	28.7	96	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	29.6	99	72-127	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25L0055-01

Analysis: EPA 625.1 SIM

Nitrobenzene-d5	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Trace ID: 25L0055-02

Analysis: EPA 624.1

1,2-Dichloroethane-d4	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.

Analysis: EPA 625.1 SIM

	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.
2-Fluorobiphenyl	Note S02 : The surrogate recovery was out of control low when compared to control limits. All results and reporting limits must be considered estimated.
Nitrobenzene-d5	Note S02 : The surrogate recovery was out of control low when compared to control limits. All results and reporting limits must be considered estimated.

Trace ID: T176800-BS1

Analysis: EPA 625.1

Phenol-d5	Note S09 : One of the acid surrogate recoveries was outside the control limit. Since the other two acid surrogates were within the control limits, no data requires qualification.
Terphenyl-d14	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

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TRACE ANALYTICAL LABORATORIES, INC.

CHAIN-OF-CUSTODY RECORD

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Page 1 of 1

Trace ID No. 25L0055

Report Results To:

Company Name: City of Cadillac	
Report To: Cindy Tomaszewski	
Mailing Address: 1121 Plett Rd	
City, State, Zip Code: Cadillac MI 49601	
Office Phone: 231-775-2368	Cell Phone: 231-878-6230
Email Address: lab@cadillac-mi.net; pmiller@cadillac-mi.net	

Bill To:

PO #:
Contact Name: Cindy Tomaszewski
Billing Address (if different): 200 N Lake St
City, State, Zip Code: Cadillac MI 49601
Phone Number:
Billing Email Address: lab@cadillac-mi.net

Trace Use:

Logged By: MJ
Checked By: CN
Sample Collection Time (Hrs):

Requested Turnaround Times (TAT)

- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:

WW = Wastewater, DW = Drinking Water, GW = Groundwater, LW = Liquid Waste
AQ = Aqueous, O = Oil, S = Solid, SL = Sludge
A = Air, WI = Wipes, U = Unknown

Project Name: 2025 IPP/MAHL Additional

Sampled By (print): Derek Goodrich

Analysis Requested

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Sample Type: Composite	Sample Type: Grab	Metals Field Filterd (Y or N)	Matrix - see above →	Number of Containers	Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes	Possible Health Hazards?
1	12-1	7:59	CCI		✓	N	WW	6	x		x		x		x		X	X	X	X		
2	12-1	8:10	Hutchinson		✓	N	WW	6	x		x		x		x		X	X	X	X		

Please Sign

	Released By	Received By	Date	Time		Released By	Received By	Date	Time
1)	[signature] /City	UPS Ground	12-1-25	1300	2)	UPS	[signature]	12/2/25	1015
3)					4)				

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

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25L0055
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

	Original Observation	Corrected Temperature	IR-9 (CF: +0.2°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.6°C)	IR-14 (CF: -1.3°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Date: 12/2/25										
Time: 10:15										
Initials: MJ										
Package Description: Cooler										
Package Temp °C	0.0	0.0		/						
Representative Sample Temp °C	3.9	3.9		/						/

Sample Receipt

Yes	No	
[x]	[ ]	Received on ice or other coolant
[x]	[ ]	Ice still present upon receipt
[ ]	[x]	Custody seals present

[ ] Yes [ ] No Custody seals intact (if applicable)

[ ] Trace Courier [ ] Client Drop-off [x] UPS [ ] Fed Ex [ ] US Mail [ ] Other

Sample Condition

Yes	No	N/A	
[x]	[ ]	[ ]	All sample containers arrived unbroken and labeled
[x]	[ ]	[ ]	Sufficient sample to run requested analyses
[x]	[ ]	[ ]	Correct chemical preservative added to samples
[ ]	[x]	[ ]	Samples preserved at Trace ______
[x]	[ ]	[ ]	Chemical preservation verified, check EMD pH test strip used (if applicable)
			[x] pH 0-2.5 (Lot: HC461264) [ ] pH 11.0-13.0 (Lot: HC022540) [ ] Other
[x]	[ ]	[ ]	Air bubbles absent from VOAs (no bubble grater than 6mm) ______

Chain of Custody (COC)

Yes	No	
[x]	[ ]	All bottle labels agree with COC
[x]	[ ]	COC filled out properly
[x]	[ ]	COC signed by client

Notes:

Form 70-A.61
Effective 12/1/25

TRACE Analytical Laboratories, Inc.

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December 15, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25L0450
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

PJLA Testing Accreditation# 1031

TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25L0450-01	Domestic #1	Wastewater	DG	12/05/25	12/05/25 13:50
Sample type: Composite					
25L0450-02	Domestic #2	Wastewater	DG	12/05/25	12/05/25 13:50
Sample type: Composite					
25L0450-03	WWTP Raw Influent	Wastewater	DG	12/05/25	12/05/25 13:50
Sample type: Composite					
25L0450-04	WWTP Primary Effluent	Wastewater	DG	12/05/25	12/05/25 13:50
Sample type: Composite					
25L0450-05	WWTP Final Effluent	Wastewater	DG	12/05/25	12/05/25 13:50
Sample type: Composite					

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ANALYTICAL RESULTS

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0450-01 | Matrix: Wastewater | Date Collected: 12/05/25
Sample ID: Domestic #1 | Date Received: 12/05/25 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T177054									
Beryllium	<0.0040 mg/L	0.0040	1	12/11/25	ejb	12/12/25	jma		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS								DL03	
Analysis Method: EPA 625.1									
Batch: T176922									
Phenol	<10 ug/L	10	4	12/08/25	kbc	12/09/25	jlh		
Surrogates:									
2-Fluorophenol	23 %	20-53	4	12/08/25	kbc	12/09/25	jlh		
Phenol-d5	16 %	11-40	4	12/08/25	kbc	12/09/25	jlh		
Nitrobenzene-d5	71 %	36-103	4	12/08/25	kbc	12/09/25	jlh		
2-Fluorobiphenyl	64 %	36-119	4	12/08/25	kbc	12/09/25	jlh	N	
2,4,6-Tribromophenol	63 %	30-105	4	12/08/25	kbc	12/09/25	jlh	N	
Terphenyl-d14	57 %	37-109	4	12/08/25	kbc	12/09/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176932									
Benzidine	**1.5 ug/L**	**0.89**	**8**	**12/08/25**	**als**	**12/12/25**	**jlh**		
3,3'-Dichlorobenzidine	<8.9 ug/L	8.9	8	12/08/25	als	12/12/25	jlh		
Surrogates:									
Nitrobenzene-d5	* **179** %	38-130	8	12/08/25	als	12/12/25	jlh	S10	
2-Fluorobiphenyl	59 %	39-118	8	12/08/25	als	12/12/25	jlh	N	
Terphenyl-d14	51 %	14-96	8	12/08/25	als	12/12/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176893									
Bromomethane	<5.0 ug/L	5.0	1	12/05/25	ats	12/06/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	98 %	68-133	1	12/05/25	ats	12/06/25	ats		
Toluene-d8	103 %	75-120	1	12/05/25	ats	12/06/25	ats		
4-Bromofluorobenzene	98 %	69-119	1	12/05/25	ats	12/06/25	ats		
1,2-Dichlorobenzene-d4	101 %	72-127	1	12/05/25	ats	12/06/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0450-01 Matrix: Wastewater Date Collected: 12/05/25
Sample ID: Domestic #1 Date Received: 12/05/25 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0450-02 | Matrix: Wastewater | Date Collected: 12/05/25
Sample ID: Domestic #2 | Date Received: 12/05/25 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T177054									
Beryllium	<0.0040 mg/L	0.0040	1	12/11/25	ejb	12/12/25	jma		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS								DL03	
Analysis Method: EPA 625.1									
Batch: T176922									
Phenol	<9.6 ug/L	9.6	4	12/08/25	kbc	12/09/25	jlh		
Surrogates:									
2-Fluorophenol	25 %	20-53	4	12/08/25	kbc	12/09/25	jlh		
Phenol-d5	18 %	11-40	4	12/08/25	kbc	12/09/25	jlh		
Nitrobenzene-d5	74 %	36-103	4	12/08/25	kbc	12/09/25	jlh		
2-Fluorobiphenyl	63 %	36-119	4	12/08/25	kbc	12/09/25	jlh	N	
2,4,6-Tribromophenol	64 %	30-105	4	12/08/25	kbc	12/09/25	jlh	N	
Terphenyl-d14	52 %	37-109	4	12/08/25	kbc	12/09/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T177123									
Benzidine	**0.56 ug/L**	**0.21**	**2**	**12/12/25**	**als**	**12/12/25**	**jlh**		
3,3'-Dichlorobenzidine	<2.1 ug/L	2.1	2	12/12/25	als	12/12/25	jlh		
Surrogates:									
Nitrobenzene-d5	96 %	38-130	2	12/12/25	als	12/12/25	jlh		
2-Fluorobiphenyl	41 %	39-118	2	12/12/25	als	12/12/25	jlh	N	
Terphenyl-d14	19 %	14-96	2	12/12/25	als	12/12/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176953									
Bromomethane	<5.0 ug/L	5.0	1	12/08/25	sb	12/08/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	126 %	68-133	1	12/08/25	sb	12/08/25	sb		
Toluene-d8	84 %	75-120	1	12/08/25	sb	12/08/25	sb		
4-Bromofluorobenzene	106 %	69-119	1	12/08/25	sb	12/08/25	sb		
1,2-Dichlorobenzene-d4	104 %	72-127	1	12/08/25	sb	12/08/25	sb		

CERTIFICATE OF ANALYSIS

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ANALYTICAL RESULTS

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0450-02 Matrix: Wastewater Date Collected: 12/05/25
Sample ID: Domestic #2 Date Received: 12/05/25 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

CERTIFICATE OF ANALYSIS

ANALYTICAL RESULTS

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0450-03 | Matrix: Wastewater | Date Collected: 12/05/25
Sample ID: WWTP Raw Influent | Date Received: 12/05/25 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T177054									
Beryllium	<0.0040 mg/L	0.0040	1	12/11/25	ejb	12/12/25	jma		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS								DL03	
Analysis Method: EPA 625.1									
Batch: T176922									
Phenol	<10 ug/L	10	4	12/08/25	kbc	12/10/25	jlh		
Surrogates:									
2-Fluorophenol	27 %	20-53	4	12/08/25	kbc	12/10/25	jlh		
Phenol-d5	17 %	11-40	4	12/08/25	kbc	12/10/25	jlh		
Nitrobenzene-d5	76 %	36-103	4	12/08/25	kbc	12/10/25	jlh		
2-Fluorobiphenyl	64 %	36-119	4	12/08/25	kbc	12/10/25	jlh	N	
2,4,6-Tribromophenol	63 %	30-105	4	12/08/25	kbc	12/10/25	jlh	N	
Terphenyl-d14	61 %	37-109	4	12/08/25	kbc	12/10/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176932									
Benzidine	**2.0 ug/L**	**0.12**	**1**	**12/08/25**	**als**	**12/10/25**	**jlh**		
3,3'-Dichlorobenzidine	<1.2 ug/L	1.2	1	12/08/25	als	12/10/25	jlh		
Surrogates:									
Nitrobenzene-d5	42 %	38-130	1	12/08/25	als	12/10/25	jlh		
2-Fluorobiphenyl	41 %	39-118	1	12/08/25	als	12/10/25	jlh	N	
Terphenyl-d14	16 %	14-96	1	12/08/25	als	12/10/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176953									
Bromomethane	<5.0 ug/L	5.0	1	12/08/25	sb	12/08/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	123 %	68-133	1	12/08/25	sb	12/08/25	sb		
Toluene-d8	87 %	75-120	1	12/08/25	sb	12/08/25	sb		
4-Bromofluorobenzene	103 %	69-119	1	12/08/25	sb	12/08/25	sb		
1,2-Dichlorobenzene-d4	105 %	72-127	1	12/08/25	sb	12/08/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0450-03 Matrix: Wastewater Date Collected: 12/05/25
Sample ID: WWTP Raw Influent Date Received: 12/05/25 13:50

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0450-04 Matrix: Wastewater Date Collected: 12/05/25
Sample ID: WWTP Primary Effluent Date Received: 12/05/25 13:50

PARAMETERS		RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL										
Analysis Method: EPA 200.8 Rev. 5.4										
Batch: T177054										
Beryllium		<0.0040 mg/L	0.0040	1	12/11/25	ejb	12/12/25	jma		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									DL03	
Analysis Method: EPA 625.1										
Batch: T176922										
Phenol		<10 ug/L	10	4	12/08/25	kbc	12/10/25	jlh		
Surrogates:										
2-Fluorophenol		26 %	20-53	4	12/08/25	kbc	12/10/25	jlh		
Phenol-d5		16 %	11-40	4	12/08/25	kbc	12/10/25	jlh		
Nitrobenzene-d5		72 %	36-103	4	12/08/25	kbc	12/10/25	jlh		
2-Fluorobiphenyl		59 %	36-119	4	12/08/25	kbc	12/10/25	jlh	N	
2,4,6-Tribromophenol		59 %	30-105	4	12/08/25	kbc	12/10/25	jlh	N	
Terphenyl-d14		52 %	37-109	4	12/08/25	kbc	12/10/25	jlh		
Analysis Method: EPA 625.1 SIM										
Batch: T177123										
Benzidine		**1.2 ug/L**	**0.24**	**2**	**12/12/25**	**als**	**12/12/25**	**jlh**		
3,3'-Dichlorobenzidine		<2.4 ug/L	2.4	2	12/12/25	als	12/12/25	jlh		
Surrogates:										
Nitrobenzene-d5	*	**32** %	38-130	2	12/12/25	als	12/12/25	jlh	S02	
2-Fluorobiphenyl	*	**26** %	39-118	2	12/12/25	als	12/12/25	jlh	S02, N	
Terphenyl-d14		14 %	14-96	2	12/12/25	als	12/12/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 624.1										
Batch: T176953										
Bromomethane		<5.0 ug/L	5.0	1	12/08/25	sb	12/08/25	sb		
Surrogates:										
1,2-Dichloroethane-d4		126 %	68-133	1	12/08/25	sb	12/08/25	sb		
Toluene-d8		86 %	75-120	1	12/08/25	sb	12/08/25	sb		
4-Bromofluorobenzene		104 %	69-119	1	12/08/25	sb	12/08/25	sb		
1,2-Dichlorobenzene-d4		108 %	72-127	1	12/08/25	sb	12/08/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0450-04 Matrix: Wastewater Date Collected: 12/05/25
Sample ID: WWTP Primary Effluent Date Received: 12/05/25 13:50

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0450-05 Matrix: Wastewater Date Collected: 12/05/25
Sample ID: WWTP Final Effluent Date Received: 12/05/25 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T177054									
Beryllium	<0.0040 mg/L	0.0040	1	12/11/25	ejb	12/12/25	jma		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176922									
Phenol	<5.0 ug/L	5.0	1	12/08/25	kbc	12/10/25	jlh		
Surrogates:									
2-Fluorophenol	30 %	20-53	1	12/08/25	kbc	12/10/25	jlh		
Phenol-d5	18 %	11-40	1	12/08/25	kbc	12/10/25	jlh		
Nitrobenzene-d5	82 %	36-103	1	12/08/25	kbc	12/10/25	jlh		
2-Fluorobiphenyl	74 %	36-119	1	12/08/25	kbc	12/10/25	jlh	N	
2,4,6-Tribromophenol	75 %	30-105	1	12/08/25	kbc	12/10/25	jlh	N	
Terphenyl-d14	59 %	37-109	1	12/08/25	kbc	12/10/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176932									
Benzidine	<0.10 ug/L	0.10	1	12/08/25	als	12/10/25	jlh		
3,3'-Dichlorobenzidine	<1.0 ug/L	1.0	1	12/08/25	als	12/10/25	jlh		
Surrogates:									
Nitrobenzene-d5	72 %	38-130	1	12/08/25	als	12/10/25	jlh		
2-Fluorobiphenyl	62 %	39-118	1	12/08/25	als	12/10/25	jlh	N	
Terphenyl-d14	25 %	14-96	1	12/08/25	als	12/10/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176953									
Bromomethane	<5.0 ug/L	5.0	1	12/08/25	sb	12/08/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	128 %	68-133	1	12/08/25	sb	12/08/25	sb		
Toluene-d8	86 %	75-120	1	12/08/25	sb	12/08/25	sb		
4-Bromofluorobenzene	106 %	69-119	1	12/08/25	sb	12/08/25	sb		
1,2-Dichlorobenzene-d4	108 %	72-127	1	12/08/25	sb	12/08/25	sb		

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Muskegon, MI 49444-2673

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ANALYTICAL RESULTS

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0450-05 Matrix: Wastewater Date Collected: 12/05/25
Sample ID: WWTP Final Effluent Date Received: 12/05/25 13:50

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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QUALITY CONTROL RESULTS

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177054
QC Batch Method: EPA 200.2
Analysis Description: Beryllium, Total
Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T177054-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.00025	0.00025	

LABORATORY CONTROL SAMPLE: T177054-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.218	109	85-115	

MATRIX SPIKE: T177054-MS1 Original: **25L0450-04**

Parameter	Units	Original Result	Spike Conc.	MS Result	MS % Rec	% Rec Unit	Notes
Beryllium	mg/L	0	0.200	0.210	105	70-130	

MATRIX SPIKE: T177054-MS2 Original: **25L0450-05**

Parameter	Units	Original Result	Spike Conc.	MS Result	MS % Rec	% Rec Unit	Notes
Beryllium	mg/L	0	0.200	0.210	105	70-130	

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176922
QC Batch Method: EPA 625 Extraction for BNAs
Analysis Description: TTO Semi-Volatiles
Analysis Method: EPA 625.1

METHOD BLANK: T176922-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	%	26	20-53	
Phenol-d5 (S)	%	15	11-40	
Nitrobenzene-d5 (S)	%	85	36-103	
2-Fluorobiphenyl (S)	%	72	36-119	
2,4,6-Tribromophenol (S)	%	85	30-105	
Terphenyl-d14 (S)	**%**	**120**	**37-109**	**S10**

LABORATORY CONTROL SAMPLE: T176922-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	19.2	19	17-120	
2-Fluorophenol (S)	%	100	30.6	31	20-53	
Phenol-d5 (S)	%	100	26.8	27	11-40	
Nitrobenzene-d5 (S)	%	100	99.2	99	36-103	
2-Fluorobiphenyl (S)	%	100	85.7	86	36-119	
2,4,6-Tribromophenol (S)	%	100	88.8	89	30-105	
Terphenyl-d14 (S)	%	100	94.4	94	37-109	

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176932
QC Batch Method: GC/MS No Prep
Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T176932-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	53	38-130	
2-Fluorobiphenyl (S)	%	58	39-118	
Terphenyl-d14 (S)	%	38	14-96	

LABORATORY CONTROL SAMPLE: T176932-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	1.02	51	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	<2.0	79	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.729	73	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.631	63	39-118	
Terphenyl-d14 (S)	%	1.00	0.390	39	14-96	

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177123
QC Batch Method: GC/MS No Prep
Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T177123-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes

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METHOD BLANK: T177123-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	66	38-130	
2-Fluorobiphenyl (S)	%	57	39-118	
Terphenyl-d14 (S)	%	34	14-96	

LABORATORY CONTROL SAMPLE: T177123-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	1.22	61	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	<2.0	96	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.783	78	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.707	71	39-118	
Terphenyl-d14 (S)	%	1.00	0.423	42	14-96	

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176893	Analysis Description: TTO Volatiles
QC Batch Method: EPA 624 Purge and Trap	Analysis Method: EPA 624.1

METHOD BLANK: T176893-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	**%**	**99**	**68-133**	**M-01**
Toluene-d8 (S)	%	102	75-120	
4-Bromofluorobenzene (S)	%	98	69-119	
1,2-Dichlorobenzene-d4 (S)	%	100	72-127	

LABORATORY CONTROL SAMPLE: T176893-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	43.6	87	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	29.8	99	68-133	
Toluene-d8 (S)	%	30.0	31.0	104	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.8	99	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	30.2	101	72-127	

Trace Project ID: 25L0450
Client Project ID: 2025 IPP/MAHL Additional

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QC Batch: T176953
QC Batch Method: EPA 624 Purge and Trap
Analysis Description: TTO Volatiles
Analysis Method: EPA 624.1

METHOD BLANK: T176953-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	109	68-133	
Toluene-d8 (S)	%	93	75-120	
4-Bromofluorobenzene (S)	%	106	69-119	
1,2-Dichlorobenzene-d4 (S)	%	104	72-127	

LABORATORY CONTROL SAMPLE: T176953-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	48.7	97	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	31.0	104	68-133	
Toluene-d8 (S)	%	30.0	27.9	93	75-120	
4-Bromofluorobenzene (S)	%	30.0	30.0	100	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	29.6	99	72-127	

MATRIX SPIKE / MATRIX SPIKE DUPLICATE: T176953-MSD1 Original: **25L0450-05**

Parameter	Units	Original Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Notes
Bromomethane	ug/L	0	50.0	54.9	56.8	110	114	1-242	3	61	
1,2-Dichloroethane-d4 (S)	%		30.0	35.5	35.1	118	117	68-133			
Toluene-d8 (S)	%		30.0	26.7	26.1	89	87	75-120			
4-Bromofluorobenzene (S)	%		30.0	29.6	29.5	99	98	69-119			
1,2-Dichlorobenzene-d4 (S)	%		30.0	29.7	29.6	99	99	72-127			

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25L0450-01

Analysis: EPA 625.1

Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Analysis: EPA 625.1 SIM

Nitrobenzene-d5 — Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Trace ID: 25L0450-02

Analysis: EPA 625.1

Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Trace ID: 25L0450-03

Analysis: EPA 625.1

Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Trace ID: 25L0450-04

Analysis: EPA 625.1

Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Trace ID: 25L0450-04RE1

Analysis: EPA 625.1 SIM

2-Fluorobiphenyl	Note S02 : The surrogate recovery was out of control low when compared to control limits. All results and reporting limits must be considered estimated.
Nitrobenzene-d5	Note S02 : The surrogate recovery was out of control low when compared to control limits. All results and reporting limits must be considered estimated.

Trace ID: T176893-BLK1

Analysis: EPA 624.1

1,2-Dichloroethane-d4	Note M-01 : An incorrect peak was integrated by the computer software. The analyst identified the correct peak and integrated it manually.

Trace ID: T176922-BLK1

Analysis: EPA 625.1

Terphenyl-d14	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

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CHAIN-OF-CUSTODY RECORD

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Page 1 of 1

Trace ID No. 25L0450

Trace Use:
Logged By: NC
Checked By: CMM
Sample Collection Time (Hrs):

Report Results To:		Bill To:	
Company Name:	City of Cadillac	PO #:	
Report To:	Cindy Tomaszewski	Contact Name:	Cindy Tomaszewski
Mailing Address:	1121 Plett Rd	Billing Address (if different):	200 N Lake St
City, State, Zip Code:	Cadillac MI 49601	City, State, Zip Code:	Cadillac MI 49601
Office Phone:	231-775-2368 Cell Phone: 231-878-6230	Phone Number:	
Email Address:	lab@cadillac-mi.net; pmiller@cadillac-mi.net	Billing Email Address:	lab@cadillac-mi.net

Requested Turnaround Times (TAT)
- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:
WW = Wastewater, DW = Drinking Water, GW = Groundwater, LW = Liquid Waste, AQ = Aqueous, O = Oil, S = Solid, SL = Sludge, A = Air, WI = Wipes, U = Unknown

Project Name: 2025 IPP/MAHL Additional

Sampled By (print): Derek Goodrich

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Composite	Grab	Metals Field Filterd (Y or N)	Matrix - see above →	Number of Containers	Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes
1	12-4		Domestic #1	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	
2			Domestic #2	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	grab VOCs to
3			WWTP Raw Influent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	be composited
4			WWTP Primary Effluent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	at the lab
5			WWTP Final Effluent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	

*All composites started on "Sample Collection Date"

Possible Health Hazards?

Please Sign	Released By	Received By	Date	Time	Released By	Received By	Date	Time
1) / 2)	[signature]	[signature]	12.5.25	13:50	[signature]	[signature]		
3) / 4)					[signature]			

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

CERTIFICATE OF ANALYSIS

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TRACE
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www.trace-labs.com

25L0450
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

	Original Observation	Corrected Temperature	IR-9 (CF: +0.2°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.6°C)	IR-14 (CF: -1.3°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Date: 12/5/25										
Time: 1600										
Initials: MJ										
Package Description: Cooler										
Package Temp °C	0.0	0.0		✓						
Representative Sample Temp °C	2.6	2.6		✓						✓

Sample Receipt

Yes	No	
☑	☐	Received on ice or other coolant
☑	☐	Ice still present upon receipt
☐	☑	Custody seals present

☐ Yes ☐ No Custody seals intact (if applicable)

☑ Trace Courier ☐ Client Drop-off ☐ UPS ☐ Fed Ex ☐ US Mail ☐ Other

Sample Condition

Yes	No	N/A	
☑	☐	☐	All sample containers arrived unbroken and labeled
☑	☐	☐	Sufficient sample to run requested analyses
☑	☐	☐	Correct chemical preservative added to samples
☐	☑	☐	Samples preserved at Trace (if yes, fill in preservation information below)

Preservation - Reagent:___________ LIMS ID:_____________ Date/Time:_________________

Yes	No	N/A	
☑	☐	☐	Chemical preservation verified, check EMD pH test strip used (if applicable)

☑ pH 0-2.5 (Lot: HC461264) ☐ pH 11.0-13.0 (Lot: HC022540) ☐ Other

Yes	No	N/A	
~~☑~~	☑	☐	Air bubbles absent from VOAs (no bubble greater than 6mm) head Space <6mm in multiple VO.

SAB 12/5

Chain of Custody (COC)

Yes	No	
☑	☐	All bottle labels agree with COC
☑	☐	COC filled out properly
☑	☐	COC signed by client

Notes:

Form 70-A.61
Effective 12/1/25

TRACE Analytical Laboratories, Inc.

CERTIFICATE OF ANALYSIS

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December 16, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25L0231
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

PJLA Testing Accreditation# 1031

TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 25L0231
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25L0231-01 Sample type: Grab	FIAMM	Wastewater	DG	12/02/25 08:08	12/03/25 10:38
25L0231-02 Sample type: Grab	MI Rubber	Wastewater	DG	12/02/25 08:50	12/03/25 10:38

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ANALYTICAL RESULTS

Trace Project ID: 25L0231
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0231-01 | Matrix: Wastewater | Date Collected: 12/02/25 08:08
Sample ID: FIAMM | Date Received: 12/03/25 10:38

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL										
Analysis Method: EPA 200.8 Rev. 5.4										
Batch: T176770										
Beryllium	<0.0040	mg/L	0.0040	1	12/04/25	ejb	12/09/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 625.1										
Batch: T176800										
Phenol	<5.0	ug/L	5.0	1	12/04/25	kbc	12/05/25	jlh		
Surrogates:										
2-Fluorophenol	25	%	20-53	1	12/04/25	kbc	12/05/25	jlh		
Phenol-d5	17	%	11-40	1	12/04/25	kbc	12/05/25	jlh		
Nitrobenzene-d5	68	%	36-103	1	12/04/25	kbc	12/05/25	jlh		
2-Fluorobiphenyl	64	%	36-119	1	12/04/25	kbc	12/05/25	jlh	N	
2,4,6-Tribromophenol	67	%	30-105	1	12/04/25	kbc	12/05/25	jlh	N	
Terphenyl-d14	71	%	37-109	1	12/04/25	kbc	12/05/25	jlh		
Analysis Method: EPA 625.1 SIM										
Batch: T176834										
Benzidine	<0.11	ug/L	0.11	1	12/05/25	als	12/10/25	jlh		
3,3'-Dichlorobenzidine	<1.1	ug/L	1.1	1	12/05/25	als	12/10/25	jlh		
Surrogates:										
Nitrobenzene-d5	73	%	38-130	1	12/05/25	als	12/10/25	jlh		
2-Fluorobiphenyl	61	%	39-118	1	12/05/25	als	12/10/25	jlh	N	
Terphenyl-d14	31	%	14-96	1	12/05/25	als	12/10/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 624.1										
Batch: T176772										
Bromomethane	<5.0	ug/L	5.0	1	12/03/25	ats	12/04/25	ats		
Surrogates:										
1,2-Dichloroethane-d4	93	%	68-133	1	12/03/25	ats	12/04/25	ats		
Toluene-d8	109	%	75-120	1	12/03/25	ats	12/04/25	ats		
4-Bromofluorobenzene	98	%	69-119	1	12/03/25	ats	12/04/25	ats		
1,2-Dichlorobenzene-d4	100	%	72-127	1	12/03/25	ats	12/04/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25L0231
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0231-01 Matrix: Wastewater Date Collected: 12/02/25 08:08
Sample ID: FIAMM Date Received: 12/03/25 10:38

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25L0231
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0231-02 Matrix: Wastewater Date Collected: 12/02/25 08:50
Sample ID: MI Rubber Date Received: 12/03/25 10:38

PARAMETERS		RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL											
Analysis Method: EPA 200.8 Rev. 5.4											
Batch: T176770											
Beryllium		<0.0040	mg/L	0.0040	1	12/04/25	ejb	12/09/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 625.1											
Batch: T176800											
Phenol		<5.0	ug/L	5.0	1	12/04/25	kbc	12/05/25	jlh		
Surrogates:											
2-Fluorophenol		31	%	20-53	1	12/04/25	kbc	12/05/25	jlh		
Phenol-d5		20	%	11-40	1	12/04/25	kbc	12/05/25	jlh		
Nitrobenzene-d5		85	%	36-103	1	12/04/25	kbc	12/05/25	jlh		
2-Fluorobiphenyl		76	%	36-119	1	12/04/25	kbc	12/05/25	jlh	N	
2,4,6-Tribromophenol		83	%	30-105	1	12/04/25	kbc	12/05/25	jlh	N	
Terphenyl-d14		79	%	37-109	1	12/04/25	kbc	12/05/25	jlh		
Analysis Method: EPA 625.1 SIM											
Batch: T176834											
Benzidine		**1.1**	**ug/L**	**0.11**	**1**	**12/05/25**	**als**	**12/10/25**	**jlh**		
3,3'-Dichlorobenzidine		<1.1	ug/L	1.1	1	12/05/25	als	12/10/25	jlh		
Surrogates:											
Nitrobenzene-d5	*	**30**	%	38-130	1	12/05/25	als	12/10/25	jlh	S11	
2-Fluorobiphenyl	*	**24**	%	39-118	1	12/05/25	als	12/10/25	jlh	S11, N	
Terphenyl-d14	*	**11**	%	14-96	1	12/05/25	als	12/10/25	jlh	S11	
VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 624.1											
Batch: T176772											
Bromomethane		<5.0	ug/L	5.0	1	12/03/25	ats	12/04/25	ats		
Surrogates:											
1,2-Dichloroethane-d4		94	%	68-133	1	12/03/25	ats	12/04/25	ats		
Toluene-d8		109	%	75-120	1	12/03/25	ats	12/04/25	ats		
4-Bromofluorobenzene		98	%	69-119	1	12/03/25	ats	12/04/25	ats		
1,2-Dichlorobenzene-d4		100	%	72-127	1	12/03/25	ats	12/04/25	ats		

ANALYTICAL RESULTS

Trace Project ID: 25L0231
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0231-02 Matrix: Wastewater Date Collected: 12/02/25 08:50
Sample ID: MI Rubber Date Received: 12/03/25 10:38

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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QUALITY CONTROL RESULTS

Trace Project ID: 25L0231
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176770
QC Batch Method: EPA 200.2
Analysis Description: Beryllium, Total
Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T176770-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.0040	0.0040	

LABORATORY CONTROL SAMPLE: T176770-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.189	94	85-115	

Trace Project ID: 25L0231
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176800
QC Batch Method: EPA 625 Extraction for BNAs
Analysis Description: TTO Semi-Volatiles
Analysis Method: EPA 625.1

METHOD BLANK: T176800-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<5.0	5.0	
2-Fluorophenol (S)	%	37	20-53	
Phenol-d5 (S)	%	24	11-40	
Nitrobenzene-d5 (S)	%	81	36-103	
2-Fluorobiphenyl (S)	%	73	36-119	
2,4,6-Tribromophenol (S)	%	81	30-105	
Terphenyl-d14 (S)	%	99	37-109	

LABORATORY CONTROL SAMPLE: T176800-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	21.5	22	17-120	
2-Fluorophenol (S)	%	100	38.6	39	20-53	
Phenol-d5 (S)	**%**	**100**	**3.57**	**4**	**11-40**	**S09**
Nitrobenzene-d5 (S)	%	100	102	102	36-103	
2-Fluorobiphenyl (S)	%	100	93.4	93	36-119	
2,4,6-Tribromophenol (S)	%	100	105	105	30-105	
Terphenyl-d14 (S)	**%**	**100**	**126**	**126**	**37-109**	**S10**

MATRIX SPIKE: T176800-MS1 Original: **25L0231-02**

Parameter	Units	Original Result	Spike Conc.	MS Result	MS % Rec	% Rec Unit	Notes
Phenol	ug/L	2.90	99.4	18.6	16	5-120	
2-Fluorophenol (S)	%		99.4	29.2	29	20-53	
Phenol-d5 (S)	%		99.4	3.47	3	11-40	
Nitrobenzene-d5 (S)	%		99.4	78.4	79	36-103	
2-Fluorobiphenyl (S)	%		99.4	75.4	76	36-119	
2,4,6-Tribromophenol (S)	%		99.4	74.4	75	30-105	
Terphenyl-d14 (S)	%		99.4	78.2	79	37-109	

Trace Project ID: 25L0231
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176834
QC Batch Method: GC/MS No Prep

Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T176834-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	70	38-130	
2-Fluorobiphenyl (S)	%	69	39-118	
Terphenyl-d14 (S)	%	41	14-96	

LABORATORY CONTROL SAMPLE: T176834-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	0.730	37	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	<2.0	70	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.666	67	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.636	64	39-118	
Terphenyl-d14 (S)	%	1.00	0.398	40	14-96	

Trace Project ID: 25L0231
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176772
QC Batch Method: EPA 624 Purge and Trap

Analysis Description: TTO Volatiles
Analysis Method: EPA 624.1

METHOD BLANK: T176772-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes

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METHOD BLANK: T176772-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	100	68-133	
Toluene-d8 (S)	%	101	75-120	
4-Bromofluorobenzene (S)	%	98	69-119	
1,2-Dichlorobenzene-d4 (S)	%	101	72-127	

LABORATORY CONTROL SAMPLE: T176772-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	**ug/L**	**50.0**	**35.4**	**71**	**15-185**	**M-04**
1,2-Dichloroethane-d4 (S)	%	30.0	29.6	99	68-133	
Toluene-d8 (S)	%	30.0	30.0	100	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.5	98	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	30.0	100	72-127	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25L0231-02

Analysis: EPA 625.1 SIM

2-Fluorobiphenyl	Note S11 : The base/neutral surrogate recoveries were out of control. The results for the base/neutral compounds must be considered estimated.
Nitrobenzene-d5	Note S11 : The base/neutral surrogate recoveries were out of control. The results for the base/neutral compounds must be considered estimated.
Terphenyl-d14	Note S11 : The base/neutral surrogate recoveries were out of control. The results for the base/neutral compounds must be considered estimated.

Trace ID: T176772-BS1

Analysis: EPA 624.1

Bromomethane	Note M-04 : The entire peak was not integrated. The analyst manually integrated the peak so as to include it's entire area.

Trace ID: T176800-BS1

Analysis: EPA 625.1

Phenol-d5	Note S09 : One of the acid surrogate recoveries was outside the control limit. Since the other two acid surrogates were within the control limits, no data requires qualification.
Terphenyl-d14	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

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CHAIN-OF-CUSTODY RECORD

TRACE ANALYTICAL LABORATORIES, INC.

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Page 1 of 1

Trace ID No. 25L0231

Report Results To:

Company Name:	City of Cadillac
Report To:	Cindy Tomaszewski
Mailing Address:	1121 Plett Rd
City, State, Zip Code:	Cadillac MI 49601
Office Phone:	231-775-2368
Cell Phone:	231-878-6230
Email Address:	lab@cadillac-mi.net; pmiller@cadillac-mi.net

Bill To:

PO #:	
Contact Name:	Cindy Tomaszewski
Billing Address (if different):	200 N Lake St
City, State, Zip Code:	Cadillac MI 49601
Phone Number:	
Billing Email Address:	lab@cadillac-mi.net

Trace Use:

Logged By:	CN
Checked By:	MCL
Sample Collection Time (Hrs):	

Requested Turnaround Times (TAT)

- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:

WW = Wastewater	AQ = Aqueous	A = Air
DW = Drinking Water	O = Oil	WI = Wipes
GW = Groundwater	S = Solid	U = Unknown
LW = Liquid Waste	SL = Sludge	

Project Name: 2025 IPP/MAHL Additional

Sampled By (print): DEREK GOODRICH

Analysis Requested

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Composite	Grab	Metals Field Filterd (Y or N)	Matrix - see above →	Number of Containers	Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes	Possible Health Hazards?
1	12-2-25	8:05	FIAMM		✓	N	WW	6	x		x		x		x		X	X	X	X		
2	12-2-25	8:50	MI RUBBER		✓	N	WW	6	x		x		x		x		X	X	X	X		

Please Sign

	Released By	Received By	Date	Time		Released By	Received By	Date	Time
1)	CAT Bunyewski	UPS Ground	12-2-25	1310	2)	UPS Groud	[illegible]	12/3/25	1103
3)					4)				

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

CERTIFICATE OF ANALYSIS

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25L0231
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

	Original Observation	Corrected Temperature	IR-9 (CF: +0.2°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.6°C)	IR-14 (CF: -1.3°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Date: 12/3/25										
Time: 10:22										
Initials: CN										
Package Description: Cooler										
Package Temp °C	-1.3	-1.1								
Representative Sample Temp °C										

Sample Receipt

Yes	No	
[X]	[ ]	Received on ice or other coolant
[X]	[ ]	Ice still present upon receipt
[ ]	[X]	Custody seals present

[ ] Yes [ ] No Custody seals intact (if applicable)

[ ] Trace Courier [ ] Client Drop-off [X] UPS [ ] Fed Ex [ ] US Mail [ ] Other

Sample Condition

Yes	No	N/A	
[X]	[ ]	[ ]	All sample containers arrived unbroken and labeled
[X]	[ ]	[ ]	Sufficient sample to run requested analyses
[X]	[ ]	[ ]	Correct chemical preservative added to samples
[ ]	[ ]	[X]	Samples preserved at Trace
[X]	[ ]	[ ]	Chemical preservation verified, check EMD pH test strip used (if applicable)

[X] pH 0-2.5 (Lot: HC461264) [ ] pH 11.0-13.0 (Lot: HC022540) [ ] Other

Yes	No	N/A	
[X]	[ ]	[X] CN	Air bubbles absent from VOAs (no bubble grater than 6mm)

Chain of Custody (COC)

Yes	No	
[X]	[ ]	All bottle labels agree with COC
[X]	[ ]	COC filled out properly
[X]	[ ]	COC signed by client

Notes:

Form 70-A.61
Effective 12/1/25

TRACE Analytical Laboratories, Inc.

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December 16, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25L0322
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

PJLA Testing Accreditation# 1031

TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 25L0322
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25L0322-01	AAR-6TH	Wastewater	DG	12/03/25 13:50	12/04/25 11:40
Sample type: Grab					
25L0322-02	AKWEL	Wastewater	DG	12/03/25 14:00	12/04/25 11:40
Sample type: Grab					

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ANALYTICAL RESULTS

Trace Project ID: 25L0322
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0322-01 | Matrix: Wastewater | Date Collected: 12/03/25 13:50
Sample ID: AAR-6TH | Date Received: 12/04/25 11:40

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T176986									
Beryllium	<0.0040 mg/L	0.0040	1	12/09/25	ejb	12/12/25	jma		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176922									
Phenol	<5.0 ug/L	5.0	1	12/08/25	kbc	12/09/25	jlh		
Surrogates:									
2-Fluorophenol	26 %	20-53	1	12/08/25	kbc	12/09/25	jlh		
Phenol-d5	18 %	11-40	1	12/08/25	kbc	12/09/25	jlh		
Nitrobenzene-d5	81 %	36-103	1	12/08/25	kbc	12/09/25	jlh		
2-Fluorobiphenyl	77 %	36-119	1	12/08/25	kbc	12/09/25	jlh	N	
2,4,6-Tribromophenol	78 %	30-105	1	12/08/25	kbc	12/09/25	jlh	N	
Terphenyl-d14	82 %	37-109	1	12/08/25	kbc	12/09/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176834									
Benzidine	<0.23 ug/L	0.23	2	12/05/25	als	12/10/25	jlh		
3,3'-Dichlorobenzidine	<2.3 ug/L	2.3	2	12/05/25	als	12/10/25	jlh		
Surrogates:									
Nitrobenzene-d5	66 %	38-130	2	12/05/25	als	12/10/25	jlh		
2-Fluorobiphenyl	48 %	39-118	2	12/05/25	als	12/10/25	jlh	N	
Terphenyl-d14	24 %	14-96	2	12/05/25	als	12/10/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176833									
Bromomethane	<5.0 ug/L	5.0	1	12/04/25	ats	12/05/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	99 %	68-133	1	12/04/25	ats	12/05/25	ats		
Toluene-d8	102 %	75-120	1	12/04/25	ats	12/05/25	ats		
4-Bromofluorobenzene	99 %	69-119	1	12/04/25	ats	12/05/25	ats		
1,2-Dichlorobenzene-d4	101 %	72-127	1	12/04/25	ats	12/05/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25L0322
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0322-01 Matrix: Wastewater Date Collected: 12/03/25 13:50
Sample ID: AAR-6TH Date Received: 12/04/25 11:40

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25L0322
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0322-02 Matrix: Wastewater Date Collected: 12/03/25 14:00
Sample ID: AKWEL Date Received: 12/04/25 11:40

PARAMETERS		RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL											
Analysis Method: EPA 200.8 Rev. 5.4											
Batch: T176986											
Beryllium		<0.0040	mg/L	0.0040	1	12/09/25	ejb	12/12/25	jma		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS										DL03	
Analysis Method: EPA 625.1											
Batch: T176922											
Phenol		<53	ug/L	53	20	12/08/25	kbc	12/09/25	jlh		
Surrogates:											
2-Fluorophenol	*	**10**	%	20-53	20	12/08/25	kbc	12/09/25	jlh	S09	
Phenol-d5		20	%	11-40	20	12/08/25	kbc	12/09/25	jlh		
Nitrobenzene-d5		48	%	36-103	20	12/08/25	kbc	12/09/25	jlh		
2-Fluorobiphenyl		51	%	36-119	20	12/08/25	kbc	12/09/25	jlh	N	
2,4,6-Tribromophenol		56	%	30-105	20	12/08/25	kbc	12/09/25	jlh	N	
Terphenyl-d14		46	%	37-109	20	12/08/25	kbc	12/09/25	jlh		
Analysis Method: EPA 625.1 SIM											
Batch: T176834											
Benzidine		<0.42	ug/L	0.42	4	12/05/25	als	12/10/25	jlh		
3,3'-Dichlorobenzidine		<4.2	ug/L	4.2	4	12/05/25	als	12/10/25	jlh		
Surrogates:											
Nitrobenzene-d5		71	%	38-130	4	12/05/25	als	12/10/25	jlh		
2-Fluorobiphenyl		56	%	39-118	4	12/05/25	als	12/10/25	jlh	N	
Terphenyl-d14		43	%	14-96	4	12/05/25	als	12/10/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 624.1											
Batch: T176833											
Bromomethane		<5.0	ug/L	5.0	1	12/04/25	ats	12/05/25	ats		
Surrogates:											
1,2-Dichloroethane-d4		96	%	68-133	1	12/04/25	ats	12/05/25	ats		
Toluene-d8		103	%	75-120	1	12/04/25	ats	12/05/25	ats		
4-Bromofluorobenzene		99	%	69-119	1	12/04/25	ats	12/05/25	ats		
1,2-Dichlorobenzene-d4		100	%	72-127	1	12/04/25	ats	12/05/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25L0322
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0322-02
Sample ID: AKWEL
Matrix: Wastewater
Date Collected: 12/03/25 14:00
Date Received: 12/04/25 11:40

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673

ANALYTICAL LABORATORIES, INC.

231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

QUALITY CONTROL RESULTS

Trace Project ID: 25L0322
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176986
QC Batch Method: EPA 200.2
Analysis Description: Beryllium, Total
Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T176986-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.0040	0.0040	

LABORATORY CONTROL SAMPLE: T176986-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.210	105	85-115	

Trace Project ID: 25L0322
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176922
QC Batch Method: EPA 625 Extraction for BNAs
Analysis Description: TTO Semi-Volatiles
Analysis Method: EPA 625.1

METHOD BLANK: T176922-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	%	26	20-53	
Phenol-d5 (S)	%	15	11-40	
Nitrobenzene-d5 (S)	%	85	36-103	
2-Fluorobiphenyl (S)	%	72	36-119	
2,4,6-Tribromophenol (S)	%	85	30-105	
Terphenyl-d14 (S)	**%**	**120**	**37-109**	**S10**

LABORATORY CONTROL SAMPLE: T176922-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	19.2	19	17-120	
2-Fluorophenol (S)	%	100	30.6	31	20-53	
Phenol-d5 (S)	%	100	26.8	27	11-40	
Nitrobenzene-d5 (S)	%	100	99.2	99	36-103	
2-Fluorobiphenyl (S)	%	100	85.7	86	36-119	
2,4,6-Tribromophenol (S)	%	100	88.8	89	30-105	
Terphenyl-d14 (S)	%	100	94.4	94	37-109	

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Trace Project ID: 25L0322
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176834
QC Batch Method: GC/MS No Prep
Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T176834-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	70	38-130	
2-Fluorobiphenyl (S)	%	69	39-118	
Terphenyl-d14 (S)	%	41	14-96	

LABORATORY CONTROL SAMPLE: T176834-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	0.730	37	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	<2.0	70	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.666	67	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.636	64	39-118	
Terphenyl-d14 (S)	%	1.00	0.398	40	14-96	

Trace Project ID: 25L0322
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176833
QC Batch Method: EPA 624 Purge and Trap
Analysis Description: TTO Volatiles
Analysis Method: EPA 624.1

METHOD BLANK: T176833-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	96	68-133	
Toluene-d8 (S)	%	105	75-120	
4-Bromofluorobenzene (S)	%	99	69-119	
1,2-Dichlorobenzene-d4 (S)	%	102	72-127	

LABORATORY CONTROL SAMPLE: T176833-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	25.4	51	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	28.8	96	68-133	

LABORATORY CONTROL SAMPLE: T176833-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Toluene-d8 (S)	%	30.0	31.4	104	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.5	98	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	30.2	101	72-127	

AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25L0322-01

Analysis: EPA 625.1 SIM

Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Trace ID: 25L0322-02

Analysis: EPA 625.1

Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

2-Fluorophenol — Note S09 : One of the acid surrogate recoveries was outside the control limit. Since the other two acid surrogates were within the control limits, no data requires qualification.

Analysis: EPA 625.1 SIM

Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Trace ID: T176922-BLK1

Analysis: EPA 625.1

Terphenyl-d14 — Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

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CHAIN-OF-CUSTODY RECORD

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Phone 231.773.5998
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Page 1 of 1

Trace ID No. 25L0322

Report Results To:

Company Name: City of Cadillac	
Report To: Cindy Tomaszewski	
Mailing Address: 1121 Plett Rd	
City, State, Zip Code: Cadillac MI 49601	
Office Phone: 231-775-2368	Cell Phone: 231-878-6230
Email Address: lab@cadillac-mi.net; pmiller@cadillac-mi.net	

Bill To:

PO #:
Contact Name: Cindy Tomaszewski
Billing Address (if different): 200 N Lake St
City, State, Zip Code: Cadillac MI 49601
Phone Number:
Billing Email Address: lab@cadillac-mi.net

Trace Use:

Logged By: CN
Checked By: SAB
Sample Collection Time (Hrs):

Requested Turnaround Times (TAT)

- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:

WW = Wastewater	AQ = Aqueous	A = Air
DW = Drinking Water	O = Oil	WI = Wipes
GW = Groundwater	S = Solid	U = Unknown
LW = Liquid Waste	SL = Sludge	

Project Name: 2025 IPP/MAHL Additional

Sampled By (print): Derek Goodrich

Trace No.	Sample Collection Date DG	Sample Collection Time	Sample ID/Name	Sample Type (check one): Composite	Grab	Metals Field Filterd (Y or N)	Matrix - see above →	Number of Containers	Preservation: Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1								Remarks/Notes	Possible Health Hazards?
1	[illegible]	1530	AAN-6+7		✓	N	WW	6	x		x		x		x		X	X	X	X									
2	[illegible]	200	AKWEL		✓	N	WW	6	x		x		x		x		X	X	X	X									
	12-3-25																												

Please Sign

	Released By	Received By	Date	Time		Released By	Received By	Date	Time
1)	C Tomaszewski	UPS Ground	12-3-25	1600	2)	UPS	[illegible]	12/4/25	11:43
3)					4)				

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

CERTIFICATE OF ANALYSIS

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673

TRACE ANALYTICAL LABORATORIES, INC.

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25L0322
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

	Original Observation	Corrected Temperature	IR-9 (CF: +0.2°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.6°C)	IR-14 (CF: -1.3°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Date: 12/4/25										
Time: 11:06										
Initials: CN										
Package Description: Cooler										
Package Temp °C	0.1	0.1		X						
Representative Sample Temp °C	4.9	4.9		X						X

Sample Receipt

Yes	No	
[X]	[ ]	Received on ice or other coolant
[X]	[ ]	Ice still present upon receipt
[ ]	[X]	Custody seals present

[ ] Yes [ ] No Custody seals intact (if applicable)

[ ] Trace Courier [ ] Client Drop-off [X] UPS [ ] Fed Ex [ ] US Mail [ ] Other

Sample Condition

Yes	No	N/A	
~~[X]~~ CN	[X]	[ ]	All sample containers arrived unbroken and labeled
[X]	[ ]	[ ]	Sufficient sample to run requested analyses
[X]	[ ]	[ ]	Correct chemical preservative added to samples
[ ]	[ ]	[X]	Samples preserved at Trace (if yes, fill in preservation information below)

Preservation - Reagent:________ LIMS ID:________ Date/Time:________

Yes	No	N/A	
[X]	[ ]	[ ]	Chemical preservation verified, check EMD pH test strip used (if applicable)

[X] pH 0-2.5 (Lot: HC461264) [ ] pH 11.0-13.0 (Lot: HC022540) [ ] Other

Yes	No	N/A	
[ ]	[ ]	[ ]	Air bubbles absent from VOAs (no bubble greater than 6mm) ________

Chain of Custody (COC)

Yes	No	
[X]	[ ]	All bottle labels agree with COC
[X]	[ ]	COC filled out properly
[X]	[ ]	COC signed by client

Notes:

AAR-6TH 25L0322 Bottle C is broken
1000 ML Sodium Thiosulfate amb

Form 70-A.61
Effective 12/1/25

TRACE Analytical Laboratories, Inc.

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December 16, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25L0525
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

TNI — ACCREDITED LABORATORY — TNI ENVIRONMENTAL STANDARD
PJLA Testing Accreditation# 1031

TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 25L0525
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25L0525-01	CRE	Wastewater	DG	12/08/25 13:40	12/09/25 10:24
Sample type: Grab					
25L0525-02	Borg Warner	Wastewater	DG	12/08/25 13:20	12/09/25 10:24
Sample type: Grab					

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ANALYTICAL RESULTS

Trace Project ID: 25L0525
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0525-01 Matrix: Wastewater Date Collected: 12/08/25 13:40
Sample ID: CRE Date Received: 12/09/25 10:24

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T177000									
Beryllium	<0.0040 mg/L	0.0040	1	12/12/25	ejb	12/13/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T177029									
Phenol	<5.0 ug/L	5.0	1	12/10/25	kbc	12/10/25	jlh		
Surrogates:									
2-Fluorophenol	33 %	20-53	1	12/10/25	kbc	12/10/25	jlh		
Phenol-d5	20 %	11-40	1	12/10/25	kbc	12/10/25	jlh		
Nitrobenzene-d5	95 %	36-103	1	12/10/25	kbc	12/10/25	jlh		
2-Fluorobiphenyl	81 %	36-119	1	12/10/25	kbc	12/10/25	jlh	N	
2,4,6-Tribromophenol	81 %	30-105	1	12/10/25	kbc	12/10/25	jlh	N	
Terphenyl-d14	93 %	37-109	1	12/10/25	kbc	12/10/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T177123									
Benzidine	<0.21 ug/L	0.21	2	12/12/25	als	12/12/25	jlh		
3,3'-Dichlorobenzidine	<2.1 ug/L	2.1	2	12/12/25	als	12/12/25	jlh		
Surrogates:									
Nitrobenzene-d5	74 %	38-130	2	12/12/25	als	12/12/25	jlh		
2-Fluorobiphenyl	61 %	39-118	2	12/12/25	als	12/12/25	jlh	N	
Terphenyl-d14	35 %	14-96	2	12/12/25	als	12/12/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T177015									
Bromomethane	<5.0 ug/L	5.0	1	12/09/25	sb	12/10/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	133 %	68-133	1	12/09/25	sb	12/10/25	sb		
Toluene-d8	85 %	75-120	1	12/09/25	sb	12/10/25	sb		
4-Bromofluorobenzene	104 %	69-119	1	12/09/25	sb	12/10/25	sb		
1,2-Dichlorobenzene-d4	109 %	72-127	1	12/09/25	sb	12/10/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25L0525
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0525-01 Matrix: Wastewater Date Collected: 12/08/25 13:40
Sample ID: CRE Date Received: 12/09/25 10:24

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25L0525
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0525-02 | Matrix: Wastewater | Date Collected: 12/08/25 13:20
Sample ID: Borg Warner | Date Received: 12/09/25 10:24

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T177000									
Beryllium	<0.0040 mg/L	0.0040	1	12/12/25	ejb	12/13/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T177029									
Phenol	<5.0 ug/L	5.0	1	12/10/25	kbc	12/10/25	jlh		
Surrogates:									
2-Fluorophenol	25 %	20-53	1	12/10/25	kbc	12/10/25	jlh		
Phenol-d5	16 %	11-40	1	12/10/25	kbc	12/10/25	jlh		
Nitrobenzene-d5	74 %	36-103	1	12/10/25	kbc	12/10/25	jlh		
2-Fluorobiphenyl	64 %	36-119	1	12/10/25	kbc	12/10/25	jlh	N	
2,4,6-Tribromophenol	54 %	30-105	1	12/10/25	kbc	12/10/25	jlh	N	
Terphenyl-d14	76 %	37-109	1	12/10/25	kbc	12/10/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T177123									
Benzidine	<0.20 ug/L	0.20	2	12/12/25	als	12/12/25	jlh		
3,3'-Dichlorobenzidine	<2.0 ug/L	2.0	2	12/12/25	als	12/12/25	jlh		
Surrogates:									
Nitrobenzene-d5	77 %	38-130	2	12/12/25	als	12/12/25	jlh		
2-Fluorobiphenyl	64 %	39-118	2	12/12/25	als	12/12/25	jlh	N	
Terphenyl-d14	38 %	14-96	2	12/12/25	als	12/12/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T177015									
Bromomethane	<5.0 ug/L	5.0	1	12/09/25	sb	12/10/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	132 %	68-133	1	12/09/25	sb	12/10/25	sb		
Toluene-d8	84 %	75-120	1	12/09/25	sb	12/10/25	sb		
4-Bromofluorobenzene	101 %	69-119	1	12/09/25	sb	12/10/25	sb		
1,2-Dichlorobenzene-d4	111 %	72-127	1	12/09/25	sb	12/10/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25L0525
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0525-02 | Matrix: Wastewater | Date Collected: 12/08/25 13:20
Sample ID: Borg Warner | Date Received: 12/09/25 10:24

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

CERTIFICATE OF ANALYSIS

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QUALITY CONTROL RESULTS

Trace Project ID: 25L0525
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177000	Analysis Description: Beryllium, Total
QC Batch Method: EPA 200.2	Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T177000-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.0040	0.0040	

LABORATORY CONTROL SAMPLE: T177000-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.177	89	85-115	

Trace Project ID: 25L0525
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177029	Analysis Description: TTO Semi-Volatiles
QC Batch Method: EPA 625 Extraction for BNAs	Analysis Method: EPA 625.1

METHOD BLANK: T177029-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	%	29	20-53	
Phenol-d5 (S)	%	18	11-40	
Nitrobenzene-d5 (S)	%	75	36-103	
2-Fluorobiphenyl (S)	%	63	36-119	
2,4,6-Tribromophenol (S)	%	61	30-105	
Terphenyl-d14 (S)	%	82	37-109	

LABORATORY CONTROL SAMPLE: T177029-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	18.2	18	17-120	
2-Fluorophenol (S)	%	100	29.0	29	20-53	
Phenol-d5 (S)	%	100	23.4	23	11-40	
Nitrobenzene-d5 (S)	%	100	79.3	79	36-103	
2-Fluorobiphenyl (S)	%	100	66.0	66	36-119	
2,4,6-Tribromophenol (S)	%	100	65.4	65	30-105	
Terphenyl-d14 (S)	%	100	73.1	73	37-109	

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MATRIX SPIKE: T177029-MS1 Original: **25L0525-02**

Parameter	Units	Original Result	Spike Conc.	MS Result	MS % Rec	% Rec Unit	Notes
Phenol	ug/L	0	101	30.2	30	5-120	
2-Fluorophenol (S)	%		101	40.7	40	20-53	
Phenol-d5 (S)	%		101	35.7	35	11-40	
Nitrobenzene-d5 (S)	%		101	73.5	73	36-103	
2-Fluorobiphenyl (S)	%		101	59.2	58	36-119	
2,4,6-Tribromophenol (S)	%		101	57.9	57	30-105	
Terphenyl-d14 (S)	%		101	63.5	63	37-109	

Trace Project ID: 25L0525
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177123	Analysis Description: 625 Benzidines SIM
QC Batch Method: GC/MS No Prep	Analysis Method: EPA 625.1 SIM

METHOD BLANK: T177123-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	66	38-130	
2-Fluorobiphenyl (S)	%	57	39-118	
Terphenyl-d14 (S)	%	34	14-96	

LABORATORY CONTROL SAMPLE: T177123-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	1.22	61	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	<2.0	96	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.783	78	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.707	71	39-118	
Terphenyl-d14 (S)	%	1.00	0.423	42	14-96	

Trace Project ID: 25L0525
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177015	Analysis Description: TTO Volatiles
QC Batch Method: EPA 624 Purge and Trap	Analysis Method: EPA 624.1

METHOD BLANK: T177015-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes

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METHOD BLANK: T177015-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	130	68-133	
Toluene-d8 (S)	%	85	75-120	
4-Bromofluorobenzene (S)	%	104	69-119	
1,2-Dichlorobenzene-d4 (S)	%	108	72-127	

LABORATORY CONTROL SAMPLE: T177015-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	42.8	86	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	35.8	119	68-133	
Toluene-d8 (S)	%	30.0	26.5	88	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.1	97	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	29.6	99	72-127	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

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CHAIN-OF-CUSTODY RECORD

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Page 1 of 1

Trace ID No. 25L0525

Report Results To:		Bill To:	
Company Name:	City of Cadillac	PO #:	
Report To:	Cindy Tomaszewski	Contact Name:	Cindy Tomaszewski
Mailing Address:	1121 Plett Rd	Billing Address (if different):	200 N Lake St
City, State, Zip Code:	Cadillac MI 49601	City, State, Zip Code:	Cadillac MI 49601
Office Phone:	231-775-2368; Cell Phone: 231-878-6230	Phone Number:	
Email Address:	lab@cadillac-mi.net; pmiller@cadillac-mi.net	Billing Email Address:	lab@cadillac-mi.net

Trace Use:
Logged By: CN
Checked By: MJ
Sample Collection Time (Hrs):

Requested Turnaround Times (TAT)
- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:
WW = Wastewater; AQ = Aqueous; A = Air
DW = Drinking Water; O = Oil; WI = Wipes
GW = Groundwater; S = Solid; U = Unknown
LW = Liquid Waste; SL = Sludge

Project Name: 2025 IPP/MAHL Additional
Sampled By (print): Derek Goodrich

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Composite	Grab	Metals Field Filterd (Y or N)	Matrix - see above →	Number of Containers	Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes	Possible Health Hazards?
1	12-8	140	CRE		✓	N	WW	6	x		x		x		x		X	X	X	X		
2	12-8	120	Borg Warner		✓	N	WW	6	x		x		x		x		X	X	X	X		

Please Sign	Released By	Received By	Date	Time	Released By	Received By	Date	Time
	CTomaszewski	UPS Ground	12-8-25	1530	2) UPS	[illegible]	12/9/25	10:24
	3)				4)			

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

CERTIFICATE OF ANALYSIS

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25L0525
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

	Original Observation	Corrected Temperature	IR-9 (CF: +0.2°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.6°C)	IR-14 (CF: -1.3°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Date: 12/9/25										
Time: 10:07										
Initials: CN										
Package Description: Cooler										
Package Temp °C	0.0	-0.6			x					
Representative Sample Temp °C	2.4	-1.8			x					x

Sample Receipt

Yes No
☒ ☐ Received on ice or other coolant
☒ ☐ Ice still present upon receipt
☐ ☒ Custody seals present
☐ Yes ☐ No Custody seals intact (if applicable)
☐ Trace Courier ☐ Client Drop-off ☒ UPS ☐ Fed Ex ☐ US Mail ☐ Other

Sample Condition

Yes No N/A
☒ ☐ ☐ All sample containers arrived unbroken and labeled
☒ ☐ ☐ Sufficient sample to run requested analyses
☒ ☐ ☐ Correct chemical preservative added to samples
☐ ☐ ☒ Samples preserved at Trace (if yes, fill in preservation information below)
Preservation - Reagent:____________ LIMS ID:______________ Date/Time:____________________
☒ ☐ ☐ Chemical preservation verified, check EMD pH test strip used (if applicable)
☒ pH 0-2.5 (Lot: HC461264) ☐ pH 11.0-13.0 (Lot: HC022540) ☐ Other
☒ ☐ ☐ Air bubbles absent from VOAs (no bubble greater than 6mm) ____________________

Chain of Custody (COC)

Yes No
☒ ☐ All bottle labels agree with COC
☒ ☐ COC filled out properly
☒ ☐ COC signed by client

Notes:

Form 70-A.61
Effective 12/1/25

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December 23, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25L0845
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 25L0845
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25L0845-01 Sample type: Grab	Hutchinson	Wastewater	DG	12/10/25 13:20	12/11/25 11:00
25L0845-02 Sample type: Grab	CCI	Wastewater	DG	12/10/25 13:30	12/11/25 11:00

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ANALYTICAL RESULTS

Trace Project ID: 25L0845
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0845-01 | Matrix: Wastewater | Date Collected: 12/10/25 13:20
Sample ID: Hutchinson | Date Received: 12/11/25 11:00

PARAMETERS		RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL										
Analysis Method: EPA 200.8 Rev. 5.4										
Batch: T177271										
Beryllium		<0.0040 mg/L	0.0040	1	12/17/25	ejb	12/17/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 625.1										
Batch: T177307										
Phenol		**36 ug/L**	**5.0**	**1**	**12/17/25**	**kbc**	**12/19/25**	**jlh**		
Surrogates:										
2-Fluorophenol	*	**65** %	20-53	1	12/17/25	kbc	12/19/25	jlh	S01	
Phenol-d5	*	**46** %	11-40	1	12/17/25	kbc	12/19/25	jlh	S01	
Nitrobenzene-d5		89 %	36-103	1	12/17/25	kbc	12/19/25	jlh		
2-Fluorobiphenyl		72 %	36-119	1	12/17/25	kbc	12/19/25	jlh	N	
2,4,6-Tribromophenol		84 %	30-105	1	12/17/25	kbc	12/19/25	jlh	N	
Terphenyl-d14		78 %	37-109	1	12/17/25	kbc	12/19/25	jlh		
Analysis Method: EPA 625.1 SIM										
Batch: T177123										
Benzidine		<0.21 ug/L	0.21	2	12/12/25	als	12/12/25	jlh		
3,3'-Dichlorobenzidine		<2.1 ug/L	2.1	2	12/12/25	als	12/12/25	jlh		
Surrogates:										
Nitrobenzene-d5		110 %	38-130	2	12/12/25	als	12/12/25	jlh		
2-Fluorobiphenyl		55 %	39-118	2	12/12/25	als	12/12/25	jlh	N	
Terphenyl-d14		31 %	14-96	2	12/12/25	als	12/12/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 624.1										
Batch: T177168										
Bromomethane		<5.0 ug/L	5.0	1	12/12/25	ats	12/14/25	ats		
Surrogates:										
1,2-Dichloroethane-d4		94 %	68-133	1	12/12/25	ats	12/14/25	ats		
Toluene-d8		106 %	75-120	1	12/12/25	ats	12/14/25	ats		
4-Bromofluorobenzene		98 %	69-119	1	12/12/25	ats	12/14/25	ats		
1,2-Dichlorobenzene-d4		101 %	72-127	1	12/12/25	ats	12/14/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25L0845
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0845-01 Matrix: Wastewater Date Collected: 12/10/25 13:20
Sample ID: Hutchinson Date Received: 12/11/25 11:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25L0845
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0845-02 Matrix: Wastewater Date Collected: 12/10/25 13:30
Sample ID: CCI Date Received: 12/11/25 11:00

PARAMETERS		RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL										
Analysis Method: EPA 200.8 Rev. 5.4										
Batch: T177271										
Beryllium		<0.0040 mg/L	0.0040	1	12/17/25	ejb	12/17/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 625.1										
Batch: T177307										
Phenol		**6.1 ug/L**	**5.0**	**1**	**12/17/25**	**kbc**	**12/19/25**	**jlh**		
Surrogates:										
2-Fluorophenol	*	**68** %	20-53	1	12/17/25	kbc	12/19/25	jlh	S01	
Phenol-d5	*	**49** %	11-40	1	12/17/25	kbc	12/19/25	jlh	S01	
Nitrobenzene-d5		92 %	36-103	1	12/17/25	kbc	12/19/25	jlh		
2-Fluorobiphenyl		69 %	36-119	1	12/17/25	kbc	12/19/25	jlh	N	
2,4,6-Tribromophenol		86 %	30-105	1	12/17/25	kbc	12/19/25	jlh	N	
Terphenyl-d14		74 %	37-109	1	12/17/25	kbc	12/19/25	jlh		
Analysis Method: EPA 625.1 SIM										
Batch: T177123										
Benzidine		<0.22 ug/L	0.22	2	12/12/25	als	12/12/25	jlh		
3,3'-Dichlorobenzidine		<2.2 ug/L	2.2	2	12/12/25	als	12/12/25	jlh		
Surrogates:										
Nitrobenzene-d5	*	**568** %	38-130	2	12/12/25	als	12/12/25	jlh	S10	
2-Fluorobiphenyl		54 %	39-118	2	12/12/25	als	12/12/25	jlh	N	
Terphenyl-d14		32 %	14-96	2	12/12/25	als	12/12/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 624.1										
Batch: T177168										
Bromomethane		<5.0 ug/L	5.0	1	12/12/25	ats	12/14/25	ats		
Surrogates:										
1,2-Dichloroethane-d4		94 %	68-133	1	12/12/25	ats	12/14/25	ats		
Toluene-d8		108 %	75-120	1	12/12/25	ats	12/14/25	ats		
4-Bromofluorobenzene		98 %	69-119	1	12/12/25	ats	12/14/25	ats		
1,2-Dichlorobenzene-d4		102 %	72-127	1	12/12/25	ats	12/14/25	ats		

CERTIFICATE OF ANALYSIS

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ANALYTICAL RESULTS

Trace Project ID: 25L0845
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0845-02
Sample ID: CCI
Matrix: Wastewater
Date Collected: 12/10/25 13:30
Date Received: 12/11/25 11:00

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
VOLATILE ORGANIC COMPOUNDS BY GC-MS										

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QUALITY CONTROL RESULTS

Trace Project ID: 25L0845
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177271	Analysis Description: Beryllium, Total
QC Batch Method: EPA 200.2	Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T177271-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.00025	0.00025	

LABORATORY CONTROL SAMPLE: T177271-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.187	94	85-115	

Trace Project ID: 25L0845
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177144	Analysis Description: TTO Semi-Volatiles
QC Batch Method: EPA 625 Extraction for BNAs	Analysis Method: EPA 625.1

METHOD BLANK: T177144-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	%	21	20-53	
Phenol-d5 (S)	%	12	11-40	
Nitrobenzene-d5 (S)	%	40	36-103	
2-Fluorobiphenyl (S)	%	52	36-119	
2,4,6-Tribromophenol (S)	%	83	30-105	
Terphenyl-d14 (S)	%	61	37-109	

LABORATORY CONTROL SAMPLE: T177144-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	**ug/L**	**100**	**13.8**	**14**	**17-120**	**BS01**
2-Fluorophenol (S)	%	100	24.1	24	20-53	
Phenol-d5 (S)	%	100	15.9	16	11-40	
Nitrobenzene-d5 (S)	%	100	48.6	49	36-103	
2-Fluorobiphenyl (S)	%	100	63.5	64	36-119	
2,4,6-Tribromophenol (S)	%	100	93.9	94	30-105	
Terphenyl-d14 (S)	%	100	75.2	75	37-109	

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Trace Project ID: 25L0845
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177307	Analysis Description: TTO Semi-Volatiles
QC Batch Method: EPA 625 Extraction for BNAs	Analysis Method: EPA 625.1

METHOD BLANK: T177307-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	**%**	**54**	**20-53**	**S01**
Phenol-d5 (S)	%	32	11-40	
Nitrobenzene-d5 (S)	%	102	36-103	
2-Fluorobiphenyl (S)	%	87	36-119	
2,4,6-Tribromophenol (S)	**%**	**110**	**30-105**	**S01**
Terphenyl-d14 (S)	**%**	**112**	**37-109**	**S10**

LABORATORY CONTROL SAMPLE: T177307-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	27.5	27	17-120	
2-Fluorophenol (S)	%	100	46.8	47	20-53	
Phenol-d5 (S)	%	100	27.6	28	11-40	
Nitrobenzene-d5 (S)	%	100	92.1	92	36-103	
2-Fluorobiphenyl (S)	%	100	73.8	74	36-119	
2,4,6-Tribromophenol (S)	%	100	83.9	84	30-105	
Terphenyl-d14 (S)	%	100	84.2	84	37-109	

Trace Project ID: 25L0845
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177123	Analysis Description: 625 Benzidines SIM
QC Batch Method: GC/MS No Prep	Analysis Method: EPA 625.1 SIM

METHOD BLANK: T177123-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	66	38-130	
2-Fluorobiphenyl (S)	%	57	39-118	
Terphenyl-d14 (S)	%	34	14-96	

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LABORATORY CONTROL SAMPLE: T177123-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	1.22	61	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	<2.0	96	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.783	78	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.707	71	39-118	
Terphenyl-d14 (S)	%	1.00	0.423	42	14-96	

Trace Project ID: 25L0845
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177168
QC Batch Method: EPA 624 Purge and Trap
Analysis Description: TTO Volatiles
Analysis Method: EPA 624.1

METHOD BLANK: T177168-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	94	68-133	
Toluene-d8 (S)	%	106	75-120	
4-Bromofluorobenzene (S)	%	99	69-119	
1,2-Dichlorobenzene-d4 (S)	%	102	72-127	

LABORATORY CONTROL SAMPLE: T177168-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	41.6	83	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	28.3	94	68-133	
Toluene-d8 (S)	%	30.0	31.8	106	75-120	
4-Bromofluorobenzene (S)	%	30.0	28.8	96	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	30.6	102	72-127	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25L0845-01RE1

Analysis: EPA 625.1

2-Fluorophenol	Note S01 : The surrogate recovery was out of control high when compared to control limits. The results must be considered estimated.
Phenol-d5	Note S01 : The surrogate recovery was out of control high when compared to control limits. The results must be considered estimated.

Trace ID: 25L0845-02

Analysis: EPA 625.1 SIM

Nitrobenzene-d5	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Trace ID: 25L0845-02RE1

Analysis: EPA 625.1

2-Fluorophenol	Note S01 : The surrogate recovery was out of control high when compared to control limits. The results must be considered estimated.
Phenol-d5	Note S01 : The surrogate recovery was out of control high when compared to control limits. The results must be considered estimated.

Trace ID: T177144-BS1

Analysis: EPA 625.1

Phenol	Note BS01 : The BS recovery was out of control low. The result and reporting limit for this analyte, in this quality control batch, must be considered estimated.

Trace ID: T177307-BLK1

Analysis: EPA 625.1

2,4,6-Tribromophenol	Note S01 : The surrogate recovery was out of control high when compared to control limits. The results must be considered estimated.
2-Fluorophenol	Note S01 : The surrogate recovery was out of control high when compared to control limits. The results must be considered estimated.
Terphenyl-d14	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

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CHAIN-OF-CUSTODY RECORD

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Page 1 of 1

Trace ID No. 25L0845

Report Results To:

Company Name: City of Cadillac	
Report To: Cindy Tomaszewski	
Mailing Address: 1121 Plett Rd	
City, State, Zip Code: Cadillac MI 49601	
Office Phone: 231-775-2368	Cell Phone: 231-878-6230
Email Address: lab@cadillac-mi.net; pmiller@cadillac-mi.net	

Bill To:

PO #:
Contact Name: Cindy Tomaszewski
Billing Address (if different): 200 N Lake St
City, State, Zip Code: Cadillac MI 49601
Phone Number:
Billing Email Address: lab@cadillac-mi.net

Trace Use:

Logged By: MC
Checked By: MJ
Sample Collection Time (Hrs):

Requested Turnaround Times (TAT)

- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:

WW = Wastewater	AQ = Aqueous	A = Air
DW = Drinking Water	O = Oil	WI = Wipes
GW = Groundwater	S = Solid	U = Unknown
LW = Liquid Waste	SL = Sludge	

Project Name: 2025 IPP/MAHL Additional

Sampled By (print): Derek Goodwin

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Sample Type: Composite	Sample Type: Grab	Metals Field Filterd (Y or N)	Matrix - see above →	Number of Containers	Preservation: Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes	Possible Health Hazards?
1	12-11	1:20	Hutchinson		✓	N	WW	6	x		x		x		x		X	X	X	X		
2	12-11	1:30	CCI		✓	N	WW	6	x		x		x		x		X	X	X	X		

Please Sign

	Released By	Received By	Date	Time		Released By	Received By	Date	Time
1)	[illegible signature]	[illegible signature]	12/11/25	1100	2)				
3)					4)				

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

CERTIFICATE OF ANALYSIS

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231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

25L0845
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

Date: 12/11/25	Original Observation	Corrected Temperature	IR-9 (CF: +0.2°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.6°C)	IR-14 (CF: -1.3°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Time: 1200										
Initials: Meg										
Package Description: cooler										
Package Temp °C	-1.3	-2.6				✓				
Representative Sample Temp °C	4.8	3.5				✓				✓

Sample Receipt

Yes	No	
☑	☐	Received on ice or other coolant
☑	☐	Ice still present upon receipt
☐	☑	Custody seals present

☐ Yes ☐ No Custody seals intact (if applicable)

☐ Trace Courier ☐ Client Drop-off ☑ UPS ☐ Fed Ex ☐ US Mail ☐ Other

Sample Condition

Yes	No	N/A	
☑	☐	☐	All sample containers arrived unbroken and labeled
☑	☐	☐	Sufficient sample to run requested analyses
☑	☐	☐	Correct chemical preservative added to samples
☐	☑	☐	Samples preserved at Trace (if yes, fill in preservation information below)

Preservation - Reagent:__________ LIMS ID:__________ Date/Time:__________

Yes	No	N/A	
☑	☐	☐	Chemical preservation verified, check EMD pH test strip used (if applicable)

☑ pH 0-2.5 (Lot: HC461264) ☐ pH 11.0-13.0 (Lot: HC022540) ☐ Other

Yes	No	N/A	
☑	☐	☐	Air bubbles absent from VOAs (no bubble greater than 6mm) __________

Chain of Custody (COC)

Yes	No	
☑	☐	All bottle labels agree with COC
☑	☐	COC filled out properly
☑	☐	COC signed by client

Notes:

Form 70-A.61
Effective 12/1/25

TRACE Analytical Laboratories, Inc.

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December 16, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE:	Trace Project	25L0449
	Client Project	2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

ACCREDITED LABORATORY TNI ENVIRONMENTAL STANDARD

PJLA Testing Accreditation# 1031

TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 25L0449
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25L0449-01	AAR FAC	Wastewater	DG	12/05/25 13:10	12/05/25 13:50
Sample type: Grab					
25L0449-02	AAR RT	Wastewater	DG	12/05/25 13:00	12/05/25 13:50
Sample type: Grab					

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ANALYTICAL RESULTS

Trace Project ID: 25L0449
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0449-01 Matrix: Wastewater Date Collected: 12/05/25 13:10
Sample ID: AAR FAC Date Received: 12/05/25 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T177054									
Beryllium	<0.0040 mg/L	0.0040	1	12/11/25	ejb	12/12/25	jma		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS								DL03	
Analysis Method: EPA 625.1									
Batch: T176922									
Phenol	<5.1 ug/L	5.1	2	12/08/25	kbc	12/09/25	jlh		
Surrogates:									
2-Fluorophenol	25 %	20-53	2	12/08/25	kbc	12/09/25	jlh		
Phenol-d5	16 %	11-40	2	12/08/25	kbc	12/09/25	jlh		
Nitrobenzene-d5	73 %	36-103	2	12/08/25	kbc	12/09/25	jlh		
2-Fluorobiphenyl	57 %	36-119	2	12/08/25	kbc	12/09/25	jlh	N	
2,4,6-Tribromophenol	60 %	30-105	2	12/08/25	kbc	12/09/25	jlh	N	
Terphenyl-d14	47 %	37-109	2	12/08/25	kbc	12/09/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176932									
Benzidine	**0.60 ug/L**	**0.11**	**1**	**12/08/25**	**als**	**12/10/25**	**jlh**		
3,3'-Dichlorobenzidine	<1.1 ug/L	1.1	1	12/08/25	als	12/10/25	jlh		
Surrogates:									
Nitrobenzene-d5	69 %	38-130	1	12/08/25	als	12/10/25	jlh		
2-Fluorobiphenyl	63 %	39-118	1	12/08/25	als	12/10/25	jlh	N	
Terphenyl-d14	36 %	14-96	1	12/08/25	als	12/10/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176893									
Bromomethane	<5.0 ug/L	5.0	1	12/05/25	ats	12/06/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	97 %	68-133	1	12/05/25	ats	12/06/25	ats		
Toluene-d8	104 %	75-120	1	12/05/25	ats	12/06/25	ats		
4-Bromofluorobenzene	98 %	69-119	1	12/05/25	ats	12/06/25	ats		
1,2-Dichlorobenzene-d4	101 %	72-127	1	12/05/25	ats	12/06/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25L0449
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0449-01 Matrix: Wastewater Date Collected: 12/05/25 13:10
Sample ID: AAR FAC Date Received: 12/05/25 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25L0449
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0449-02 | Matrix: Wastewater | Date Collected: 12/05/25 13:00
Sample ID: AAR RT | Date Received: 12/05/25 13:50

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T177054									
Beryllium	<0.0040 mg/L	0.0040	1	12/11/25	ejb	12/12/25	jma		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T176922									
Phenol	<5.0 ug/L	5.0	1	12/08/25	kbc	12/09/25	jlh		
Surrogates:									
2-Fluorophenol	31 %	20-53	1	12/08/25	kbc	12/09/25	jlh		
Phenol-d5	18 %	11-40	1	12/08/25	kbc	12/09/25	jlh		
Nitrobenzene-d5	92 %	36-103	1	12/08/25	kbc	12/09/25	jlh		
2-Fluorobiphenyl	78 %	36-119	1	12/08/25	kbc	12/09/25	jlh	N	
2,4,6-Tribromophenol	76 %	30-105	1	12/08/25	kbc	12/09/25	jlh	N	
Terphenyl-d14	88 %	37-109	1	12/08/25	kbc	12/09/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T176932									
Benzidine	<0.10 ug/L	0.10	1	12/08/25	als	12/10/25	jlh		
3,3'-Dichlorobenzidine	<1.0 ug/L	1.0	1	12/08/25	als	12/10/25	jlh		
Surrogates:									
Nitrobenzene-d5	81 %	38-130	1	12/08/25	als	12/10/25	jlh		
2-Fluorobiphenyl	67 %	39-118	1	12/08/25	als	12/10/25	jlh	N	
Terphenyl-d14	41 %	14-96	1	12/08/25	als	12/10/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T176893									
Bromomethane	<5.0 ug/L	5.0	1	12/05/25	ats	12/06/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	97 %	68-133	1	12/05/25	ats	12/06/25	ats		
Toluene-d8	104 %	75-120	1	12/05/25	ats	12/06/25	ats		
4-Bromofluorobenzene	98 %	69-119	1	12/05/25	ats	12/06/25	ats		
1,2-Dichlorobenzene-d4	100 %	72-127	1	12/05/25	ats	12/06/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25L0449
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0449-02 Matrix: Wastewater Date Collected: 12/05/25 13:00
Sample ID: AAR RT Date Received: 12/05/25 13:50

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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QUALITY CONTROL RESULTS

Trace Project ID: 25L0449
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177054	Analysis Description: Beryllium, Total
QC Batch Method: EPA 200.2	Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T177054-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.00025	0.00025	

LABORATORY CONTROL SAMPLE: T177054-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.218	109	85-115	

Trace Project ID: 25L0449
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176922	Analysis Description: TTO Semi-Volatiles
QC Batch Method: EPA 625 Extraction for BNAs	Analysis Method: EPA 625.1

METHOD BLANK: T176922-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	%	26	20-53	
Phenol-d5 (S)	%	15	11-40	
Nitrobenzene-d5 (S)	%	85	36-103	
2-Fluorobiphenyl (S)	%	72	36-119	
2,4,6-Tribromophenol (S)	%	85	30-105	
Terphenyl-d14 (S)	**%**	**120**	**37-109**	**S10**

LABORATORY CONTROL SAMPLE: T176922-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	19.2	19	17-120	
2-Fluorophenol (S)	%	100	30.6	31	20-53	
Phenol-d5 (S)	%	100	26.8	27	11-40	
Nitrobenzene-d5 (S)	%	100	99.2	99	36-103	
2-Fluorobiphenyl (S)	%	100	85.7	86	36-119	
2,4,6-Tribromophenol (S)	%	100	88.8	89	30-105	
Terphenyl-d14 (S)	%	100	94.4	94	37-109	

MATRIX SPIKE: T176922-MS1 Original: **25L0449-02**

Parameter	Units	Original Result	Spike Conc.	MS Result	MS % Rec	% Rec Unit	Notes
Phenol	ug/L	0	100	21.0	21	5-120	
2-Fluorophenol (S)	%		100	30.9	31	20-53	
Phenol-d5 (S)	%		100	27.0	27	11-40	
Nitrobenzene-d5 (S)	%		100	99.7	100	36-103	
2-Fluorobiphenyl (S)	%		100	86.6	86	36-119	
2,4,6-Tribromophenol (S)	%		100	97.7	98	30-105	
Terphenyl-d14 (S)	%		100	104	104	37-109	

Trace Project ID: 25L0449
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176932
QC Batch Method: GC/MS No Prep
Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T176932-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	53	38-130	
2-Fluorobiphenyl (S)	%	58	39-118	
Terphenyl-d14 (S)	%	38	14-96	

LABORATORY CONTROL SAMPLE: T176932-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	1.02	51	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	<2.0	79	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.729	73	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.631	63	39-118	
Terphenyl-d14 (S)	%	1.00	0.390	39	14-96	

Trace Project ID: 25L0449
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T176893
QC Batch Method: EPA 624 Purge and Trap
Analysis Description: TTO Volatiles
Analysis Method: EPA 624.1

METHOD BLANK: T176893-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes

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Muskegon, MI 49444-2673

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METHOD BLANK: T176893-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	**%**	**99**	**68-133**	**M-01**
Toluene-d8 (S)	%	102	75-120	
4-Bromofluorobenzene (S)	%	98	69-119	
1,2-Dichlorobenzene-d4 (S)	%	100	72-127	

LABORATORY CONTROL SAMPLE: T176893-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	43.6	87	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	29.8	99	68-133	
Toluene-d8 (S)	%	30.0	31.0	104	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.8	99	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	30.2	101	72-127	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25L0449-01

Analysis: EPA 625.1

Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Trace ID: T176893-BLK1

Analysis: EPA 624.1

1,2-Dichloroethane-d4 — Note M-01 : An incorrect peak was integrated by the computer software. The analyst identified the correct peak and integrated it manually.

Trace ID: T176922-BLK1

Analysis: EPA 625.1

Terphenyl-d14 — Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

CHAIN-OF-CUSTODY RECORD

TRACE ANALYTICAL LABORATORIES, INC.

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Page 1 of 1

Trace ID No. 25L0449

Report Results To:

Company Name: City of Cadillac
Report To: Cindy Tomaszewski
Mailing Address: 1121 Plett Rd
City, State, Zip Code: Cadillac MI 49601
Office Phone: 231-775-2368 Cell Phone: 231-878-6230
Email Address: lab@cadillac-mi.net; pmiller@cadillac-mi.net

Bill To:

PO #:
Contact Name: Cindy Tomaszewski
Billing Address (if different): 200 N Lake St
City, State, Zip Code: Cadillac MI 49601
Phone Number:
Billing Email Address: lab@cadillac-mi.net

Trace Use:

Logged By: MS
Checked By: NC
Sample Collection Time (Hrs):

Requested Turnaround Times (TAT)

- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:

WW = Wastewater, DW = Drinking Water, GW = Groundwater, LW = Liquid Waste, AQ = Aqueous, O = Oil, S = Solid, SL = Sludge, A = Air, WI = Wipes, U = Unknown

Project Name: 2025 IPP/MAHL Additional
Sampled By (print): Derek G

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Composite	Grab	Metals Field Filterd (Y or N)	Matrix - see above →	Number of Containers	Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes	Possible Health Hazards?
1	12-5	110	AAR FAC		✓	N	WW	6	x		x		x		x		X	X	X	X		
2	12-5	100	AAR RT		✓	N	WW	6	x		x		x		x		X	X	X	X		

Please Sign	Released By	Received By	Date	Time		Released By	Received By	Date	Time
1)	[illegible signature]	[illegible signature]	12-5-25	13:35	2)	[illegible signature]	[illegible signature]	12/5/25	1600
3)					4)				

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673

TRACE
ANALYTICAL LABORATORIES, INC.

231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

25L0449
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

	Original Observation	Corrected Temperature	IR-9 (CF: +0.2°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.6°C)	IR-14 (CF: -1.3°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Date: 12/5/25										
Time: 1600										
Initials: MJ										
Package Description: Cooler										
Package Temp °C	0.0	0.0		/						
Representative Sample Temp °C	2.6	2.6		/						/

Sample Receipt

Yes	No	
[x]	[ ]	Received on ice or other coolant
[x]	[ ]	Ice still present upon receipt
[ ]	[x]	Custody seals present

[ ] Yes [ ] No Custody seals intact (if applicable)

[x] Trace Courier [ ] Client Drop-off [ ] UPS [ ] Fed Ex [ ] US Mail [ ] Other

Sample Condition

Yes	No	N/A	
[x]	[ ]	[ ]	All sample containers arrived unbroken and labeled
[x]	[ ]	[ ]	Sufficient sample to run requested analyses
[x]	[ ]	[ ]	Correct chemical preservative added to samples
[ ]	[x]	[ ]	Samples preserved at Trace (if yes, fill in preservation information below)

Preservation - Reagent:____________ LIMS ID:______________ Date/Time:______________________

Yes	No	N/A	
[x]	[ ]	[ ]	Chemical preservation verified, check EMD pH test strip used (if applicable)

[x] pH 0-2.5 (Lot: HC461264) [ ] pH 11.0-13.0 (Lot: HC022540) [ ] Other

Yes	No	N/A	
[x]	[ ]	[ ]	Air bubbles absent from VOAs (no bubble greater than 6mm) ____________________

Chain of Custody (COC)

Yes	No	
[x]	[ ]	All bottle labels agree with COC
[x]	[ ]	COC filled out properly
[x]	[ ]	COC signed by client

Notes:

Form 70-A.61
Effective 12/1/25

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December 23, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25L1024
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

PJLA Testing Accreditation# 1031

TNI EL V1:2016

Trace Analytical Laboratories, Inc.
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SAMPLE SUMMARY

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25L1024-01	Domestic #1	Wastewater	DG	12/14/25	12/15/25 13:00
Sample type: Composite					
25L1024-02	Domestic #2	Wastewater	DG	12/14/25	12/15/25 13:00
Sample type: Composite					
25L1024-03	WWTP Raw Influent	Wastewater	DG	12/14/25	12/15/25 13:00
Sample type: Composite					
25L1024-04	WWTP Primary Effluent	Wastewater	DG	12/14/25	12/15/25 13:00
Sample type: Composite					
25L1024-05	WWTP Final Effluent	Wastewater	DG	12/14/25	12/15/25 13:00
Sample type: Composite					

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ANALYTICAL RESULTS

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1024-01 | Matrix: Wastewater | Date Collected: 12/14/25
Sample ID: Domestic #1 | Date Received: 12/15/25 13:00

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL										
Analysis Method: EPA 200.8 Rev. 5.4										
Batch: T177324										
Beryllium	<0.0040	mg/L	0.0040	1	12/18/25	ejb	12/23/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 625.1										
Batch: T177256										
Phenol	<10	ug/L	10	4	12/16/25	kbc	12/19/25	jlh	DL03	
Surrogates:										
2-Fluorophenol	45	%	20-53	4	12/16/25	kbc	12/19/25	jlh		
Phenol-d5	28	%	11-40	4	12/16/25	kbc	12/19/25	jlh		
Nitrobenzene-d5	* **114**	%	36-103	4	12/16/25	kbc	12/19/25	jlh	S10	
2-Fluorobiphenyl	75	%	36-119	4	12/16/25	kbc	12/19/25	jlh	N	
2,4,6-Tribromophenol	96	%	30-105	4	12/16/25	kbc	12/19/25	jlh	N	
Terphenyl-d14	76	%	37-109	4	12/16/25	kbc	12/19/25	jlh		
Analysis Method: EPA 625.1 SIM										
Batch: T177269										
Benzidine	<0.22	ug/L	0.22	2	12/16/25	als	12/18/25	jlh		
3,3'-Dichlorobenzidine	<2.2	ug/L	2.2	2	12/16/25	als	12/18/25	jlh		
Surrogates:										
Nitrobenzene-d5	* **207**	%	38-130	2	12/16/25	als	12/18/25	jlh	S10	
2-Fluorobiphenyl	39	%	39-118	2	12/16/25	als	12/18/25	jlh	N	
Terphenyl-d14	31	%	14-96	2	12/16/25	als	12/18/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 624.1										
Batch: T177285										
Bromomethane	<5.0	ug/L	5.0	1	12/16/25	ats	12/16/25	ats		
Surrogates:										
1,2-Dichloroethane-d4	88	%	68-133	1	12/16/25	ats	12/16/25	ats		
Toluene-d8	113	%	75-120	1	12/16/25	ats	12/16/25	ats		
4-Bromofluorobenzene	99	%	69-119	1	12/16/25	ats	12/16/25	ats		
1,2-Dichlorobenzene-d4	101	%	72-127	1	12/16/25	ats	12/16/25	ats		

CERTIFICATE OF ANALYSIS

Trace Analytical Laboratories, Inc.
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ANALYTICAL RESULTS

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1024-01 Matrix: Wastewater Date Collected: 12/14/25
Sample ID: Domestic #1 Date Received: 12/15/25 13:00

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1024-02 Matrix: Wastewater Date Collected: 12/14/25
Sample ID: Domestic #2 Date Received: 12/15/25 13:00

PARAMETERS		RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL											
Analysis Method: EPA 200.8 Rev. 5.4											
Batch: T177324											
Beryllium		<0.0040	mg/L	0.0040	1	12/18/25	ejb	12/23/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 625.1											
Batch: T177256											
Phenol		**14**	**ug/L**	**9.9**	**4**	**12/16/25**	**kbc**	**12/22/25**	**jlh**		
Surrogates:											
2-Fluorophenol		39	%	20-53	4	12/16/25	kbc	12/22/25	jlh		
Phenol-d5		24	%	11-40	4	12/16/25	kbc	12/22/25	jlh		
Nitrobenzene-d5	*	**112**	%	36-103	4	12/16/25	kbc	12/22/25	jlh	S10	
2-Fluorobiphenyl		76	%	36-119	4	12/16/25	kbc	12/22/25	jlh	N	
2,4,6-Tribromophenol		99	%	30-105	4	12/16/25	kbc	12/22/25	jlh	N	
Terphenyl-d14		78	%	37-109	4	12/16/25	kbc	12/22/25	jlh		
Analysis Method: EPA 625.1 SIM											
Batch: T177269											
Benzidine		<0.22	ug/L	0.22	2	12/16/25	als	12/18/25	jlh		
3,3'-Dichlorobenzidine		<2.2	ug/L	2.2	2	12/16/25	als	12/18/25	jlh		
Surrogates:											
Nitrobenzene-d5	*	**212**	%	38-130	2	12/16/25	als	12/18/25	jlh	S10	
2-Fluorobiphenyl		40	%	39-118	2	12/16/25	als	12/18/25	jlh	N	
Terphenyl-d14		27	%	14-96	2	12/16/25	als	12/18/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 624.1											
Batch: T177285											
Bromomethane		<5.0	ug/L	5.0	1	12/16/25	ats	12/16/25	ats		
Surrogates:											
1,2-Dichloroethane-d4		89	%	68-133	1	12/16/25	ats	12/16/25	ats		
Toluene-d8		111	%	75-120	1	12/16/25	ats	12/16/25	ats		
4-Bromofluorobenzene		99	%	69-119	1	12/16/25	ats	12/16/25	ats		
1,2-Dichlorobenzene-d4		102	%	72-127	1	12/16/25	ats	12/16/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1024-02 Matrix: Wastewater Date Collected: 12/14/25
Sample ID: Domestic #2 Date Received: 12/15/25 13:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

ANALYTICAL RESULTS

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1024-03 | Matrix: Wastewater | Date Collected: 12/14/25
Sample ID: WWTP Raw Influent | Date Received: 12/15/25 13:00

PARAMETERS		RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL											
Analysis Method: EPA 200.8 Rev. 5.4											
Batch: T177324											
Beryllium		<0.0040	mg/L	0.0040	1	12/18/25	ejb	12/23/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 625.1											
Batch: T177256											
Phenol		<5.0	ug/L	5.0	2	12/16/25	kbc	12/22/25	jlh	DL03	
Surrogates:											
2-Fluorophenol		27	%	20-53	2	12/16/25	kbc	12/22/25	jlh		
Phenol-d5		15	%	11-40	2	12/16/25	kbc	12/22/25	jlh		
Nitrobenzene-d5		92	%	36-103	2	12/16/25	kbc	12/22/25	jlh		
2-Fluorobiphenyl		64	%	36-119	2	12/16/25	kbc	12/22/25	jlh	N	
2,4,6-Tribromophenol		76	%	30-105	2	12/16/25	kbc	12/22/25	jlh	N	
Terphenyl-d14		71	%	37-109	2	12/16/25	kbc	12/22/25	jlh		
Analysis Method: EPA 625.1 SIM											
Batch: T177269											
Benzidine		<0.23	ug/L	0.23	2	12/16/25	als	12/18/25	jlh		
3,3'-Dichlorobenzidine		<2.3	ug/L	2.3	2	12/16/25	als	12/18/25	jlh		
Surrogates:											
Nitrobenzene-d5		110	%	38-130	2	12/16/25	als	12/18/25	jlh		
2-Fluorobiphenyl	*	**33**	%	39-118	2	12/16/25	als	12/18/25	jlh	S10, N	
Terphenyl-d14		22	%	14-96	2	12/16/25	als	12/18/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 624.1											
Batch: T177285											
Bromomethane		<5.0	ug/L	5.0	1	12/16/25	ats	12/16/25	ats		
Surrogates:											
1,2-Dichloroethane-d4		89	%	68-133	1	12/16/25	ats	12/16/25	ats		
Toluene-d8		112	%	75-120	1	12/16/25	ats	12/16/25	ats		
4-Bromofluorobenzene		98	%	69-119	1	12/16/25	ats	12/16/25	ats		
1,2-Dichlorobenzene-d4		102	%	72-127	1	12/16/25	ats	12/16/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1024-03 Matrix: Wastewater Date Collected: 12/14/25
Sample ID: WWTP Raw Influent Date Received: 12/15/25 13:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

CERTIFICATE OF ANALYSIS

Page 8 of 19 25L1024 TRACE_QC FINAL 12 23 25 1548

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ANALYTICAL RESULTS

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1024-04 Matrix: Wastewater Date Collected: 12/14/25
Sample ID: WWTP Primary Effluent Date Received: 12/15/25 13:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T177324									
Beryllium	<0.0040 mg/L	0.0040	1	12/18/25	ejb	12/23/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T177256									
Phenol	<10 ug/L	10	4	12/16/25	kbc	12/22/25	jlh	DL03	
Surrogates:									
2-Fluorophenol	26 %	20-53	4	12/16/25	kbc	12/22/25	jlh		
Phenol-d5	16 %	11-40	4	12/16/25	kbc	12/22/25	jlh		
Nitrobenzene-d5	82 %	36-103	4	12/16/25	kbc	12/22/25	jlh		
2-Fluorobiphenyl	53 %	36-119	4	12/16/25	kbc	12/22/25	jlh	N	
2,4,6-Tribromophenol	63 %	30-105	4	12/16/25	kbc	12/22/25	jlh	N	
Terphenyl-d14	45 %	37-109	4	12/16/25	kbc	12/22/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T177269									
Benzidine	<0.26 ug/L	0.26	2	12/16/25	als	12/18/25	jlh		
3,3'-Dichlorobenzidine	<2.6 ug/L	2.6	2	12/16/25	als	12/18/25	jlh		
Surrogates:									
Nitrobenzene-d5	50 %	38-130	2	12/16/25	als	12/18/25	jlh		
2-Fluorobiphenyl	46 %	39-118	2	12/16/25	als	12/18/25	jlh	N	
Terphenyl-d14	26 %	14-96	2	12/16/25	als	12/18/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T177285									
Bromomethane	<5.0 ug/L	5.0	1	12/16/25	ats	12/17/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	88 %	68-133	1	12/16/25	ats	12/17/25	ats		
Toluene-d8	112 %	75-120	1	12/16/25	ats	12/17/25	ats		
4-Bromofluorobenzene	99 %	69-119	1	12/16/25	ats	12/17/25	ats		
1,2-Dichlorobenzene-d4	101 %	72-127	1	12/16/25	ats	12/17/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1024-04 Matrix: Wastewater Date Collected: 12/14/25
Sample ID: WWTP Primary Effluent Date Received: 12/15/25 13:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

CERTIFICATE OF ANALYSIS

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ANALYTICAL RESULTS

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1024-05 | Matrix: Wastewater | Date Collected: 12/14/25
Sample ID: WWTP Final Effluent | Date Received: 12/15/25 13:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T177324									
Beryllium	<0.0040 mg/L	0.0040	1	12/18/25	ejb	12/23/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T177256									
Phenol	<5.0 ug/L	5.0	1	12/16/25	kbc	12/22/25	jlh		
Surrogates:									
2-Fluorophenol	38 %	20-53	1	12/16/25	kbc	12/22/25	jlh		
Phenol-d5	21 %	11-40	1	12/16/25	kbc	12/22/25	jlh		
Nitrobenzene-d5	94 %	36-103	1	12/16/25	kbc	12/22/25	jlh		
2-Fluorobiphenyl	75 %	36-119	1	12/16/25	kbc	12/22/25	jlh	N	
2,4,6-Tribromophenol	84 %	30-105	1	12/16/25	kbc	12/22/25	jlh	N	
Terphenyl-d14	52 %	37-109	1	12/16/25	kbc	12/22/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T177269									
Benzidine	<0.10 ug/L	0.10	1	12/16/25	als	12/18/25	jlh		
3,3'-Dichlorobenzidine	<1.0 ug/L	1.0	1	12/16/25	als	12/18/25	jlh		
Surrogates:									
Nitrobenzene-d5	78 %	38-130	1	12/16/25	als	12/18/25	jlh		
2-Fluorobiphenyl	68 %	39-118	1	12/16/25	als	12/18/25	jlh	N	
Terphenyl-d14	31 %	14-96	1	12/16/25	als	12/18/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T177285									
Bromomethane	<5.0 ug/L	5.0	1	12/16/25	ats	12/16/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	90 %	68-133	1	12/16/25	ats	12/16/25	ats		
Toluene-d8	112 %	75-120	1	12/16/25	ats	12/16/25	ats		
4-Bromofluorobenzene	99 %	69-119	1	12/16/25	ats	12/16/25	ats		
1,2-Dichlorobenzene-d4	102 %	72-127	1	12/16/25	ats	12/16/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1024-05 Matrix: Wastewater Date Collected: 12/14/25
Sample ID: WWTP Final Effluent Date Received: 12/15/25 13:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

QUALITY CONTROL RESULTS

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177324
QC Batch Method: EPA 200.2
Analysis Description: Beryllium, Total
Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T177324-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.0040	0.0040	

LABORATORY CONTROL SAMPLE: T177324-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.200	100	85-115	

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177256
QC Batch Method: EPA 625 Extraction for BNAs
Analysis Description: TTO Semi-Volatiles
Analysis Method: EPA 625.1

METHOD BLANK: T177256-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	%	40	20-53	
Phenol-d5 (S)	%	22	11-40	
Nitrobenzene-d5 (S)	%	87	36-103	
2-Fluorobiphenyl (S)	%	78	36-119	
2,4,6-Tribromophenol (S)	%	92	30-105	
Terphenyl-d14 (S)	%	99	37-109	

LABORATORY CONTROL SAMPLE: T177256-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	25.6	26	17-120	
2-Fluorophenol (S)	%	100	44.9	45	20-53	
Phenol-d5 (S)	%	100	27.0	27	11-40	
Nitrobenzene-d5 (S)	%	100	91.4	91	36-103	
2-Fluorobiphenyl (S)	%	100	72.6	73	36-119	
2,4,6-Tribromophenol (S)	%	100	86.6	87	30-105	
Terphenyl-d14 (S)	%	100	82.9	83	37-109	

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177269	Analysis Description: 625 Benzidines SIM
QC Batch Method: GC/MS No Prep	Analysis Method: EPA 625.1 SIM

METHOD BLANK: T177269-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	64	38-130	
2-Fluorobiphenyl (S)	%	73	39-118	
Terphenyl-d14 (S)	%	44	14-96	

LABORATORY CONTROL SAMPLE: T177269-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	1.19	59	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	<2.0	99	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.805	80	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.706	71	39-118	
Terphenyl-d14 (S)	%	1.00	0.497	50	14-96	

Trace Project ID: 25L1024
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177285	Analysis Description: TTO Volatiles
QC Batch Method: EPA 624 Purge and Trap	Analysis Method: EPA 624.1

METHOD BLANK: T177285-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	90	68-133	
Toluene-d8 (S)	%	111	75-120	
4-Bromofluorobenzene (S)	%	99	69-119	
1,2-Dichlorobenzene-d4 (S)	%	102	72-127	

LABORATORY CONTROL SAMPLE: T177285-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	34.0	68	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	27.4	91	68-133	

LABORATORY CONTROL SAMPLE: T177285-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Toluene-d8 (S)	%	30.0	32.6	109	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.4	98	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	30.3	101	72-127	

MATRIX SPIKE / MATRIX SPIKE DUPLICATE: T177285-MSD1 Original: **25L1024-05**

Parameter	Units	Original Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Notes
Bromomethane	ug/L	0	50.0	41.3	40.8	83	82	1-242	1	61	
1,2-Dichloroethane-d4 (S)	%		30.0	26.2	26.6	87	88	68-133			
Toluene-d8 (S)	%		30.0	34.4	33.6	115	112	75-120			
4-Bromofluorobenzene (S)	%		30.0	30.1	29.6	100	99	69-119			
1,2-Dichlorobenzene-d4 (S)	%		30.0	30.8	30.4	103	101	72-127			

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25L1024-01

Analysis: EPA 625.1

Nitrobenzene-d5	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.
Phenol	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Analysis: EPA 625.1 SIM

	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.
Nitrobenzene-d5	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Trace ID: 25L1024-02

Analysis: EPA 625.1

Nitrobenzene-d5	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Analysis: EPA 625.1 SIM

	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.
Nitrobenzene-d5	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

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Trace ID: 25L1024-03

Analysis: EPA 625.1

Phenol	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Analysis: EPA 625.1 SIM

	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.
2-Fluorobiphenyl	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Trace ID: 25L1024-04

Analysis: EPA 625.1

Phenol	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Analysis: EPA 625.1 SIM

	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

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CHAIN-OF-CUSTODY RECORD

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Page 1 of 1

Trace ID No. 25L1024

Report Results To:

Company Name: City of Cadillac
Report To: Cindy Tomaszewski
Mailing Address: 1121 Plett Rd
City, State, Zip Code: Cadillac MI 49601
Office Phone: 231-775-2368 Cell Phone: 231-878-6230
Email Address: lab@cadillac-mi.net; pmiller@cadillac-mi.net

Bill To:

PO #:
Contact Name: Cindy Tomaszewski
Billing Address (if different): 200 N Lake St
City, State, Zip Code: Cadillac MI 49601
Phone Number:
Billing Email Address: lab@cadillac-mi.net

Trace Use:

Logged By: NC
Checked By: MJ
Sample Collection Time (Hrs):

Requested Turnaround Times (TAT)

- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:
WW = Wastewater; DW = Drinking Water; GW = Groundwater; LW = Liquid Waste
AQ = Aqueous; O = Oil; S = Solid; SL = Sludge
A = Air; WI = Wipes; U = Unknown

Project Name: 2025 IPP/MAHL Additional
Sampled By (print): Derek Goodrich

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Composite	Grab	Metals Field Filtered (Y or N)	Matrix - see above	Number of Containers	Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes
1	12-13	—	Domestic #1	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	
2	12-13	—	Domestic #2	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	* grab VOCs to
3	12-13	—	WWTP Raw Influent	✓	✓	N	WW	~~6~~ 8 ok	x		x		x		x		X	X	X	X	be composited
4	12-13	—	WWTP Primary Effluent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	at the lab
5	12-13	—	WWTP Final Effluent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	

*All composites started on "Sample Collection Date"

Possible Health Hazards?

Please Sign	Released By	Received By	Date	Time	Released By	Received By	Date	Time
	1) [signature]	STORAGE			2) [signature]	[signature]	12-15-25	1308
	3) [signature]				4)			

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

CERTIFICATE OF ANALYSIS

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Sample Log In Checklist

25L1024
Cadillac, City of
Project Manager: Tim Brewer

	Original Observation	Corrected Temperature	IR-9 (CF: +0.2°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.6°C)	IR-14 (CF: -1.3°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Date: 12/15/25										
Time: 16:07										
Initials: CN										
Package Description: Cooler										
Package Temp °C	0.0	0.2	X							
Representative Sample Temp °C	1.9	2.1	X							X

Sample Receipt

Yes No
- [X] Yes [ ] No — Received on ice or other coolant
- [X] Yes [ ] No — Ice still present upon receipt
- [ ] Yes [X] No — Custody seals present — [ ] Yes [ ] No Custody seals intact (if applicable)
- [X] Trace Courier [ ] Client Drop-off [ ] UPS [ ] Fed Ex [ ] US Mail [ ] Other

Sample Condition

Yes No N/A
CN
- [X] Yes [X] No [ ] N/A — All sample containers arrived unbroken and labeled
- [X] Yes [ ] No [ ] N/A — Sufficient sample to run requested analyses
- [X] Yes [ ] No [ ] N/A — Correct chemical preservative added to samples
- [ ] Yes [ ] No [X] N/A — Samples preserved at Trace (if yes, fill in preservation information below)
  Preservation - Reagent:__________ LIMS ID:____________ Date/Time:______________
- [X] Yes [ ] No [ ] N/A — Chemical preservation verified, check EMD pH test strip used (if applicable)
  [X] pH 0-2.5 (Lot: HC461264) [ ] pH 11.0-13.0 (Lot: HC022540) [ ] Other
- [X] Yes [ ] No [ ] N/A — Air bubbles absent from VOAs (no bubble greater than 6mm) ______________

Chain of Custody (COC)

Yes No
- [X] Yes [ ] No — All bottle labels agree with COC
- [X] Yes [ ] No — COC filled out properly
- [X] Yes [ ] No — COC signed by client

Notes:

Form 70-A.61
Effective 12/1/25

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December 23, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25L1026
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

PJLA Testing Accreditation# 1031

TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25L1026-01	Domestic #1	Wastewater	DG	12/13/25	12/15/25 13:00
Sample type: Composite					
25L1026-02	Domestic #2	Wastewater	DG	12/13/25	12/15/25 13:00
Sample type: Composite					
25L1026-03	WWTP Raw Influent	Wastewater	DG	12/13/25	12/15/25 13:00
Sample type: Composite					
25L1026-04	WWTP Primary Effluent	Wastewater	DG	12/13/25	12/15/25 13:00
Sample type: Composite					
25L1026-05	WWTP Final Effluent	Wastewater	DG	12/13/25	12/15/25 13:00
Sample type: Composite					

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ANALYTICAL RESULTS

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1026-01 Matrix: Wastewater Date Collected: 12/13/25
Sample ID: Domestic #1 Date Received: 12/15/25 13:00

PARAMETERS		RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL											
Analysis Method: EPA 200.8 Rev. 5.4											
Batch: T177324											
Beryllium		<0.0040	mg/L	0.0040	1	12/18/25	ejb	12/23/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 625.1											
Batch: T177256											
Phenol		**12**	**ug/L**	**10**	**4**	**12/16/25**	**kbc**	**12/22/25**	**jlh**		
Surrogates:											
2-Fluorophenol		43	%	20-53	4	12/16/25	kbc	12/22/25	jlh		
Phenol-d5		25	%	11-40	4	12/16/25	kbc	12/22/25	jlh		
Nitrobenzene-d5	*	**117**	%	36-103	4	12/16/25	kbc	12/22/25	jlh	S10	
2-Fluorobiphenyl		78	%	36-119	4	12/16/25	kbc	12/22/25	jlh	N	
2,4,6-Tribromophenol		96	%	30-105	4	12/16/25	kbc	12/22/25	jlh	N	
Terphenyl-d14		76	%	37-109	4	12/16/25	kbc	12/22/25	jlh		
Analysis Method: EPA 625.1 SIM											
Batch: T177313											
Benzidine		<0.24	ug/L	0.24	2	12/17/25	als	12/18/25	jlh		
3,3'-Dichlorobenzidine		<2.4	ug/L	2.4	2	12/17/25	als	12/18/25	jlh		
Surrogates:											
Nitrobenzene-d5	*	**170**	%	38-130	2	12/17/25	als	12/18/25	jlh	S10	
2-Fluorobiphenyl		42	%	39-118	2	12/17/25	als	12/18/25	jlh	N	
Terphenyl-d14		29	%	14-96	2	12/17/25	als	12/18/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 624.1											
Batch: T177239											
Bromomethane		<5.0	ug/L	5.0	1	12/15/25	sb	12/16/25	sb		
Surrogates:											
1,2-Dichloroethane-d4	*	**138**	%	68-133	1	12/15/25	sb	12/16/25	sb	S03	
Toluene-d8		84	%	75-120	1	12/15/25	sb	12/16/25	sb		
4-Bromofluorobenzene		102	%	69-119	1	12/15/25	sb	12/16/25	sb		
1,2-Dichlorobenzene-d4		114	%	72-127	1	12/15/25	sb	12/16/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1026-01 Matrix: Wastewater Date Collected: 12/13/25
Sample ID: Domestic #1 Date Received: 12/15/25 13:00

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1026-02 Matrix: Wastewater Date Collected: 12/13/25
Sample ID: Domestic #2 Date Received: 12/15/25 13:00

PARAMETERS		RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL											
Analysis Method: EPA 200.8 Rev. 5.4											
Batch: T177324											
Beryllium		<0.0040	mg/L	0.0040	1	12/18/25	ejb	12/23/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 625.1											
Batch: T177256											
Phenol		<10	ug/L	10	4	12/16/25	kbc	12/22/25	jlh	DL03	
Surrogates:											
2-Fluorophenol		45	%	20-53	4	12/16/25	kbc	12/22/25	jlh		
Phenol-d5		27	%	11-40	4	12/16/25	kbc	12/22/25	jlh		
Nitrobenzene-d5	*	**121**	%	36-103	4	12/16/25	kbc	12/22/25	jlh	S10	
2-Fluorobiphenyl		78	%	36-119	4	12/16/25	kbc	12/22/25	jlh	N	
2,4,6-Tribromophenol		98	%	30-105	4	12/16/25	kbc	12/22/25	jlh	N	
Terphenyl-d14		69	%	37-109	4	12/16/25	kbc	12/22/25	jlh		
Analysis Method: EPA 625.1 SIM											
Batch: T177313											
Benzidine		<0.23	ug/L	0.23	2	12/17/25	als	12/18/25	jlh		
3,3'-Dichlorobenzidine		<2.3	ug/L	2.3	2	12/17/25	als	12/18/25	jlh		
Surrogates:											
Nitrobenzene-d5		66	%	38-130	2	12/17/25	als	12/18/25	jlh		
2-Fluorobiphenyl		54	%	39-118	2	12/17/25	als	12/18/25	jlh	N	
Terphenyl-d14		32	%	14-96	2	12/17/25	als	12/18/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 624.1											
Batch: T177239											
Bromomethane		<5.0	ug/L	5.0	1	12/15/25	sb	12/16/25	sb		
Surrogates:											
1,2-Dichloroethane-d4	*	**137**	%	68-133	1	12/15/25	sb	12/16/25	sb	S03	
Toluene-d8		84	%	75-120	1	12/15/25	sb	12/16/25	sb		
4-Bromofluorobenzene		98	%	69-119	1	12/15/25	sb	12/16/25	sb		
1,2-Dichlorobenzene-d4		111	%	72-127	1	12/15/25	sb	12/16/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1026-02 Matrix: Wastewater Date Collected: 12/13/25
Sample ID: Domestic #2 Date Received: 12/15/25 13:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1026-03 Matrix: Wastewater Date Collected: 12/13/25
Sample ID: WWTP Raw Influent Date Received: 12/15/25 13:00

PARAMETERS		RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL											
Analysis Method: EPA 200.8 Rev. 5.4											
Batch: T177339											
Beryllium		<0.0040	mg/L	0.0040	1	12/18/25	ejb	12/23/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 625.1											
Batch: T177256											
Phenol		<10	ug/L	10	4	12/16/25	kbc	12/22/25	jlh	DL03	
Surrogates:											
2-Fluorophenol		45	%	20-53	4	12/16/25	kbc	12/22/25	jlh		
Phenol-d5		29	%	11-40	4	12/16/25	kbc	12/22/25	jlh		
Nitrobenzene-d5	*	**113**	%	36-103	4	12/16/25	kbc	12/22/25	jlh	S10	
2-Fluorobiphenyl		73	%	36-119	4	12/16/25	kbc	12/22/25	jlh	N	
2,4,6-Tribromophenol		93	%	30-105	4	12/16/25	kbc	12/22/25	jlh	N	
Terphenyl-d14		61	%	37-109	4	12/16/25	kbc	12/22/25	jlh		
Analysis Method: EPA 625.1 SIM											
Batch: T177313											
Benzidine		<0.23	ug/L	0.23	2	12/17/25	als	12/18/25	jlh		
3,3'-Dichlorobenzidine		<2.3	ug/L	2.3	2	12/17/25	als	12/18/25	jlh		
Surrogates:											
Nitrobenzene-d5		84	%	38-130	2	12/17/25	als	12/18/25	jlh		
2-Fluorobiphenyl	*	**38**	%	39-118	2	12/17/25	als	12/18/25	jlh	S10, N	
Terphenyl-d14		20	%	14-96	2	12/17/25	als	12/18/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 624.1											
Batch: T177239											
Bromomethane		<5.0	ug/L	5.0	1	12/15/25	sb	12/16/25	sb		
Surrogates:											
1,2-Dichloroethane-d4	*	**134**	%	68-133	1	12/15/25	sb	12/16/25	sb	S03	
Toluene-d8		86	%	75-120	1	12/15/25	sb	12/16/25	sb		
4-Bromofluorobenzene		100	%	69-119	1	12/15/25	sb	12/16/25	sb		
1,2-Dichlorobenzene-d4		111	%	72-127	1	12/15/25	sb	12/16/25	sb		

CERTIFICATE OF ANALYSIS

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888-979-4469 Fax
www.trace-labs.com

ANALYTICAL RESULTS

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1026-03
Matrix: Wastewater
Date Collected: 12/13/25
Sample ID: WWTP Raw Influent
Date Received: 12/15/25 13:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1026-04 Matrix: Wastewater Date Collected: 12/13/25
Sample ID: WWTP Primary Effluent Date Received: 12/15/25 13:00

PARAMETERS		RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL											
Analysis Method: EPA 200.8 Rev. 5.4											
Batch: T177339											
Beryllium		<0.0040	mg/L	0.0040	1	12/18/25	ejb	12/23/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 625.1											
Batch: T177256											
Phenol		<10	ug/L	10	4	12/16/25	kbc	12/22/25	jlh	DL03	
Surrogates:											
2-Fluorophenol		42	%	20-53	4	12/16/25	kbc	12/22/25	jlh		
Phenol-d5		25	%	11-40	4	12/16/25	kbc	12/22/25	jlh		
Nitrobenzene-d5	*	**118**	%	36-103	4	12/16/25	kbc	12/22/25	jlh	S10	
2-Fluorobiphenyl		77	%	36-119	4	12/16/25	kbc	12/22/25	jlh	N	
2,4,6-Tribromophenol		93	%	30-105	4	12/16/25	kbc	12/22/25	jlh	N	
Terphenyl-d14		63	%	37-109	4	12/16/25	kbc	12/22/25	jlh		
Analysis Method: EPA 625.1 SIM											
Batch: T177313											
Benzidine		<0.23	ug/L	0.23	2	12/17/25	als	12/18/25	jlh		
3,3'-Dichlorobenzidine		<2.3	ug/L	2.3	2	12/17/25	als	12/18/25	jlh		
Surrogates:											
Nitrobenzene-d5		42	%	38-130	2	12/17/25	als	12/18/25	jlh		
2-Fluorobiphenyl	*	**38**	%	39-118	2	12/17/25	als	12/18/25	jlh	S10, N	
Terphenyl-d14		20	%	14-96	2	12/17/25	als	12/18/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 624.1											
Batch: T177239											
Bromomethane		<5.0	ug/L	5.0	1	12/15/25	sb	12/16/25	sb		
Surrogates:											
1,2-Dichloroethane-d4	*	**134**	%	68-133	1	12/15/25	sb	12/16/25	sb	S03	
Toluene-d8		86	%	75-120	1	12/15/25	sb	12/16/25	sb		
4-Bromofluorobenzene		98	%	69-119	1	12/15/25	sb	12/16/25	sb		
1,2-Dichlorobenzene-d4		110	%	72-127	1	12/15/25	sb	12/16/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1026-04 | Matrix: Wastewater | Date Collected: 12/13/25
Sample ID: WWTP Primary Effluent | Date Received: 12/15/25 13:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

CERTIFICATE OF ANALYSIS

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ANALYTICAL RESULTS

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1026-05 Matrix: Wastewater Date Collected: 12/13/25
Sample ID: WWTP Final Effluent Date Received: 12/15/25 13:00

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T177339									
Beryllium	<0.0040 mg/L	0.0040	1	12/18/25	ejb	12/23/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T177256									
Phenol	<5.0 ug/L	5.0	1	12/16/25	kbc	12/23/25	jlh		
Surrogates:									
2-Fluorophenol	43 %	20-53	1	12/16/25	kbc	12/23/25	jlh		
Phenol-d5	24 %	11-40	1	12/16/25	kbc	12/23/25	jlh		
Nitrobenzene-d5	101 %	36-103	1	12/16/25	kbc	12/23/25	jlh		
2-Fluorobiphenyl	81 %	36-119	1	12/16/25	kbc	12/23/25	jlh	N	
2,4,6-Tribromophenol	87 %	30-105	1	12/16/25	kbc	12/23/25	jlh	N	
Terphenyl-d14	56 %	37-109	1	12/16/25	kbc	12/23/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T177313									
Benzidine	<0.10 ug/L	0.10	1	12/17/25	als	12/18/25	jlh		
3,3'-Dichlorobenzidine	<1.0 ug/L	1.0	1	12/17/25	als	12/18/25	jlh		
Surrogates:									
Nitrobenzene-d5	87 %	38-130	1	12/17/25	als	12/18/25	jlh		
2-Fluorobiphenyl	75 %	39-118	1	12/17/25	als	12/18/25	jlh	N	
Terphenyl-d14	32 %	14-96	1	12/17/25	als	12/18/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T177239									
Bromomethane	<5.0 ug/L	5.0	1	12/15/25	sb	12/16/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	* **137** %	68-133	1	12/15/25	sb	12/16/25	sb	S03	
Toluene-d8	85 %	75-120	1	12/15/25	sb	12/16/25	sb		
4-Bromofluorobenzene	100 %	69-119	1	12/15/25	sb	12/16/25	sb		
1,2-Dichlorobenzene-d4	112 %	72-127	1	12/15/25	sb	12/16/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L1026-05 Matrix: Wastewater Date Collected: 12/13/25
Sample ID: WWTP Final Effluent Date Received: 12/15/25 13:00

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

CERTIFICATE OF ANALYSIS

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QUALITY CONTROL RESULTS

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177324	Analysis Description: Beryllium, Total
QC Batch Method: EPA 200.2	Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T177324-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.0040	0.0040	

LABORATORY CONTROL SAMPLE: T177324-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.200	100	85-115	

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177339	Analysis Description: Beryllium, Total
QC Batch Method: EPA 200.2	Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T177339-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.00025	0.00025	

LABORATORY CONTROL SAMPLE: T177339-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.195	97	85-115	

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177256	Analysis Description: TTO Semi-Volatiles
QC Batch Method: EPA 625 Extraction for BNAs	Analysis Method: EPA 625.1

METHOD BLANK: T177256-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	%	40	20-53	
Phenol-d5 (S)	%	22	11-40	

METHOD BLANK: T177256-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Nitrobenzene-d5 (S)	%	87	36-103	
2-Fluorobiphenyl (S)	%	78	36-119	
2,4,6-Tribromophenol (S)	%	92	30-105	
Terphenyl-d14 (S)	%	99	37-109	

LABORATORY CONTROL SAMPLE: T177256-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	25.6	26	17-120	
2-Fluorophenol (S)	%	100	44.9	45	20-53	
Phenol-d5 (S)	%	100	27.0	27	11-40	
Nitrobenzene-d5 (S)	%	100	91.4	91	36-103	
2-Fluorobiphenyl (S)	%	100	72.6	73	36-119	
2,4,6-Tribromophenol (S)	%	100	86.6	87	30-105	
Terphenyl-d14 (S)	%	100	82.9	83	37-109	

MATRIX SPIKE: T177256-MS1 Original: **25L1026-05**

Parameter	Units	Original Result	Spike Conc.	MS Result	MS % Rec	% Rec Unit	Notes
Phenol	ug/L	0	101	22.2	22	5-120	
2-Fluorophenol (S)	%		101	37.8	38	20-53	
Phenol-d5 (S)	%		101	22.3	22	11-40	
Nitrobenzene-d5 (S)	%		101	98.4	98	36-103	
2-Fluorobiphenyl (S)	%		101	81.2	81	36-119	
2,4,6-Tribromophenol (S)	%		101	91.8	91	30-105	
Terphenyl-d14 (S)	%		101	63.4	63	37-109	

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177313
QC Batch Method: GC/MS No Prep
Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T177313-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	78	38-130	
2-Fluorobiphenyl (S)	%	73	39-118	
Terphenyl-d14 (S)	%	41	14-96	

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LABORATORY CONTROL SAMPLE: T177313-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	1.23	62	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	2.19	110	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.838	84	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.760	76	39-118	
Terphenyl-d14 (S)	%	1.00	0.482	48	14-96	

Trace Project ID: 25L1026
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177239
QC Batch Method: EPA 624 Purge and Trap

Analysis Description: TTO Volatiles
Analysis Method: EPA 624.1

METHOD BLANK: T177239-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	133	68-133	
Toluene-d8 (S)	%	84	75-120	
4-Bromofluorobenzene (S)	%	103	69-119	
1,2-Dichlorobenzene-d4 (S)	%	107	72-127	

LABORATORY CONTROL SAMPLE: T177239-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	44.6	89	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	36.6	122	68-133	
Toluene-d8 (S)	%	30.0	26.3	88	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.0	97	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	29.5	98	72-127	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25L1026-01

Analysis: EPA 624.1

1,2-Dichloroethane-d4	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.

Analysis: EPA 625.1

Nitrobenzene-d5	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Analysis: EPA 625.1 SIM

	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.
Nitrobenzene-d5	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Trace ID: 25L1026-02

Analysis: EPA 624.1

1,2-Dichloroethane-d4	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.

Analysis: EPA 625.1

Nitrobenzene-d5	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.
Phenol	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Analysis: EPA 625.1 SIM

	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Trace ID: 25L1026-03

Analysis: EPA 624.1

1,2-Dichloroethane-d4	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.

Analysis: EPA 625.1

Nitrobenzene-d5	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.
Phenol	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Analysis: EPA 625.1 SIM

	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.
2-Fluorobiphenyl	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Trace ID: 25L1026-04

Analysis: EPA 624.1

1,2-Dichloroethane-d4	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.

Analysis: EPA 625.1

Nitrobenzene-d5	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.
Phenol	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Analysis: EPA 625.1 SIM

	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.
2-Fluorobiphenyl	Note S10 : One of the base/neutral surrogate recoveries was outside the control limit. Since the other two base/neutral surrogates were within the control limits, no data requires qualification.

Trace ID: 25L1026-05

Analysis: EPA 624.1

1,2-Dichloroethane-d4	Note S03 : The surrogate recovery was out of control high when compared to control limits. As there are no positive results, no data requires qualification.

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CHAIN-OF-CUSTODY RECORD

Page 1 of 1

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Trace ID No. 25L1026

Report Results To:
Company Name: City of Cadillac
Report To: Cindy Tomaszewski
Mailing Address: 1121 Plett Rd
City, State, Zip Code: Cadillac MI 49601
Office Phone: 231-775-2368 Cell Phone: 231-878-6230
Email Address: lab@cadillac-mi.net; pmiller@cadillac-mi.net

Bill To:
PO #:
Contact Name: Cindy Tomaszewski
Billing Address (if different): 200 N Lake St
City, State, Zip Code: Cadillac MI 49601
Phone Number:
Billing Email Address: lab@cadillac-mi.net

Trace Use:
Logged By: NC
Checked By: MJ
Sample Collection Time (Hrs):

Requested Turnaround Times (TAT)
- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:
WW = Wastewater, DW = Drinking Water, GW = Groundwater, LW = Liquid Waste, AQ = Aqueous, O = Oil, S = Solid, SL = Sludge, A = Air, WI = Wipes, U = Unknown

Project Name: 2025 IPP/MAHL Additional
Sampled By (print): Derek Goodrich

Analysis Requested

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Composite	Grab	Metals Field Filterd (Y or N)	Matrix - see above	Number of Containers	Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes
1	12/12	—	Domestic #1	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	
2	12/12	—	Domestic #2	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	* grab VOCs to
3	12/12	—	WWTP Raw Influent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	be composited
4	12/12	—	WWTP Primary Effluent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	at the lab
5	12/12	—	WWTP Final Effluent	✓	✓	N	WW	8	x		x		x		x		X	X	X	X	

* All composites started on "Sample Collection Date"

Possible Health Hazards?

Please Sign

	Released By	Received By	Date	Time	Released By	Received By	Date	Time
	1) [signature]	STORAGE			2) C. Tomaszewski	[signature]	12-15-25	1300
	3) [signature]	[signature]	12/15/25	10:07	4)			

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

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25L1026
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

	Original Observation	Corrected Temperature	IR-9 (CF: +0.2°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.6°C)	IR-14 (CF: -1.3°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Date: 12/15/25										
Time: 16:07										
Initials: CN										
Package Description: Cooler										
Package Temp °C	0.0	0.2	X							
Representative Sample Temp °C	1.9	2.1	X							X

Sample Receipt

Yes	No	
[X]	[ ]	Received on ice or other coolant
[X]	[ ]	Ice still present upon receipt
[ ]	[X]	Custody seals present

[ ] Yes [ ] No Custody seals intact (if applicable)

[X] Trace Courier [ ] Client Drop-off [ ] UPS [ ] Fed Ex [ ] US Mail [ ] Other

Sample Condition

Yes	No	N/A	
[ ]	[ ]	[ ]	All sample containers arrived unbroken and labeled
[ ]	[ ]	[ ]	Sufficient sample to run requested analyses
[ ]	[ ]	[ ]	Correct chemical preservative added to samples
[ ]	[ ]	[ ]	Samples preserved at Trace (if yes, fill in preservation information below)

Preservation - Reagent:____________ LIMS ID:______________ Date/Time:____________________

Yes	No	N/A	
[ ]	[ ]	[ ]	Chemical preservation verified, check EMD pH test strip used (if applicable)

[ ] pH 0-2.5 (Lot: HC461264) [ ] pH 11.0-13.0 (Lot: HC022540) [ ] Other

Yes	No	N/A	
[ ]	[ ]	[ ]	Air bubbles absent from VOAs (no bubble greater than 6mm) ______________________

Chain of Custody (COC)

Yes	No	
[ ]	[ ]	All bottle labels agree with COC
[ ]	[ ]	COC filled out properly
[ ]	[ ]	COC signed by client

Notes:

Form 70-A.61
Effective 12/1/25

TRACE Analytical Laboratories, Inc.

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673

TRACE ANALYTICAL LABORATORIES, INC.

231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

December 23, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25L0905
Client Project 2025 IPP/MAHL Additional 12/11/25

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

PJLA
Testing
Accreditation# 1031

TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 25L0905
Client Project ID: 2025 IPP/MAHL Additional 12/11/25

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25L0905-01 Sample type: Grab	Fiamm	Wastewater	DG	12/11/25	12/12/25 09:45
25L0905-02 Sample type: Grab	MI Rubber	Wastewater	DG	12/11/25	12/12/25 09:45

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ANALYTICAL RESULTS

Trace Project ID: 25L0905
Client Project ID: 2025 IPP/MAHL Additional 12/11/25

Trace ID: 25L0905-01 Matrix: Wastewater Date Collected: 12/11/25
Sample ID: Fiamm Date Received: 12/12/25 09:45

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL										
Analysis Method: EPA 200.8 Rev. 5.4										
Batch: T177324										
Beryllium	<0.0040	mg/L	0.0040	1	12/18/25	ejb	12/23/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 625.1										
Batch: T177256										
Phenol	<5.0	ug/L	5.0	1	12/16/25	kbc	12/19/25	jlh		
Surrogates:										
2-Fluorophenol	41	%	20-53	1	12/16/25	kbc	12/19/25	jlh		
Phenol-d5	24	%	11-40	1	12/16/25	kbc	12/19/25	jlh		
Nitrobenzene-d5	86	%	36-103	1	12/16/25	kbc	12/19/25	jlh		
2-Fluorobiphenyl	67	%	36-119	1	12/16/25	kbc	12/19/25	jlh	N	
2,4,6-Tribromophenol	79	%	30-105	1	12/16/25	kbc	12/19/25	jlh	N	
Terphenyl-d14	67	%	37-109	1	12/16/25	kbc	12/19/25	jlh		
Analysis Method: EPA 625.1 SIM										
Batch: T177269										
Benzidine	<0.24	ug/L	0.24	2	12/16/25	als	12/18/25	jlh		
3,3'-Dichlorobenzidine	<2.4	ug/L	2.4	2	12/16/25	als	12/18/25	jlh		
Surrogates:										
Nitrobenzene-d5	71	%	38-130	2	12/16/25	als	12/18/25	jlh		
2-Fluorobiphenyl	63	%	39-118	2	12/16/25	als	12/18/25	jlh	N	
Terphenyl-d14	29	%	14-96	2	12/16/25	als	12/18/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS										
Analysis Method: EPA 624.1										
Batch: T177168										
Bromomethane	<5.0	ug/L	5.0	1	12/12/25	ats	12/14/25	ats		
Surrogates:										
1,2-Dichloroethane-d4	94	%	68-133	1	12/12/25	ats	12/14/25	ats		
Toluene-d8	108	%	75-120	1	12/12/25	ats	12/14/25	ats		
4-Bromofluorobenzene	99	%	69-119	1	12/12/25	ats	12/14/25	ats		
1,2-Dichlorobenzene-d4	103	%	72-127	1	12/12/25	ats	12/14/25	ats		

CERTIFICATE OF ANALYSIS

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ANALYTICAL RESULTS

Trace Project ID: 25L0905
Client Project ID: 2025 IPP/MAHL Additional 12/11/25

Trace ID: 25L0905-01 Matrix: Wastewater Date Collected: 12/11/25
Sample ID: Fiamm Date Received: 12/12/25 09:45

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25L0905
Client Project ID: 2025 IPP/MAHL Additional 12/11/25

Trace ID: 25L0905-02 | Matrix: Wastewater | Date Collected: 12/11/25
Sample ID: MI Rubber | Date Received: 12/12/25 09:45

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T177324									
Beryllium	<0.0040 mg/L	0.0040	1	12/18/25	ejb	12/23/25	drm		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T177256									
Phenol	<5.0 ug/L	5.0	1	12/16/25	kbc	12/19/25	jlh		
Surrogates:									
2-Fluorophenol	30 %	20-53	1	12/16/25	kbc	12/19/25	jlh		
Phenol-d5	17 %	11-40	1	12/16/25	kbc	12/19/25	jlh		
Nitrobenzene-d5	84 %	36-103	1	12/16/25	kbc	12/19/25	jlh		
2-Fluorobiphenyl	64 %	36-119	1	12/16/25	kbc	12/19/25	jlh	N	
2,4,6-Tribromophenol	77 %	30-105	1	12/16/25	kbc	12/19/25	jlh	N	
Terphenyl-d14	72 %	37-109	1	12/16/25	kbc	12/19/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T177269									
Benzidine	<0.24 ug/L	0.24	2	12/16/25	als	12/18/25	jlh		
3,3'-Dichlorobenzidine	<2.4 ug/L	2.4	2	12/16/25	als	12/18/25	jlh		
Surrogates:									
Nitrobenzene-d5	44 %	38-130	2	12/16/25	als	12/18/25	jlh		
2-Fluorobiphenyl	49 %	39-118	2	12/16/25	als	12/18/25	jlh	N	
Terphenyl-d14	30 %	14-96	2	12/16/25	als	12/18/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T177168									
Bromomethane	<5.0 ug/L	5.0	1	12/12/25	ats	12/14/25	ats		
Surrogates:									
1,2-Dichloroethane-d4	93 %	68-133	1	12/12/25	ats	12/14/25	ats		
Toluene-d8	108 %	75-120	1	12/12/25	ats	12/14/25	ats		
4-Bromofluorobenzene	99 %	69-119	1	12/12/25	ats	12/14/25	ats		
1,2-Dichlorobenzene-d4	102 %	72-127	1	12/12/25	ats	12/14/25	ats		

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ANALYTICAL RESULTS

Trace Project ID: 25L0905
Client Project ID: 2025 IPP/MAHL Additional 12/11/25

Trace ID: 25L0905-02 | Matrix: Wastewater | Date Collected: 12/11/25
Sample ID: MI Rubber | Date Received: 12/12/25 09:45

PARAMETERS	RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
VOLATILE ORGANIC COMPOUNDS BY GC-MS										

CERTIFICATE OF ANALYSIS

QUALITY CONTROL RESULTS

Trace Project ID: 25L0905
Client Project ID: 2025 IPP/MAHL Additional 12/11/25

QC Batch: T177324	Analysis Description: Beryllium, Total
QC Batch Method: EPA 200.2	Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T177324-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.0040	0.0040	

LABORATORY CONTROL SAMPLE: T177324-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.200	100	85-115	

Trace Project ID: 25L0905
Client Project ID: 2025 IPP/MAHL Additional 12/11/25

QC Batch: T177256	Analysis Description: TTO Semi-Volatiles
QC Batch Method: EPA 625 Extraction for BNAs	Analysis Method: EPA 625.1

METHOD BLANK: T177256-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	%	40	20-53	
Phenol-d5 (S)	%	22	11-40	
Nitrobenzene-d5 (S)	%	87	36-103	
2-Fluorobiphenyl (S)	%	78	36-119	
2,4,6-Tribromophenol (S)	%	92	30-105	
Terphenyl-d14 (S)	%	99	37-109	

LABORATORY CONTROL SAMPLE: T177256-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	25.6	26	17-120	
2-Fluorophenol (S)	%	100	44.9	45	20-53	
Phenol-d5 (S)	%	100	27.0	27	11-40	
Nitrobenzene-d5 (S)	%	100	91.4	91	36-103	
2-Fluorobiphenyl (S)	%	100	72.6	73	36-119	
2,4,6-Tribromophenol (S)	%	100	86.6	87	30-105	
Terphenyl-d14 (S)	%	100	82.9	83	37-109	

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Trace Project ID: 25L0905
Client Project ID: 2025 IPP/MAHL Additional 12/11/25

QC Batch: T177269
QC Batch Method: GC/MS No Prep
Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T177269-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	64	38-130	
2-Fluorobiphenyl (S)	%	73	39-118	
Terphenyl-d14 (S)	%	44	14-96	

LABORATORY CONTROL SAMPLE: T177269-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	1.19	59	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	<2.0	99	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.805	80	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.706	71	39-118	
Terphenyl-d14 (S)	%	1.00	0.497	50	14-96	

Trace Project ID: 25L0905
Client Project ID: 2025 IPP/MAHL Additional 12/11/25

QC Batch: T177168
QC Batch Method: EPA 624 Purge and Trap
Analysis Description: TTO Volatiles
Analysis Method: EPA 624.1

METHOD BLANK: T177168-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	94	68-133	
Toluene-d8 (S)	%	106	75-120	
4-Bromofluorobenzene (S)	%	99	69-119	
1,2-Dichlorobenzene-d4 (S)	%	102	72-127	

LABORATORY CONTROL SAMPLE: T177168-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	41.6	83	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	28.3	94	68-133	

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LABORATORY CONTROL SAMPLE: T177168-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Toluene-d8 (S)	%	30.0	31.8	106	75-120	
4-Bromofluorobenzene (S)	%	30.0	28.8	96	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	30.6	102	72-127	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25L0905-01

Analysis: EPA 625.1 SIM

Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

Trace ID: 25L0905-02

Analysis: EPA 625.1 SIM

Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.

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CHAIN-OF-CUSTODY RECORD

TRACE ANALYTICAL LABORATORIES, INC.

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Phone 231.773.5998
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Page 1 of 1

Trace ID No. 25L0905

Report Results To:

Company Name:	City of Cadillac
Report To:	Cindy Tomaszewski
Mailing Address:	1121 Plett Rd
City, State, Zip Code:	Cadillac MI 49601
Office Phone:	231-775-2368
Cell Phone:	231-878-6230
Email Address:	lab@cadillac-mi.net; pmiller@cadillac-mi.net

Bill To:

PO #:	
Contact Name:	Cindy Tomaszewski
Billing Address (if different):	200 N Lake St
City, State, Zip Code:	Cadillac MI 49601
Phone Number:	
Billing Email Address:	lab@cadillac-mi.net

Trace Use:

Logged By:	MJ
Checked By:	CN
Sample Collection Time (Hrs):	

Requested Turnaround Times (TAT)

- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

** Rush TAT Requires Prior Approval and Incurs Rush Fees*

Matrix Key:

WW = Wastewater, DW = Drinking Water, GW = Groundwater, LW = Liquid Waste, AQ = Aqueous, O = Oil, S = Solid, SL = Sludge, A = Air, WI = Wipes, U = Unknown

Project Name: 2025 IPP/MAHL Additional

Sampled By (print): Derek Goodness

Analysis Requested

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Composite	Grab	Metals Field Filterd (Y or N)	Matrix - see above →	Number of Containers	Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes	Possible Health Hazards?
1	12-11		Fiamm		✓	N	WW	6	x		x		x		x		X	X	X	X		
2	12-11		MI Rubber		✓	N	WW	6	x		x		x		x		X	X	X	X		

Please Sign

	Released By	Received By	Date	Time		Released By	Received By	Date	Time
1)	[signature]	[signature]	12/12/25	945	2)				
3)					4)				

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

CERTIFICATE OF ANALYSIS

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673

TRACE
ANALYTICAL LABORATORIES, INC.

231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

25L0905
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

	Original Observation	Corrected Temperature	IR-9 (CF: +0.2°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.6°C)	IR-14 (CF: -1.3°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Date: 12/12/25										
Time: 945										
Initials: MJ										
Package Description: Cooler										
Package Temp °C	0.0	0.0		✓						
Representative Sample Temp °C	3.7	3.7		✓						✓

Sample Receipt

Yes No
- [x] Yes [ ] No — Received on ice or other coolant
- [x] Yes [ ] No — Ice still present upon receipt
- [ ] Yes [x] No — Custody seals present

[ ] Yes [ ] No Custody seals intact (if applicable)

[ ] Trace Courier [ ] Client Drop-off [x] UPS [ ] Fed Ex [ ] US Mail [ ] Other

Sample Condition

Yes	No	N/A	
[x] (MJ)	[x]	[ ]	All sample containers arrived unbroken and labeled
[x]	[ ]	[ ]	Sufficient sample to run requested analyses
[x]	[ ]	[ ]	Correct chemical preservative added to samples
[ ]	[x]	[ ]	Samples preserved at Trace (if yes, fill in preservation information below)

Preservation - Reagent:__________ LIMS ID:__________ Date/Time:__________

Yes	No	N/A	
[x]	[ ]	[ ]	Chemical preservation verified, check EMD pH test strip used (if applicable)
			[x] pH 0-2.5 (Lot: HC461264) [ ] pH 11.0-13.0 (Lot: HC022540) [ ] Other
[x]	[ ]	[ ]	Air bubbles absent from VOAs (no bubble greater than 6mm) __________

Chain of Custody (COC)

- [x] Yes [ ] No — All bottle labels agree with COC
- [x] Yes [ ] No — COC filled out properly
- [x] Yes [ ] No — COC signed by client

Notes:

One Voa from sample point #2 arrived broken

Form 70-A.61
Effective 12/1/25

TRACE Analytical Laboratories, Inc.

CERTIFICATE OF ANALYSIS

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December 29, 2025

Cindy Tomaszewski
Cadillac, City of
200 North Lake Street
Cadillac, MI 49601

RE: Trace Project 25L0618
Client Project 2025 IPP/MAHL Additional

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

Tim Brewer
Project Manager
Enclosures

TNI EL V1:2016

Trace Analytical Laboratories, Inc.
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Muskegon, MI 49444-2673

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888-979-4469 Fax
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SAMPLE SUMMARY

Trace Project ID: 25L0618
Client Project ID: 2025 IPP/MAHL Additional

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25L0618-01 Sample type: Grab	Piranha	Wastewater	DG	12/09/25 13:25	12/10/25 11:42
25L0618-02 Sample type: Grab	ARVCO	Wastewater	DG	12/09/25 13:15	12/10/25 11:42

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ANALYTICAL RESULTS

Trace Project ID: 25L0618
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0618-01 | Matrix: Wastewater | Date Collected: 12/09/25 13:25
Sample ID: Piranha | Date Received: 12/10/25 11:42

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL									
Analysis Method: EPA 200.8 Rev. 5.4									
Batch: T177155									
Beryllium	<0.0040 mg/L	0.0040	1	12/15/25	ejb	12/15/25	jma		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 625.1									
Batch: T177082									
Phenol	<5.0 ug/L	5.0	1	12/11/25	kbc	12/12/25	jlh		
Surrogates:									
2-Fluorophenol	21 %	20-53	1	12/11/25	kbc	12/12/25	jlh		
Phenol-d5	12 %	11-40	1	12/11/25	kbc	12/12/25	jlh		
Nitrobenzene-d5	45 %	36-103	1	12/11/25	kbc	12/12/25	jlh		
2-Fluorobiphenyl	58 %	36-119	1	12/11/25	kbc	12/12/25	jlh	N	
2,4,6-Tribromophenol	90 %	30-105	1	12/11/25	kbc	12/12/25	jlh	N	
Terphenyl-d14	66 %	37-109	1	12/11/25	kbc	12/12/25	jlh		
Analysis Method: EPA 625.1 SIM									
Batch: T177123									
Benzidine	<0.21 ug/L	0.21	2	12/12/25	als	12/12/25	jlh		
3,3'-Dichlorobenzidine	<2.1 ug/L	2.1	2	12/12/25	als	12/12/25	jlh		
Surrogates:									
Nitrobenzene-d5	70 %	38-130	2	12/12/25	als	12/12/25	jlh		
2-Fluorobiphenyl	58 %	39-118	2	12/12/25	als	12/12/25	jlh	N	
Terphenyl-d14	33 %	14-96	2	12/12/25	als	12/12/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
Batch: T177107									
Bromomethane	<5.0 ug/L	5.0	1	12/11/25	sb	12/11/25	sb		
Surrogates:									
1,2-Dichloroethane-d4	108 %	68-133	1	12/11/25	sb	12/11/25	sb		
Toluene-d8	93 %	75-120	1	12/11/25	sb	12/11/25	sb		
4-Bromofluorobenzene	104 %	69-119	1	12/11/25	sb	12/11/25	sb		
1,2-Dichlorobenzene-d4	104 %	72-127	1	12/11/25	sb	12/11/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25L0618
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0618-01 | Matrix: Wastewater | Date Collected: 12/09/25 13:25
Sample ID: Piranha | Date Received: 12/10/25 11:42

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

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ANALYTICAL RESULTS

Trace Project ID: 25L0618
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0618-02 | Matrix: Wastewater | Date Collected: 12/09/25 13:15
Sample ID: ARVCO | | Date Received: 12/10/25 11:42

PARAMETERS		RESULTS	UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
METALS, TOTAL											
Analysis Method: EPA 200.8 Rev. 5.4											
Batch: T177155											
Beryllium		<0.0040	mg/L	0.0040	1	12/15/25	ejb	12/15/25	jma		
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS										DL03	
Analysis Method: EPA 625.1											
Batch: T177082											
Phenol		<11	ug/L	11	4	12/11/25	kbc	12/12/25	jlh		
Surrogates:											
2-Fluorophenol	*	**17**	%	20-53	4	12/11/25	kbc	12/12/25	jlh	S02	
Phenol-d5	*	**9**	%	11-40	4	12/11/25	kbc	12/12/25	jlh	S02	
Nitrobenzene-d5		38	%	36-103	4	12/11/25	kbc	12/12/25	jlh		
2-Fluorobiphenyl		52	%	36-119	4	12/11/25	kbc	12/12/25	jlh	N	
2,4,6-Tribromophenol		72	%	30-105	4	12/11/25	kbc	12/12/25	jlh	N	
Terphenyl-d14		58	%	37-109	4	12/11/25	kbc	12/12/25	jlh		
Analysis Method: EPA 625.1 SIM											
Batch: T177123											
Benzidine		<0.39	ug/L	0.39	2	12/12/25	als	12/12/25	jlh		
3,3'-Dichlorobenzidine		<3.9	ug/L	3.9	2	12/12/25	als	12/12/25	jlh		
Surrogates:											
Nitrobenzene-d5		100	%	38-130	2	12/12/25	als	12/12/25	jlh		
2-Fluorobiphenyl	*	**25**	%	39-118	2	12/12/25	als	12/12/25	jlh	S02, N	
Terphenyl-d14		19	%	14-96	2	12/12/25	als	12/12/25	jlh		
VOLATILE ORGANIC COMPOUNDS BY GC-MS											
Analysis Method: EPA 624.1											
Batch: T177107											
Bromomethane		<5.0	ug/L	5.0	1	12/11/25	sb	12/11/25	sb		
Surrogates:											
1,2-Dichloroethane-d4		108	%	68-133	1	12/11/25	sb	12/11/25	sb		
Toluene-d8		93	%	75-120	1	12/11/25	sb	12/11/25	sb		
4-Bromofluorobenzene		103	%	69-119	1	12/11/25	sb	12/11/25	sb		
1,2-Dichlorobenzene-d4		105	%	72-127	1	12/11/25	sb	12/11/25	sb		

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ANALYTICAL RESULTS

Trace Project ID: 25L0618
Client Project ID: 2025 IPP/MAHL Additional

Trace ID: 25L0618-02 Matrix: Wastewater Date Collected: 12/09/25 13:15
Sample ID: ARVCO Date Received: 12/10/25 11:42

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL

VOLATILE ORGANIC COMPOUNDS BY GC-MS

CERTIFICATE OF ANALYSIS

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QUALITY CONTROL RESULTS

Trace Project ID: 25L0618
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177155	Analysis Description: Beryllium, Total
QC Batch Method: EPA 200.2	Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T177155-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Beryllium	mg/L	<0.0040	0.0040	

LABORATORY CONTROL SAMPLE: T177155-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Beryllium	mg/L	0.200	0.197	98	85-115	

Trace Project ID: 25L0618
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177082	Analysis Description: TTO Semi-Volatiles
QC Batch Method: EPA 625 Extraction for BNAs	Analysis Method: EPA 625.1

METHOD BLANK: T177082-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenol	ug/L	<2.5	2.5	
2-Fluorophenol (S)	%	25	20-53	
Phenol-d5 (S)	%	14	11-40	
Nitrobenzene-d5 (S)	%	49	36-103	
2-Fluorobiphenyl (S)	%	66	36-119	
2,4,6-Tribromophenol (S)	%	95	30-105	
Terphenyl-d14 (S)	%	83	37-109	

LABORATORY CONTROL SAMPLE: T177082-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenol	ug/L	100	18.6	19	17-120	
2-Fluorophenol (S)	%	100	27.1	27	20-53	
Phenol-d5 (S)	%	100	17.8	18	11-40	
Nitrobenzene-d5 (S)	%	100	47.9	48	36-103	
2-Fluorobiphenyl (S)	%	100	63.4	63	36-119	
2,4,6-Tribromophenol (S)	%	100	86.8	87	30-105	
Terphenyl-d14 (S)	%	100	77.6	78	37-109	

MATRIX SPIKE: T177082-MS1 Original: **25L0618-01**

Parameter	Units	Original Result	Spike Conc.	MS Result	MS % Rec	% Rec Unit	Notes
Phenol	ug/L	0	105	16.1	15	5-120	
2-Fluorophenol (S)	%		105	25.0	24	20-53	
Phenol-d5 (S)	%		105	15.4	15	11-40	
Nitrobenzene-d5 (S)	%		105	57.9	55	36-103	
2-Fluorobiphenyl (S)	%		105	80.1	76	36-119	
2,4,6-Tribromophenol (S)	**%**		**105**	117	**112**	**30-105**	**S09**
Terphenyl-d14 (S)	%		105	94.3	90	37-109	

Trace Project ID: 25L0618
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177123
QC Batch Method: GC/MS No Prep
Analysis Description: 625 Benzidines SIM
Analysis Method: EPA 625.1 SIM

METHOD BLANK: T177123-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	66	38-130	
2-Fluorobiphenyl (S)	%	57	39-118	
Terphenyl-d14 (S)	%	34	14-96	

LABORATORY CONTROL SAMPLE: T177123-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	1.22	61	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	<2.0	96	53-133	
Nitrobenzene-d5 (S)	%	1.00	0.783	78	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.707	71	39-118	
Terphenyl-d14 (S)	%	1.00	0.423	42	14-96	

Trace Project ID: 25L0618
Client Project ID: 2025 IPP/MAHL Additional

QC Batch: T177107
QC Batch Method: EPA 624 Purge and Trap
Analysis Description: TTO Volatiles
Analysis Method: EPA 624.1

METHOD BLANK: T177107-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes

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METHOD BLANK: T177107-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<5.0	5.0	
1,2-Dichloroethane-d4 (S)	%	105	68-133	
Toluene-d8 (S)	%	96	75-120	
4-Bromofluorobenzene (S)	%	104	69-119	
1,2-Dichlorobenzene-d4 (S)	%	103	72-127	

LABORATORY CONTROL SAMPLE: T177107-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.0	44.2	88	15-185	
1,2-Dichloroethane-d4 (S)	%	30.0	29.9	100	68-133	
Toluene-d8 (S)	%	30.0	29.5	98	75-120	
4-Bromofluorobenzene (S)	%	30.0	29.9	100	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	29.7	99	72-127	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25L0618-02

Analysis: EPA 625.1

	Note DL03 : The reporting limit was raised due to a dilution required because of matrix interference and/or analyte interference.
2-Fluorophenol	Note S02 : The surrogate recovery was out of control low when compared to control limits. All results and reporting limits must be considered estimated.
Phenol-d5	Note S02 : The surrogate recovery was out of control low when compared to control limits. All results and reporting limits must be considered estimated.

Analysis: EPA 625.1 SIM

2-Fluorobiphenyl	Note S02 : The surrogate recovery was out of control low when compared to control limits. All results and reporting limits must be considered estimated.

Trace ID: T177082-MS1

Analysis: EPA 625.1

2,4,6-Tribromophenol	Note S09 : One of the acid surrogate recoveries was outside the control limit. Since the other two acid surrogates were within the control limits, no data requires qualification.

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CHAIN-OF-CUSTODY RECORD

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Page 1 of 1

Trace ID No. 25L0618

Report Results To:		Bill To:	
Company Name:	City of Cadillac	PO #:	
Report To:	Cindy Tomaszewski	Contact Name:	Cindy Tomaszewski
Mailing Address:	1121 Plett Rd	Billing Address (if different):	200 N Lake St
City, State, Zip Code:	Cadillac MI 49601	City, State, Zip Code:	Cadillac MI 49601
Office Phone:	231-775-2368 Cell Phone: 231-878-6230	Phone Number:	
Email Address:	lab@cadillac-mi.net; pmiller@cadillac-mi.net	Billing Email Address:	lab@cadillac-mi.net

Trace Use:
Logged By: MCR
Checked By: NC
Sample Collection Time (Hrs):

Requested Turnaround Times (TAT)
- ■ Standard: 7-10 Business days
- ☐ 5-6 Business Days*
- ☐ 3-4 Business Day*
- ☐ 1-2 Business days*

Rush TAT Requires Prior Approval and Incurs Rush Fees

Matrix Key:
WW = Wastewater; DW = Drinking Water; GW = Groundwater; LW = Liquid Waste; AQ = Aqueous; O = Oil; S = Solid; SL = Sludge; A = Air; WI = Wipes; U = Unknown

Project Name: 2025 IPP/MAHL Additional
Sampled By (print): Derek Goodrich

Trace No.	Sample Collection Date	Sample Collection Time	Sample ID/Name	Composite	Grab	Metals Field Filterd (Y or N)	Matrix - see above →	Number of Containers	Cool ≤ 4°C	Hydrochloric Acid (HCl)	Nitric Acid (HNO3)	Sulfuric Acid (H2SO4)	Sodium Thiosulfate	Sodium Hydroxide (NaOH)	Ascorbic Acid	Other	phenol - EPA 625.1	beryllium - EPA 200.8	benzidine - EPA 625.1 SIMS	bromomethane - EPA 624.1	Remarks/Notes	Possible Health Hazards?
1	12-9	1125	Pincusta		✓	N	WW	6	x		x		x		x		X	X	X	X		
2	12-?	1115	ARUCO		✓	N	WW	6	x		x		x		x		X	X	X	X		

Please Sign

	Released By	Received By	Date	Time		Released By	Received By	Date	Time
1)	[signature]	UPS			2)	UPS	McKenna Rhoades	12/10/25	1142
3)					4)				

In executing this Chain of Custody, the client acknowledges the terms as set forth at www.trace-labs.com/terms-of-agreement.

☐ Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-Z.3

CERTIFICATE OF ANALYSIS

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25L0618
Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

	Original Observation	Corrected Temperature	IR-9 (CF: +0.2°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.6°C)	IR-14 (CF: -1.3°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Date: 12/10/25										
Time: 1141										
Initials: MCE										
Package Description: cooler										
Package Temp °C	2.7	1.4				✓				
Representative Sample Temp °C	5.9	4.6				✓				✓

Sample Receipt

Yes	No	
☒	☐	Received on ice or other coolant
☒	☐	Ice still present upon receipt
☐	☒	Custody seals present

☐ Yes ☐ No Custody seals intact (if applicable)

☐ Trace Courier ☐ Client Drop-off ☒ UPS ☐ Fed Ex ☐ US Mail ☐ Other

Sample Condition

Yes	No	N/A	
☒	☐	☐	All sample containers arrived unbroken and labeled
☒	☐	☐	Sufficient sample to run requested analyses
☒	☐	☐	Correct chemical preservative added to samples
☐	☒	☐	Samples preserved at Trace (if yes, fill in preservation information below)

Preservation - Reagent:__________ LIMS ID:__________ Date/Time:__________

Yes	No	N/A	
☒	☐	☐	Chemical preservation verified, check EMD pH test strip used (if applicable)

☒ pH 0-2.5 (Lot: HC461264) ☐ pH 11.0-13.0 (Lot: HC022540) ☐ Other

Yes	No	N/A	
☐	☐	☐	Air bubbles absent from VOAs (no bubble greater than 6mm) __________

Chain of Custody (COC)

Yes	No	
☒	☐	All bottle labels agree with COC
☒	☐	COC filled out properly
☒	☐	COC signed by client

Notes:

CERTIFICATE OF ANALYSIS